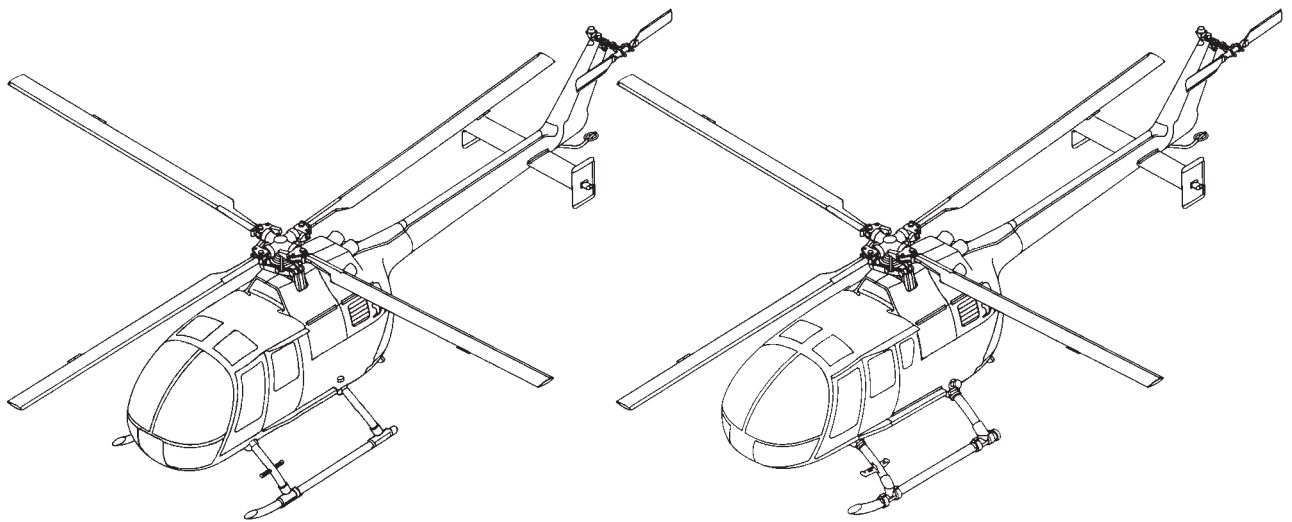


BO 105 CB-5/CBS-5

APPROVED ROTORCRAFT

FLIGHT MANUAL



Valid for Model BO 105 C Variants: **CB-5**
and Model BO 105 S Variants: **CBS-5**

Registration No. _____

Serial No. _____

LBA Type Certificate No. 3025

THIS MANUAL MUST BE CARRIED IN THE HELICOPTER AT ALL TIMES.

IT IS THE OPERATOR'S RESPONSIBILITY TO MAINTAIN THIS MANUAL IN A CURRENT STATUS IN ACCORDANCE WITH THE LIST OF EFFECTIVE PAGES.

Published by:

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LUFTFAHRT-BUNDESAMT (LBA)

This Flight Manual is required by LBA Type Certificate Data Sheet No. 3025.

NOTE This FLM CB-5/CBS-5 issue supercedes the previous FLM CBS-5 issue and all revisions thereof.

FEDERAL AVIATION ADMINISTRATION (FAA)

This Flight Manual, required by FAA Type Certificate Data Sheet No. H3EU, is FAA approved as of March 28, 1996.

INTERSTATE AVIATION COMMITTEE - AVIATION REGISTER (IAC-AR)

This Flight Manual, required by IAC-AR Type Certificate Data Sheet No. 82-105, is IAC-AR approved as of December 09, 1995.

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REVISION	4	(see entry below)

REVISION 4

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RECORD OF REVISIONS

[illegible]

RECORD OF REVISIONS

RECORD OF REVISIONS (Continued)

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SECTION 1

GENERAL

1.1 GENERAL

This manual contains the information necessary for safe and efficient operation of the BO-105 helicopter.

The user is assumed to have flying experience, therefore elementary instructions and basic principles have been omitted.

1.2 ORGANIZATION OF THE MANUAL

This flight manual is largely organized in the standardized format recommended by the Helicopter Association International.

It is divided into an **approved part** consisting of sections 1 thru 5, 10 and 11, and into a **non-approved part** consisting of sections 6 thru 9 and 12.

1.2.1 Approved Data

This manual meets all certificating authority requirements for approved data.

1.2.2 Description of the Sections

The sections of the manual are fairly independent with each section beginning with its own table of contents.

Approved Part

SECTION 1 GENERAL

presents information of general interest to the pilot, basic helicopter data and conversion tables. In addition it provides definitions and explanations of symbols, abbreviations, and terminology used in the manual.

SECTION 2 LIMITATIONS

contains those limitations required by regulation or necessary for safe operation of the helicopter and approved by the regulatory authority.

SECTION 3 EMERGENCY AND MALFUNCTION PROCEDURES

contains the recommended procedures for coping with various types of emergencies, malfunctions or critical situations.

SECTION 4 NORMAL PROCEDURES

contains the recommended procedures for normal ground and air operation of the helicopter.

SECTION 5 PERFORMANCE DATA

contains airworthiness and performance information necessary for preflight and inflight mission planning.

SECTION 10 OPTIONAL EQUIPMENT

contains Flight Manual Supplements (FMS) providing operating instructions of approved optional equipment.

SECTION 11 SPECIAL OPERATIONS

contains Flight Manual Supplements (FMS) with instructions for special operations (e.g. IFR, operation with opened doors, etc.).

Non-approved Part**SECTION 6 MASS AND BALANCE**

contains the definitions for various mass and balance locations and the procedure for the determination of the center of gravity.

Appended to this section are the **Mass and Balance Record** form for maintaining a continuous record of changes in structure and equipment affecting the mass and balance, and the **Equipment List** form for the listing of optional equipment with data necessary for mass and balance computations.

SECTION 7 SYSTEM DESCRIPTION

contains a brief description of the helicopter, its systems and the various standard equipment with information considered most important to the flight crew.

SECTION 8 HANDLING, SERVICING, AND MAINTENANCE

contains servicing data, cleaning and care procedures as well as information for ground handling.

SECTION 9 OPERATIONAL INFORMATION

contains information for flight planning such as fuel consumption and max cruising speed diagrams.

SECTION 12 APPENDIX

contains Flight Manual Appendices (FMA) for non-approved optional equipment and special operations.

1.2.3 Page Numbering

The numbering of pages within each section consists of the section number or designation, a dash and the consecutive number of the page beginning with "1"; e.g. for Section 3: 3 - 1, 3 - 2, etc.

Figures are likewise numbered consecutively by section, such as Fig. 3-1, Fig. 3-2, etc.

Exceptions:

- The numbering of the **Table of Contents** pages preceding each section in this manual consists of the section number, a dash and the consecutive Roman numeral (lower case) of the page, beginning with "i"; e.g. for Section 3: 3 - i, 3 - ii, etc.
- The page numbers of the flight manual supplements and appendices consist of the section number, a dash, the number of the supplement/appendix, a dash and the consecutive number of the page; e.g. 10 - 17 - 1, 10 - 17 - 2, etc.
- Figures within a flight manual supplement/appendix are numbered consecutively, such as Fig. 1, Fig. 2, etc.

The numbers of blank reversed pages are printed on the preceding page by using dual page numbering; e.g. 3 - 9/(3 - 10 blank).

If, at a later date, pages have to be added to the initial printing, the new pages may carry the number of the preceding page plus a letter suffix; e.g. 2 - 6A, 2 - 6B, etc.

1.2.4 **Flight Manual Supplements (FMS)**

Each FMS is self-contained and corresponds in its general arrangement to the basic flight manual, but only additional information or different data will be the subject of an FMS.

Each FMS, although complete in nature, shall therefore be used in conjunction with the basic flight manual.

A Log of Supplements is provided for each applicable section as an index listing the current supplements.

The manufacturer retains the right to convert optional equipment to standard equipment at any time as a product improvement program. FLM coverage of the converted optional equipment, however, will remain as an FMS in Section 10 and also as an optional equipment item entry in the Equipment List. An entry in the STD EQPT EFFECTIVITY column of the Log of Supplements is used to indicate the converted optional equipment by providing an effectivity statement of the affected helicopters.

1.2.5 **Flight Manual Appendices (FMA)**

Each FMA is self-contained and corresponds in its general arrangement to the basic flight manual, but only additional information or different data will be the subject of an FMA.

Each FMA, although complete in nature, shall therefore be used in conjunction with the approved flight manual.

When two or more appendices are issued, a Log of Appendices will be provided as an index listing the current appendices.

1.3 EFFECTIVITY STATEMENTS

1.3.1 Definition of Effectivity Identification

The contents of this manual applies to all the BO-105 helicopter model variants listed on the title page and defined in this section. However, certain portions of the manual may apply to only specific models, variants, serial numbered helicopters, etc. Therefore, an effectivity identification system is used to indicate where differences brought about by helicopter modifications, Service Bulletins, customer options, variations of the basic model helicopter, etc. occur within the manual.

This system comprises effectivity statements, identified by the word **EFFECTIVITY** followed by the definition of applicability. The applicability may be defined using:

- a specified model or variant type,
- an eight-digit numeric indicator,
- a Service Bulletin (SB) modification number, and/or
- a clear text definition.

The eight-digit numeric indicator begins with the four digits of the lowest assigned manufacturer's serial number, to indicate first effectivity, and ends with the four digits of the highest assigned manufacturer's serial number, to indicate last effectivity, of an unbroken sequence of assigned serial numbered helicopters. A hyphen is shown between the two numbers. Open ended effectivity is indicated by "9999" in the last effectivity, e.g. 0415-9999 indicates helicopter serial number S-415 and subsequent.

An SB effectivity is indicated as either "Before SB....." or "After SB....." meaning respectively before and after incorporation of the specific numbered Service Bulletin on the helicopter.

1.3.2 Method of Indicating Effectivity

The amount of manual coverage required by the differences brought about by model, variants or modification differences (effectivity) varies from small or minor (portions of text) to extensive (entire pages). Therefore, the effectivity statements may apply to portions of a page or an entire page.

Effectivity statements applying to portions of a page immediately precede and are located in the extreme left hand margin directly above the text to which they relate. When the defined applicability ends, an effectivity statement defined as "ALL" is used to mean that the information following applies again to all models, variants, etc. as listed on the FLM title page or, when applicable, as defined by the page effectivity statement.

Effectivity statements applying to the entire page content will be located at the bottom of the page adjacent to the page number.

1.3.3 Incorporating Pages Having Entire Page Effectivity Statements

Compare the effectivity statements of those pages having identical page numbers and determine which page(s) applies. Insert that page(s) in the FLM binder. Discard the page(s) that does not apply.

NOTE Do not discard pages having a page effectivity statement “After SB.....” that at present may not apply to the helicopter unless however, it is absolutely certain that the stated Service Bulletin will never be incorporated.

1.4 REVISION SERVICE

This manual is kept up-to-date by revisions and temporary revisions.

1.4.1 Revisions

Revisions are issued periodically. They are printed on white paper and are incorporated into the manual in accordance with a “Letter of Transmittal” sheet which need not be inserted in the manual. Revisions are numbered consecutively beginning with the number 1.

1.4.2 Temporary Revisions

Temporary revisions are provided to transmit information between revisions. They are printed on yellow paper and are accompanied by a “Letter of Transmittal” sheet and an updated “List of Effective Temporary Revisions” page.

Temporary revisions are numbered consecutively beginning with the number 1; e.g. Temp. Rev. 1, Temp. Rev. 2, etc.

1.4.3 Identifying Revised Material

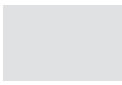
Changes (except as noted below) to the text and tables (including new material on added pages) are indicated by a vertical line in the outer margin.

Change symbols will not be shown for:

- Introductory material.
- Blank space resulting from the deletion of text, or an illustration or a part of an illustration, or table.
- Correction of minor inaccuracies, such as spelling, punctuation, relocation of material, etc, unless such correction changes the meaning of instructive information and procedures.

Changes to illustrations (except diagrams and schematics) are indicated by a miniature pointing hand. A vertical line next to changed text and callouts on illustrations may be used in lieu of a pointing hand. Shading and screening are used for diagrams and sche-

matics to highlight the area containing the changed information. Extensively changed presentations may be indicated by a screen border around the affected area.



1.5 FLIGHT MANUAL APPLICABILITY

This flight manual is applicable for the following listed BO-105 models and variants.

BO-105 Model	Variants and Characteristics	Approval Authority
C	CB-5 Max GM: 2500 kg (5512 lb) Engines: Allison 250-C20B TOP per engine: 257 kW (345 SHP) Main transmission: ZF FS 72 B	LBA FAA
S	CBS-5 (BO-105 CB-5 Stretched) Max GM: 2500 kg (5512 lb) Engines: Allison 250-C20B TOP per engine: 257 kW (345 SHP) Main transmission: ZF FS 72 B	LBA FAA

1.6 GENERAL DESCRIPTION OF THE HELICOPTER

The BO 105 is a five-place, multi-purpose, utility helicopter having a hingeless, four-bladed main rotor and a semi-rigid, two-bladed tail rotor. The main and tail rotor blades are constructed of fiber-reinforced composite materials.

Engines

The helicopter is powered by two Allison 250-C20 series engines of the freewheel turbine type. The engine power is transmitted to the main rotor transmission via independent drive systems.

Transmission

The main transmission consists of three major subassemblies in which two bevel gear input stages with freewheel clutches, a spur collector gear stage, and a planetary reduction stage are housed.

Fuel Tanks

The total fuel quantity is stored in three flexible bladder-type tanks, two main tanks (forward and aft) and one supply tank, all installed in the tank compartment beneath the cabin floor.

1.7 HELICOPTER DIMENSIONS

Figure NO TAG provides a three-view drawing of the helicopter with the respective principal exterior dimensions.

Locations on and within the helicopter can be determined in relation to fuselage stations (F.S. or STA.), waterlines (W.L.), and buttock lines (B.L.) measured in millimeters from known reference points.

1.8 CONVERSION CHARTS

See Fig. NO TAG thru NO TAG.



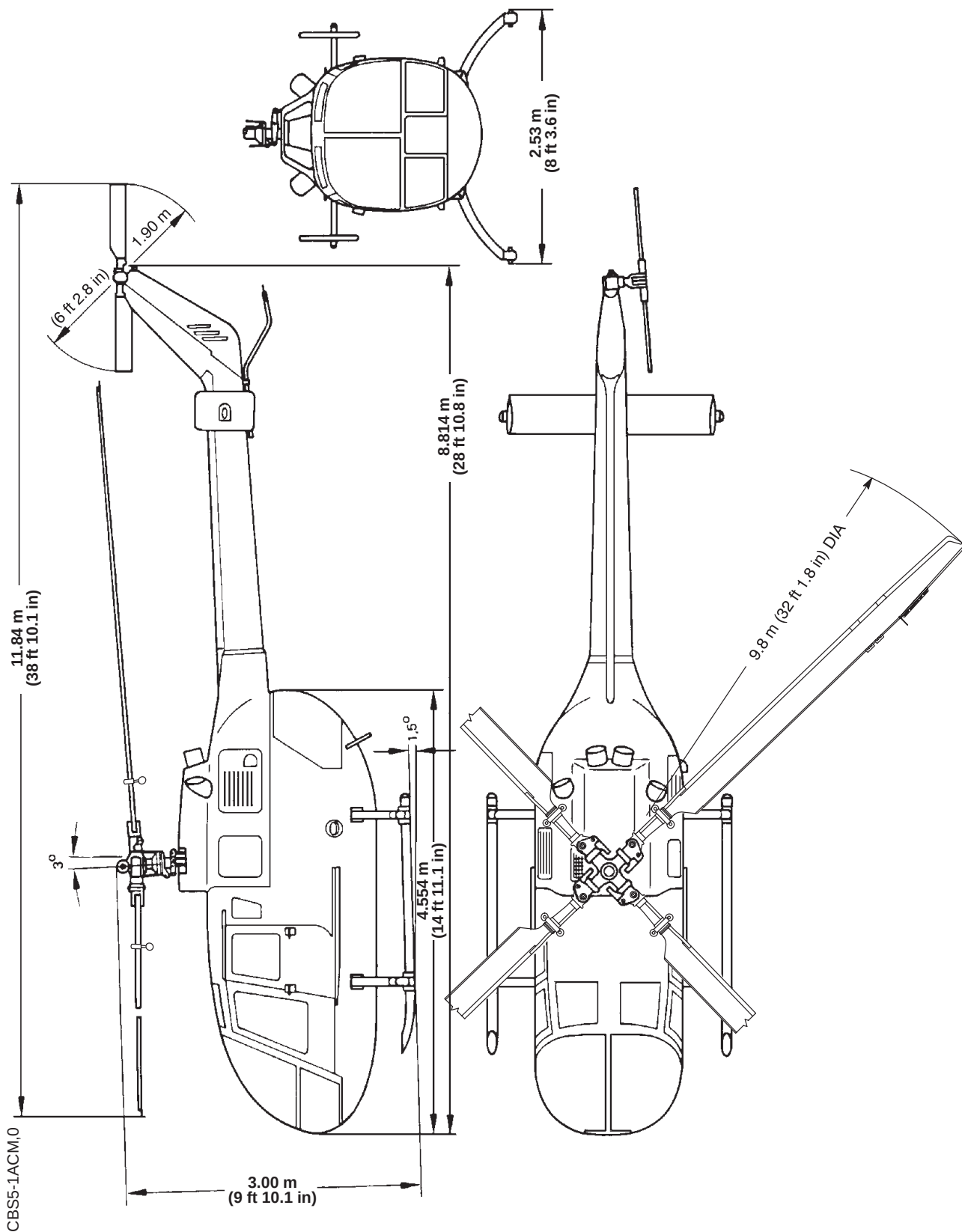


Fig. 1-1 Principal Exterior Dimensions

CELSIUS - FAHRENHEIT

$$^{\circ}\text{F} = 9/5 \times ^{\circ}\text{C} + 32 = 1.8 (^{\circ}\text{C} + 17.8)$$

$$^{\circ}\text{C} = 5/9 \times (^{\circ}\text{F} - 32)$$

EXAMPLES: 225 $^{\circ}\text{F}$ = 107 $^{\circ}\text{C}$

845 $^{\circ}\text{C}$ = 1553 $^{\circ}\text{F}$

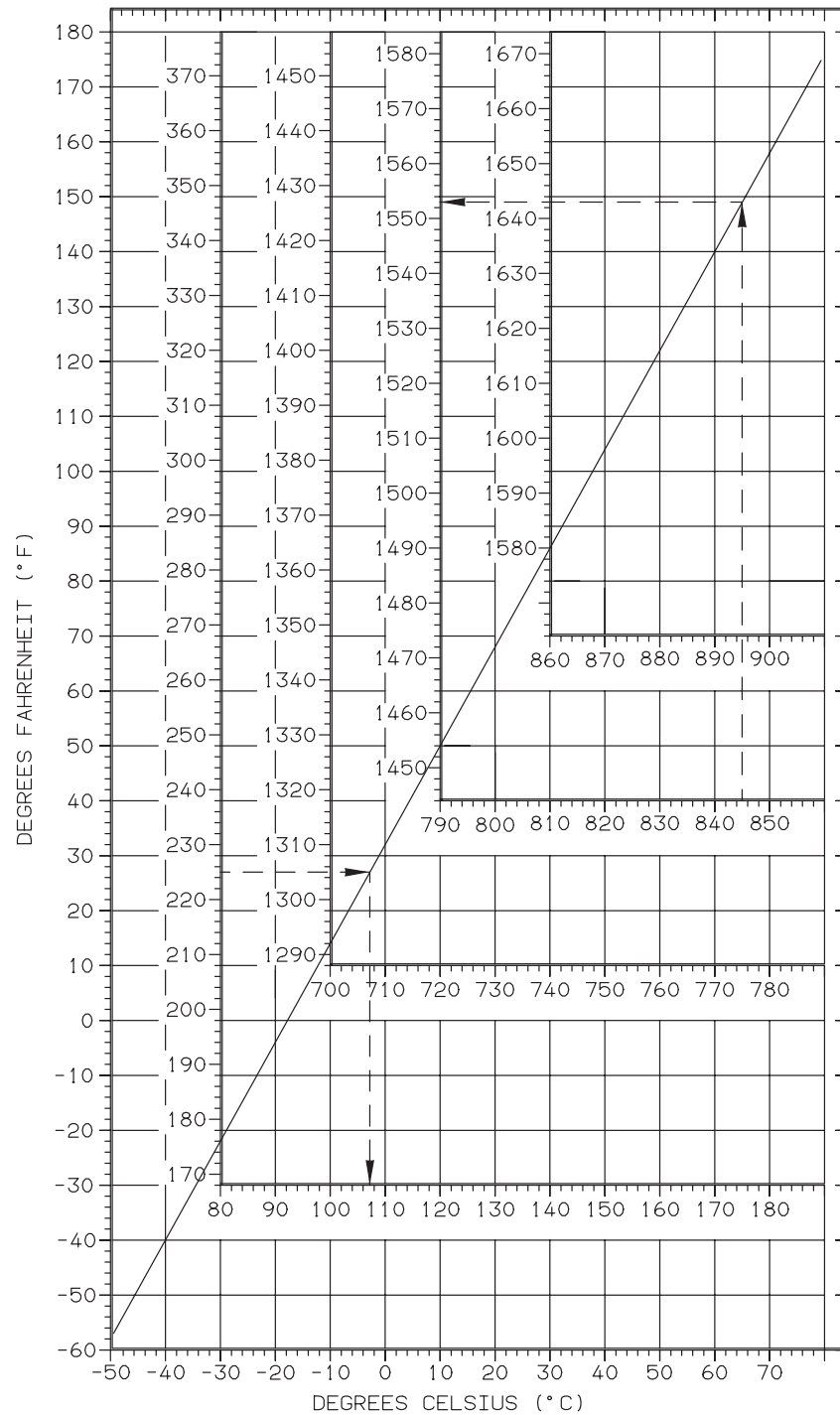


Fig. 1-2 Conversion Chart: Celsius - Fahrenheit

KILOMETERS PER HOUR - KNOTS

$$1 \text{ km/h} = 0.54 \text{ kt}$$

$$1 \text{ kt} = 1.853 \text{ km/h}$$

EXAMPLES: 42 kt = 78 km/h
266 km/h = 143.6 kt

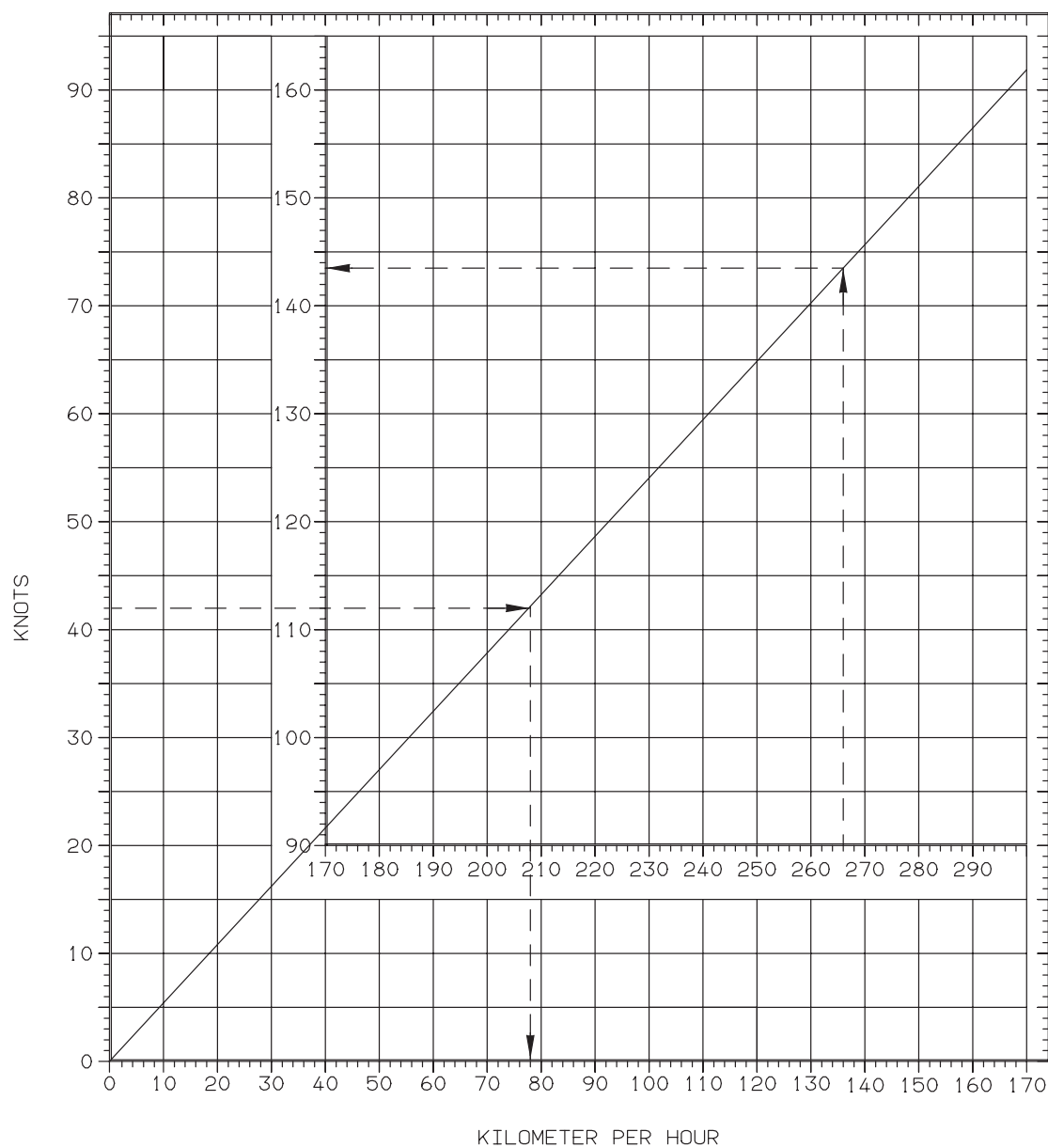


Fig. 1-3 Conversion Chart: Kilometers/Hour - Knots

METERS PER SECOND - FEET PER MINUTE

$$1 \text{ m/s} = 196.85 \text{ ft/min}$$

$$100 \text{ ft/min} = 0.508 \text{ m/s}$$

EXAMPLES: $4.7 \text{ m/s} = 925.2 \text{ ft/min}$
 $3447 \text{ ft/min} = 17.5 \text{ m/s}$

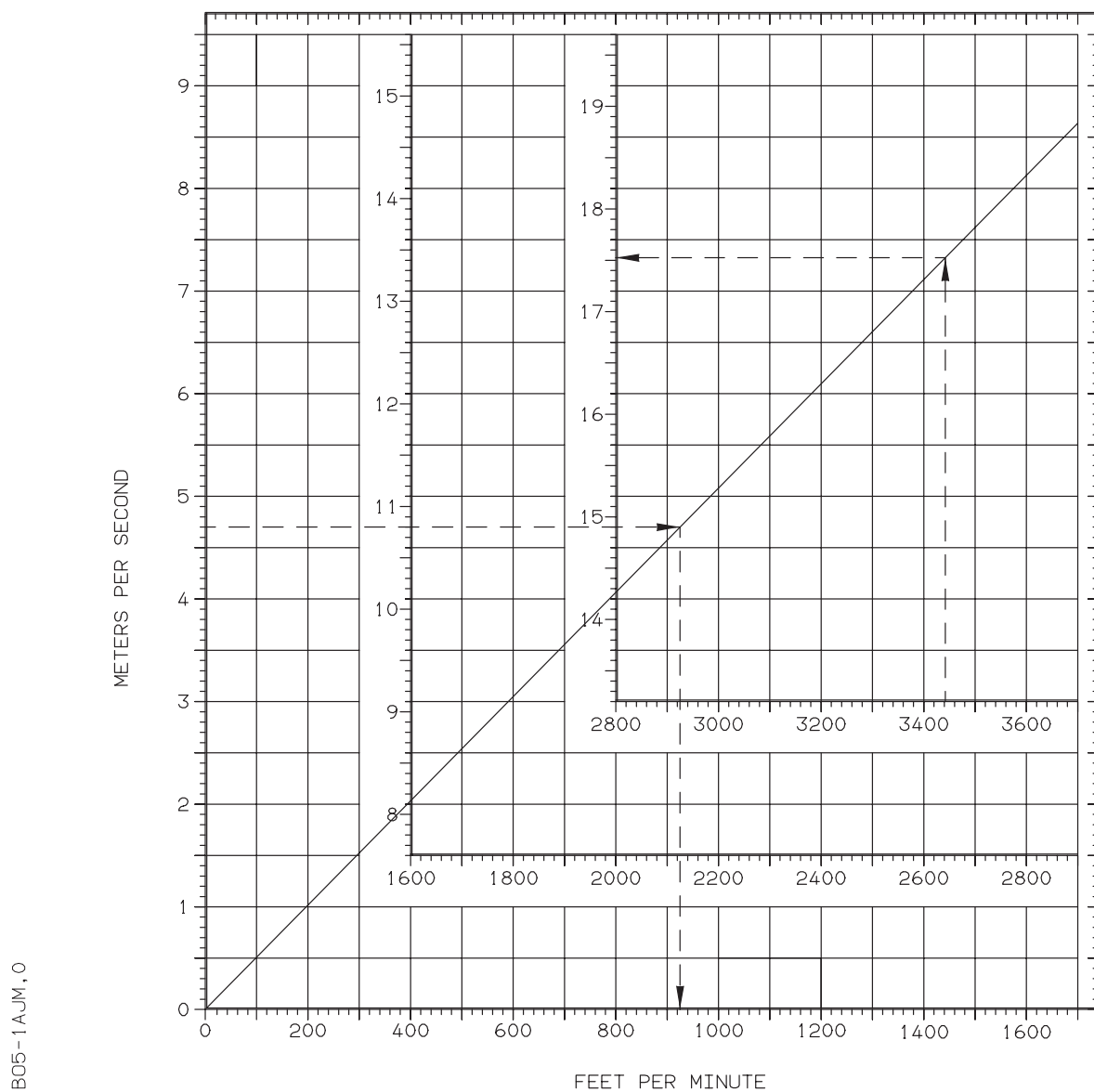


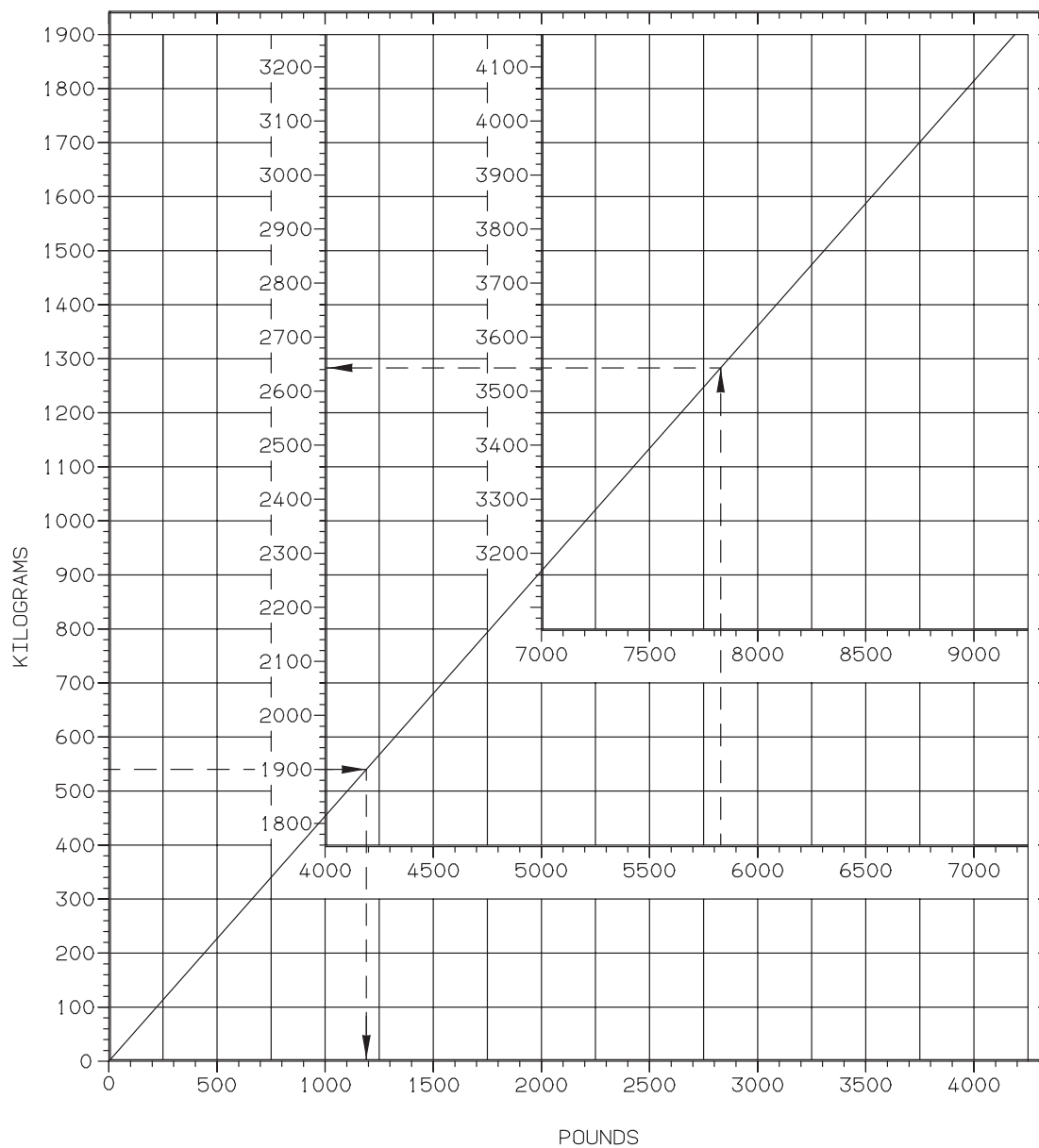
Fig. 1-4 Conversion Chart: Meters/Second - Feet/Minute

KILOGRAMS - POUNDS

$$1 \text{ kg} = 2.205 \text{ lb}$$

$$1 \text{ lb} = 0.454 \text{ kg}$$

EXAMPLES: 540 kg = 1190.7 lb
 5821 lb = 2643 kg



B05-1ALM,0

Fig. 1-5 Conversion Chart: Kilograms - Pounds

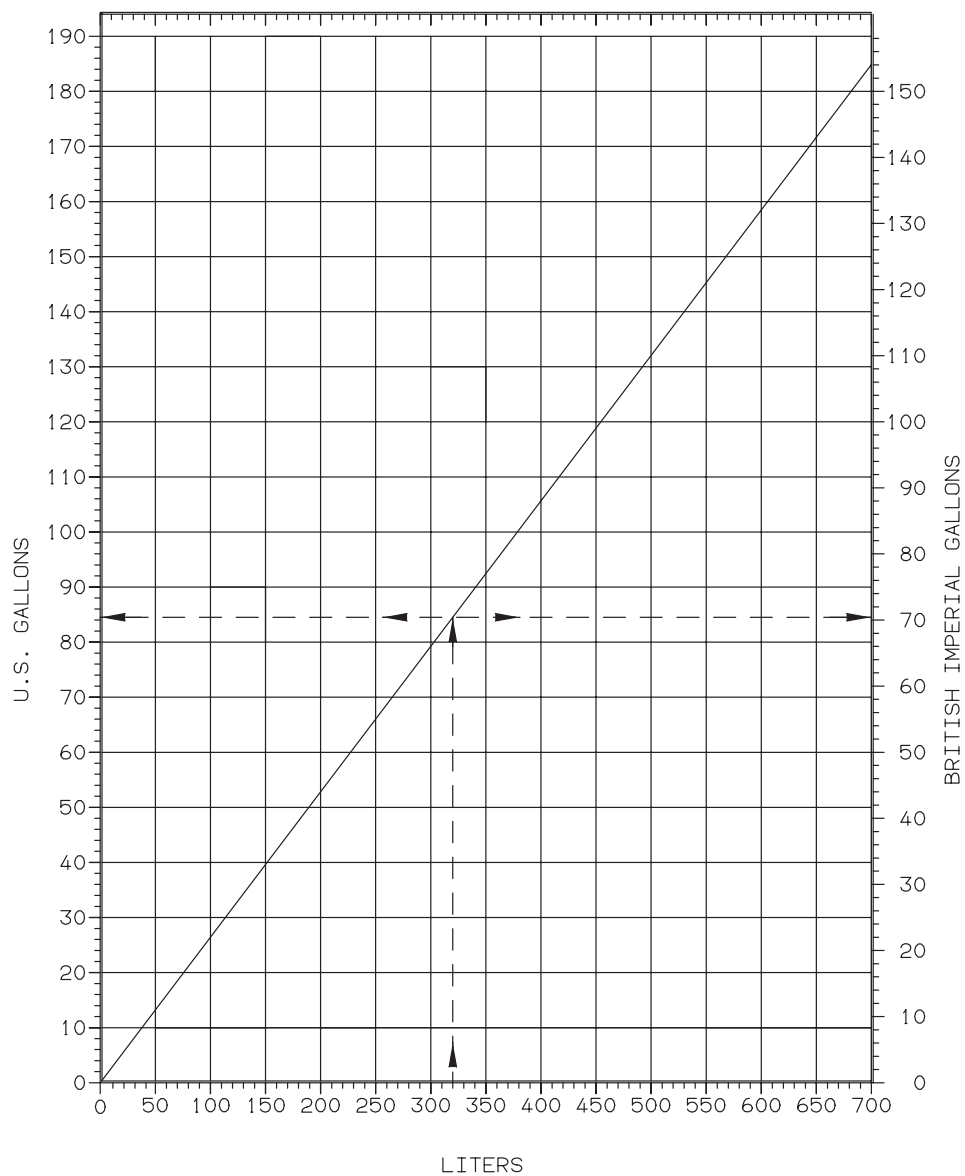
LITERS - U.S. GALLONS - IMPERIAL GALLONS

1 U.S. gal	= 3.785 l
1 l	= 0.264 U.S. gal



1 Imp. gal	= 4.546 l
1 l	= 0.220 Imp. gal

EXAMPLES: 320 l = 84.5 U.S. gal
 320 l = 70.4 Imp. gal
 70.4 Imp. gal = 84.5 U.S. gal



B05-1ANM,0

Fig. 1-6 Conversion Chart: Liters - US Gal - Imp. Gal

HECTOPASCALS (MILLIBARS) - INCHES OF MERCURY

$$1 \text{ hPa(mbar)} = 0.02953 \text{ in. Hg}$$

$$1 \text{ in.Hg} = 33.865 \text{ hPa(mbar)}$$

EXAMPLES: $1007.5 \text{ hPa(mbar)} = 29.75 \text{ in. Hg}$
 $30.96 \text{ in. Hg} = 1048.4 \text{ hPa(mbar)}$

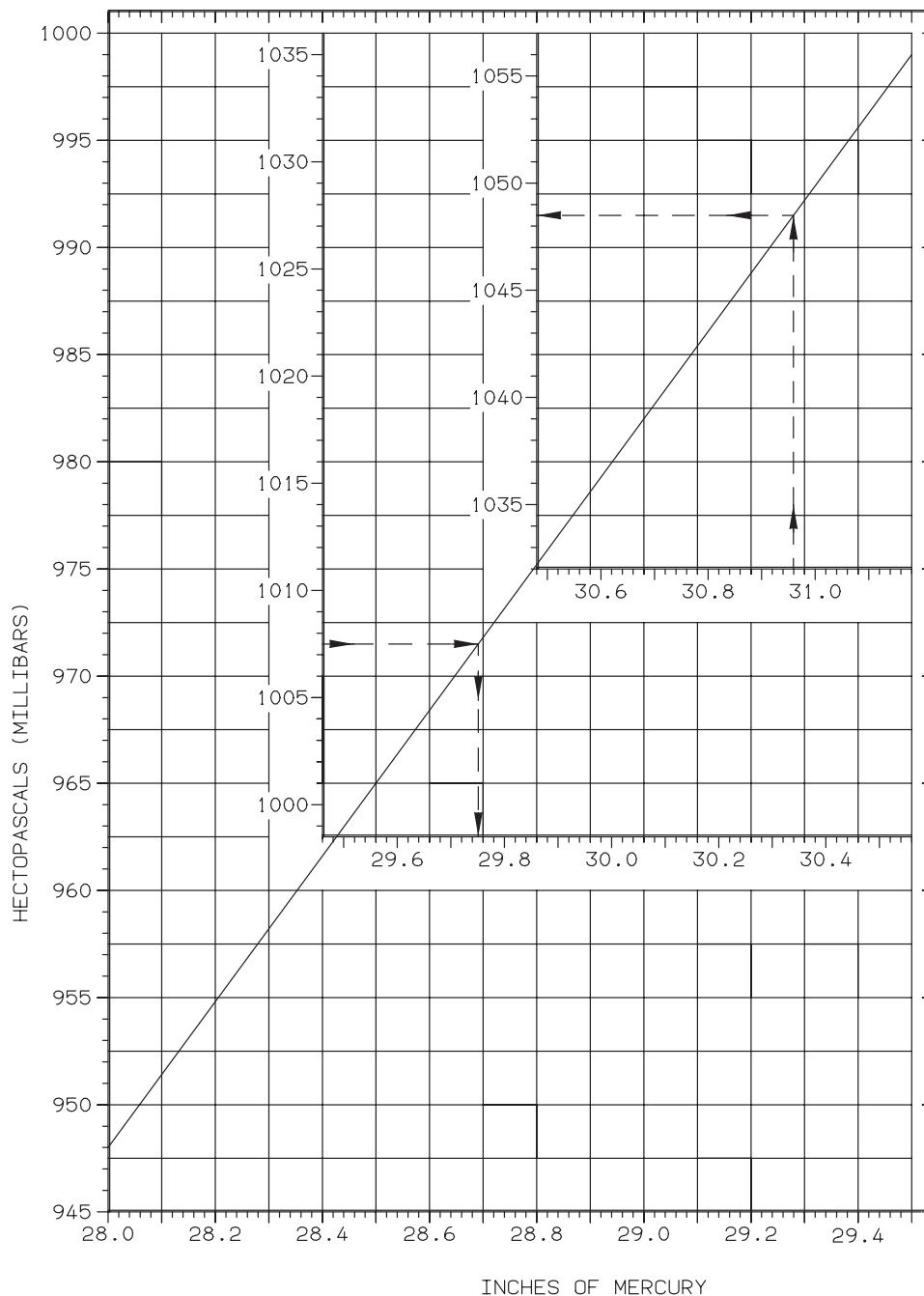


Fig. 1-7 Conversion Chart: Hectopascals - Inches of Mercury

1.9 TERMINOLOGY AND DEFINITIONS OF TERMS

1.9.1 Warnings, Cautions and Notes

Throughout this manual WARNINGS, CAUTIONS and NOTES are used to emphasize important and critical instructions.

WARNING AN OPERATING PROCEDURE, TECHNIQUE, ETC WHICH, IF NOT STRICTLY OBSERVED, COULD RESULT IN PERSONAL INJURY OR LOSS OF LIFE.

CAUTION AN OPERATING PROCEDURE, TECHNIQUE, ETC WHICH, IF NOT STRICTLY OBSERVED, COULD RESULT IN DAMAGE TO OR DESTRUCTION OF EQUIPMENT.

NOTE An operating procedure, technique, condition, etc that is essential to emphasize.

WARNINGS and CAUTIONS immediately precede and are located directly above the text to which they relate.

NOTES follow the text to which they relate and are located directly below the text to which they apply.

1.9.2 Use of Procedural Terms

The procedural term usage and meaning are as follows:

"Shall" and "Must"	have been used to express a mandatory requirement.
"Should"	has been used to express non-mandatory provisions.
"May"	has been used to express permissiveness.
"Will"	has been used only to indicate futurity, never to express a mandatory requirement.

1.10 ABBREVIATIONS AND SYMBOLS

A	- ampere
a/c, acft	- aircraft
AC	- alternating current
AEO	- all engines operating
AGL	- above ground level
approx	- approximate
assy	- assembly
ASTM	- American Society for Testing Materials
BAT, BATT	- battery
BO	- Bölkow
CAS	- calibrated airspeed
CFR	- Code of Federal Regulations
CG	- center of gravity
CIS	- Community of Independent States
cm	- centimeter
coll.	- collision
COMM	- communication (radio)
compt	- compartment
CTR	- center
cu ft	- cubic feet
DA	- density altitude
DBL	- double
DC	- direct current
e.g.	- for example
EL	- equipment list
EMER	- emergency
ENG	- engine
EPU	- external power unit
EPWR	- emergency power
eqpt	- equipment
etc	- et cetera (and so forth)
EXT	- external, extinguisher
FAA	- Federal Aviation Administration, United States
FAR	- Federal Aviation Regulation
Fig.	- figure
FLM	- flight manual
FMA	- flight manual appendix
FMS	- flight manual supplement
fpm	- feet per minute
ft	- foot (feet)
GAL	- gallon
GEN	- generator
GM	- gross mass
h, hr	- hours of time
Hg	- mercury (hydrargyrum)
HIGE	- hover in ground effect

HOGE	-	hover out of ground effect
hPa	-	hectopascal
HV	-	height velocity
HY, HYD, HYDR	-	hydraulic
IAC-AR	-	Interstate Aviation Committee - Aviation Register, CIS
IAS	-	indicated airspeed
ICS	-	intercommunication system
i.e.	-	that is (to say)
IFR	-	instrument flight rules
IGE	-	in ground effect
Imp.	-	Imperial
in.	-	inch
IND	-	indicator
ISA	-	International Standard Atmosphere
JAR	-	Joint Airworthiness Regulation
KCAS	-	knots calibrated airspeed
kg	-	kilogram
KIAS	-	knots indicated airspeed
km	-	kilometer
km/h	-	kilometers per hour
kp	-	kilopond
kt	-	knot
KTAS	-	knots true airspeed
kW	-	kilowatt
L	-	left
L, l, ltr	-	liter
lb	-	pound
LBA	-	Luftfahrt-Bundesamt, Federal Republic of Germany
LEP	-	list of effective pages
LH	-	left hand
m	-	meter
max	-	maximum, maximal
MCP	-	maximum continuous power
MEL	-	minimum equipment list
MIL	-	military standard
min	-	minimum, minimal, minutes of time
MM	-	mast moment
mm	-	millimeter
MSL	-	mean sea level
N ₁ , n1	-	gas generator speed
N ₂ , n2	-	power turbine speed
NAV	-	navigation (radio)
NM	-	nautical mile
No.	-	number
N _{Ro}	-	rotor speed
OAT	-	outside air temperature
OEI	-	one engine inoperative
OGE	-	out of ground effect

Pa	- Pascal
PA	- pressure altitude
PAX	- passenger
P/N	- part number
PRESS	- pressure
psig	- pound-force per square inch gage
PU	- polyurethane
PWR	- power
R	- right
R/C	- rate of climb
R/D	- rate of descent
RD	- reference datum
Rev.	- revision
RH	- right hand
RPM	- revolutions per minute
S-	- serial (manufacturer's number)
SB	- service bulletin
sec	- seconds of time
SEL	- selector
SGL	- single
SL	- sea level
S/N	- serial number
sq	- square
STBY	- standby
std	- standard
sw	- switch
T.	- transmission
TAS	- true airspeed
TEMP	- temperature
Temp. Rev.	- temporary revision
T/O	- takeoff
TOP	- takeoff power
TOT	- turbine outlet temperature
U.S., US	- United States
V	- volt
VFR	- visual flight rules
VHF	- very high frequency
V _{NE}	- never-exceed speed (velocity never exceed)
VOR	- very high frequency omnirange station
V _Y	- best rate-of-climb speed
XMSN	- transmission
°C	- degrees Celsius (centigrade)
°F	- degrees Fahrenheit
σ	- density ratio
$\frac{1}{\sqrt{\sigma}}$	- true airspeed factor (the reciprocal of the square root of the density ratio, at the density altitude)
ΔPA	- pressure altitude correction

SECTION 2

LIMITATIONS

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SECTION 2

LIMITATIONS

WARNING WHEN ANY LIMITATION IS EXCEEDED, MAINTENANCE ACTION MAY BE REQUIRED AND NECESSARY BEFORE NEXT FLIGHT. ENTER DURATION AND AMOUNT OF EXCESS IN LOGBOOK.

2.1 GENERAL

This helicopter shall be operated in compliance with the limitations of this section.

The "Airworthiness Limitations" in Chapter 101 of the BO-105 Rotorcraft Maintenance Manual must be complied with.

For definitions of terms, abbreviations and symbols used in this section refer to Section 1.

2.2 KINDS OF OPERATION

The helicopter in its basic configuration is approved for operation under day and night Visual Flight Rules.

With special equipment installed and operative, and under observance of the procedures and limitations described in FMS 11-2, the helicopter is also certified for operation according to Instrument Flight Rules.

2.3 BASIS OF CERTIFICATION

This helicopter is certified by the LBA in the Normal Category based on FAR PART 27, Amendments 27-1 through 27-3.

NOTE The performance data are based on Joint Airworthiness Regulations (JAR) PART 29, Second Draft.

2.4 OCCUPANTS

EFFECTIVITY *Model C Variants*

The helicopter in its basic configuration is approved as a five-place rotorcraft.

EFFECTIVITY *Model S Variants*

The helicopter in its basic configuration is approved as a five-place rotorcraft with an optional six-place configuration.

EFFECTIVITY *ALL*

LBA APPROVED

Rev. 0

2.4.1 Flight Crew

The minimum flight crew consists of one pilot in the right crew seat.

2.5 FLIGHT WITH OPTIONAL EQUIPMENT INSTALLED

Refer to Sections 10 and 11 for additional limitations, procedures and performance data.

2.6 MASS AND LOAD LIMITATIONS

2.6.1 Maximum Gross Mass

Maximum approved gross mass 2500 kg

2.6.2 Minimum Gross Mass

Minimum approved mass 1140 kg

2.6.3 Loading

Maximum cargo load is limited by Mass and Balance consideration.

Maximum allowable floor loading 600 kg/m²

Max allowable load avionics equipment bay 20 kg

CAUTION CARGO, BAGGAGE AND LOOSE ITEMS MUST BE PROPERLY STOWED AND TIED DOWN IN ORDER TO MAKE IN-FLIGHT SHIFTING IMPOSSIBLE (REFER TO SECTION 6).

2.6.4 Cargo Securing Points

Maximum allowable load per:

– Tie-down ring 100 kg

– Insert 200 kg

2.7 CENTER OF GRAVITY LIMITATIONS

2.7.1 Longitudinal Center of Gravity

Station zero (datum) is an imaginary vertical plane, perpendicular to the aircraft centerline and located 3000 mm forward of the leveling point (see Section 6).

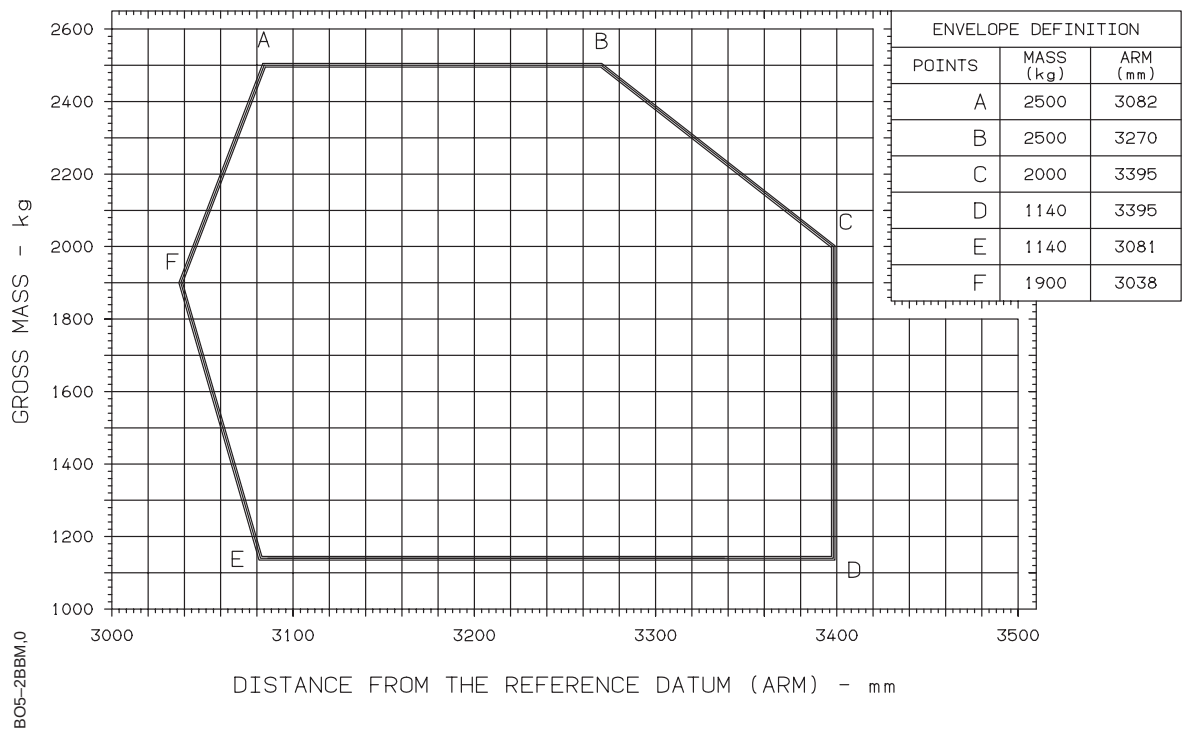


Fig. 2-1 Allowable CG Envelope (Longitudinal)

2.7.2 Lateral Center of Gravity

Lateral center of gravity limits left and right of the fuselage centerline are:

for a gross mass up to 2400 kg	100 mm
for a gross mass above 2400 kg	80 mm

2.8 AIRSPEED LIMITATIONS

2.8.1 Forward Flight

The following table (Fig.NO TAG) is used in determining the airspeed limits for any gross mass up to 2500 kg.

AIRSPEED LIMITS (V_{NE}) - KIAS GM 2300 kg AT ANY GROSS MASS ABOVE 2300 kg, DECREASE V_{NE} BY 10 KIAS											
PA - FT	OAT - °C										
	-45	-30	-20	-10	0	+10	+20	+30	+40	+50	+54
SL	145	145	145	145	145	145	145	140	140	135	135
2000	140	145	145	145	145	140	140	135	135	130	130
4000	135	145	145	140	140	135	135	130	130	125	—
6000	130	140	140	135	135	130	130	125	120	115	—
8000	125	135	135	130	130	125	120	115	110	—	—
10000	120	130	130	125	120	115	110	105	100	—	—
12000	115	125	120	115	110	105	100	95	90	—	—
14000	110	115	110	105	100	95	90	85	—	—	—
16000	105	105	100	95	90	85	80	80	—	—	—
17000	100	100	95	90	85	80	80	—	—	—	—

Fig. 2-2 Never-exceed Speed (V_{NE})

The max airspeed for steady autorotation is 100 KIAS or that in accordance with the V_{NE} table (see Fig.NO TAG) whichever value is less.

The max airspeed for OEI operation is 100 KIAS or that in accordance with the V_{NE} table (see Fig.NO TAG) whichever value is less.

2.8.2 Sideward Flight

Sideward flight or crosswind hover has been demonstrated up to 25 kt airspeed.

2.8.3 Rearward Flight

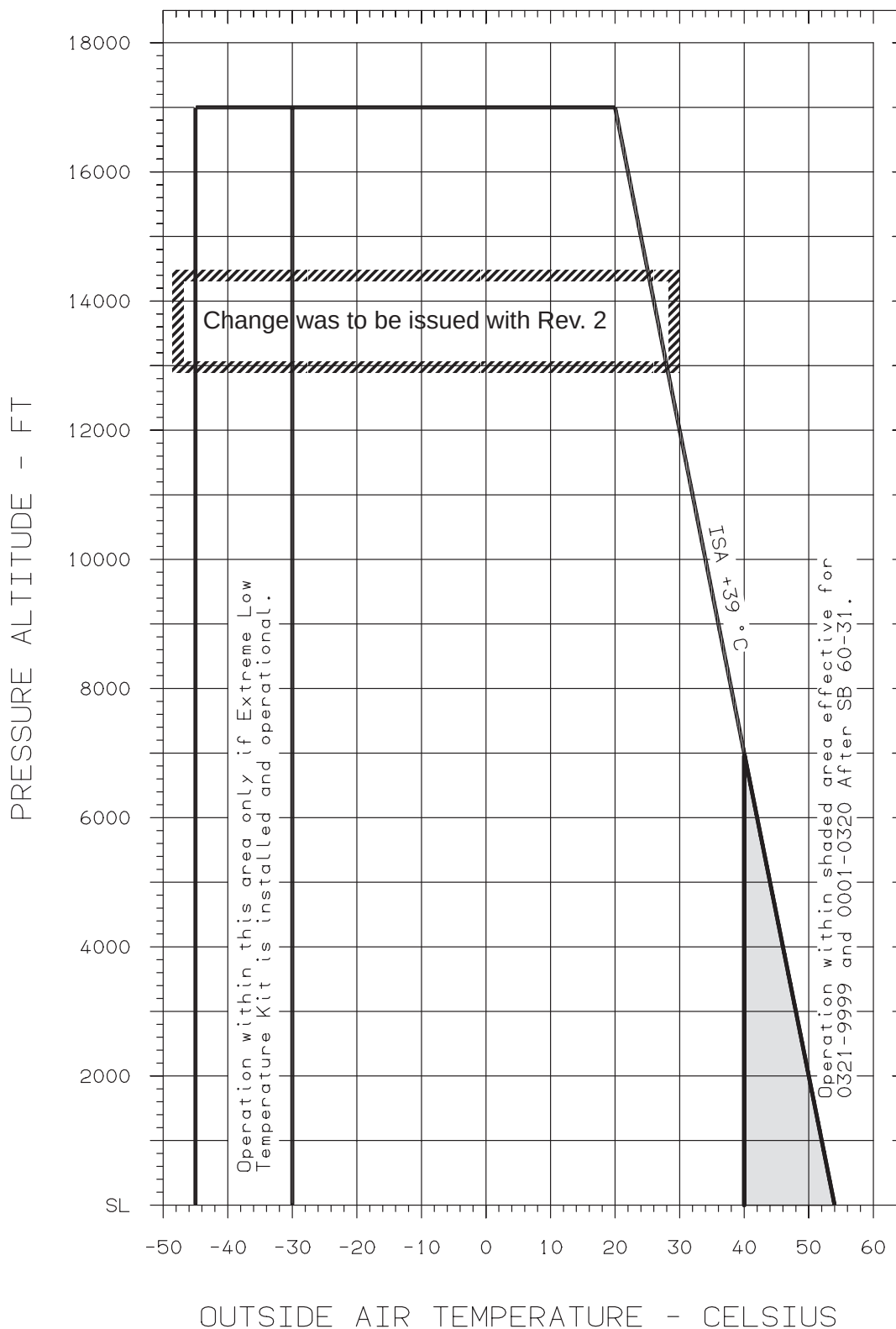
Rearward flight or tailwind hover has been demonstrated up to 20 kt airspeed.

2.9 ALTITUDE LIMITATIONS (see Fig.NO TAG)

NOTE Altitude values used throughout this manual are Pressure Altitude unless stated otherwise.

Maximum altitude:

- when operating on MIL-T-5624, grade JP-5
or ASTM D-1655 Jet A and Jet A1 17,000 ft
- when operating on MIL-T-5624, grade JP-4;
MIL-T-83133, grade JP-8; ASTM D-1655 Jet B;
or alternate fuel (AVGAS/Jet fuel mixture) 13,700 ft



CBS5-2BEM,2

Fig. 2-3 Altitude/Temperature Limit Diagram

2.10 ENVIRONMENTAL OPERATING CONDITIONS

2.10.1 Altitude/Ambient Air Temperature Limitations (see Fig.NO TAG)

Min ambient air temperature –30 °C

EFFECTIVITY *If extreme low temperature kit (P/N 105-80006) is installed and operational*

Min ambient air temperature –45 °C

EFFECTIVITY *0001-0320 Before SB 60-31*

Max ambient air temperature +40 °C

EFFECTIVITY *0321-9999, and 0001-0320 After SB 60-31*

Max ambient air temperature +54 °C

When operating in high ambient temperatures (+45 to +54 °C) at sea level, the oil temperatures must be closely monitored in particular during hover. The oil temperatures may be reduced at all times by either landing or transition to cruise flight.

EFFECTIVITY *ALL*

2.10.2 Engine Anti-icing

Engine anti-icing must be on when the ambient temperature is below +4 °C and visible moisture is present, except for takeoff and landing.

2.10.3 Pitot Heating

Pitot heating must be on when the ambient temperature is below +4 °C and visible moisture is present.

2.10.4 Icing Conditions

Flight into known icing conditions is prohibited.

2.10.5 Snow Conditions

Operation in snow is prohibited except when the Continuous Ignition System (see FMS 11-3) is installed and switched on.

The engine inlet deflector shield must be installed when operation in snow is expected.

After operation in snow, a post flight visual check of the compressor inlet must be conducted in accordance with Section 11, FMS 11-3, paragraph 4.4.

NOTE Record engine operation in snow in the journey logbook.

2.10.6 Battery Temperature Limitations

When the engines are started using the onboard battery at an OAT of below –30 °C, the battery shall be preheated to at least –30 °C.

When the engines are started using an external power unit at an OAT of –40 °C or below, the battery shall be preheated to at least –40 °C.

2.11 ROTOR RPM LIMITATIONS

EFFECTIVITY *Model C Variants*

	Power ON	Power OFF
Minimum Transient	95%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	110%
Maximum Transient (15-sec limited)	105%	—

EFFECTIVITY *Model S Variants*

	Power ON	Power OFF
Minimum Transient	95%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	104%
Maximum Transient (15-sec limited)	105%	110%

EFFECTIVITY *ALL*

2.12 ENGINE AND TRANSMISSION POWER LIMITATIONS

Engines: Allison 250-C20 B

Transmission: Zahnradfabrik Friedrichshafen AG (ZF) FS 72B

CAUTION • OEI EMERGENCY POWER RATING IS LIMITED TO USE ONLY AFTER ACTUAL FAILURE OF AN ENGINE.

- FOR OEI TRAINING OR DEMONSTRATION PURPOSES MAXIMUM CONTINUOUS POWER RATING (OEI) SHALL NOT BE EXCEEDED.

CONDITION	TRANSMISSION (HELICOPTER) LIMITS MAX TORQUE %	ENGINE OPERATING LIMITS		
		MAX TOT °C	MAX N ₁ %	MAX N ₂ %
Start/shutdown Transient (Max 10 sec above 810 °C)		927		
Transient (Max 6 sec above 810 °C)		843		
Transient (Max 15 sec)			106	105
All Engines Operating				
Max Continuous Power	2 x 86	779	105	102
Takeoff Power	2 x 86	810 ¹⁾	105	102
One Engine Inoperative				
Max Continuous Power	95	779	105	102
Emergency Power	95	810 ²⁾	105	102
NOTE ¹⁾ Takeoff Power TOT range of 779-810 °C has a 5-min limit.				
²⁾ Emergency Power TOT range of 779-810 °C has a 30-min limit.				

2.13 OTHER ENGINE AND TRANSMISSION LIMITATIONS

2.13.1 Engine Oil Pressure

Below 79% N_1 speed 3.5 to 9.1 kp/cm²

From 79 to 94% N_1 speed 6.3 to 9.1 kp/cm²

At 94% N_1 speed and above 8.1 to 9.1 kp/cm²

During engine starting, a positive oil pressure indication must be obtained upon reaching 59% N_1 (Idle) speed.

NOTE During cold weather operation, 10 kp/cm² engine oil pressure is allowable following an engine start. When the 9.1 kp/cm² limit is exceeded, operate engine at minimum power (IDLE) until normal oil pressure limits are attained.

2.13.2 Engine Oil Temperature

Minimum during start -30 °C

Normal operating range +30 to +107 °C

NOTE Engines may be operated at oil temperatures from 0 to 107 °C provided engine oil pressure is within specified limits.

EFFECTIVITY *If extreme low temperature kit (P/N 105-80006) is installed and operational*

Normal operating range 0 to +107 °C

EFFECTIVITY ALL

Maximum +107 °C

2.13.3 Engine Starter

To prevent starter overheat damage, limit starter energize time to the following:

External Power	Onboard Battery
20 seconds - ON	30 seconds - ON
60 seconds - OFF	60 seconds - OFF
20 seconds - ON	30 seconds - ON
60 seconds - OFF	60 seconds - OFF
20 seconds - ON	30 seconds - ON
30 minutes - OFF	30 minutes - OFF

The 30-minute cooling period is required before beginning another starting cycle.

The starter energize time is the time that elapses between initiation of the starter and ignition in the turbine.

Motoring condition

External Power	Onboard Battery
15 seconds - ON	20 seconds - ON
60 seconds - OFF	60 seconds - OFF
15 seconds - ON	20 seconds - ON
23 minutes - OFF	23 minutes - OFF

2.13.4 Generator

Maximum load per generator 150 A

2.13.5 Ground Power Starts

The current flow during starting engines shall not exceed 600 A when using 28V DC ground power units.

2.13.6 Main Transmission Oil Pressure

Minimum 0.5 kp/cm²

Normal range 0.8 to 2.0 kp/cm²

NOTE During engine starting, the oil pressure may exceed the normal range.

2.13.7 Main Transmission Oil Temperature

Minimum during start -30 °C

Normal operation range +30 to +105 °C

EFFECTIVITY *If extreme low temperature kit (P/N 105-80006) is installed and operational*

Normal operation range 0 to +105 °C

EFFECTIVITY ALL

Maximum +105 °C

Transient (10-min limited) +120 °C

2.14 FUEL LIMITATIONS

2.14.1 Fuel Pressure

Normal operating range 0.6 to 1.7 kp/cm²

Maximum 1.7 kp/cm²

2.14.2 Fuel Specifications

Fuel conforming to the following specification is authorized for use:

Type Fuels (Primary)

Wide-cut Type	- MIL-T-5624, grade JP-4	*)
	- ASTM D-1655, Jet B	
	- CAN 2-3.22	
Kerosene Type	- MIL-T-5624, grade JP-5	*)
	- MIL-T-83133, grade JP-8	*)
	- ASTM D-1655, Jet A or Jet A1	
	- CAN 2-3.23	
	- GOST #10227, TS-1 (Regular or Premium)	
	- GOST #16564, RT	

*) Contains an anti-icing additive which conforms to MIL-I-27686 and does not require additional anti-icing additives.

Type Fuels (Emergency)

CAUTION MIL-G-5572 FUEL CONTAINING TRICRESYLPHOSPHATE (TCP) ADDITIVE SHALL NOT BE USED.

MIL-G-5572E and subsequent, all grades without TCP

NOTE Maximum of 6 hours operation per turbine overhaul period.

2.14.3 Fuel Specifications - Cold Weather

WARNING AT AMBIENT TEMPERATURES BELOW 4 °C, SOME TYPE OF FUEL ANTI-ICING ADDITIVE IS REQUIRED.

To assure consistent starts at ambient temperatures below 4 °C, the following fuels are recommended:

Type Fuels (Primary)

- | | | |
|---------------|--------------------------|----|
| Wide-cut Type | - MIL-T-5624, grade JP-4 | *) |
| | - ASTM D-1655, Jet B | |

*) Contains an anti-icing additive which conforms to MIL-I-27686 and does not require additional anti-icing additives.

NOTE • Jet A, Jet A1, JP-5 or JP-8 may start the engines at temperatures below 4 °C; however, when cold soaked, marginal starts may result due to viscosity changes. Once started, the engines will operate satisfactorily on JP-5, JP-8, Jet A and Jet A1 at fuel temperature and OAT down to -32 °C.

- GOST specification fuels must have a minimum viscosity of 6 centistokes when starting cold engines under low temperature conditions.

Type Fuels (Alternate)

CAUTION • JP-4 OR JET B FUEL SHALL NOT BE MIXED WITH AVGAS.

- THIS HELICOPTER MAY NOT BE OPERATED WITH AN AVGAS/JET FUEL MIXTURE WHEN THE OAT IS 4 °C OR HIGHER.

AVGAS/Jet A, Jet A1, JP-5 or JP-8 (MIL-T-83133) mixture

The AVGAS/Jet fuel mixture is defined as:

- One part by volume AVGAS (MIL-G-5572) of either grade 80/87 or grade 100/130 (100L) having 0.53 ml/l maximum lead content. Do not use grade 100/130 having 4.6 ml/l lead content.
- Two parts by volume ASTM D-1655, Jet A or Jet A1, or MIL-T-5624, grade JP-5.

NOTE Use of 100/130 (100L) grade AVGAS/Jet fuel mixture shall be restricted to 300 hours in one engine overhaul period due to the high lead content of the fuel.

2.14.4 Fuel Additives

2.14.4.1 Anti-icing

Icing protection is required when operating with primary fuels at temperatures below +4 °C (40 °F).

- MIL-I-27686; Concentration by volume: Max 0.15 %, min 0.035 %.
- Fluid I (GOST 8313); Concentration by volume: Max 0.3 %, min 0.10 %. *)
- Fluid IM (TU 6-10-1458); Concentration by volume: Max 0.3 %, min 0.10 %. *)

*) Only for use with GOST specification fuels.

2.14.5 Fuel Temperature

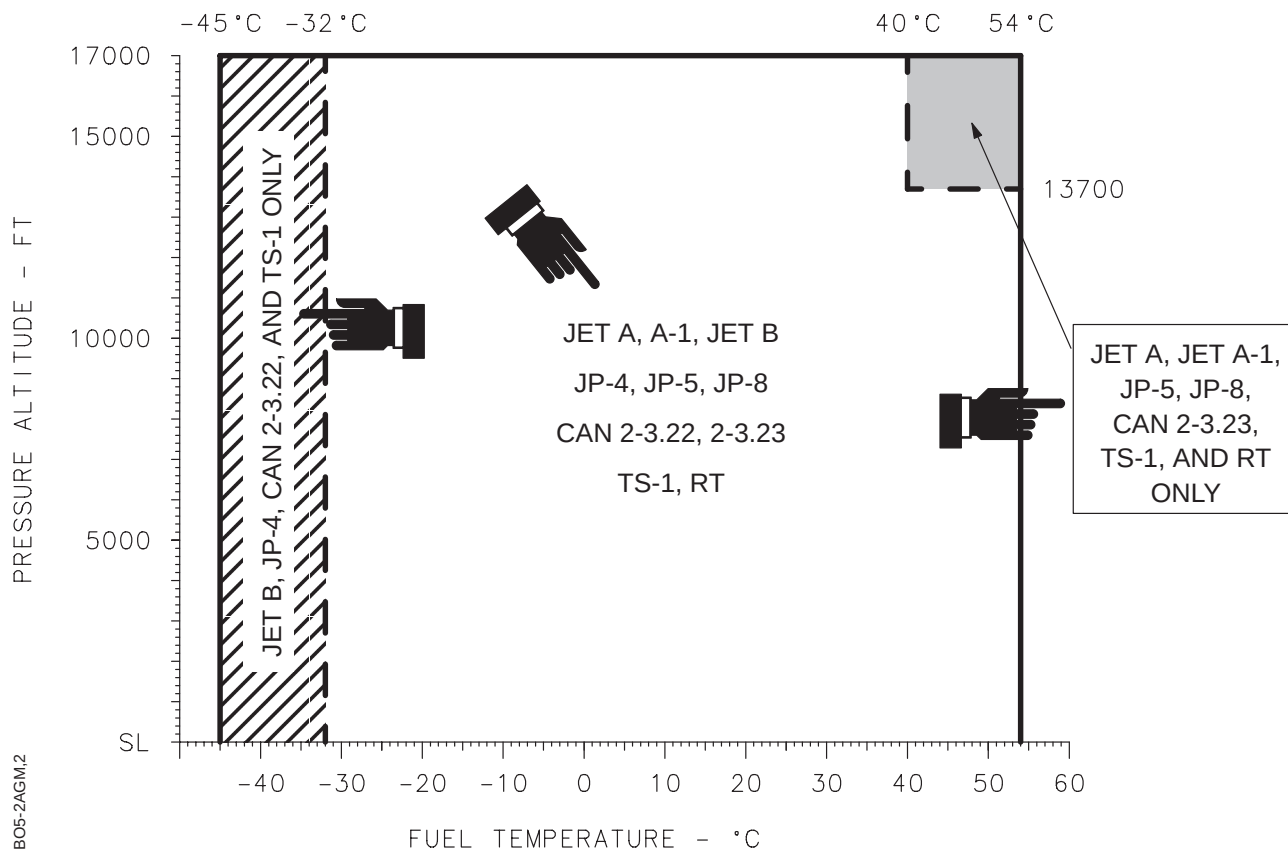


Fig. 2-4 Temperature Limits for Approved Primary Fuels

2.14.6 Fuel Quantities

TANK	USABLE FUEL		UNUSABLE FUEL	
	liters	kilograms	liters	kilograms
Main	477	382	7.5	6.0
Supply	93	74	2.5	2.0
Totals	570	456	10.0	8.0

Fuel mass values are based on a fuel density of 0.8 kg/l.

The usable fuel quantity is empty when zero is indicated on the relevant fuel quantity gauge.

2.15 OIL LIMITATIONS

2.15.1 Oil Specifications

	Oil Type	OAT Limit
Engines	MIL-L-23699 MIL-L-7808	−30 °C −30 °C
Main Transmission	MIL-L-23699	−30 °C
Intermediate Gearbox	MIL-L-23699	−45 °C
Tail Rotor Gearbox	MIL-L-23699	−45 °C
Main Rotor Hub	MIL-L-23699	−45 °C

2.15.2 Oil Quantities

	Liters	Kilograms
Engines (each tank)	4.5	4.5
Main Transmission	10.3	10.3
Intermediate Gearbox	0.6	0.6
Tail Rotor Gearbox	0.4	0.4
Main Rotor Hub	1.8	1.8
Oil mass values are based on an oil density of 1 kg/l.		

2.16 HYDRAULIC SYSTEM LIMITATIONS

2.16.1 Operational Hydraulic System

For takeoff the operational hydraulic system must be system 1. A change-over to system 2, by operating the hydraulic test switch during flight, is prohibited. The hydraulic test switch is for ground and preflight test purposes only.

2.16.2 Hydraulic Fluid

Hydraulic fluid type MIL-H-5606 is authorized for use at all ambient temperatures.

Hydraulic fluid quantity per system 1.2 liters

2.17 OPERATIONAL LIMITATIONS

2.17.1 Rotor Starting and Stopping in Wind

Rotor starting and stopping has been demonstrated with up to 45 KTAS headwinds.

2.17.2 Hover Turn

Maximum rate of turns are:

for a gross mass up to 2400 kg	90° per sec
for a gross mass above 2400 kg	45° per sec

2.17.3 Main Rotor Mast Moment

The mast moment indicator indicates the bending moment of the main rotor mast. The red radial marking shall not be exceeded either in flight or during ground operation.

Maximum	red radial
Caution range	yellow arc
Normal range	green arc
Calibration point	white radial

NOTE • The **LIMIT** warning light comes on momentarily if the pointer reaches the upper end of the yellow arc.

- A logbook entry and maintenance action are required whenever:
 - the **LIMIT** warning light remains on (even after pointer returns to a lower mast moment), and
 - the helicopter is operated with a defective mast moment system.
- Periodical maintenance actions are required according to Maintenance Manual, if the helicopter is operated without the mast moment system.

2.17.4 Slope Operations

Slope operations (takeoff/landing) are limited in the degree of sloping terrain upon which the maneuver may be performed:

- **Mast moment indicator installed and operative**

Ground sloping down to the left	Max 15°
Ground sloping down to the right	Max 8°
Ground sloping down forward	Max 8°
Ground sloping down rearward	Max 10°
- **Mast moment indicator inoperative or removed**

Ground sloping in all directions	Max 5°
--	--------

2.17.5 Prohibited Flight Maneuvers

Aerobatic maneuvers are prohibited.

2.17.6 Mass-altitude-temperature

For mass-altitude-temperature limitations at takeoff, landing, and during ground effect maneuvers refer to Section 5.

2.17.7 Extreme Low Temperature Operation

Operation at extreme low ambient temperatures (below -30°C) is permitted only under the following conditions:

- Extreme low temperature kit must be installed and operational
- Cockpit/cabin temperatures must be above -30°C
- Engine and transmission oil temperatures must be above 0°C prior to takeoff.

Refer to BO105 Maintenance Manual, Chapter 101, "Measures to be taken when operating at extremely low temperatures".

2.17.8 Engine Inlet Deflector Shield

EFFECTIVITY 0001-0450 Before SB 60-37

The engine inlet deflector shield must be removed when the OAT is above $+30^{\circ}\text{C}$.

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

The engine inlet deflector shield must be installed at all times.

EFFECTIVITY ALL

2.18 INSTRUMENT MARKINGS

The pointers and scales of the instruments are marked as follows:

Left engine	1
Right engine	2
Main rotor/transmission	R
Main fuel tank	M
Supply fuel tank	S
Minimum and maximum limits	red radial
Calibration point	white radial
Start and transient limits	red dot
Normal/continuous range	green arc
Start and caution range	yellow arc
Never exceed speed – power-off	blue radial

2.18.1 Airspeed Indicator

10 to 45 kt	yellow arc
45 to 145 kt	green arc
145 kt	red radial
100 kt	blue radial

2.18.2 Engine RPM (N₁) Indicator

59 to 105%	green arc
105%	red radial

2.18.3 Triple Tachometer

Engine RPM (N₂) (power-on limits)

95%		red dot
98%		red radial
98 to 102%		green arc
102%		red radial
105%		red dot

EFFECTIVITY *Model C Variants***Rotor RPM (power-off limits)**

85%		red radial
85 to 110%		green arc
110%		red radial

EFFECTIVITY *Model S Variants***Rotor RPM (power-off limits)**

85%		red radial
85 to 104%		green arc
104%		red radial
110%		red dot

EFFECTIVITY *ALL*

2.18.4 Turbine Outlet Temperature (TOT) Indicator

0 to 779 °C		green arc
779 to 810 °C		yellow arc
810 °C		red radial
843 °C		red dot
927 °C		red dot

2.18.5 Triple Oil Pressure Indicator**Engine oil pressure**

3.5 kp/cm ²	red radial
3.5 to 6.3 kp/cm ²	yellow arc
6.3 to 9.1 kp/cm ²	green arc
9.1 kp/cm ²	red radial

Transmission oil pressure

0.5 kp/cm ²	red radial
0.5 to 0.8 kp/cm ²	yellow arc
0.8 to 2.0 kp/cm ²	green arc

2.18.6 Triple Oil Temperature Indicator**Engine oil temperature**

30 to 107 °C	green arc
107 °C	red radial

Transmission oil temperature

30 to 105 °C	green arc
105 °C	red radial

EFFECTIVITY *If extreme low temperature kit (P/N 105-80006) is installed and operational*

Engine oil temperature

−30 to 0 °C	yellow arc
0 to 107 °C	green arc
107 °C	red radial

Transmission oil temperature

−30 to 0 °C	yellow arc
0 to 105 °C	green arc
105 °C	red radial

EFFECTIVITY *ALL*

2.18.7 Dual Torque Indicator

0 to 86% green arc
86% red radial
95% red radial

2.18.8 Fuel Pressure Indicator

0 to 0.6 kp/cm² yellow arc
0.6 to 1.7 kp/cm² green arc
1.7 kp/cm² red radial

2.19 PLACARDS AND DECALS

The placards and decals depicted herein are typical and presented in either the English language or bilingual (German-English). Other languages may be used as a customer option. Slight formal differences from the installed placards and decals do not affect the information presented therein.

EFFECTIVITY *For IAC-AR registered helicopters only*

During flights in metric airspace, check the altitude using the conversion table placard (ft → m and hPa → in. Hg) located on the windshield center post in pilot's field-of-view.

EFFECTIVITY *All*

Text:

**THIS HELICOPTER MUST BE OPERATED
IN COMPLIANCE WITH THE OPERATING
LIMITATIONS SPECIFIED IN THE LBA
APPROVED ROTORCRAFT FLIGHT MANUAL**

**THE "AIRWORTHINESS
LIMITATIONS" SECTION OF THE
ROTORCRAFT MAINTENANCE
MANUAL MUST BE COMPLIED WITH.**

Location: In pilot's field-of-view

Text:

Pressure Altitude -ft-	OAT -°C-										
	-45	-30	-20	-10	0	+10	+20	+30	+40	+50	+54
0	145	145	145	145	145	145	145	140	140	135	135
2000	140	145	145	145	145	140	140	135	135	130	130
4000	135	145	145	140	140	135	135	130	130	125	
6000	130	140	140	135	135	130	130	125	120	115	
8000	125	135	135	130	130	125	120	115	110		
10000	120	130	130	125	120	115	110	105	100		
12000	115	125	120	115	110	105	100	95	90		
14000	110	115	110	105	100	95	90	85			
16000	105	105	100	95	90	85	80	80			
17000	100	100	95	90	85	80	80				
VNE -KNOTS IAS-											
VNE is reduced by 10 kt for any gross mass above 2300 kg											

Location: In pilot's field-of-view

Text:

**DURING GROUND OPERATION ONLY
SMALL CYCLIC STICK DISPLACE-
MENTS FOR FUNCTIONAL TESTS**

Location: In pilot's field-of-view

Text:

MAX. BODENLAST 600 kg/m² LADEGUT VERZURREN ZUL. BELASTUNG PRO VERZURRÖSE 100 kg ZUL. BELASTUNG PRO GEWINDEEINSATZ 200 kg MAX. WEIGHT PER sq. ft. 120 lbs CARGO TO BE SECURED MAX LOAD PER EYE 220 lbs MAX LOAD PER INSERT 440 lbs
--

Location: Cargo compartment

Text:

MAX ZULADUNG 20kg MAX LOAD 44lbs

Location: Avionics equipment bay

Text:

BEI FASS-ODER KANISTERBETANKUNG EINGEBAUTES SIEB VERWENDEN FOR BARREL OR GAS CAN REFUELING USE INSTALLED SCREEN BEI BETRIEB UNTER +4°C ENTEISUNGSZUSATZ NACH MIL-I-27686 VOLUMENANTEIL: MAX. 0,15%; MIN. 0,035% FOR OPERATION BELOW +4°C (40°F) ADD ANTI-ICING ADDITIVE MIL-I-27686 CONCENTRATION BY VOLUME: MAX. 0,15%; MIN. 0,035%

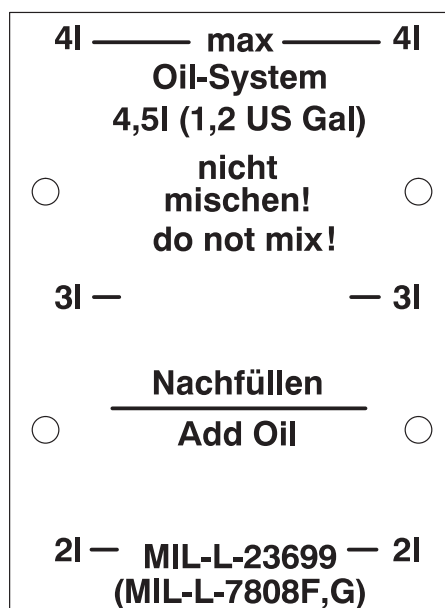
Location: Fuselage exterior, adjacent fuel tank filler neck

Text:

KRAFTSTOFF 570 LITER FUEL CAP. 150 US GAL MIL-T-5624: JP-4, JP-5 MIL-T-83133: JP-8 ASTM-D-1655: JET A, A-1, B

Location: Fuselage exterior, adjacent fuel tank filler neck

Text:



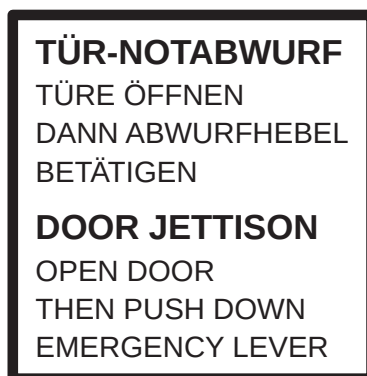
Location: Engine oil tank

Text:

ÖL 10,3 LITER
OIL 2.7 US GAL
MIL-L-23699

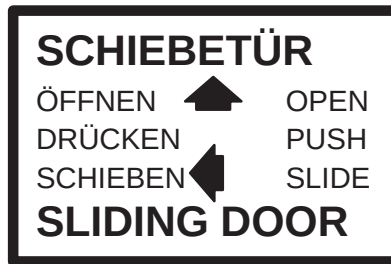
Location: Adjacent main transmission filler neck

Text:



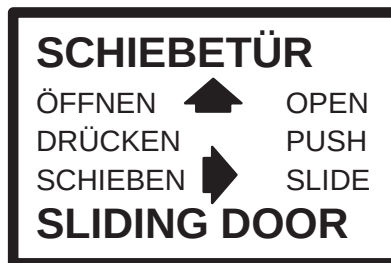
Location: Interior each cockpit door – above door handle

Text:



Location: Interior LH sliding door – above door handle

Text:



Location: Interior RH sliding door – above door handle

SECTION 3

EMERGENCY AND MALFUNCTION PROCEDURES

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SECTION 3

EMERGENCY AND MALFUNCTION PROCEDURES

3.1 GENERAL

This section contains the recommended procedures for coping with various types of emergencies, malfunctions and critical situations.

WARNING AFTER AN ACTUAL EMERGENCY OR MALFUNCTION MAKE AN ENTRY IN THE HELICOPTER LOGBOOK AND, WHEN NECESSARY, THE EFFECTED SYSTEM LOGBOOK (E.G. ENGINE LOGBOOK). MAINTENANCE ACTION MAY BE REQUIRED AND NECESSARY BEFORE THE NEXT FLIGHT.

For definitions of terms, abbreviations and symbols used in this section, refer to Section 1.

3.1.1 Basic Rules

These procedures deal with common emergencies. However, they do not prevent the pilot from taking additional action necessary to recover the emergency situation.

Although the procedures contained in this section are considered the best available, the pilot's sound judgement is of paramount importance when confronted with an emergency.

To assist the pilot during an inflight emergency, three basic rules have been established:

1. Maintain aircraft control
2. Analyse the situation
3. Take proper action.

NOTE It is impossible to establish a predetermined set of instructions that would provide a ready-made decision applicable to all situations.

3.1.2 Memory Items

Emergency procedure steps which shall be performed immediately without reference to either this manual or the pilot's checklist are written in boldface letters on a gray background (as shown here) and shall be committed to memory.

Therefore, those emergency procedure steps appearing without boldface letters may be accomplished referring to either this manual or the pilot's checklist, and when time and situation permit.

3.1.3 Operating Condition

The following terms are used in emergency procedures to describe the operating condition of a system, subsystem, assembly or component:

Affected Fails to operate in the normal or usual manner.

Normal Operates in the normal or usual manner.

3.1.4 Urgency of Landing

NOTE The type of emergency and the emergency conditions combined with the pilot's analysis of the condition of the helicopter and his proficiency are of prime importance in determining the urgency of a landing.

The following terms are used to reflect the degree of urgency of an emergency landing:

LAND IMMEDIATELY

The urgency of landing is paramount. Primary consideration is to assure survival of the occupants. Landing in water, trees or other unsafe areas should be considered only as a last resort.

LAND AS SOON AS POSSIBLE

Land without delay at the nearest adequate site (i.e. open field) at which a safe approach and landing can be made.

LAND AS SOON AS PRACTICABLE

The landing site and duration of flight are at the discretion of the pilot. Extended flight beyond the nearest approved landing area where appropriate assistance can be expected is not recommended.

3.2 WARNING AND CAUTION LIGHTS

Most emergency situations will be indicated by either a red warning light or a yellow caution light coming on.

A red warning light indicates a hazard which may require immediate corrective action.

A yellow caution light indicates the possible need for future corrective action.

It is always possible that a warning/caution light will unnecessarily come on. Whenever possible, check the light against its associated instrument to verify that an emergency condition has actually occurred.

Following is an alphabetical listing of the warning and caution light indications with the respective conditions, any further indications, and the emergency procedure.

3.2.1 Warning Light Indications

WARNING LIGHT INDICATIONS

BAT 60

Conditions/Indications

Battery temperature high (above 60°C)

Procedure

• ON GROUND

- | | |
|---------------|-------------------------|
| 1. Battery sw | – Off |
| 2. Engines | – Leave running in idle |

If warning light remains on for more than 5 minutes:

- | | |
|------------|-------------|
| 3. Engines | – Shut down |
|------------|-------------|

CAUTION BATTERY MUST BE INSPECTED OR REPLACED PRIOR TO NEXT FLIGHT.

• IN FLIGHT

- | | |
|--------------------------------|-------|
| 1. Battery sw | – Off |
| 2. LAND AS SOON AS PRACTICABLE | |

CAUTION BATTERY MUST BE INSPECTED OR REPLACED AT DESTINATION PRIOR TO NEXT FLIGHT.

WARNING LIGHT INDICATIONS**BAT 70****Conditions/Indications**

Battery overtemperature (above 70°C)

Procedure**• ON GROUND**

- | | |
|---------------|-------------|
| 1. Battery sw | – Off |
| 2. Engines | – Shut down |

CAUTION BATTERY MUST BE INSPECTED OR REPLACED PRIOR TO NEXT FLIGHT.

• IN FLIGHT

- | | |
|---------------|-------|
| 1. Battery sw | – Off |
|---------------|-------|

2. LAND AS SOON AS POSSIBLE

After landing:

- | | |
|--------------------------------------|-------------------------|
| 3. Engines | – Leave running in idle |
| 4. Collective lever and cyclic stick | – Lock |

NOTE Continue flight only if visual inspection reveals no indication of battery overheating. Leave battery off or disconnect battery.

CAUTION BATTERY MUST BE INSPECTED OR REPLACED AT DESTINATION PRIOR TO NEXT FLIGHT.

WARNING LIGHT INDICATIONS

(Engine 1)

or



(Engine 2)

Conditions/Indications

Overtemperature in engine compartment

Procedure**• ON GROUND**

- | | |
|-----------------------------------|--------------------------|
| 1. Both EMERG FUEL VALVE sw | – CLOSE |
| 2. Both FUEL PUMPS/SUPPLY TANK sw | – Off |
| 3. Both power levers | – OFF |
| 4. Battery sw | – Off |
| 5. Passengers | – Alert/Evacuate |
| 6. Fire | – Extinguish if possible |

• IN FLIGHT

- | | |
|--|---------|
| 1. EMERG FUEL VALVE sw (affected engine) | – CLOSE |
| 2. FUEL PUMPS/SUPPLY TANK sw (affected engine) | – Off |
| 3. Power lever (affected engine) | – OFF |
| 4. Passengers | – Alert |
| 5. LAND AS SOON AS POSSIBLE | |

After landing:

- | | |
|---------|--------------------------|
| 6. Fire | – Extinguish if possible |
|---------|--------------------------|

WARNING LIGHT INDICATIONS**FILT 1**

and/or

FILT 2**Conditions/Indications**

Engine fuel filter(s) clogged

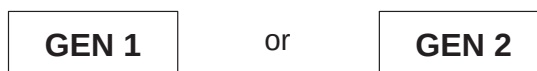
Procedure**• ONE WARNING LIGHT ON**

LAND AS SOON AS PRACTICABLE

Be prepared for single engine failure

• BOTH WARNING LIGHTS ON**LAND AS SOON AS POSSIBLE**

Be prepared for double engine failure

WARNING LIGHT INDICATIONS**Conditions/Indications**

Generator failure

- Voltmeter indication normal (27 - 29 V)
- **GEN** warning light (affected generator) on

Generator overvoltage

- Voltmeter indication above normal (30 V or more)
- **GEN** warning light (normal generator) on

NOTE Normal generator is disconnected by reverse current of affected generator.

Procedure

NOTE One generator alone will provide sufficient power for normal services. Non-essential Auxiliary Bus loads are shed automatically.

● GENERATOR FAILURE

- | | |
|--|---|
| 1. Generator field sw (affected generator) | – GEN TRIP, then hold in GEN RES for approx 2 sec and release |
|--|---|

If **GEN** warning light remains on:

- | | |
|--|---------------------------|
| 2. Generator field sw (affected generator) | – GEN TRIP |
| 3. STARTER/GENERATOR sw (affected generator) | – Off |
| 4. CURRENT IND sw | – Select normal generator |
| 5. Ammeter and voltmeter | – Monitor |

● GENERATOR OVERVOLTAGE

- | | |
|--|---------------------------|
| 1. Generator field sw (affected generator) | – GEN TRIP |
| 2. STARTER/GENERATOR sw (affected generator) | – Off |
| 3. GEN warning light (affected generator) | – Check on |
| 4. GEN warning light (normal generator) | – Check off |
| 5. CURRENT IND sw | – Select normal generator |
| 6. Ammeter and voltmeter | – Monitor |

WARNING LIGHT INDICATIONS**GEN 1**

and

GEN 2**Conditions/Indications**

Both generators have failed or are disconnected from the power distribution circuit.

Procedure

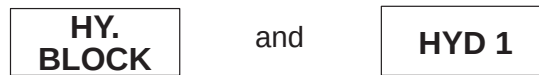
1. Each generator field sw in turn – GEN TRIP, then hold in GEN RES for approx 2 sec and release

If both **GEN** warning lights remain on:

2. Both generator field sw – GEN TRIP
3. Both STARTER/GENERATOR sw – Off
4. CURRENT IND sw – BUS BAR (BATT)
5. Electrical consumption – Reduce
6. Ammeter and voltmeter – Monitor
7. LAND AS SOON AS PRACTICABLE

Residual Battery Endurance					
Continuous load [A]	15	20	25	30	40
Time [min]	60	45	35	30	22
NOTE Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.					

WARNING A TOTAL ELECTRICAL FAILURE (LOSS OF BOTH GENERATORS AND ONBOARD BATTERY POWER) WILL LIMIT THE FUEL AVAILABLE TO THAT QUANTITY CONTAINED IN THE SUPPLY TANK AT THE TIME OF FAILURE.

WARNING LIGHT INDICATIONS**Conditions/Indications**

Servo jam in system 1 with automatic cross-over to system 2

- Increased force required to move the affected control axis

Procedure

- CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.
 - DURING ENGINE SHUTDOWN, BATTERY SHALL REMAIN ON UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS**HYD 1****Conditions/Indications**

System 1 pressure loss with automatic cross-over to system 2

Procedure

- CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.
 - DURING ENGINE SHUTDOWN, BATTERY SHALL REMAIN ON UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS

HYD 2

Conditions/Indications

System 2 pressure loss, system 1 retains power

Procedure

CAUTION • DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.

- AVOID ABRUPT MANEUVERS AND BANK ANGLES GREATER THAN 30°.

LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS**LIMIT**

MAST MOMENT INDICATOR

Conditions/Indications

MM indicator **red radial** exceeded

- **LIMIT** warning light on, however, goes off when pointer returns to within yellow arc

MM indicator **white radial** exceeded

- **LIMIT** warning light on and remains on

Procedure**● RED RADIAL EXCEEDED**

1. Control input – Reduce
2. **LIMIT** warning light – Off

● WHITE RADIAL EXCEEDED

1. Control input – Reduce
2. MM indicator – Maintain within green arc
3. LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS

LOW FUEL

Conditions/Indications

Failure of both main tank pumps, or
Main tank empty

- Supply tank fuel quantity below 60 kg (75 l)

Procedure

- | | |
|---|------------|
| 1. Supply tank fuel level | – Check |
| 2. Main tank fuel level | – Check |
| 3. Both fuel main pump circuit breakers | – Check in |
| 4. Both fuel main pump switches | – Check on |

If **LOW FUEL** warning light remains on:

5. LAND WITHIN 10 MINUTES

WARNING LIGHT INDICATIONS**MAG
PLUG 1**

or

**MAG
PLUG 2****Conditions/Indications**

Metal particles detected in engine oil

Procedure

WARNING A **MAG PLUG** WARNING LIGHT COMING ON TOGETHER WITH EITHER A TORQUE FLUCTUATION OR LOSS OF OIL PRESSURE SIGNIFIES AN IMPENDING ENGINE FAILURE. PERFORM SINGLE ENGINE EMERGENCY SHUTDOWN OF AFFECTED ENGINE IMMEDIATELY.

● ON GROUND

1. Single engine emergency shutdown – Perform

● IN FLIGHT

1. Single engine emergency shutdown – Perform

2. LAND AS SOON AS PRACTICABLE

If OEI operation is not possible:

3. LAND AS SOON AS POSSIBLE

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

NOTE For procedures applicable to helicopters 0001-0450 Before SB 60-37 see para 3.6.5, Oil Cooler Fan Failure.

WARNING LIGHT INDICATIONS

OIL COOL

Conditions/Indications

Failure of oil cooler fan

Procedure

• AT HOVER (IGE/OGE)

1. Engine and transmission oil temperatures – Monitor

2. LAND AS SOON AS POSSIBLE

• IN FLIGHT

1. Engine and transmission oil temperatures – Monitor

2. Airspeed – Attain as high as possible with minimum power required

3. LAND AS SOON AS PRACTICABLE

If any oil temperature nears limit:

4. LAND AS SOON AS POSSIBLE

NOTE Transmission oil temperature transient range of 105-120 °C has a 10-min limit.

EFFECTIVITY ALL

WARNING LIGHT INDICATIONS**R P M**

flashing

Conditions/Indications**N₁ Split**

- N₁ indicators show difference of 12% or more
- Audio signal - beeping tone

N_{R0} Low

- N_{R0} 95% or less
- Audio signal - beeping tone

N_{R0} High

- N_{R0} 103% or more
- Audio signal - none

NOTE If a two-stage warning unit is installed, a steady tone will come on with N_{R0} 108% or more.

Procedure● **N₁ S P L I T**

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Engine condition | – Analyze |

NOTE RPM warning light may come on during beep trim operation. Try to match N₁ until warning light goes off.

If residual torque is available:

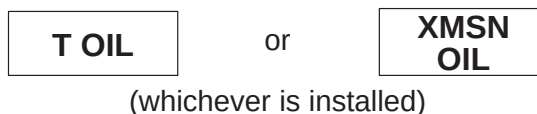
- | | |
|--------------------|-----------------|
| 3. Affected engine | – Leave running |
|--------------------|-----------------|

If residual torque is not available:

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

● **N_{R0} L O W / N_{R0} H I G H**

- | | |
|---------------------|---|
| 1. Collective lever | – Adjust as necessary to maintain N _{R0} within normal operating range |
|---------------------|---|

WARNING LIGHT INDICATIONS**Conditions/Indications**

Transmission oil **pressure** below minimum

- Transmission oil pressure indication 0.5 kp/cm² or less

Transmission oil **temperature** above maximum

- Transmission oil temperature indication 105 °C or more

Procedure

• **OIL PRESSURE BELOW MINIMUM**

1. LAND IMMEDIATELY

NOTE Descend with minimum power.

• **OIL TEMPERATURE ABOVE MAXIMUM**

NOTE Transmission oil temperature transient range of 105 - 120 °C has a 10-min limit.

- | | |
|------------------------------------|---|
| 1. Power | – Reduce |
| 2. Airspeed | – Attain as high as possible with minimum power required |
| 3. LAND AS SOON AS POSSIBLE | |

3.2.2 Caution Light Indications

EFFECTIVITY 0001-0160

CAUTION LIGHT INDICATIONS



EPU ON

EFFECTIVITY 0161-9999

CAUTION LIGHT INDICATIONS



EFFECTIVITY ALL

Conditions/Indications

External power is applied to the electrical distribution system

NOTE **EPU ON** caution light going off does not indicate that the EPU cable is disconnected.

Procedure

After EPU starts:

- | | |
|--------------------|----------------|
| 1. EPU cable | – Disconnected |
| 2. EPU access door | – Closed |

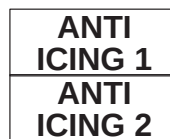
EFFECTIVITY 0001-0160

CAUTION LIGHT INDICATIONS



EFFECTIVITY 0161-9999

CAUTION LIGHT INDICATIONS



EFFECTIVITY ALL

Conditions/Indications

Engine anti-icing system in operation

- TOT increases by approximately 15 °C

Procedure

No corrective action necessary.

ENGINE EMERGENCY CONDITIONS

3.3 ENGINE EMERGENCY CONDITIONS

3.3.1 Single Engine Failure - Hover IGE

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

1. Collective lever	– Adjust to OEI-limits or below
2. Landing attitude	– Establish
3. Collective lever	– Raise as necessary to stop descent and cushion landing

After landing:

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

ENGINE EMERGENCY CONDITIONS

3.3.2 Single Engine Failure - Hover OGE

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
|---------------------|---------------------------------|

- | | |
|-------------|------------------------|
| 2. Airspeed | – Increase if possible |
|-------------|------------------------|

● FORCED LANDING

- | | |
|---------------------|-------------|
| 3. Landing attitude | – Establish |
|---------------------|-------------|

- | | |
|---------------------|--|
| 4. Collective lever | – Raise as necessary to stop descent and cushion landing |
|---------------------|--|

After landing:

- | | |
|--------------------------------------|-----------|
| 5. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

● TRANSITION TO OEI-FLIGHT

- | | |
|-------------|-----------------------------|
| 3. Airspeed | – 60 KIAS (V _Y) |
|-------------|-----------------------------|

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

5. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS

3.3.3 Single Engine Failure - Takeoff

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
|---------------------|---------------------------------|

● REJECTED TAKEOFF

- | | |
|---------------------|-------------|
| 2. Landing attitude | – Establish |
|---------------------|-------------|

- | | |
|---------------------|--|
| 3. Collective lever | – Raise as necessary to stop descent and cushion landing |
|---------------------|--|

After landing:

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

● TRANSITION TO OEI-FLIGHT

- | | |
|-------------|-----------------------------|
| 2. Airspeed | – 60 KIAS (V _Y) |
|-------------|-----------------------------|

- | | |
|--------------------------------------|-----------|
| 3. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

4. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS

3.3.4 Single Engine Failure - Flight

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|--|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Airspeed | – 60 KIAS (V _Y) |
| 3. Single engine emergency shutdown | – Perform |
| 4. LAND AS SOON AS PRACTICABLE | |

ENGINE EMERGENCY CONDITIONS

3.3.5 Single Engine Landing

Conditions/Indications

One engine inoperative (OEI)

Procedure

LANDING APPROACH

1. Airspeed – 60 KIAS (V_Y)
2. Shallow approach – Establish

ON FINAL

3. Airspeed – 40 KIAS
4. Rate of descent – 300 fpm

TOUCHDOWN

5. Airspeed – Reduce, depending on power available
6. Landing attitude – Establish
7. Collective lever – Raise as necessary to stop descent and cushion landing

ENGINE EMERGENCY CONDITIONS

3.3.6 Single Engine Emergency Shutdown

NOTE Before performing an inflight single engine emergency shutdown, determine if the situation will allow for OEI flight.

Make certain that:

- the controls of the affected engine are selected, and
- the collective lever is adjusted to maintain the normal engine within OEI-limits.

Procedure

1. Power lever (affected engine)	– IDLE, then OFF
2. EMER FUEL VALVE switch (affected engine)	– CLOSE
3. FUEL PUMPS/SUPPLY TANK switch (affected engine)	– Off

ENGINE EMERGENCY CONDITIONS

3.3.7 Inflight Restart

NOTE An inflight restart may be attempted after a flameout or shutdown subject to the pilot's evaluation of the cause of flameout.

Procedure

CAUTION DO NOT ATTEMPT INFLIGHT RESTART IF CAUSE OF ENGINE FAILURE IS OBVIOUSLY MECHANICAL.

- | | |
|------------------------------|--|
| 1. Collective lever friction | – Adjust to maintain position of lever when released |
| 2. Electrical consumption | – Reduce |
| 3. Power lever | – OFF |
| 4. STARTER/GENERATOR sw | – Off |
| 5. EMERG FUEL VALVE sw | – OPEN |
| 6. FUEL PUMPS/SUPPLY TANK sw | – On |
| 7. IGNITION sw | – On |
| 8. STARTER/GENERATOR sw | – START and hold |

CAUTION ENGINE GAS PRODUCER SECTION MAY LOCKUP AFTER AN INFLIGHT FLAMEOUT OR SHUTDOWN. THEREFORE, EXCEPT DURING AN EMERGENCY, INFLIGHT RESTART SHOULD NOT BE ATTEMPTED DURING THE TIME PERIOD BETWEEN ONE MINUTE AND TEN MINUTES AFTER FLAMEOUT OR SHUTDOWN.

At 15% N_1 :

- | | |
|-------------------------|-------------------------------------|
| 9. Power lever | – IDLE |
| 10. TOT and N_1 | – Monitor closely, observe TOT peak |
| 11. N_2 increase | – Monitor |
| 12. Engine oil pressure | – Positive indication |

At 58% N_1 :

- | | |
|---------------------------------|---------------|
| 13. STARTER/GENERATOR sw | – Release |
| 14. IGNITION sw | – Off |
| 15. STARTER/GENERATOR sw | – GEN ON |
| 16. Power lever | – FLIGHT |
| 17. Electrical consumers | – As required |
| 18. LAND AS SOON AS PRACTICABLE | |

If restart is not successful:

- | | |
|--------------------------------------|-----------|
| 19. Single engine emergency shutdown | – Perform |
|--------------------------------------|-----------|

ENGINE EMERGENCY CONDITIONS

3.3.8 Engine Overspeed - Driveshaft Failure

Conditions/Indications

- **RPM** warning (N_1 SPLIT) on

Affected engine:

- Torque decreases to zero
- N_1 decreases
- N_2 increases above N_{R0}

Normal engine:

- Torque, N_1 and TOT increase
- N_{R0} and N_2 decrease

Procedure

- | | |
|--|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Affected engine | – Identify |
| 3. Single engine emergency shutdown | – Perform |
| 4. LAND AS SOON AS PRACTICABLE | |

ENGINE EMERGENCY CONDITIONS

3.3.9 Engine Overspeed - Governor Failure

Conditions/Indications

- **RPM** warning may come on
- N_{R0} and both N_2 increase

Affected engine:

- Torque, N_1 and TOT increase

Normal engine:

- Torque, N_1 and TOT decrease

Procedure

- | | |
|----------------------------------|--|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Affected engine | – Identify |
| 3. Power lever (affected engine) | – Move towards IDLE until N_2 and N_{R0} stabilize in normal range |
| 4. LAND AS SOON AS PRACTICABLE | |

ENGINE EMERGENCY CONDITIONS

3.3.10 Engine Underspeed - Governor Failure

Conditions/Indications

- **RPM** warning may come on

Affected engine:

- Torque, N_1 and TOT decrease
- N_2 may decrease

Normal engine:

- Torque, N_1 and TOT increase
- N_{R0} and N_2 decrease

Procedure

1. Collective lever	– Adjust to OEI-limits or below
----------------------------	--

2. Affected engine	– Identify
---------------------------	-------------------

3. ENGINE TRIM switch	– Compensate N_2/N_{R0} drop
------------------------------	--

4. LAND AS SOON AS PRACTICABLE

If no residual torque (affected engine) is available:

5. Single engine emergency shut-down	– Perform
---	------------------

ENGINE EMERGENCY CONDITIONS

3.3.11 Compressor Stall

Conditions/Indications

- Dull humming or popping sounds
- Torque and N_1 indications erratic
- TOT fluctuates
- Possible light yaw jerk

Procedure

1. Collective lever	– Adjust to OEI-limits or below
2. Affected engine	– Identify
3. Power lever	– Move slowly to IDLE

If compressor stall is still evident:

4. Single engine emergency shut-down	– Perform
5. LAND AS SOON AS PRACTICABLE	

If compressor stall ends:

- | | |
|----------------|-------------------------|
| 4. Power lever | – Move slowly to FLIGHT |
|----------------|-------------------------|

- NOTE**
- Severe stalls can cause damage to engine and drive system components and must be handled as an emergency condition.
 - Stalls of less severe nature (one or two low intensity pops) may permit continued operation of the engine at a reduced power level thus avoiding the condition that caused the compressor stall.

ENGINE EMERGENCY CONDITIONS

3.3.12 Droop Compensation Failure

Conditions/Indications

- N_{R0} and both N_2 decrease when collective lever is raised
- N_{R0} and both N_2 increase when collective lever is lowered

Procedure

NOTE Avoid large collective lever changes.

1. Collective lever – Maintain N_{R0} within limits
2. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS

3.3.13 Engine Oil Pressure Low

Conditions/Indications

Affected engine:

- Oil pressure indication 3.5 kp/cm² or less

Procedure

1. Single engine emergency shutdown – Perform
2. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS**3.3.14 Engine Oil Temperature High****Conditions/Indications**

Affected engine:

- Oil temperature indication 107 °C or more

Procedure

- | | |
|---|------------------|
| 1. Power (affected engine) | – Reduce |
| 2. Oil temperature (affected engine) | – Monitor |
| 3. LAND AS SOON AS PRACTICABLE | |

If oil temperature remains above limit:

- | | |
|---|------------------|
| 4. Single engine emergency shut-down | – Perform |
|---|------------------|

ENGINE EMERGENCY CONDITIONS

3.3.15 Double Engine Failure - Hover IGE

Conditions/Indications

- Left yawing motion
- N_{R0} and both N_2 decrease
- **RPM** warning (N_{R0} LOW) on
- Both **GEN** warning lights on
- Engine instruments (both engines) indicate power loss

Procedure

1. Right pedal	– Apply as necessary to stop yaw
2. Landing attitude	– Establish
3. Collective lever	– Raise as necessary to cushion landing

ENGINE EMERGENCY CONDITIONS

3.3.16 Double Engine Failure - Flight

Conditions/Indications

- Left yawing motion
- N_{R0} and both N_2 decrease
- **RPM** warning (N_{R0} LOW) on
- Both **GEN** warning lights on
- Engine instruments (both engines) indicate power loss

Procedure

- | | |
|-----------------|-----------|
| 1. Autorotation | – Perform |
|-----------------|-----------|

ENGINE EMERGENCY CONDITIONS**3.3.17 Double Engine Emergency Shutdown****Procedure****• ON GROUND**

- | | |
|--------------------------------------|---------|
| 1. Both power levers | – OFF |
| 2. Both EMERG FUEL VALVE sw | – CLOSE |
| 3. Both FUEL PUMPS/SUPPLY TANK
SW | – Off |
| 4. Battery sw | – Off |

• IN FLIGHT

- | | |
|--------------------------------------|---------|
| 1. Both power levers | – OFF |
| 2. Both EMERG FUEL VALVE sw | – CLOSE |
| 3. Both FUEL PUMPS/SUPPLY TANK
SW | – Off |

ENGINE EMERGENCY CONDITIONS

3.3.18 Autorotation

Procedure

1. Collective lever	– Full down, maintain N_{R0} within limits
----------------------------	--

2. Airspeed	– 75 KIAS recommended
--------------------	------------------------------

NOTE Max range airspeed 100 KIAS at 85% N_{R0}
 Min rate-of-descent airspeed 60 KIAS at 100% N_{R0}

3. Double engine emergency shut-down	– Perform
---	------------------

AT APPROX 100 FT AGL:

4. Flare attitude	– Establish (approx 15° to 25°) to reduce forward speed and rate of descent; control N_{R0}
-------------------	---

AT APPROX 6 TO 10 FT AGL:

5. Flare attitude	– Reduce to approx 5°
6. Heading	– Maintain
7. Collective lever	– Raise to stop descent and cushion landing

8. Battery switch	– Off
--------------------------	--------------

NOTE The appropriate values must be adjusted according to prevailing conditions of gross mass, wind and terrain.

FIRE EMERGENCY CONDITIONS

3.4 FIRE EMERGENCY CONDITIONS

3.4.1 Cabin/Cargo Compartment Fire

Conditions/Indications

- Smoke, burning odor, flames

Procedure

● ON GROUND

- | | |
|--------------------------------------|--------------------------|
| 1. Passengers | – Alert/Evacuate |
| 2. Double engine emergency shut-down | – Perform |
| 3. Fire | – Extinguish if possible |

● IN FLIGHT

- | | |
|-----------------------------|--|
| 1. Passengers | – Alert |
| 2. Fire | – Extinguish if possible |
| 3. Toxic fumes, smoke | – Eliminate; Open sliding doors, windows and vents |
| 4. LAND AS SOON AS POSSIBLE | |

After landing:

- | | |
|--------------------------------------|-----------|
| 5. Double engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

FIRE EMERGENCY CONDITIONS

3.4.2 Electrical Fire

Conditions/Indications

- Odor of burning insulation and/or acrid smoke

Procedure

● ON GROUND

1. Passengers	– Alert/Evacuate
2. Double engine emergency shut-down	– Perform
3. EPU, if connected	– Disconnect
4. Fire	– Extinguish if possible

● IN FLIGHT

1. Passengers	– Alert
2. Both generator field sw	– GEN TRIP
3. Both STARTER/GENERATOR sw	– Off
4. Battery sw	– Off
5. Fire	– Extinguish if possible
6. Electrical consumption	– Reduce
7. Battery sw	– On
8. CURRENT IND sw	– BUS BAR (BATT)
9. Ammeter and voltmeter	– Monitor
10. LAND AS SOON AS PRACTICABLE	

Residual Battery Endurance					
Continuous load [A]	15	20	25	30	40
Time [min]	60	45	35	30	22
NOTE Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.					

If indications of electrical fire continue:

11. Battery sw	– Off if flight conditions permit
12. LAND AS SOON AS POSSIBLE	

TAIL ROTOR FAILURE CONDITIONS

3.5 TAIL ROTOR FAILURE CONDITIONS

3.5.1 Tail Rotor Drive Failure - Hover

Conditions/Indications

Complete loss of tail rotor thrust

- Rapid uncontrolled right yawing motion

Procedure

• HOVER IGE

- | | |
|---------------------|-------------|
| 1. Collective lever | – Full down |
|---------------------|-------------|

and simultaneously

- | | |
|---------------------|-------------|
| 2. Landing attitude | – Establish |
|---------------------|-------------|

After landing:

- | | |
|--------------------------------------|-----------|
| 3. Double engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

• HOVER OGE

- | | |
|---------------------|-------------|
| 1. Collective lever | – Full down |
|---------------------|-------------|

If height permits:

- | | |
|-------------|--------|
| 2. Airspeed | – Gain |
|-------------|--------|

- | | |
|--------------------------------------|-----------|
| 3. Double engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

- | | |
|---------------------|---|
| 4. Collective lever | – Raise to stop descent and cushion landing |
|---------------------|---|

TAIL ROTOR FAILURE CONDITIONS

3.5.2 Tail Rotor Drive Failure - Flight

Conditions/Indications

Complete loss of tail rotor thrust which may be caused by a severed drive shaft and/or separation of the tail rotor assembly from the helicopter.

- Rapid right yawing motion eventually combined with a right roll and downward pitch.

Procedure

WARNING EXTENDED FLIGHT IS NOT POSSIBLE AFTER TAIL ROTOR DRIVE SYSTEM FAILURE. AUTOROTATION MUST BE ENTERED IMMEDIATELY.

1. Collective lever	– Full down immediately, maintain N_{RO} within limits
----------------------------	--

2. Airspeed	– 75 KIAS recommended
--------------------	------------------------------

NOTE Airspeed indication may be unreliable due to right yaw flight condition.

3. Double engine emergency shut-down	– Perform
---	------------------

WARNING LANDING WITHOUT THE TAIL ROTOR OPERATING IS HAZARDOUS AS THE HELICOPTER WILL TOUCH DOWN IN AN EXTREME YAW ATTITUDE AND PROBABLY ROLL OVER.

4. Modified side flare	– As necessary to obtain zero groundspeed upon touchdown
-------------------------------	---

NOTE The rate of descent will be higher than normal due to the extreme yaw attitude and will require a modified side flare to reduce the groundspeed to as near zero as possible to prevent the helicopter from rolling over upon touchdown.

5. Collective lever	– Raise to stop descent and cushion landing
----------------------------	--

TAIL ROTOR FAILURE CONDITIONS

3.5.3 Fixed-pitch Tail Rotor Control Failures

Conditions/Indications

Fixed-pitch tail rotor failure

- No directional response after pedal movement
- Locked pedals

Procedure

The emergency procedures for a tail rotor control failure will vary depending on flight conditions, power setting and mass of the helicopter.

Lowering or raising the collective lever will result in a left or right yawing motion to a degree dependent upon the tail rotor thrust remaining. Additionally, at high airspeeds, an ever increasing yaw oscillation may occur, however, is dampened by reducing airspeed to approx 60 KIAS and, if possible, increasing the power setting.

Approach and landing should be made with a suitable rate of descent, controlling sideslip angle with the collective lever. If the situation permits, a crosswind landing should be performed to minimize the amount of sideslip. Before touchdown, the ground speed should be reduced to a minimum. Touchdown so that the skids are aligned with the direction of landing.

Right Pedal Fixed-pitch Settings

Raising collective lever will result in a right yawing motion. Lower collective lever to control sideslip and touch down with forward speed.

Left Pedal Fixed-pitch Settings

Lowering collective lever will result in a left yawing motion. Approach with a right sideslip and reduce ground speed to allow for raising the collective lever as necessary to touchdown.

TAIL ROTOR FAILURE CONDITIONS**3.5.4 Pedal Vibrations****Conditions/Indications**

Impending tail rotor system failure

- Unusual pedal vibrations

Procedure

LAND AS SOON AS POSSIBLE

SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

3.6 SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

3.6.1 Electrical Short Circuit

Conditions/Indications

- Voltmeter indicates low voltage
- Ammeter may indicate excessive current
- Failure of equipment powered by affected bus
- **GEN 1** and/or **GEN 2** warning light may come on

Procedure

1. Both generator field sw – **GEN TRIP**
2. Both STARTER/GENERATOR sw – **Off**
3. CURRENT IND sw – BUS BAR (BATT)
4. Electrical consumption – Reduce
5. Ammeter and voltmeter – Monitor
6. LAND AS SOON AS PRACTICABLE

Residual Battery Endurance					
Continuous load [A]	15	20	25	30	40
Time [min]	60	45	35	30	22
NOTE Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.					

If electrical short circuit indications continue:

7. Battery sw – **Off if flight conditions permit**
8. LAND AS SOON AS POSSIBLE

SYSTEM EMERGENCY/MALFUNCTION CONDITIONS**3.6.2 Cyclic Trim Actuator Failure/Runaway****Conditions/Indications**

- Unsymmetrical cyclic stick forces

NOTE Cyclic stick full travel remains available.

Procedure

CAUTION DURING ENGINE SHUTDOWN, HOLD OR LOCK CYCLIC STICK IN NEUTRAL POSITION UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE

SYSTEM EMERGENCY/MALFUNCTION CONDITIONS**3.6.3 Supply Tank Fuel Pump Failure****Conditions/Indications**

Affected engine:

- Fuel pressure indication below 0.6 kp/cm²

Procedure

1. FUEL PUMPS/SUPPLY TANK sw (affected engine) – Off
2. Altitude – 13,700 ft or less
3. Bank angles above 45° and abrupt maneuvers – Avoid
4. LAND AS SOON AS PRACTICABLE

SYSTEM EMERGENCY/MALFUNCTION CONDITIONS**3.6.4 Servo Jam - Hydraulic System 2****Conditions/Indications**

- Increased force required to move the affected control axis

Procedure

CAUTION • DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.

- AVOID ABRUPT MANEUVERS AND BANK ANGLES GREATER THAN 30°.

LAND AS SOON AS PRACTICABLE

EFFECTIVITY 0001-0450 Before SB 60-37

SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

3.6.5 Oil Cooler Fan Failure

Conditions/Indications

Failure of oil cooler fan

- Rapid oil temperature increase of both engines followed by transmission oil temperature increase.

Procedure

• AT HOVER (IGE/OGE)

1. LAND AS SOON AS POSSIBLE

• IN FLIGHT

1. Airspeed
 - Attain as high as possible with minimum power required
2. LAND AS SOON AS PRACTICABLE

If any oil temperature nears limit:

3. LAND AS SOON AS POSSIBLE

NOTE Transmission oil temperature transient range of 105-120 °C has a 10-min limit.

EFFECTIVITY ALL

SECTION 4

NORMAL PROCEDURES

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SECTION 4

NORMAL PROCEDURES

4.1 GENERAL

This section contains instructions and recommended procedures that are peculiar to the operation of this helicopter.

For definition of terms, abbreviations and symbols used in this section refer to Section 1.

4.2 PREPARATION FOR FLIGHT

4.2.1 Flight Restrictions

The minimum, normal, maximum and cautionary operation ranges for the helicopter and its subsystems are indicated by instrument markings, placards and decals.

For helicopter and subsystem restrictions refer to Section 2, Limitations.

4.2.2 Flight Planning

Refer to Sections 5 and 9 to determine required fuel, airspeeds and power settings for takeoff, climb, cruise, and hovering and landing data necessary to accomplish the mission.

4.2.3 Mass and Balance

The takeoff and anticipated landing gross mass and balance should be obtained before takeoff and checked against mass and load limits, and center of gravity limitations (see Section 2).

4.3 PREFLIGHT CHECK

4.3.1 General

The preflight check shall be accomplished, at the latest, before the first flight of the day and according to either the Flight Manual, the Maintenance Manual or the Pilot's Checklist.

The preflight check is not a detailed mechanical inspection, but essentially a visual check of the helicopter for correct condition.

Items marked with a star (★) need to be checked each flight before takeoff.

When unusual local conditions dictate, the extent and/or frequency of this check shall be increased as necessary to promote safe operation.

NOTE • The following list contains only check items for the clean configuration.

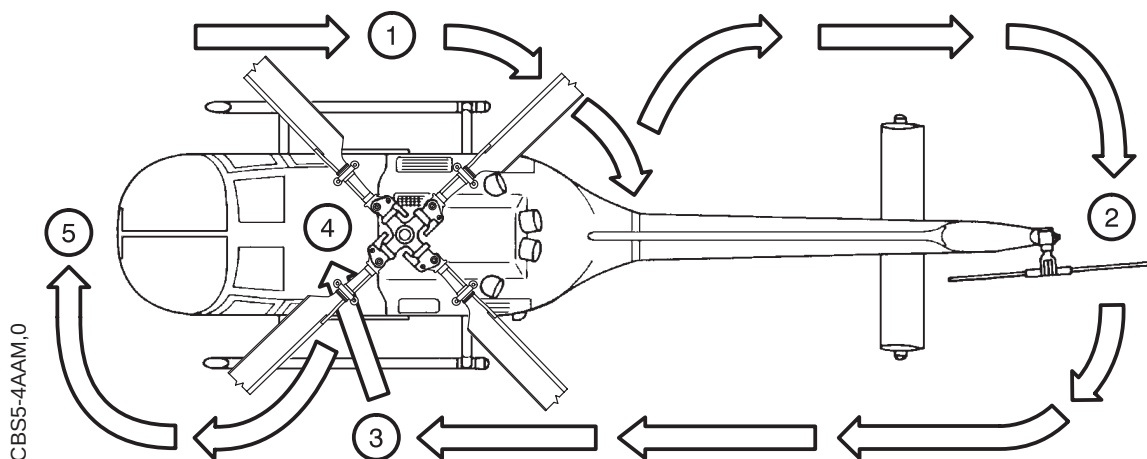
- In addition to these items, check antennas and all installed optional equipment. Make certain that relevant intermediate and special inspections, in accordance with the Maintenance Manual, are complied with.
- For optional equipment check items refer to the respective Flight Manual Supplement or Maintenance Manual Chapter 800.

4.3.2 Exterior Check

The exterior check is laid out as a walk-around check, starting forward right at the pilot's door, proceeding clockwise to the tail boom, to the left hand side (including the upper and lower areas of the helicopter) and being completed at the helicopter nose area.

NOTE • The helicopter should be headed into the wind before starting engines.

- The area around the helicopter should be clear of all foreign objects.
- To avoid excessive drain on the helicopter battery, particularly during cold weather, all ground operations should be conducted using an external power unit (EPU).
- When the battery is used, the operation of electrical equipment should be kept to a minimum.



- | | |
|--------------------------|--------------|
| 1. FUSELAGE - RIGHT SIDE | 4. CABIN TOP |
| 2. TAIL BOOM | 5. NOSE |
| 3. FUSELAGE - LEFT SIDE | |

Fig. 4-1 Exterior Check Sequence

- ★ Helicopter forms and documents
 - Check, complete
- Fuselage underside
 - Condition, no fuel leaks
- Each fuel tank (3)
 - Drain sample
- When OAT is above 40 °C or below –32 °C:
 - Fuel temperature
 - Measure; determine fuel type to be used and/or the permissible altitude in accordance with Section 2.
- ★ Covers and tie-downs
 - Removed
- ★ Ice and snow (if any)
 - Removed
- ★ Ground handling wheels
 - Removed

Cockpit

- All switches
 - Off
- Power levers
 - OFF
- Rotor brake lever (if installed)
 - Full down
- Circuit breakers
 - As required
- Battery switch
 - On
- Voltmeter indication
 - 24V DC nominal
- If EPU is used:
 - Battery switch
 - Off
 - Pitot tube heating
 - Function

Lighting:

- Position lights
 - Function
- Anti-collision lights
 - Function
- Landing light(s)
 - Function

Before Night Flights:

- Dome and instrument lights
 - Function
- Utility lamp
 - Function
- Hand lamp
 - Function
- Battery switch
 - Off
- Hand fire extinguisher
 - On board, check pressure
- Flashlight
 - On board, function

First aid kit (if installed here)

– On board

Fuselage - Right Side

Cockpit overhead window

– Condition

★ OAT indicator probe

– Condition

Cockpit door

– Condition, function, jettison handle secured

Pilot seat and safety belts

– Condition

★ Static port

– Clean, unobstructed

★ Pitot tube(s)

– Condition, clean, unobstructed

Landing gear and step

– Condition

Cabin door

– Condition, function

★ Baggage, cargo, loose items

– Stowed, secured

Passenger seats and seat belts

– Condition

★ Transmission compartment

– No foreign objects, no leakage, free of ice/snow

Oil cooler

– No leakage

★ Engine oil levels

– Above ADD OIL mark

Engine oil tanks

– Security of mounting, no leakage, filler caps secured

★ Transmission oil level

– Above MIN mark

Transmission

– Condition, security of mounting, no leakage, filler cap closed

Rotor brake (if installed)

– Condition

★ Engine air intake

– Condition, no dirt, no foreign objects, free of ice/snow

Compressor

– Rotate by hand, check for smooth rotation

★ Transmission access door

– Closed, secured

★ Engine compartment

– No foreign objects, no fuel/oil leakage

★ Engine compartment ventilation air intake

– Clear

Engine

– Condition

Wiring, linkages and lines

– Condition, no leakage, no chafing

Engine mounts	– Condition, security of mounting
N ₁ -governor and connections	– Security of mounting
Starter/Generator	– Security of mounting
★ Engine access door	– Closed, secured
Drain and vent ports	– Clear
Battery (<i>Model S Variants</i>)	– Condition, no electrolyte spillage, security of mounting
Battery connectors (2)	– Connected
Battery vents	– Clear
★ Battery compartment door	– Closed, secured
Engine cowling	– Secured
Engine exhaust pipe	– Condition, security of mounting
★ Fuselage exterior	– Condition
Clam shell doors and latches	– Condition, function
First aid kit (if installed here)	– On board
Avionics (in avionics equipment bay)	– Security of mounting
★ Baggage, cargo, loose items	– Stowed, secured
Avionics bay door	– Closed
★ Clam shell doors	– Closed, secured
Spoiler	– Condition
Tail Boom	
★ Tail boom - RH	– Condition
★ Tail-rotor driveshaft fairing	– Secured
Drain ports	– Clear
★ RH horizontal and vertical stabilizer, position light (green)	– Condition
Vertical fin, position light (white), anti-collision light	– Condition
Vertical fin cowling	– Secured
★ Tail rotor	– Condition
PU Erosion-protective film (if fitted)	– Condition, no separation

- | | |
|------------------------|------------------------------------|
| ★ Intermediate gearbox | – Oil level, no leakage |
| ★ Tail-rotor gearbox | – Condition, oil level, no leakage |

NOTE Oil levels may be easier to verify by grasping tail skid and shortly shaking tail boom.

- | | |
|---|-------------------|
| ★ Intermediate gearbox access door (if installed) | – Closed, secured |
| Tail skid | – Condition |
| ★ LH horizontal and vertical stabilizer, position light (red) | – Condition |
| ★ Tail boom - LH | – Condition |
| ★ Tail-rotor driveshaft fairing | – Secured |
| Drain ports | – Clear |

Fuselage - Left Side

- | | |
|--|--|
| ★ Fuel filler cap | – Closed, secured |
| Drain and vent ports | – Clear |
| ★ Fuselage exterior | – Condition |
| Engine exhaust pipe | – Condition, security of mounting |
| Engine cowling | – Secured |
| ★ Engine compartment | – No foreign objects, no fuel/oil leakage |
| ★ Engine compartment ventilation air intake | – Clear |
| Engine | – Condition |
| Wiring, linkages and lines | – Condition, no leakage, no chafing |
| Engine mounts | – Condition, security of mounting |
| N ₂ -governor and actuator | – Condition, security of mounting |
| Starter/Generator | – Security of mounting |
| ★ Engine access door | – Closed, secured |
| ★ Hydraulic compartment | – No foreign objects, no leakage, free of ice/snow |
| ★ Hydraulic module | – No leakage |
| ★ Area between hydraulic module and XMSN floor, especially control rod bellows | – Free of ice/snow |
| Water drain ports | – Clear |

Hydraulic filter clogging indicator pins (4)	– Check in
★ Hydraulic reservoirs (2)	
Oil level	– Above FILL mark
Trapped air indicator	– Not showing
★ Service-fitting caps	– Installed
Transmission	– Condition, security of mounting, no leakage
★ Engine air intake	– Condition, no dirt, no foreign objects, free of ice/snow
Compressor	– Rotate by hand, check for smooth rotation
★ Hydraulic compartment access door	– Closed, secured
Main rotor	– Rotate by hand in direction of rotation, check for smooth rotation
★ Rotor hub oil level	– Between MIN and MAX marks
★ Rotor blades	– Condition
★ Rotor blade attachment bolts	– Secured
PU Erosion-protective film (if fitted)	– Condition, no separation
★ Driving link assy, mixing lever assy	– Condition, secured
★ Rotating control rods	– Condition, free movement, secured
Rotating control rod spherical bearings	– Smooth operation
Swashplate and boot	– Condition, secured
Pendulum absorbers	– Condition, free movement
★ Air intake	– Condition, no foreign objects, free of ice/snow
Engine inlet deflector shield (if installed)	– Condition, no foreign objects
Cockpit overhead window	– Condition
Cabin door	– Condition, function
★ Baggage, cargo, loose items	– Stowed, secured
Passenger seats and seat belts	– Condition
Fuel vent port	– Clear

- Landing gear and step – Condition
- ★ Static port – Clean, unobstructed
- Cockpit door – Condition, function, jettison handle secured
- Copilot cyclic stick (if installed) – Secured, safety-wired
- Copilot collective lever (if installed) – Secured
- Copilot seat and safety belts – Condition
- ★ If copilot seat is unoccupied:
 - Copilot safety belts – Fastened, tightened

Nose

- Windshields – Condition
- Battery (*Model C Variants*) – Condition, no electrolyte spillage, security of mounting
 - Battery connectors (2) – Connected
 - Battery vents – Clear
- Nose compartment – No foreign objects
- ★ Nose compartment door – Closed, secured
- Cockpit air intake screens – Clear
- Drain ports – Clear
- Lower cockpit windows – Condition
- Fuselage bottom – Condition
- Anti-collision light – Condition
- Landing light(s) – Condition

4.3.3 Interior Check

- ★ Seat and pedals – Adjust
- ★ Safety belts – Fasten, adjust
- ★ Shoulder harness locks – Function
- ★ Instrument panel:
 - All instruments – Check and set

★ Switch panel:

- | | |
|---------------------------|---------------|
| EMERG FUEL VALVE switches | – Guarded |
| Circuit breakers | – As required |
| All switches | – Off |
| COMM/NAV equipment | – Off |

★ Overhead panel:

- | | |
|------------------|---------------|
| Power levers | – OFF |
| All switches | – Off |
| Circuit breakers | – As required |

★ Collective lever panel:

- | | |
|--------------|-------|
| All switches | – Off |
|--------------|-------|

Pre-start Check

★ Battery switch

- | |
|------|
| – On |
|------|

For EPU starts:

- | | |
|-----------------------------|-------------|
| Battery switch | – Off |
| EPU | – Connected |
| EPU ON caution light | – On |

★ Voltmeter indication :

- | | |
|---------|------------------|
| Battery | – Minimum 24V |
| EPU | – Between 24-28V |

★ Warning/caution lights

- | |
|---|
| – Test; all warning and caution lights (except both engine Fire, MM indicator LIMIT and LOW FUEL warning lights) must come on |
|---|

★ Both engine Fire warning lights

- | |
|--------|
| – Test |
|--------|

★ Mast moment indicator

- | |
|--------|
| – Test |
|--------|

★ LOW FUEL TEST pushbutton

- | |
|---|
| – Press; LOW FUEL warning light must come on |
|---|

★ Fuel quantity indicator

- | |
|------------------|
| – Check readings |
|------------------|

★ Fuel pumps (4)

- | |
|------------------|
| – Check function |
|------------------|

★ Cyclic stick lock (if fitted)

- | |
|----------|
| – Stowed |
|----------|

★ Collective lever lock

- | |
|-----------|
| – Release |
|-----------|

★ Flight controls

- | |
|--|
| – Check for freedom of movement, full travel |
|--|

Hydraulic override (HY BLOCK)– Check function

NOTE To achieve correct results, cyclic stick must be moved exactly along the longitudinal and lateral axes of the helicopter.

- Collective lever – Raise slightly
- HYDR TEST switch – RESET and release
- HY BLOCK** warning light – Check off
- Collective lever – Raise until **HY BLOCK** warning light comes on, then lower slightly and release
- HYDR TEST switch – RESET and release
- HY BLOCK** warning light – Check off
- Collective lever – Lower until **HY BLOCK** warning light comes on, then raise slightly and release
- HYDR TEST switch – RESET and release
- HY BLOCK** warning light – Check off
- Cyclic stick – Move forward until **HY BLOCK** warning light comes on and release
- HYDR TEST switch – RESET and release
- HY BLOCK** warning light – Check off
- Each remaining cyclic control direction – Check as above
- Collective lever – Full down

- ★ Collective lever lock – Engage
- ★ **HY BLOCK** warning light – Check on

Cyclic trim system– Check function

NOTE This check ensures that cyclic stick full travel is still guaranteed after the trim actuators have reached their full travel stop.

- Cyclic trim switch – Press forward and hold until actuator audibly stops
- Cyclic stick – Move fore and aft to limit stops; check for full fore/aft travel and free movement
- Each remaining cyclic control direction – Check as above

- ★ Cyclic stick – Trim neutral

- ★ Cyclic stick lock (if fitted)
- ★ Rotor brake lever (if fitted)
- Engage
- Full down

4.4 STARTING ENGINES

- NOTE**
- At ambient temperatures below +4 °C, refer to “Cold weather starting information” this section.
 - Either engine may be started first.
 - If, for any reason, a starting attempt is discontinued the entire starting sequence must be repeated from the beginning.
 - When a starting attempt is aborted but TOT limits were not exceeded, wait 15 seconds after N_1 has returned to zero before attempting restart to permit excess fuel to drain from combustion chamber.
 - If a start is not successful after three attempts, maintenance action is required.
 - Before starting engines, the operational hydraulic system shall always be pre-selected to system 2 (collective lever lock engaged).

Fire guard (if available)	– Posted
Rotor area	– Clear
Anti-collision light	– On

4.4.1 Starting First Engine

NOTE Operate only the controls of the corresponding engine.

CAUTION IMMEDIATELY ABORT START FOR ANY OF THE FOLLOWING:

- MAIN ROTOR IS NOT ROTATING AT 25% N_1 . (A SECOND AND THIRD START ATTEMPT MAY BE MADE. IF POWER TURBINE STILL FAILS TO ROTATE, DISCONTINUE ATTEMPT).
- STAGNATED START (FUEL STARVATION).
- TOT RISES ABNORMALLY AND RAPIDLY APPROACHES START LIMIT (927 °C).
- TOT EXCEEDS 10-SECOND LIMIT WITHIN TEMPERATURE RANGE (811-927 °C).
- NO POSITIVE ENGINE OR TRANSMISSION OIL PRESSURE INDICATIONS UPON REACHING IDLE SPEED (N_1 59-65%).
- IDLE SPEED IS NOT REACHED WITHIN 60 SECONDS UNLESS N_1 AND N_2 ARE ACCELERATING AND TOT IS WITHIN LIMITS.
- N_2 INCREASES BEYOND N_{R0} . A FREEWHEELING UNIT SLIPPAGE IS SUSPECTED AND MAINTENANCE ACTION IS REQUIRED.
- AUTO-ACCELERATION (BENDIX FUEL CONTROL SYSTEM INSTALLED). SEE “COLD WEATHER STARTING INFORMATION” THIS SECTION.
- **HYD 2** WARNING LIGHT REMAINS ON UPON REACHING GROUND IDLE SPEED (60% N_{R0}).

ABORT START PROCEDURE

POWER LEVER	– OFF
FUEL PUMPS	– OFF
IGNITION SWITCH	– OFF
STARTER SWITCH	– ON, AS NECESSARY TO LOWER TOT
STARTER SWITCH	– OFF
TOT	– Less than 150 °C, otherwise ventilate engine
Fuel pumps (4)	– On
Ignition switch	– On
Starter switch	– On; start clock (stopwatch)
N ₁	– Check increasing

CAUTION POWER LEVER MUST NOT BE MOVED FROM OFF POSITION UNTIL REQUIRED N₁ HAS BEEN REACHED. STARTING AT N₁ SPEEDS LESS THAN 12% INCREASES THE POSSIBILITY OF EXCEEDING ENGINE TOT LIMITS.

As N₁ accelerates through 12-15%:

If CECO fuel control installed:

Power lever	– <u>Slowly</u> advance to a position slightly before IDLE (trim start)
-------------	---

If BENDIX fuel control installed:

Power lever	– Move <u>direct</u> to a position slightly before IDLE
-------------	---

NOTE Even small displacements of power lever from IDLE to OFF position may cause engine to flameout.

NOTE Do not wait for N₁ to peak out. Move power lever towards IDLE immediately upon reaching the desired N₁ speed. Delay in moving power lever may diminish battery capacity early in the start cycle.

TOT and N ₁	– Monitor closely, observe TOT peak
Power lever	– Lock in IDLE
N ₂ /N _{R0}	– Check increasing
Engine oil pressure indicators	– Check positive indication
At 58% N ₁ :	
Starter switch	– Off
Ignition switch	– Off

When N_2/N_{R0} are above 40%:

Generator switch — On; monitor N_1

If N_1 decreases to below 59%:

Generator — Off
Power lever — Advance, N_1 max 70%
Generator — On

4.4.2 Starting Second Engine

CURRENT IND switch — Select operating generator
Ammeter (battery start) — Check below 100 A
Second engine — Start following procedure for first engine
After EPU starts:
Battery switch — On
EPU — Check disconnected

4.5 SYSTEM CHECKS

4.5.1 Hydraulic Checks

WARNING BOTH HYDR SYSTEMS SHALL BE OPERATIONAL. TAKEOFF WITH A KNOWN OR SUSPECTED HYDR SYSTEM FAILURE IS PROHIBITED.

TAKEOFF WITH SYSTEM 2 SELECTED AS THE OPERATING SYSTEM IS PROHIBITED (THERE IS NO AUTOMATIC CROSS-OVER FROM SYSTEM 2 TO SYSTEM 1).

DO NOT OPERATE THE HYDRAULIC TEST SWITCH DURING FLIGHT.

NOTE Hydraulic system checks may be performed between starting first and second engine at pilot's discretion.

HYD 2 warning light — Off

System 2 Check

HYD 1 and **HY BLOCK** warning lights — Check on
Cyclic stick lock (if fitted) — Release, stow
Cyclic stick — Check ease of movement (not more than 3 cm from neutral) all four directions, return to neutral
Collective lever lock — Release

- Collective lever – Check ease of movement, return to full down

Cross-over Check

- HYDR TEST switch – Place in RESET, then release
- HYD 1** and **HY BLOCK** warning lights – Off

NOTE Helicopters having optional hydraulic-powered equipment or systems; the **HYD 2** warning light may come on for approx 2 seconds.

- HYDR TEST switch guard – Close

System 1 Check

- Cyclic stick – Check ease of movement all four directions, return to neutral
- Collective lever – Check ease of movement, return to full down and lock
- HYD 1**, **HYD 2** and **HY BLOCK** warning lights – Check off

4.5.2 Electrical and Avionic Checks

Electrical system

- Functional check
- Both generator field switches – GEN TRIP; both **GEN** warning lights come on
- Gen 1 generator field switch – Hold in GEN RES for approx 2 sec; **GEN 1** warning light goes off
- Gen 2 generator field switch – Hold in GEN RES for approx 2 sec; **GEN 2** warning light goes off
- CURRENT IND switch – Select each GEN position and check that the loads are approx equal, leave switch in BUS BAR (BAT) position
- ★ COMM/NAV equipment – On and check
- ★ All other instruments and equipment – Check and set

4.5.3 Miscellaneous Checks

- ENG ANTI-ICING system – Check as required
- ENG. 1/ENG. 2 switches – On
- ANTI ICING 1/ANTI ICING 2** or **L/R** caution lights – On
- TOT – Note increase
- ENG. 1/ENG. 2 switches – Off

- ANTI ICING 1/ANTI ICING 2 or L/R** – Off
caution lights
- TOT** – Note decrease

4.6 ★ **PRE-TAKEOFF CHECK**

4.6.1 ★ **On Ground**

- Crew and passengers – Briefed
- Passenger safety belts – Fastened
- Doors – Closed, locked
- Warning/caution lights – Off

NOTE **OIL COOL** warning light (if installed) may remain on until after placing power levers full forward.

- Cyclic stick lock (if fitted) – Check stowed

CAUTION IF ENGINES HAVE BEEN SHUT DOWN FOR MORE THAN 15 MINUTES, STABILIZE AT IDLE SPEED FOR 1 MINUTE.

- Power levers – Smoothly to FLIGHT
- OIL COOL** warning light (if installed) – Check off
- N₂ – Trim to 98%
- Torque – Synchronize
- Engine and XMSN instruments – Check in green range
- Fuel quantity indicator – Check readings
- Bleed air consumers – As required
- Optional equipment controls – As required
- Collective lever lock – Check released
- Collective lever friction – Adjust as required

4.6.2 ★ **At Hover**

- Torque – Synchronize
- Hover power – Check
- N₂/N_{R0} – Adjust to 98-102%

NOTE 100% rotor RPM recommended for steady flight condition.

- Engine and XMSN instruments – Check in green range
- All warning/caution lights – Off

4.7 CRUISE CHECK

NOTE Recommended maximum values for use under normal cruise conditions are 82% Torque and 738 °C TOT.

Power check	– Perform as required or when deemed necessary (see Section 5).
-------------	---

4.8 PRE-LANDING CHECK

N_2/N_{R0}	– Adjust to 98-102%
--------------	---------------------

NOTE 100% rotor RPM recommended during landing.

All instruments	– Check
-----------------	---------

Bleed air consumers	– As required
---------------------	---------------

All warning/caution lights	– Check
----------------------------	---------

4.9 ENGINE SHUTDOWN

4.9.1 Normal

NOTE After first flight of the day perform **Deceleration Check**.

Collective lever lock	– Engage
-----------------------	----------

Power levers	– IDLE; start clock (stopwatch)
--------------	---------------------------------

Cyclic stick	– Trim neutral
--------------	----------------

Cyclic stick lock (if fitted)	– Engage
-------------------------------	----------

All electrical consumers	– Off except anti-collision lights and fuel pumps
--------------------------	---

After 2-min IDLE:

Power levers	– OFF
--------------	-------

CAUTION IF TOT DOES NOT DECREASE WITH N_1 , CLOSE EMERG FUEL VALVE AND, WHILE HOLDING POWER LEVER OFF, VENTILATE AFFECTED ENGINE UNTIL TOT SUBSTANTIALLY DECREASES. MAINTENANCE ACTION IS REQUIRED BEFORE NEXT START.

TOT and N_1	– Monitor decreasing
---------------	----------------------

Fuel pumps (4)	– Off
----------------	-------

Generator switches	– Off
--------------------	-------

OIL COOL warning light (if installed)	– Check on below 20% N_{R0}
--	-------------------------------

Circuit breakers	– As required
------------------	---------------

All switches	– Off
--------------	-------

4.9.2 Deceleration Check

NOTE Perform this check after first flight of each day and for each engine separately.

Cyclic stick	– Trim neutral
Collective lever lock	– Engage
All electrical consumers	– Off except anti-collision lights and fuel pumps
Generator switches	– Off
N ₂ (BENDIX fuel controls only)	– Stabilize at 100% for 15 seconds
Power lever of check engine	– Snap to IDLE, start clock (stopwatch)
When N ₁ of check engine passes 65%:	
Clock	– Stop, note time

NOTE • Minimum allowable deceleration time is 2 seconds. If deceleration time is less than 2 seconds, perform two more checks to confirm time.

- Maintenance action is required when:
 - confirmed deceleration time is less than 2 sec,
 - N₁ drops below 59%,
 - flameout is experienced.

Power lever of check engine	– FLIGHT
Remaining engine	– Perform deceleration check
Both power levers	– IDLE
Cyclic stick lock (if fitted)	– Engage

After 2 min:

Power levers	– OFF
--------------	-------

CAUTION IF TOT DOES NOT DECREASE WITH N₁, CLOSE EMERG FUEL VALVE AND, WHILE HOLDING POWER LEVER OFF, VENTILATE AFFECTED ENGINE UNTIL TOT SUBSTANTIALLY DECREASES. MAINTENANCE ACTION IS REQUIRED BEFORE NEXT START.

TOT and N1	– Monitor decreasing
Fuel pumps (4)	– Off
OIL COOL warning light (if installed)	– Check on below 20% N _{R0}
Circuit breakers	– As required
All switches	– Off
Engine compressor rinse	– Perform as required

4.10 COLD WEATHER STARTING INFORMATION

Following are general practices recommended for improved cold weather starting:

EFFECTIVITY *CECO fuel control system*

If engines equipped with CECO fuel control system start and then flame out at altitudes above 5000-ft MSL during cold ambient conditions, attempt another start with fuel supply pumps off. After reaching idle RPM, switch fuel supply pumps on.

EFFECTIVITY *BENDIX fuel control system*

CAUTION CONDENSATION AND FREEZING OF MOISTURE IN THE FUEL CONTROL PNEUMATIC CIRCUITS CAN OCCUR WHEN WEATHER CONDITIONS OF LOW TEMPERATURE AND HIGH RELATIVE HUMIDITY ARE ENCOUNTERED. THIS CONDITION CAN CAUSE AUTO (SPONTANEOUS) ACCELERATION OF THE AFFECTED ENGINE.

Auto acceleration resulting from frozen moisture in the fuel system air circuits does not repeat unless the aircraft is again subjected to cold soak. To prevent icing/auto acceleration, make a 10-minute ground warm-up run at IDLE before flight. This warm-up is recommended when the aircraft has been allowed to cold soak (remain out of hangar overnight) in low ambient temperature of -12°C or below and high relative humidity 45% or greater.

If auto acceleration occurs, perform the abort start procedures. Subsequently, restart the engine and resume the warm-up.

EFFECTIVITY *ALL*

When engine and main-rotor transmission oil temperatures are below -25°C and OAT is below -30°C preheat the oils to a temperature of -25°C or higher.

If available and conditions allow when the aircraft has been cold soaked at temperatures below 4°C , use an EPU to assist for faster more satisfactory starts.

If the aircraft has been cold soaked at temperatures below -18°C and a battery start must be made, preheat the engine fuel control area and battery if equipment is available and conditions allow.

If the aircraft has been cold soaked and a battery start must be made without preheating the battery, remove and store the battery until it is required if conditions allow. Store the battery in an area where it can be maintained or warmed to a temperature above ambient outside conditions or to approx 20°C .

If stagnated starts are encountered, maintenance action is required.

Due to variations in jet fuels available for commercial operation during cold weather starting, the engines may experience a short delay before ignition after each power lever is

moved to just before the IDLE position. This delay should be less than 3 seconds regardless of the type of fuel used. If the ignition delay exceeds 3 seconds, return the power lever to OFF and ventilate the engine for 20 seconds to remove excessive fuel from the combustion section.

In some instances N_1 may accelerate slowly through the 25-30% speed range on a battery start after an engine has been cold soaked and not preheated. If the start is not completed within the starter engagement time limits, shut down the engine. Before attempting the next start, wait for at least one minute to pass. This wait will allow residual heat from the previous start attempt to soak back into the engine and battery and improve conditions for the next start attempt.

At temperatures below $-30\text{ }^{\circ}\text{C}$ OAT, initial oil pressure indication may be delayed for up to 10 seconds on each engine. The final pressure build-up indication of the LH engine may be delayed for up to 50 seconds.

Depending upon the OAT, a ground idle time of up to 20 minutes may be necessary to bring engine and transmission oils up to operating temperatures.

4.11 FLIGHT CHARACTERISTICS

4.11.1 Flight Controls

CAUTION AVOID EXTREME CYCLIC STICK DISPLACEMENTS WHEN ON THE GROUND WITH ROTOR TURNING.

During ground operations with rotor turning, the cyclic stick must remain in the neutral position and the collective pitch lever in the full down position; however, for functional test purposes, minimum control movements (not more than 3 cm from neutral) are allowed.

Avoid extreme pedal movements during ground operations.

4.11.2 Lateral Control Characteristics

WARNING AVOID STEEP RIGHT TURNS AT AIRSPEEDS BELOW 70 KIAS CLOSE TO THE GROUND IN ORDER TO MAINTAIN SUFFICIENT LATERAL CONTROL MARGIN FOR RECOVERY.

Lateral control characteristics of single-rotor helicopters are significantly influenced by forward speed (airspeed), bank angles and relating g-loads.

During high load factor maneuvers at airspeeds below 70 KIAS the cyclic stick can reach the left lateral stop before retreating-blade stall entry.

Reduction of the load factor and/or nose-left side slip will improve left lateral control margin.

The recommended maximum bank angles for left and right turns at various airspeeds are as follows:

AIRSPEED (KIAS)	MAX ANGLE OF BANK
0 - 30	30°
30 - 60	45°
60 - ($V_{NE}^* - 30$)	60°
($V_{NE}^* - 30$) - V_{NE}^*	30°

* For V_{NE} values, refer to the V_{NE} table in Section 2.

SECTION 5

PERFORMANCE DATA

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SECTION 5

PERFORMANCE DATA

5.1 GENERAL

This section contains the performance information necessary for preflight and inflight planning. The information is primarily intended for mission planning and is most useful when planning operations in unfamiliar areas or at extreme conditions. Although maximum performance is not always required, regular use of the information in this section is recommended for the following reasons:

- Knowledge of performance margins will allow better decisions when unexpected conditions or alternate missions are encountered.
- Situations requiring maximum performance will be more readily recognized.
- Familiarity with the data will allow performance to be calculated more easily and quickly.
- Experience will be gained in accurately estimating the effects of conditions for which data is not presented.

5.1.1 Definitions

Some terms which are used particularly in this section are defined as follows:

5.1.1.1 Airspeeds

IAS The indicated airspeed is the reading obtained on a pitot-static airspeed indicator having no instrument calibration error. If the calibration error of the particular instrument is not known, then the actual reading may be considered as equal to IAS because the tolerances permitted on the instrument are small.

CAS The calibrated airspeed is the IAS-value corrected for pitot-static position errors only.

TAS The true airspeed of the helicopter, relative to undisturbed air, can be obtained by multiplying the CAS value by the true airspeed factor, if compressibility is neglected.

V_{NE} The never exceed speed, IAS.

V_Y The speed for best rate of climb = 60 KIAS.

5.1.1.2 Height

The true vertical clearance between the lowest part of the helicopter and the ground.

5.1.1.3 Outside Air Temperature (OAT)

The ambient temperature of the free airstream, expressed in °C.

5.1.1.4 Pressure Altitude

The expression of atmospheric pressure in terms of altitude, according to the interrelation of these factors in the international standard atmosphere. Pressure altitude may be obtained by setting the subscale of an accurate barometric altimeter to 1013.2 hPa (29.92 inches or 760 mm of mercury).

NOTE Altitude values used throughout this manual are Pressure Altitude unless stated otherwise.

5.1.2 Standard Performance Conditions

The data presented cover the maximum range of performance that can be expected and are based on the following conditions:

- Engine power does not exceed limits (see Section 2)
- Rotor RPM - 100 %.

In each area of performance, the effects of altitude, temperature, gross mass, and other parameters relating to that phase of flight are presented. In addition to the presented data, judgment and experience will be necessary to accurately determine performance under a given set of circumstances.

5.1.3 Conditions Affecting Performance

Conditions affecting performance are listed in the heading of the respective charts. The data presented are accurate only for these specific conditions.

Other general conditions applicable to the performance data are:

- Airframe and engine control rigging (assumed within allowable tolerances)
- Pilot technique (assumed normal, control movements smooth and continuous)
- Variations in performance between individual helicopters
- Instrument variation (inaccuracies or malfunctions are not accounted for).

5.1.4 Reading the Charts

None of the curves presented should be extrapolated, but interpolation between given data is permissible.

5.2 INFLIGHT POWER CHECKS

The power assurance check is provided to ensure that the engines are capable of developing the power used in establishing the performance data presented in this manual.

The engine trend check is provided to periodically monitor the long-term engine behavior.

5.2.1 Power Assurance Check

This check is performed in cruise flight with one engine operating at IDLE; each engine being checked separately, in turn. The generator and non-essential bleed air consumers (if installed) of the check engine must be switched off during its check. Engine anti-icing shall be on if ambient conditions require its use.

The power assurance check must be performed at intervals not to exceed 100 flight hours or whenever abnormal engine function is noted.

1. Note altimeter pressure setting; set for pressure altitude (29.92 in. Hg or 1013 hPa).

WARNING MAINTAIN A SUITABLE MINIMUM SAFE HEIGHT IN CASE OF ENGINE FAILURE.

2. Select a convenient pressure altitude in increments of 1000 ft from sea level up to the maximum check altitude of 10,000 ft. Establish and maintain level flight with 60 KIAS (V_Y).

WARNING SYSTEMS OR COMPONENTS REQUIRED FOR SAFE FLIGHT SHALL REMAIN ON.

3. Switch off all non-essential bleed air consumers. If engine anti-icing is frequently required for safe flight, it should be operated during each power check.
4. Reduce electrical load as much as possible.
5. Switch off generator of the engine to be checked.
6. Set power lever of non-check engine slowly to IDLE.

NOTE Observe limits (N_1 , TOT, Torque) of check engine.

7. Set N_2/N_{RO} to 100%.

WARNING DO NOT EXCEED ENGINE OR HELICOPTER LIMITS.

8. Adjust collective lever as necessary to achieve an indicated torque of 95% **or** an indicated TOT of 810 °C, **whichever is reached first**. Adjust airspeed as necessary to maintain level flight.
9. Allow check engine to stabilize at the adjusted power setting for at least 1 minute.

10. Record the following values (avoid parallax errors):
 - Pressure Altitude (FT)
 - Outside Air Temp (°C)
 - Torque (%)
 - TOT (°C)
 - N1 (%)
 - Engine Anti-icing (on or off)
11. Set power lever of non-check engine to FLIGHT.
12. Switch on generator of checked engine.
13. Repeat above steps 5. through 12. to check other engine.
14. Reset altimeter and, if necessary, switch on bleed air consumers.
15. After flight, use the appropriate Inflight Power Check diagram (see Fig. 5-1 or 5-2) to determine either maximum TOT or minimum required torque available under the prevailing ambient conditions.
16. Power is assured and all performance data contained in this manual can be achieved when each engine's:
 - indicated TOT is equal to or less than the chart derived maximum TOT value, or
 - indicated torque is equal to or greater than the chart derived minimum required torque value.

NOTE If power is not assured on one or both engines then the total power available is less than the minimum specification and all performance data contained in this manual cannot be achieved. Refer to appropriate maintenance manual to determine cause of power loss.

EXAMPLE: (see Fig. 5-1)

Determine: **Engine power assurance - Torque limit reached first**

Known:	Indicated torque	95 %
	Pressure altitude	4000 ft
	OAT	-30 °C
	Indicated TOT	730 °C
	Anti-icing	OFF

- Solution:
1. Enter chart at known torque (95 %).
 2. Move vertically up to known pressure altitude (4000 ft) and then trace horizontally left through the TOT field.
 3. Reenter chart at known OAT (-30 °C) and move vertically up to intersect tracing from above.
 4. At point of intersection read maximum TOT = 740 °C.

Since the indicated TOT (730 °C) is lower than the obtained chart TOT limit (740 °C), the power assurance check for this engine is acceptable.

EXAMPLE: (see Fig. 5-1)

Determine: **Engine power assurance - TOT limit reached first**

Known:	Indicated TOT	810 °C
	Pressure altitude	4000 ft
	OAT	20 °C
	Indicated torque	90 %
	Anti-icing	OFF

- Solution:
1. Enter chart at known OAT (20 °C).
 2. Move vertically up to known indicated TOT (810 °C) and trace horizontally right to intercept known PA (4000 ft).
 3. From this intercept point move vertically down to read minimum required torque = 86 %.

Since the indicated torque (90 %) is higher than the obtained chart minimum required torque (86 %), the power assurance check for this engine is acceptable.

INFLIGHT POWER CHECK

ALLISON 250-C20B

 N_2 – 100% N_1 – 95% OR ABOVE

GENERATOR – OFF

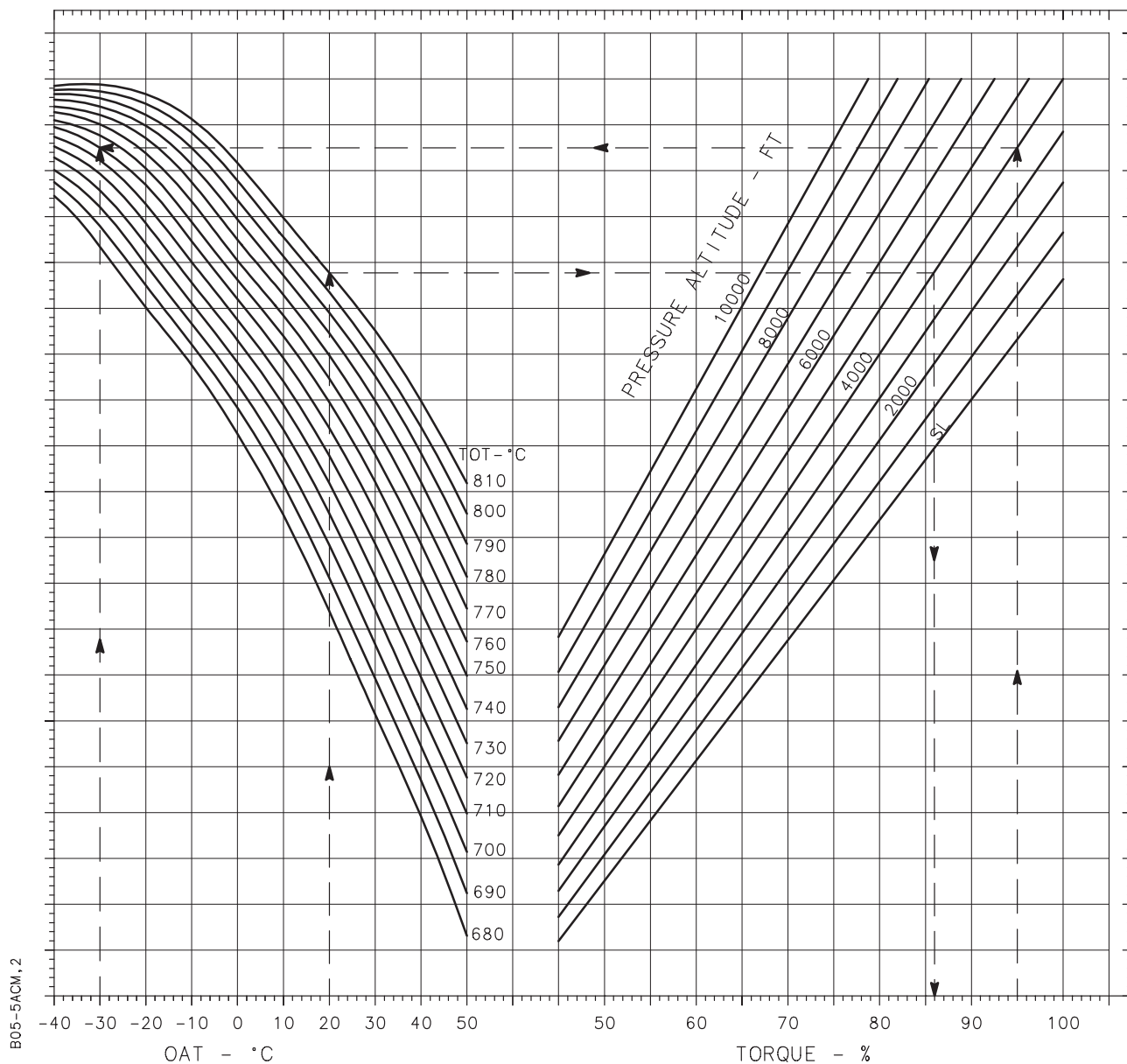
ANTI-ICING – OFF**BLEED AIR CONSUMERS – OFF**

Fig. 5-1 Inflight Power Check (Anti-icing - OFF)

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INFLIGHT POWER CHECK

ALLISON 250-C20B

N₂ – 100%N₁ – 95% OR ABOVE

GENERATOR – OFF

ANTI-ICING – ON

BLEED AIR CONSUMERS – OFF

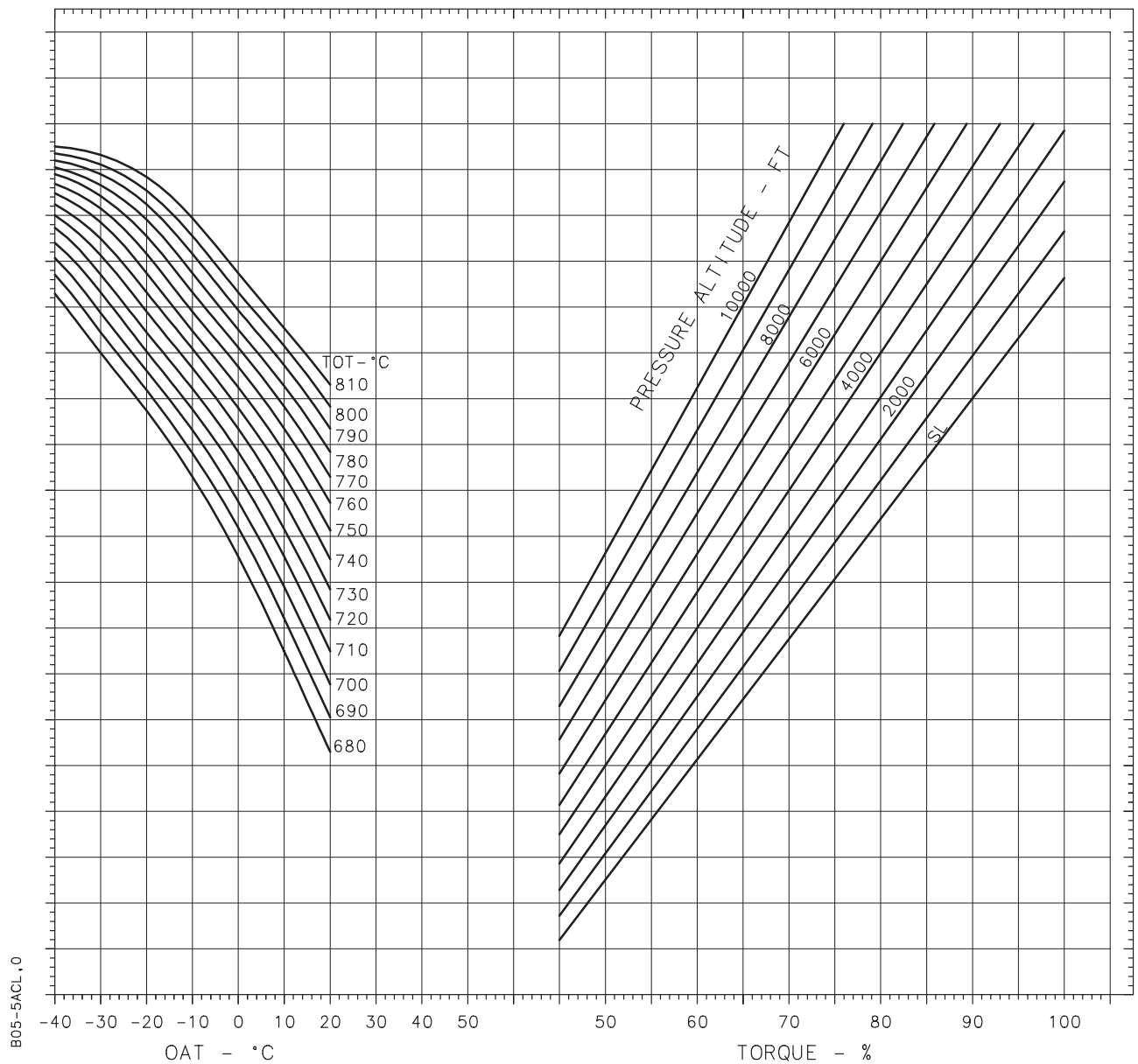


Fig. 5-2 Inflight Power Check (Anti-icing - ON)

5.2.2 Engine Trend Check

NOTE The engine trend check is not an alternate procedure to the power assurance check. Only the results of the Power Assurance Check (see Para 5.2.1) shall be used to ensure that the engines are capable of developing the power necessary to meet the presented performance data.

The engine trend check is provided for the purpose of monitoring the long-term behaviour and recognizing sudden significant trend-line deviations of the engine performance at minimum operator expense. This check, which should be repeated on a regular basis based upon operator experience, is performed in cruise flight with both engines operating; each engine being checked separately, in turn. The non-essential bleed air consumers (if installed) of the check engine must be switched off during its check. Engine anti-icing shall be on if ambient conditions require its use.

1. Note altimeter pressure setting; set for pressure altitude (29.92 in. Hg or 1013 hPa).
2. Select a convenient pressure altitude in increments of 1000 ft from sea level up to the maximum check altitude of 10,000 ft. Establish and maintain level flight with 60 KIAS (V_Y) or more.

WARNING SYSTEMS OR COMPONENTS REQUIRED FOR SAFE FLIGHT SHALL REMAIN ON.

3. Switch off all non-essential bleed air consumers. If engine anti-icing is frequently required for safe flight, it should be operated during each power check.
4. Reduce electrical consumption as much as possible.
5. Set N_2 at 100%.
6. Adjust collective lever as necessary to achieve an indicated torque of 86% **or** an indicated TOT of 779 °C, **whichever is reached first**.

NOTE If N_1 is not at least 95%, the engine bleed valve is open and the check result is not valid. In this case the check altitude should be increased.

7. Allow check engine to stabilize at the adjusted power setting for at least 1 minute.
8. Record the following values (avoid parallax errors):
 - Pressure Altitude (FT)
 - Outside Air Temp (°C)
 - Torque (%)
 - TOT (°C)
 - N_1 (%)
 - Engine Anti-icing (on or off)
9. Repeat above steps 4. through 8. to check other engine.

10. Reset altimeter and, if necessary, switch on bleed air consumers.
11. Following flight, use the appropriate Inflight Power Check diagram (see Fig. 5-1 or 5-2) to determine the minimum required torque available under the prevailing ambient conditions.

5.2.3 Power Trend Monitoring

A power trend monitoring should be established to observe the deterioration of the engines with time in service as well as potential engine malfunctions.

NOTE The power trend check preconditions and the power trend check procedure have to be followed very accurately, otherwise the trend monitoring will not be reliable.

To establish a power trend chart, the values of Δ Torque (defined as measured torque minus chart derived minimum required torque) have to be recorded versus time in service (see Fig. 5-3).

CAUTION MAINTENANCE ACTION MUST BE CONSIDERED IF A Δ TORQUE VALUE DEVIATES SIGNIFICANTLY FROM THE ESTABLISHED TREND LINE.

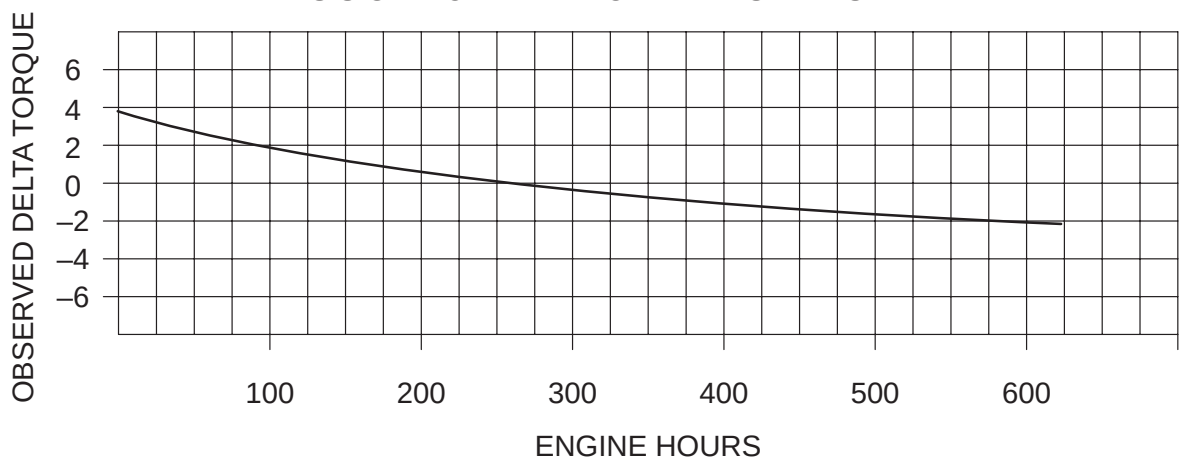


Fig. 5-3 Plotting Example for Power Trend Monitoring

5.3 DENSITY ALTITUDE

The density altitude chart (see Fig. 5-4) expresses density altitude in terms of pressure altitude and temperature. The less dense the air the higher the density altitude. For standard conditions of temperature and pressure, density altitude is identical to pressure altitude.

A high density altitude affects the performance of both rotors and engines. When density altitude is high, less lift is developed by the rotor blades for any given power setting than at standard conditions and the power output of the engines is reduced below the output for standard conditions.

Each takeoff and landing must be separately evaluated as density altitude may change considerably in a short period of time.

The true airspeed factor value $\frac{1}{\sqrt{\sigma}}$ is a conversion factor used to obtain true airspeed from calibrated airspeed by correcting for density altitude.

EXAMPLE: (see Fig. 5-4)

Determine: **Density altitude (DA)**

True airspeed factor

True airspeed (TAS)

Known:	OAT	−14 °C
	Pressure altitude	5000 ft
	CAS	100 kt

Solution:

1. Enter chart at known OAT (−14 °C).
2. Move vertically up to known pressure altitude (5000 ft).
3. Move horizontally left and read density altitude = 2630 ft.
4. Move horizontally right and read true airspeed factor = 1.04.
5. Multiply the known calibrated airspeed (100 kt) by true airspeed factor ($\frac{1}{\sqrt{\sigma}} = 1.04$) to obtain true airspeed.

$$\text{TAS} = \text{CAS} \times \frac{1}{\sqrt{\sigma}} = 100 \times 1.04 = \underline{104 \text{ KTAS}}$$

DENSITY ALTITUDE CHART

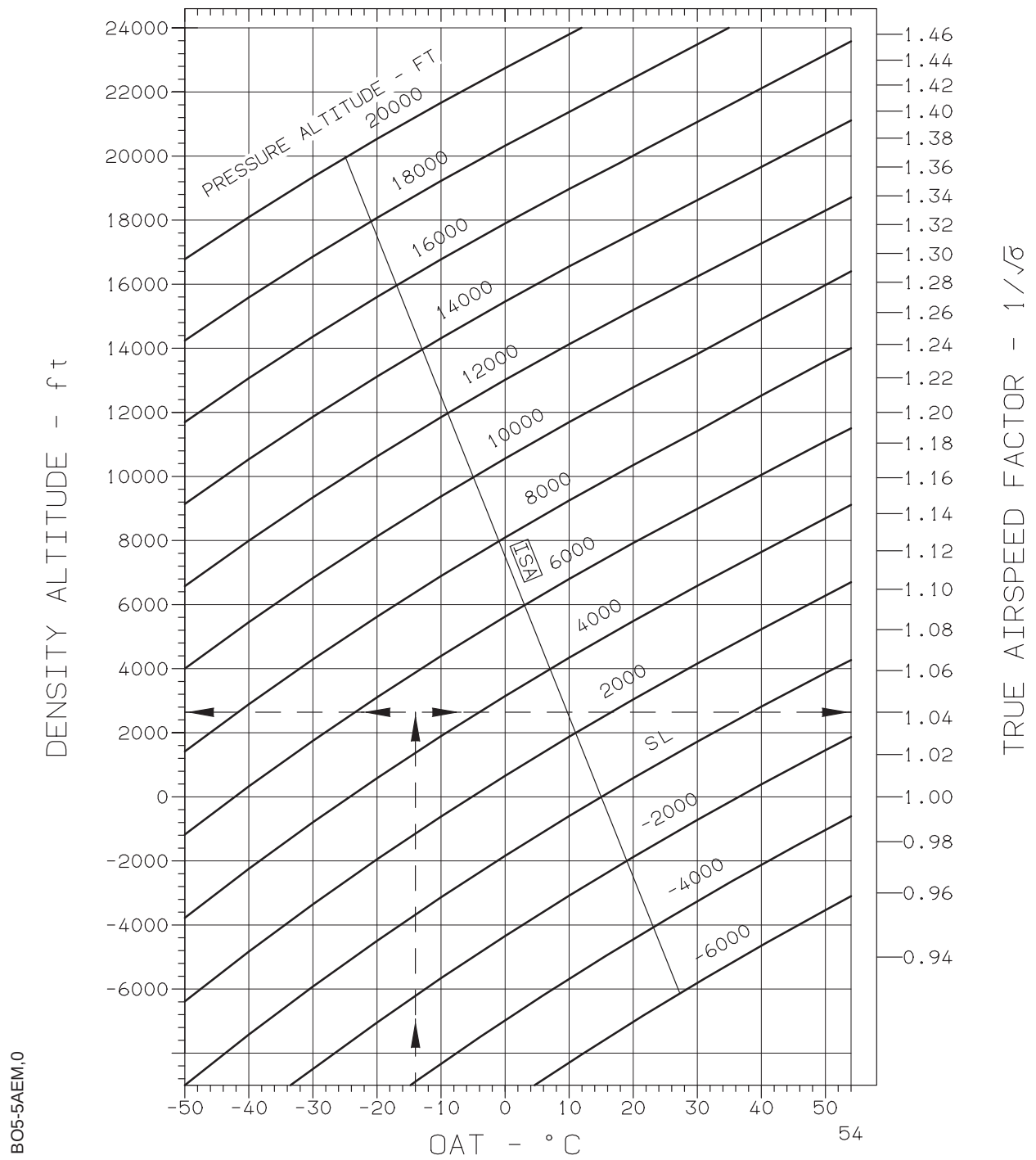


Fig. 5-4 Density Altitude Chart

5.4 AIRSPEED CALIBRATION

The airspeed calibration chart (see Fig. 5-5) is provided to determine Calibrated Airspeed (CAS) from Indicated Airspeed (IAS) using the pitot static system installation-error correction for level flight.

NOTE Indicated airspeed assumes zero instrument error.

The corrections apply for all practical altitudes and gross masses, and with the helicopter flown with no appreciable yaw.

EXAMPLE: (see Fig. 5-5)

Determine: **Calibrated airspeed (CAS), level flight**

Known: Indicated airspeed 90 KIAS

Solution: 1. Enter chart at known indicated airspeed (90 KIAS).
2. Move vertically up to LEVEL FLIGHT line.
3. Move horizontally left and read calibrated airspeed = 91 KCAS.

AIRSPEED CALIBRATION CHART

CLEAN CONFIGURATION

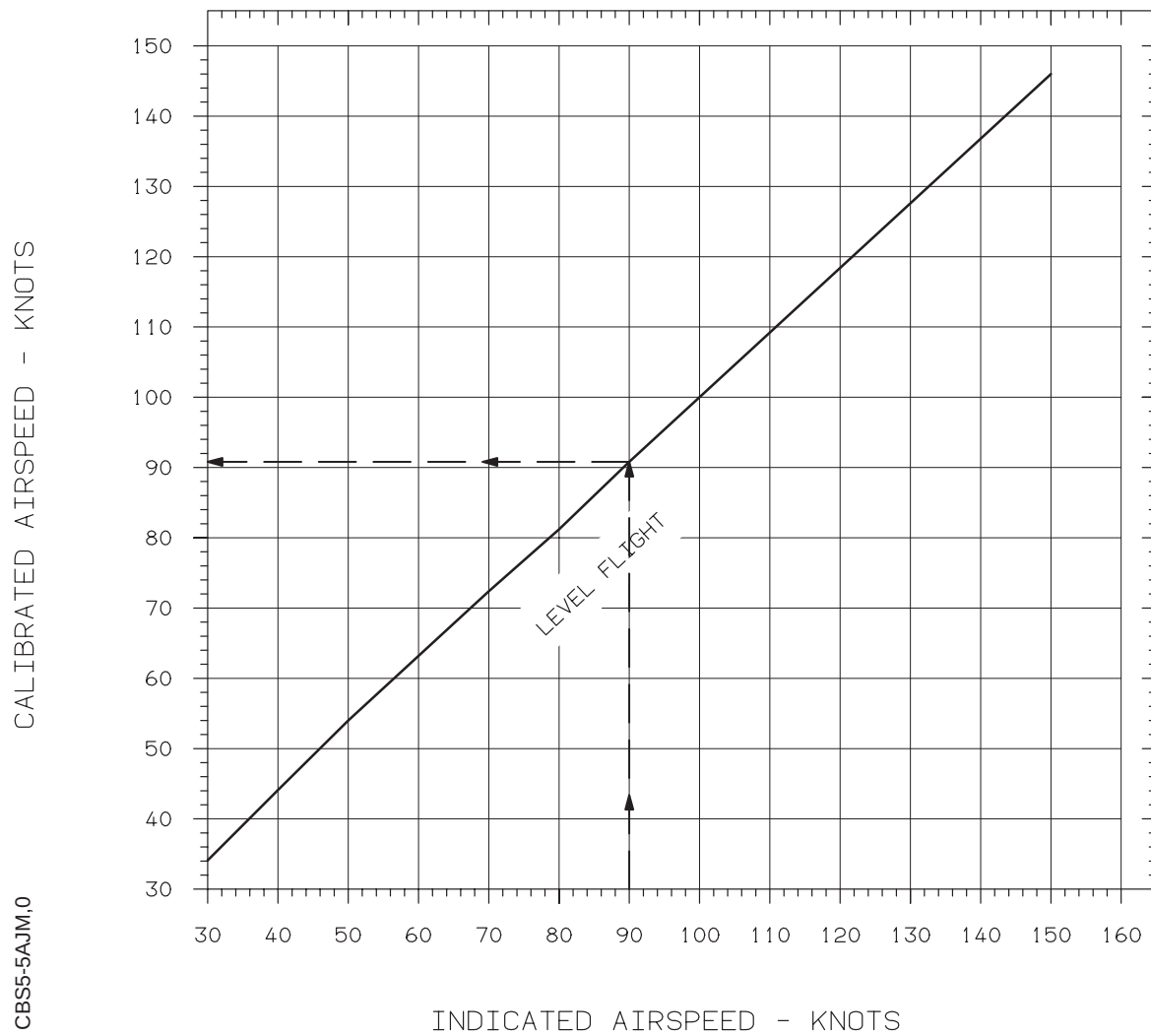
NOTE INDICATED AIRSPEED ASSUMES ZERO INSTRUMENT ERROR.

Fig. 5-5 Airspeed Calibration Chart

5.5 HEIGHT-VELOCITY DIAGRAMS

CAUTION AVOID OPERATION WITHIN THE CAUTION AREA - POWER LOSS OR ENGINE FAILURE WHILE OPERATING WITHIN THIS AREA IS LIKELY TO RESULT IN DAMAGE TO THE HELICOPTER.

The operator must become familiar with the height-velocity diagrams (see Fig. 5-6 and 5-7) to determine what altitudes and airspeeds are required to safely make an emergency landing in case of a power loss or engine failure. The altitude-airspeed combinations where a safe emergency landing on a smooth level firm surface may not be possible are represented by the caution (cross-hatched) area of the diagrams.

Two groups of height-velocity diagrams are presented; the first for **single engine failure** represented in varying gross mass and pressure altitude, and the second for **double engine failure** also represented in varying gross mass and pressure altitude.

HEIGHT-VELOCITY DIAGRAMS**SINGLE ENGINE FAILURE**

ISA CONDITIONS

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

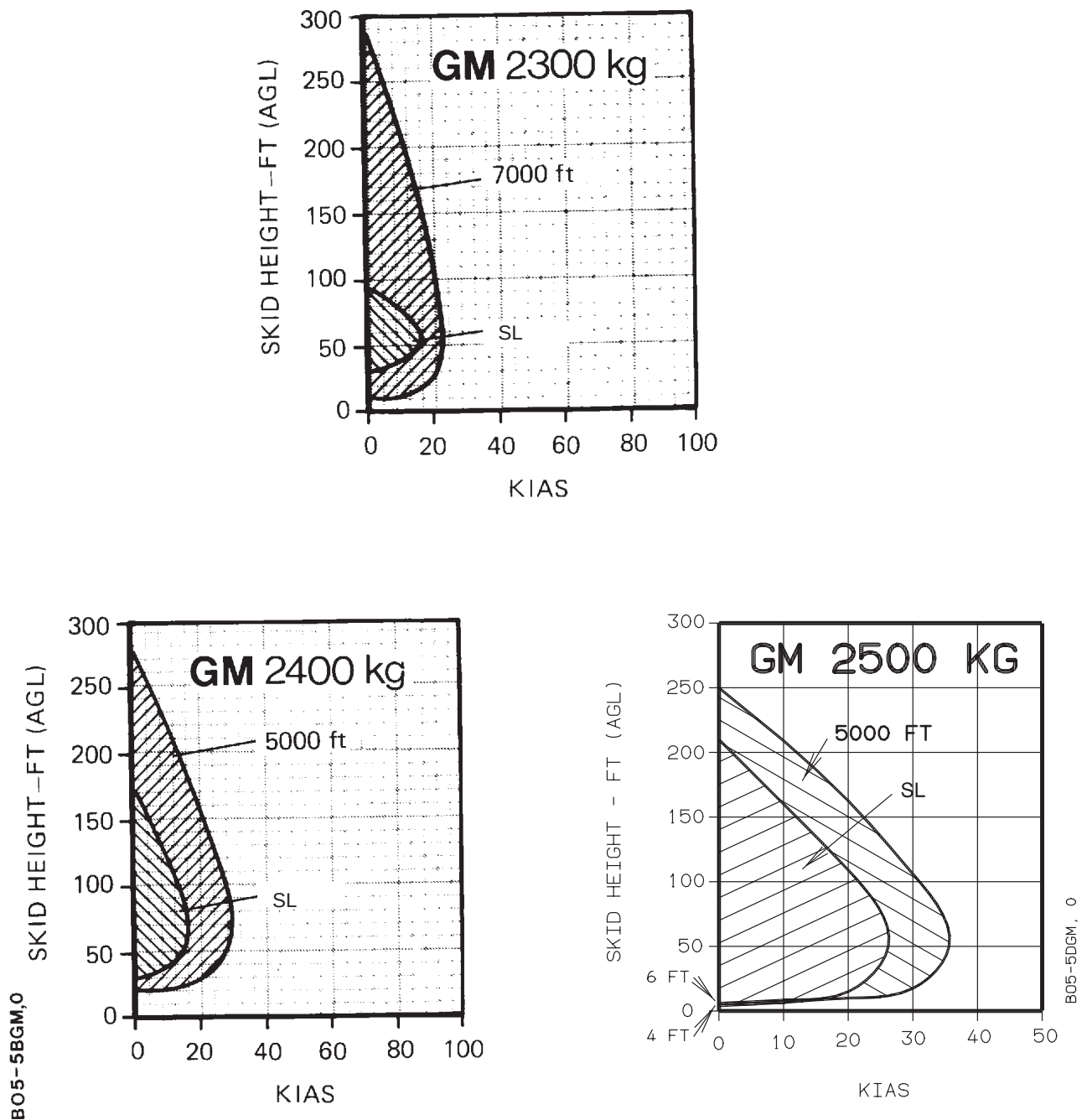
CAUTION AVOID OPERATION WITHIN CROSS-HATCHED AREAS.

Fig. 5-6 Height-velocity Diagrams - Single Engine Failure

HEIGHT-VELOCITY DIAGRAMS**DOUBLE ENGINE FAILURE**

ISA CONDITIONS

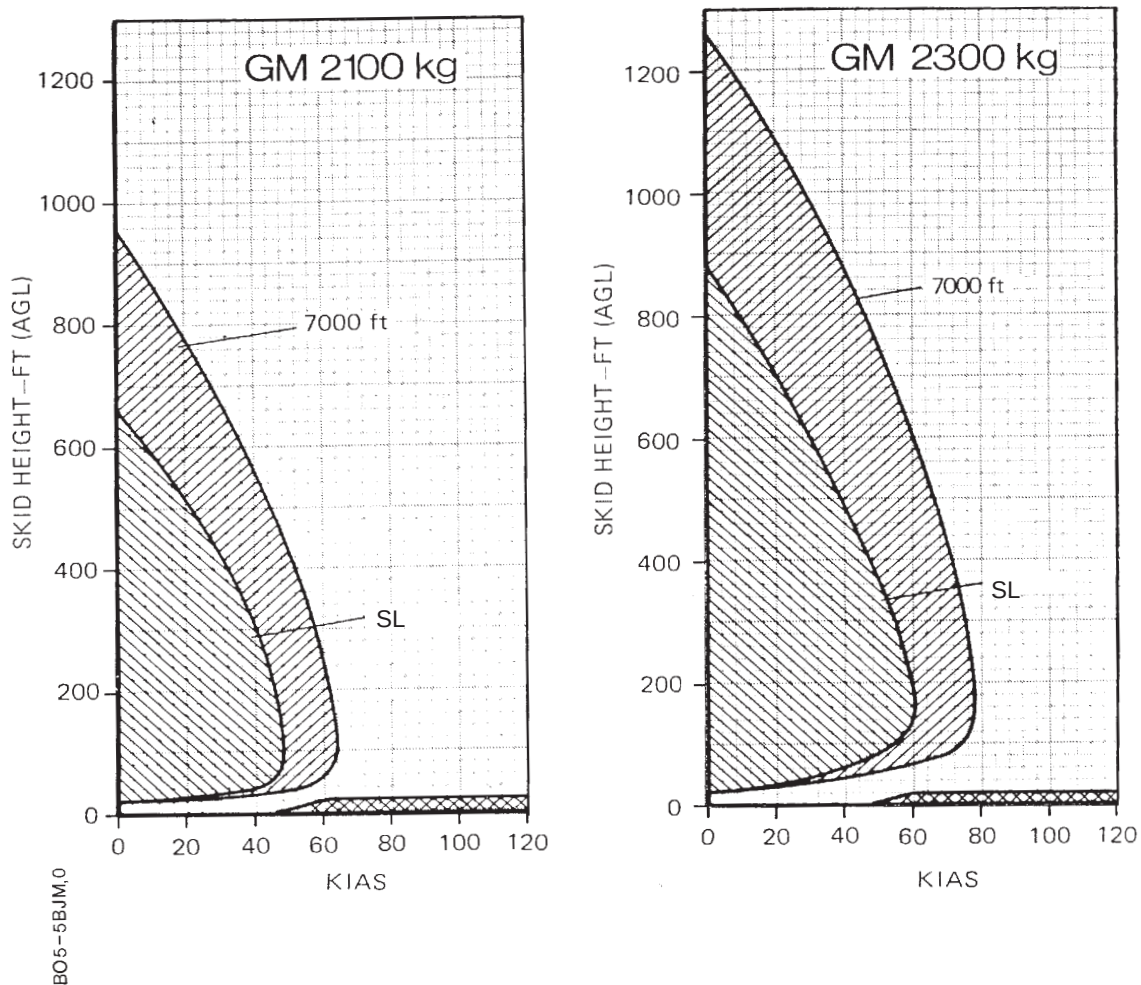
CAUTION AVOID OPERATION WITHIN CROSS-HATCHED AREAS.

Fig. 5-7 Height-velocity Diagrams - Double Engine Failure (Sheet 1 of 2)

HEIGHT-VELOCITY DIAGRAMS**DOUBLE ENGINE FAILURE**

ISA CONDITIONS

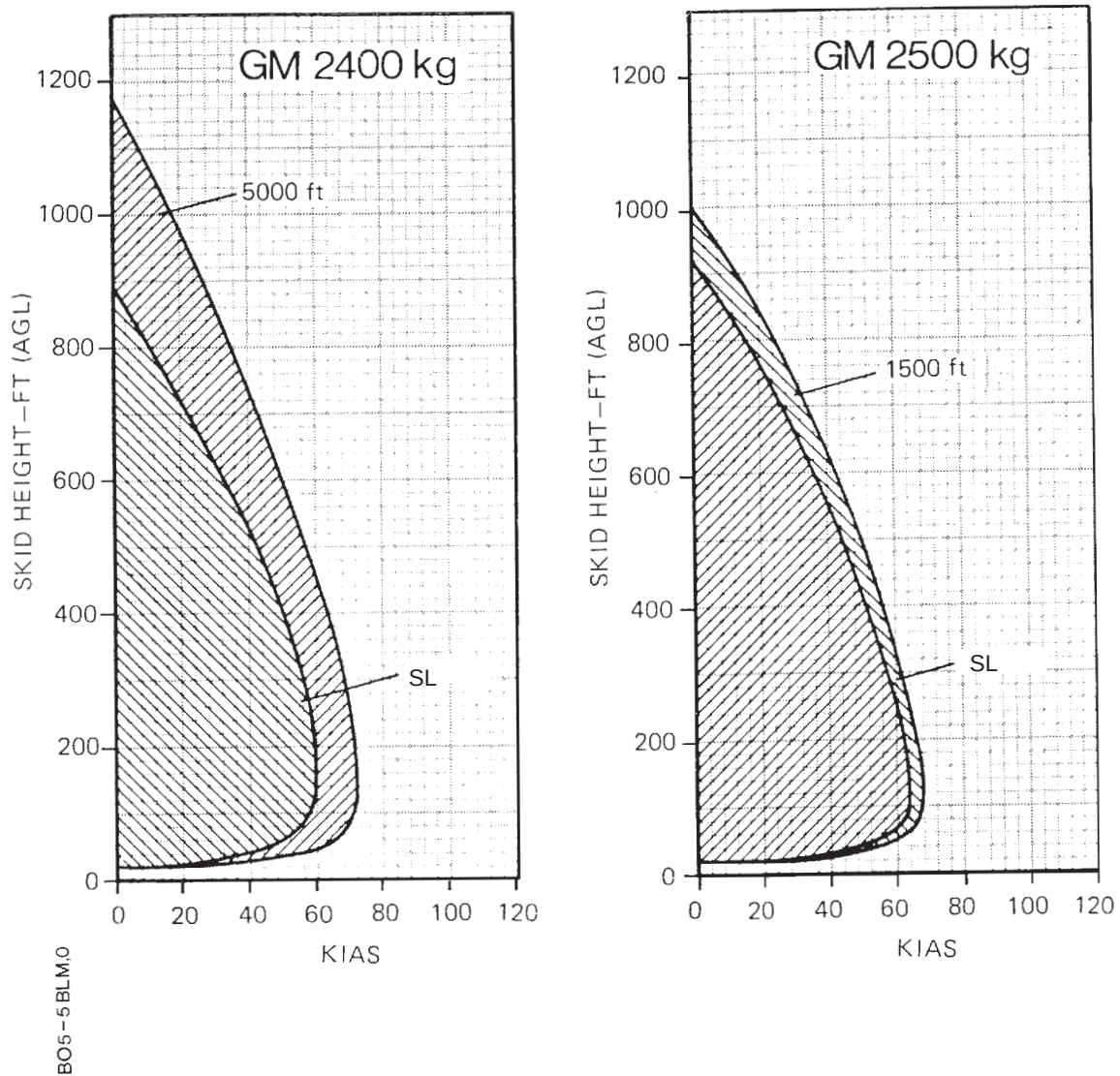
CAUTION AVOID OPERATION WITHIN CROSS-HATCHED AREAS.

Fig. 5-7 Height-velocity Diagrams - Double Engine Failure (Sheet 2 of 2)

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5.6 HOVER CEILING (see Fig. 5-8 thru 5-11)

The hover ceiling charts present the maximum gross mass for hover in and out of ground effect at various conditions of pressure altitude and outside air temperature, and under different power settings. The charts for hover in ground effect are based on a skid height of 3 ft and are calculated for hovering with engine anti-icing OFF.

- NOTE** • For operations requiring engine anti-icing ON, reduce the chart derived gross mass value by 110 kg. If the present atmospheric condition (PA, OAT) is represented as a point projected as being higher than the maximum gross mass limit, the 110 kg may be subtracted from the gross mass value of that point, however, the resultant value must not exceed the maximum gross mass limit.
- OGE hover ceiling diagrams are valid for no wind or headwind.

EXAMPLE: (see Fig. 5-8)

Determine: ① **Maximum gross mass for hovering, engine anti-icing OFF**

Known: OAT 17 °C
Pressure altitude 8000 ft

- Solution:
1. Enter chart at known OAT (17 °C).
 2. Move vertically up to known pressure altitude (8000 ft).
 3. Move horizontally left and read maximum gross mass = 2325 kg.

Determine: ② **Maximum gross mass for hovering, engine anti-icing ON**

Known: OAT -8 °C
Pressure altitude 7000 ft

- Solution:
1. Enter chart at known OAT (-8 °C).
 2. Move vertically up to known pressure altitude (7000 ft).
 3. Move horizontally left and read maximum gross mass 2660 kg.
 4. Subtract 110 kg from chart derived value (2660 kg) for 2550 kg, however, the gross mass limit applies = 2500 kg for hovering with engine anti-icing ON.

Determine: ③ **Maximum gross mass for hovering, engine anti-icing ON**

Known: OAT -13 °C
Pressure altitude 8000 ft

- Solution:
1. Enter chart at known OAT (-13 °C).
 2. Move vertically up to known pressure altitude (8000 ft).
 3. Move horizontally left and read maximum gross mass 2585 kg.
 4. Subtract 110 kg from chart derived value (2585 kg) = 2475 kg maximum gross mass for hovering with engine anti-icing ON.

HOVER CEILING IN GROUND EFFECT

2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

SKID HEIGHT – 3 FT

BLEED AIR CONSUMERS – OFF

- NOTE**
- FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE **GROSS MASS LIMIT** LINE SHALL APPLY.
 - WITH ENGINE ANTI-ICING ON, THE GROSS MASS IS DECREASED BY 110 KG.

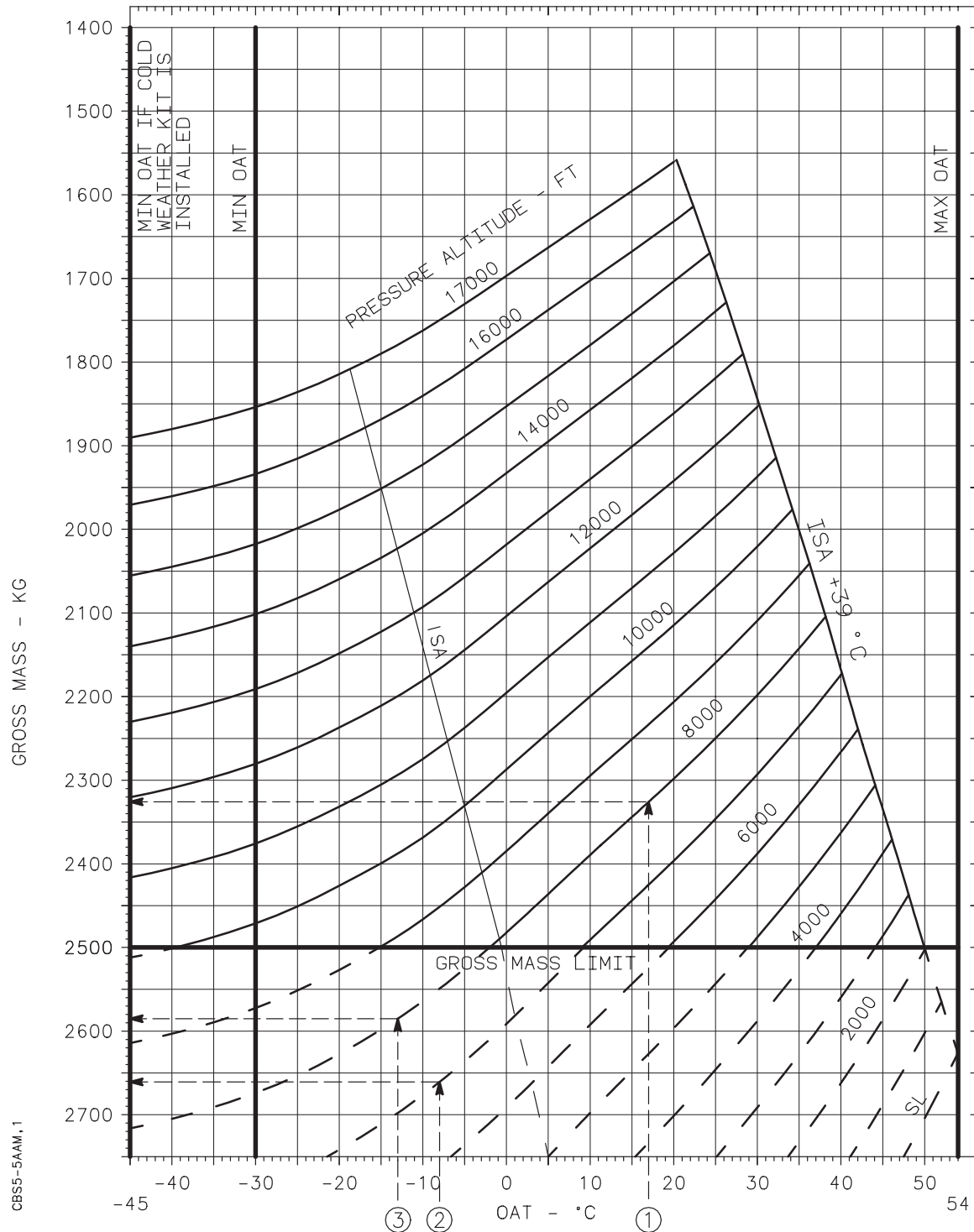


Fig. 5-8 Hover Ceiling I G E - TOP

HOVER CEILING IN GROUND EFFECT

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

SKID HEIGHT – 3 FT

BLEED AIR CONSUMERS – OFF

- NOTE**
- FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE **GROSS MASS LIMIT** LINE SHALL APPLY.
 - WITH ENGINE ANTI-ICING ON, THE GROSS MASS IS DECREASED BY 110 KG.

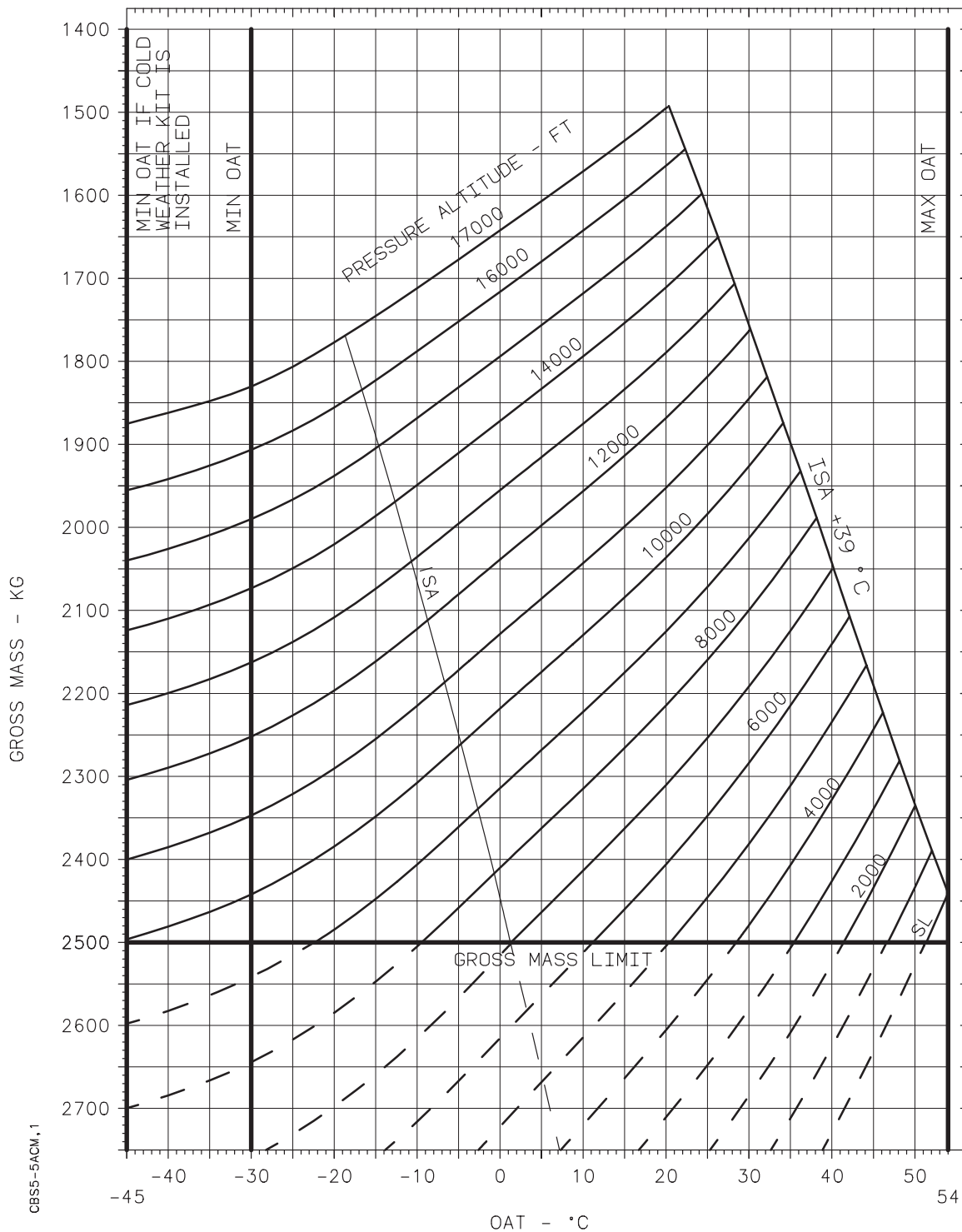


Fig. 5-9 Hover Ceiling I G E - MCP

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HOVER CEILING OUT OF GROUND EFFECT

2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)
 ZERO WIND OR HEADWIND

ANTI-ICING – OFF
 BLEED AIR CONSUMERS – OFF

- NOTE**
- FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE **GROSS MASS LIMIT** LINE SHALL APPLY.
 - WITH ENGINE ANTI-ICING ON, THE GROSS MASS IS DECREASED BY 110 KG.

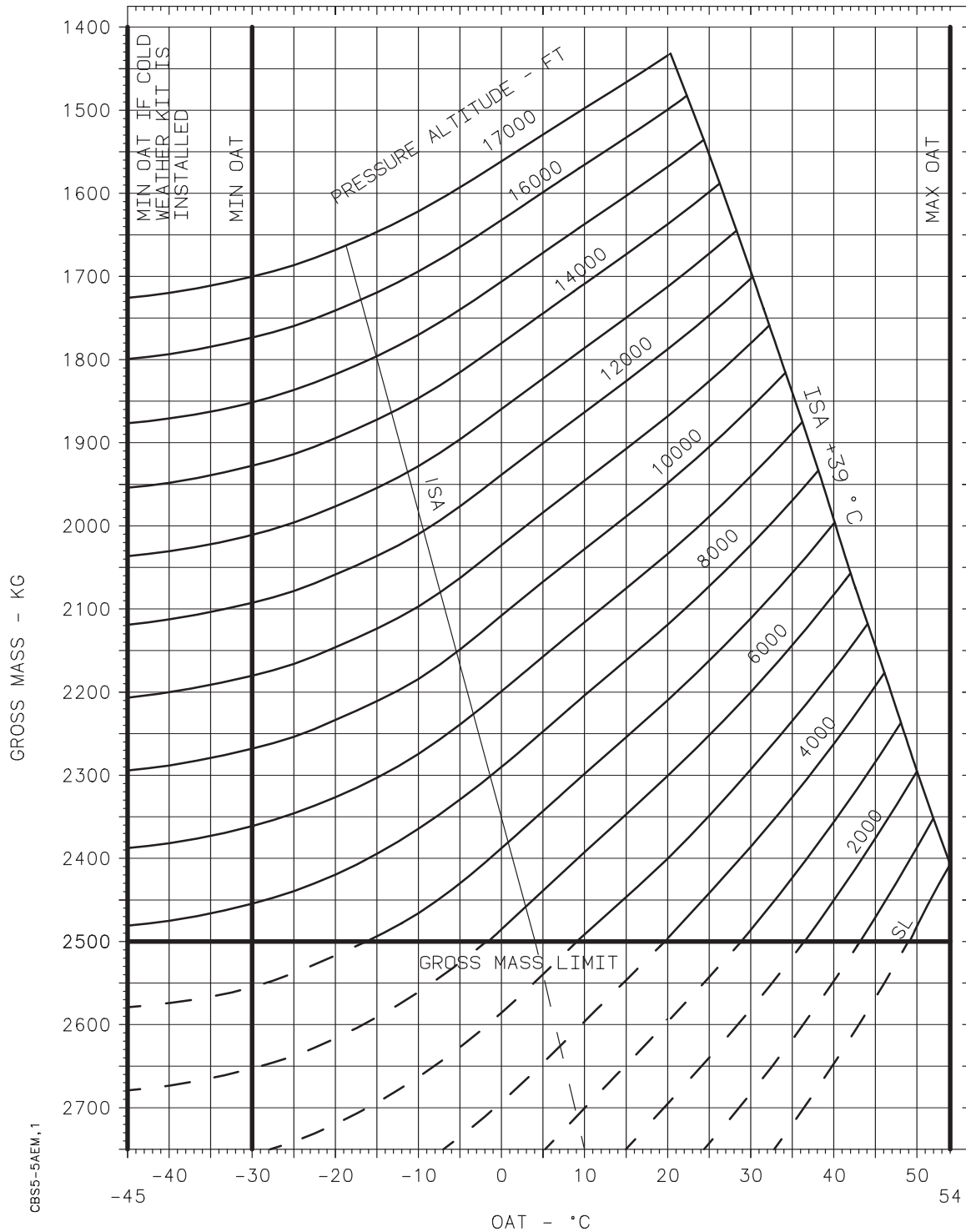


Fig. 5-10 Hover Ceiling O G E - TOP

HOVER CEILING OUT OF GROUND EFFECT

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

ZERO WIND OR HEADWIND

BLEED AIR CONSUMERS – OFF

- NOTE**
- FOR ALTITUDE/OAT COMBINATIONS NOT SHOWN, THE **GROSS MASS LIMIT** LINE SHALL APPLY.
 - WITH ENGINE ANTI-ICING ON, THE GROSS MASS IS DECREASED BY 110 KG.

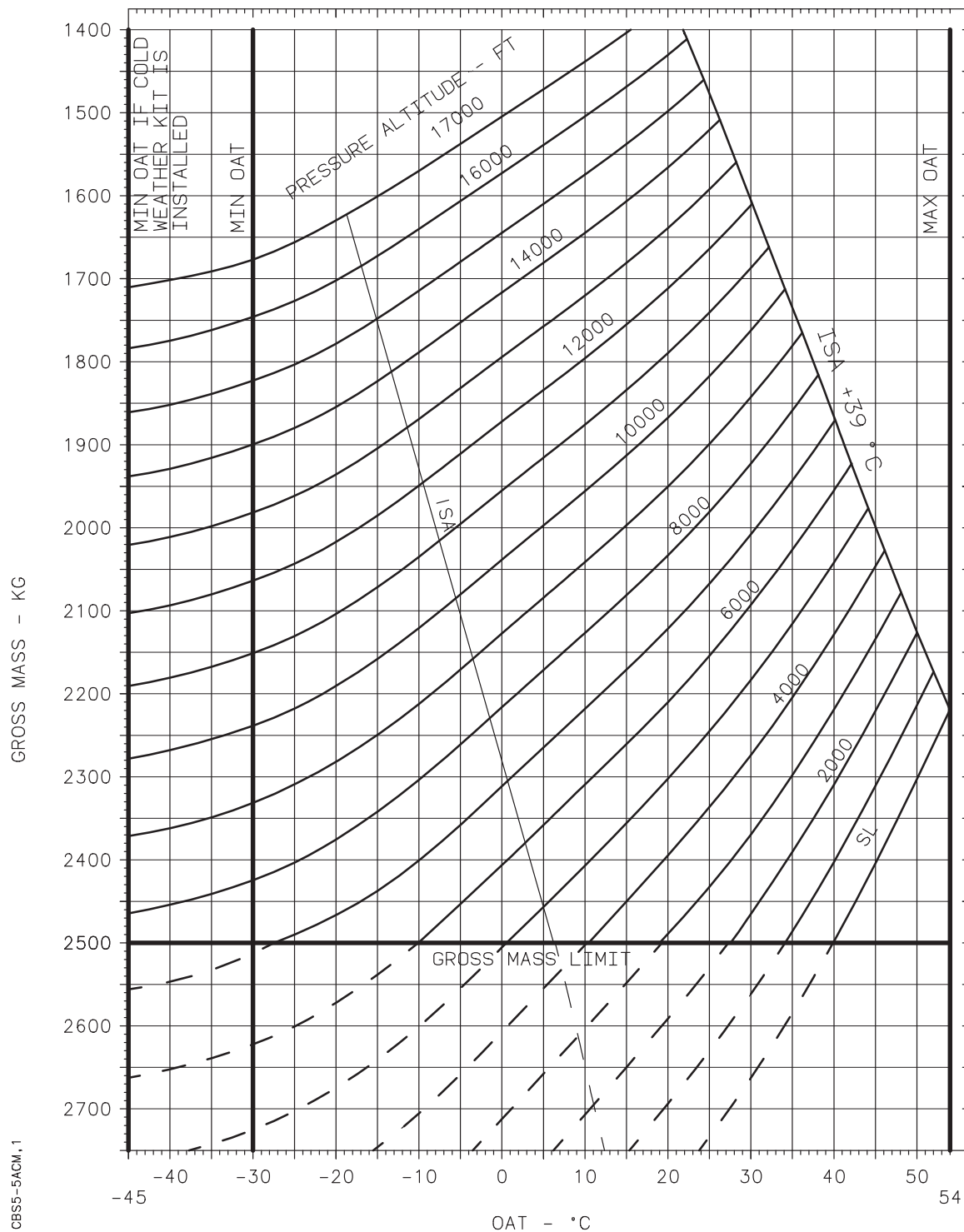


Fig. 5-11 Hover Ceiling O G E - MCP

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5.7 TAKEOFF PERFORMANCE

5.7.1 Takeoff Procedure

Use of the Category B takeoff procedure (see Fig. 5-12) assures the capability to land safely (on a smooth level surface) should a single-engine failure occur at any point during the takeoff.

NOTE Continued takeoff capability is NOT assured for OEI operation when using this takeoff procedure.

Pre-takeoff check	– Perform
Altimeter	– Set
Hover	– 3-ft skid height
Rotor RPM	– 100 % set with matched torque
All instruments	– Normal operating ranges
Acceleration and climb	– Start nose-down pitch rotation and simultaneously increase power smoothly to accelerate in a shallow climb. Adjust pitch attitude at about 25 KIAS to achieve 50 KIAS at 50 ft AGL
When 50 ft AGL is reached:	
Collective pitch	– Adjust to Takeoff Power
Airspeed	– Accelerate to V_Y
Collective pitch	– Adjust to Max Continuous Power
Climbout	– Continue to desired altitude

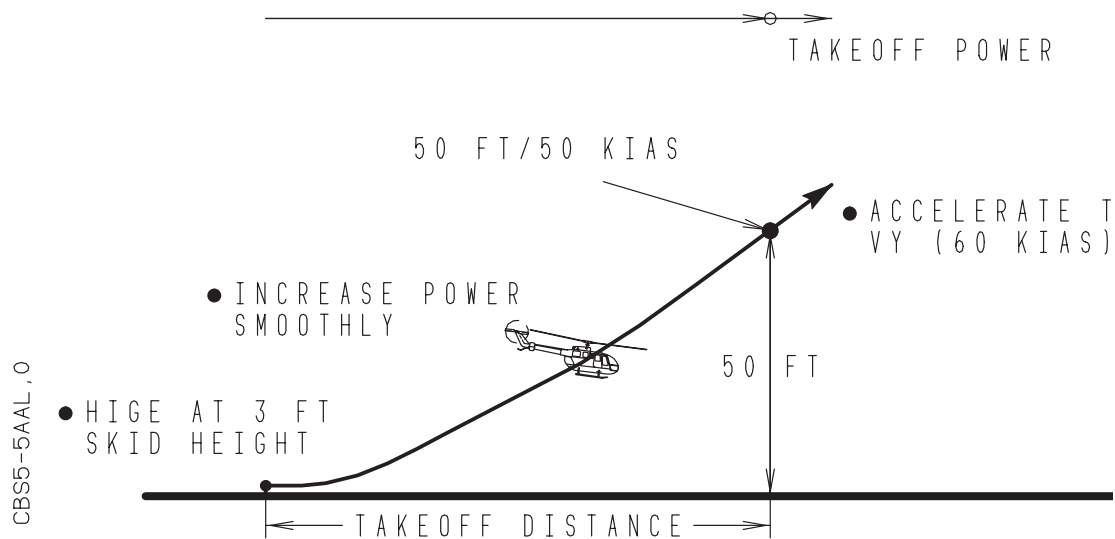


Fig. 5-12 Takeoff Profile

5.7.2 Takeoff Distance Required

The takeoff distance required is that distance necessary to travel from a 3-ft hover to a 50 ft height while accelerating to 50 KIAS.

The takeoff distance required under zero wind conditions 550 ft (170 m)

NOTE Above distance is valid for engine anti-icing ON or OFF and for all combinations of gross mass, OAT and pressure altitude.

To determine the takeoff distance required with headwind (see Fig. 5-13)

EXAMPLE: (see Fig. 5-13)

Determine: **Takeoff distance required with headwind**

Known: Headwind 18 kt

Solution:

1. Enter chart at known headwind (18 kt).
2. Move horizontally right to intersect diagonal line.
3. From point of intersection above move vertically down and read Takeoff Distance Required = 355 ft (108 m).

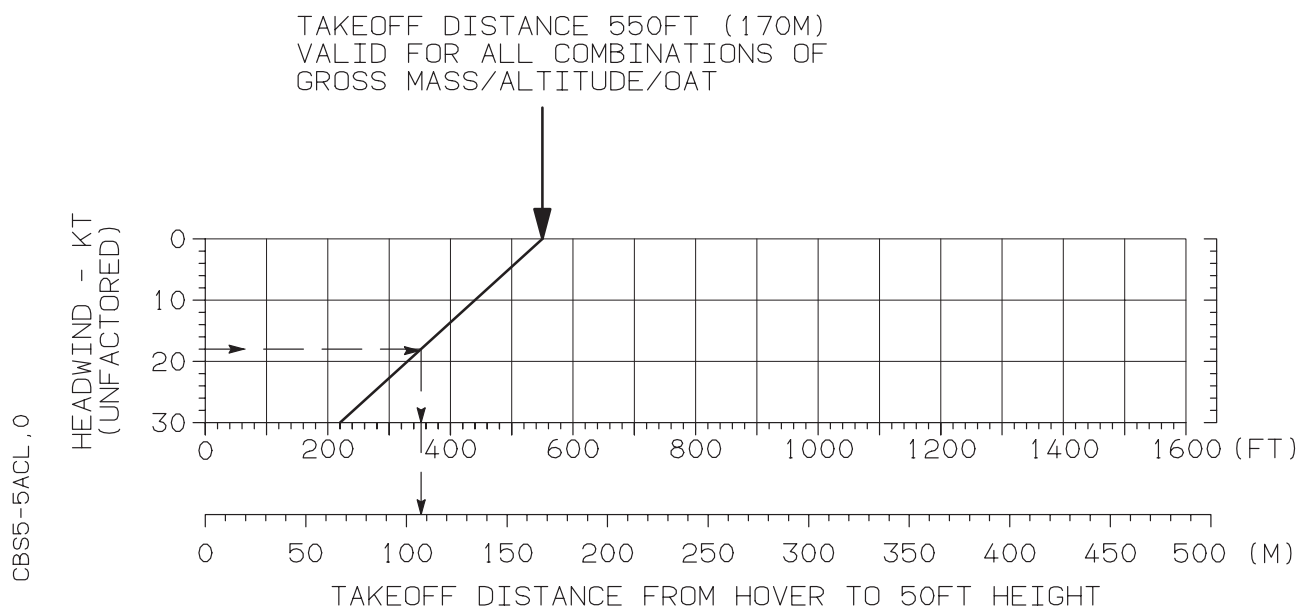


Fig. 5-13 Takeoff Distance Required

5.8 RATE OF CLIMB

The rate-of-climb charts are presented in two groups; the first group for operations with all engines operating (AEO, Fig. 5-14 and 5-15) and the second group for operations with one engine inoperative (OEI, Fig. 5-16 and 5-17). Climb rates are functions of power, pressure altitude, outside air temperature, and gross mass.

NOTE • Chart data are based on the clean configuration.

- Chart derived rate-of-climb values are established using the best rate-of-climb speed ($V_Y = 60$ KIAS).
- For AEO operations requiring engine anti-icing ON, reduce the chart derived rate-of-climb value by 250 fpm.
- For OEI operations requiring engine anti-icing ON, reduce the chart derived rate-of-climb value by 150 fpm.

EXAMPLE: (see Fig. 5-14)

Determine: **Rate of climb, OAT 10 °C or above**

Known:	Pressure altitude	12500 ft
	OAT	20 °C
	Gross mass	1850 kg

- Solution:
1. Enter chart at known pressure altitude (12500 ft).
 2. Move horizontally right to known OAT (20 °C).
 3. Move vertically down to Base Line and trace further down following parallel to dashed guide lines.
 4. Reenter chart at known gross mass (1850 kg) and move horizontally right to intersect tracing from above.
 5. From point of intersection move vertically down and read rate of climb = 1330 fpm.

Determine: **Rate of climb, OAT below 10 °C**

Known:	Pressure altitude	9500 ft
	OAT	0 °C
	Gross mass	1950 kg

- Solution:
1. Enter chart at known pressure altitude (9500 ft).
 2. Move horizontally right to known OAT (0 °C).
 3. Move vertically down to Base Line and trace further down following parallel to solid guide lines.
 4. Reenter chart at known gross mass (1950 kg) and move horizontally right to intersect tracing from above.
 5. From point of intersection move vertically down and read rate of climb = 1860 fpm.

RATE OF CLIMB

2 X ALLISON 250-C20B

AEO – TAKEOFF POWER (810 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

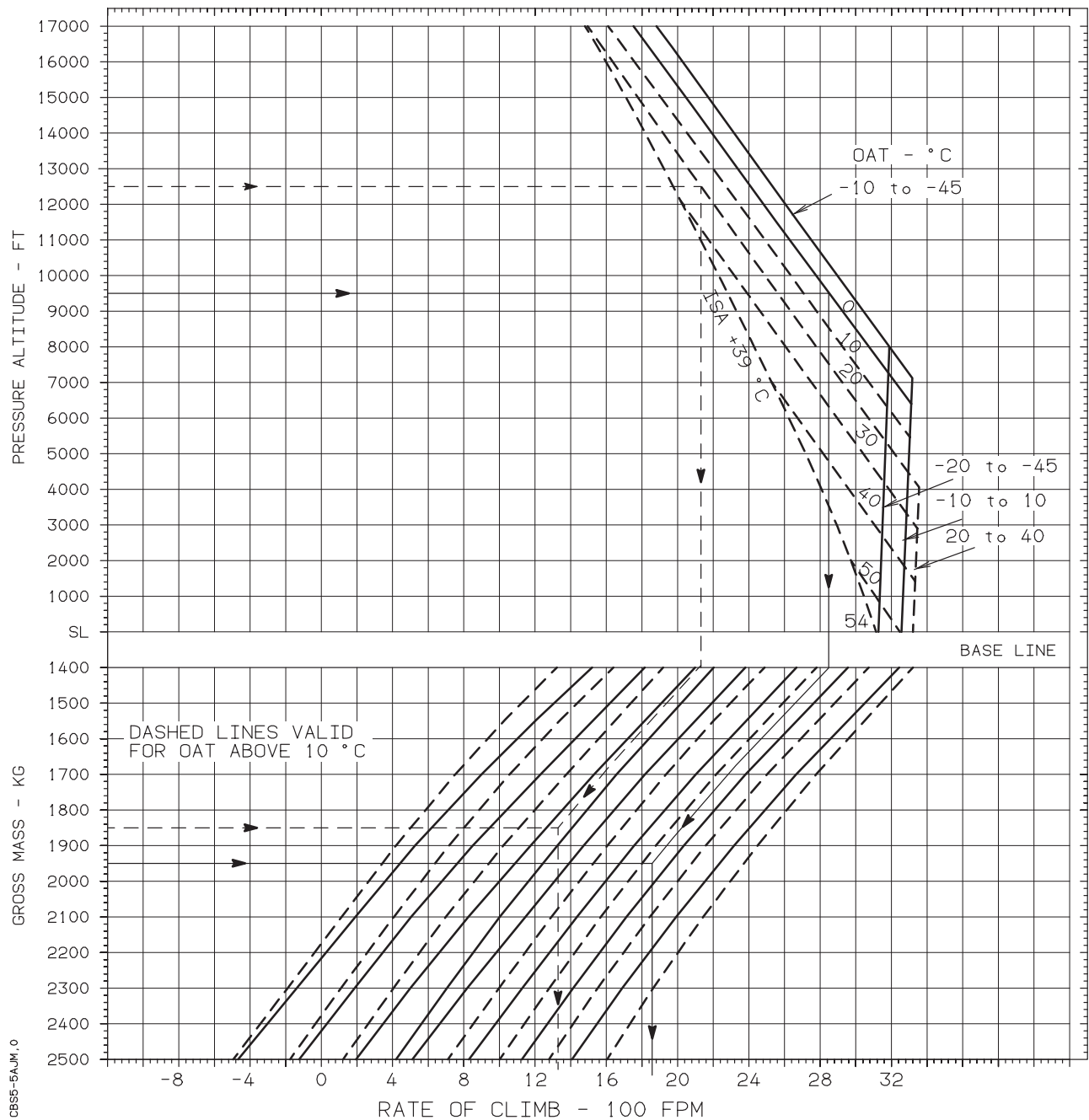
NOTE WITH ENGINE ANTI-ICING ON, THE RATE OF CLIMB IS DECREASED BY 250 FPM.

Fig. 5-14 Rate of Climb - AEO, TOP

RATE OF CLIMB

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

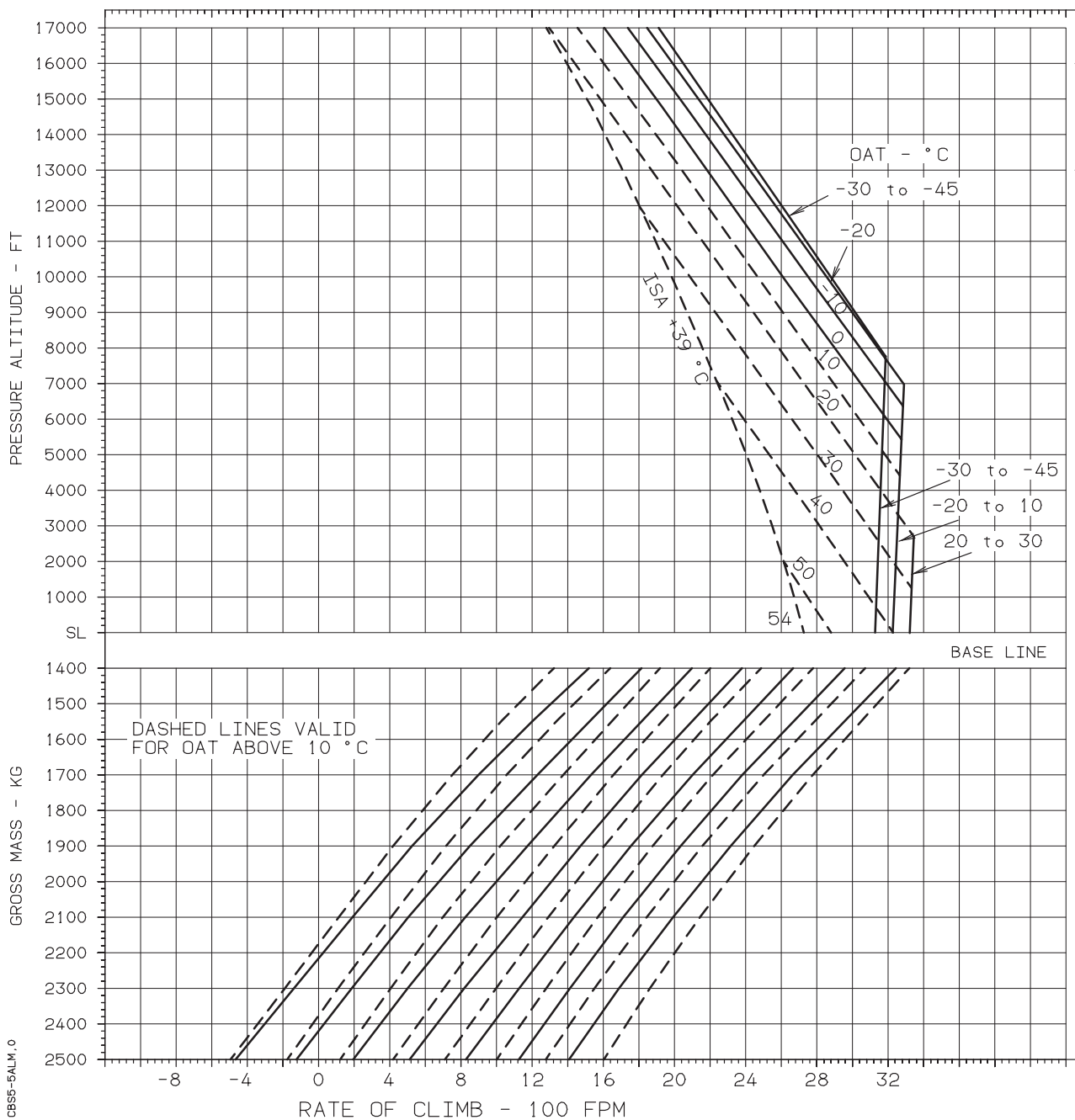
NOTE WITH ENGINE ANTI-ICING ON, THE RATE OF CLIMB IS DECREASED BY 250 FPM.

Fig. 5-15 Rate of Climb - AEO, MCP

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RATE OF CLIMB

1 X ALLISON 250-C20B

OEI – EMERGENCY POWER (810 °C TOT, 95% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

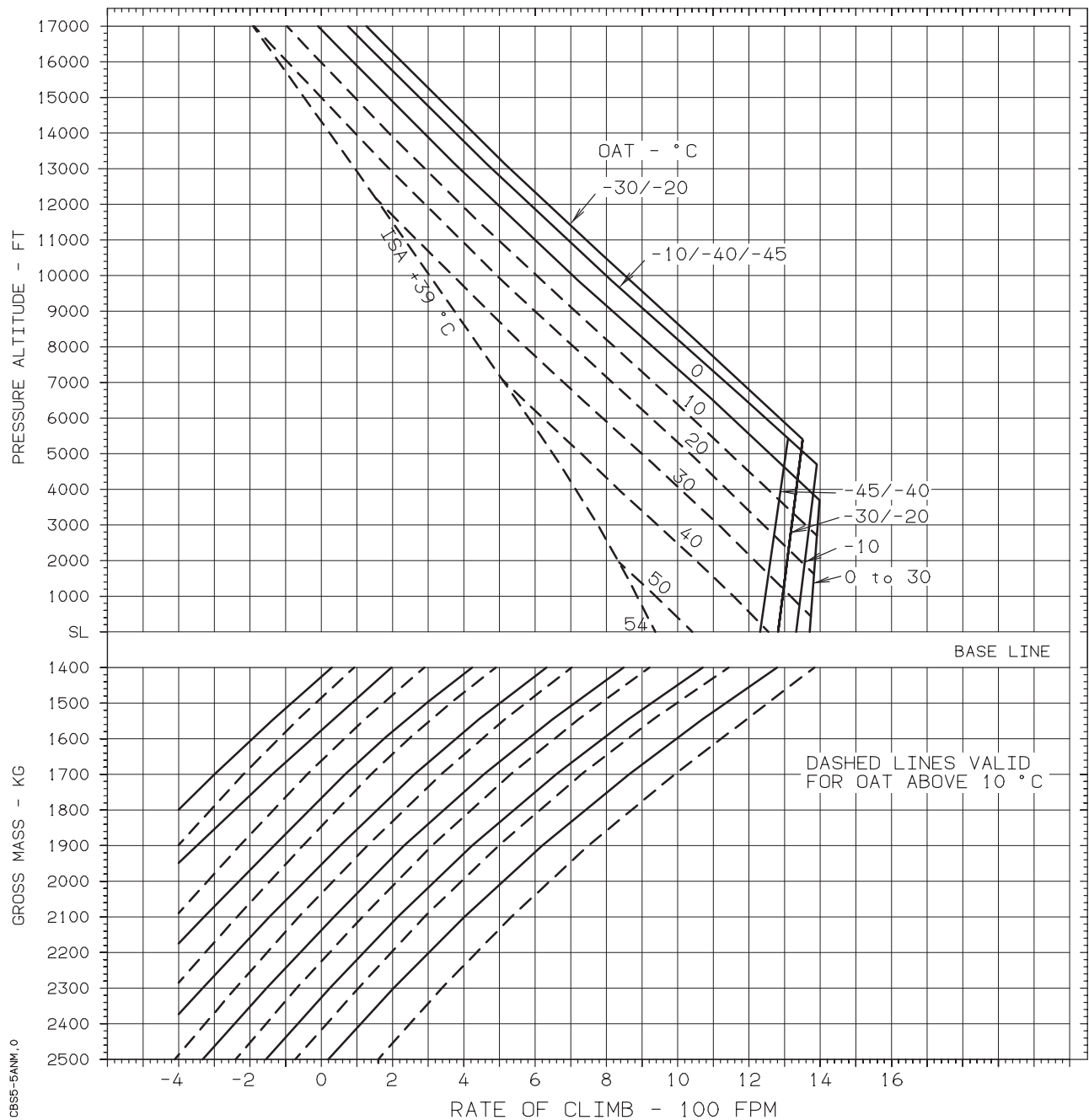
NOTE WITH ENGINE ANTI-ICING ON, THE RATE OF CLIMB IS DECREASED BY 150 FPM.

Fig. 5-16 Rate of Climb - OEI, EPWR

RATE OF CLIMB

1 X ALLISON 250-C20B

OEI – MAX CONTINUOUS POWER (779 °C TOT, 95% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

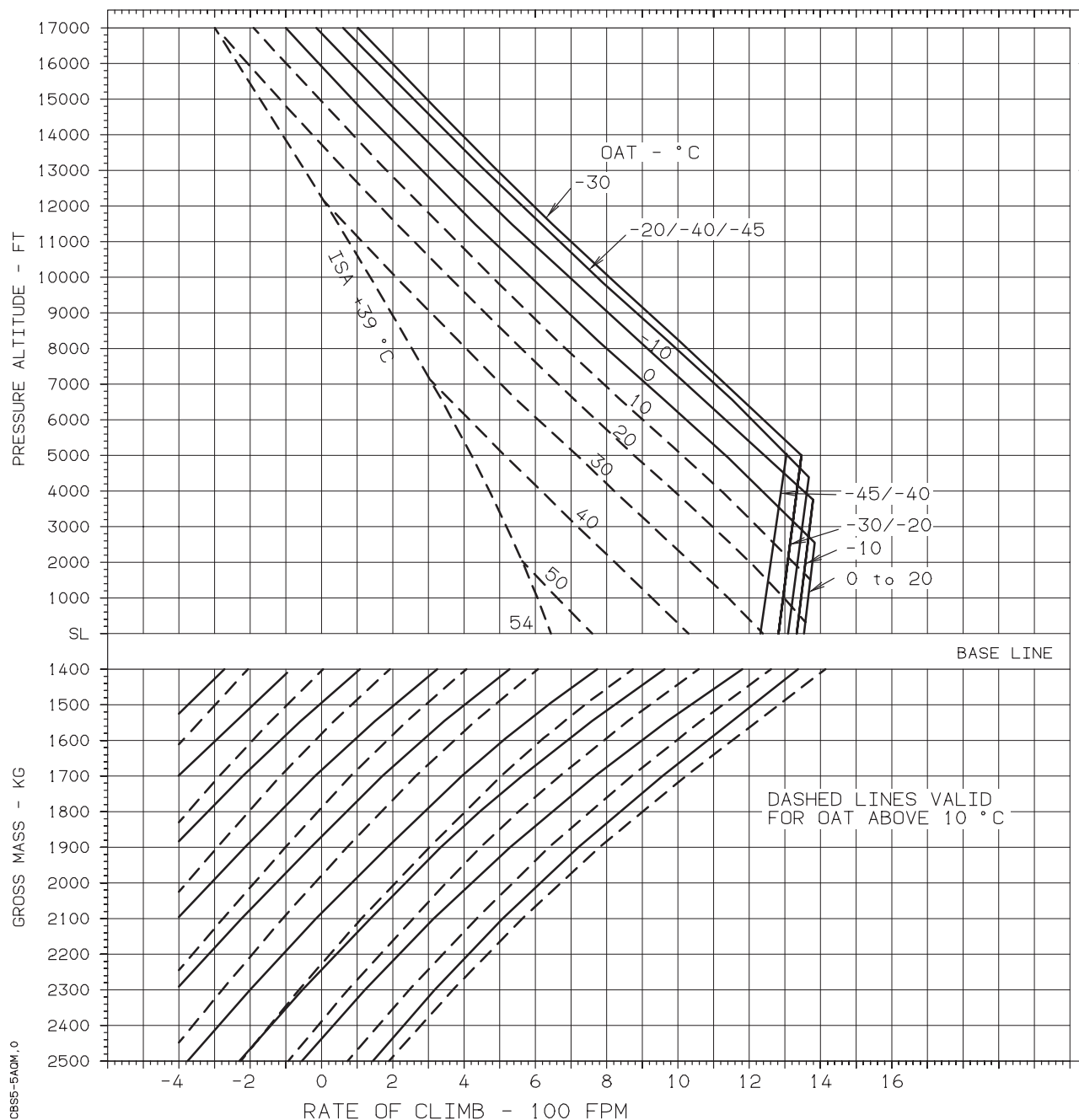
NOTE WITH ENGINE ANTI-ICING ON, THE RATE OF CLIMB IS DECREASED BY 150 FPM.

Fig. 5-17 Rate of Climb - OEI, MCP

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5.9 LANDING PERFORMANCE

5.9.1 Landing Procedure

Use of the following landing procedure (see Fig. 5-18) assures the capability to land safely (on a smooth level surface) should a single-engine failure occur any time prior to or during the approach.

NOTE Go-around capability is NOT assured for OEI operation when using this landing procedure.

- | | |
|-------------------|--|
| Pre-landing check | – Perform |
| Landing approach | – Initiate to arrive at 50 ft AGL with a speed of 40 KIAS and a rate of descent of not more than 300 fpm (entry point) |

NOTE Approach airspeed can be increased by half of the wind speed.

After reaching entry point:

- | | |
|------------------|--|
| Collective pitch | – Adjust to maintain desired rate of descent |
| Ground speed | – Reduce (near-to-ground flare) |

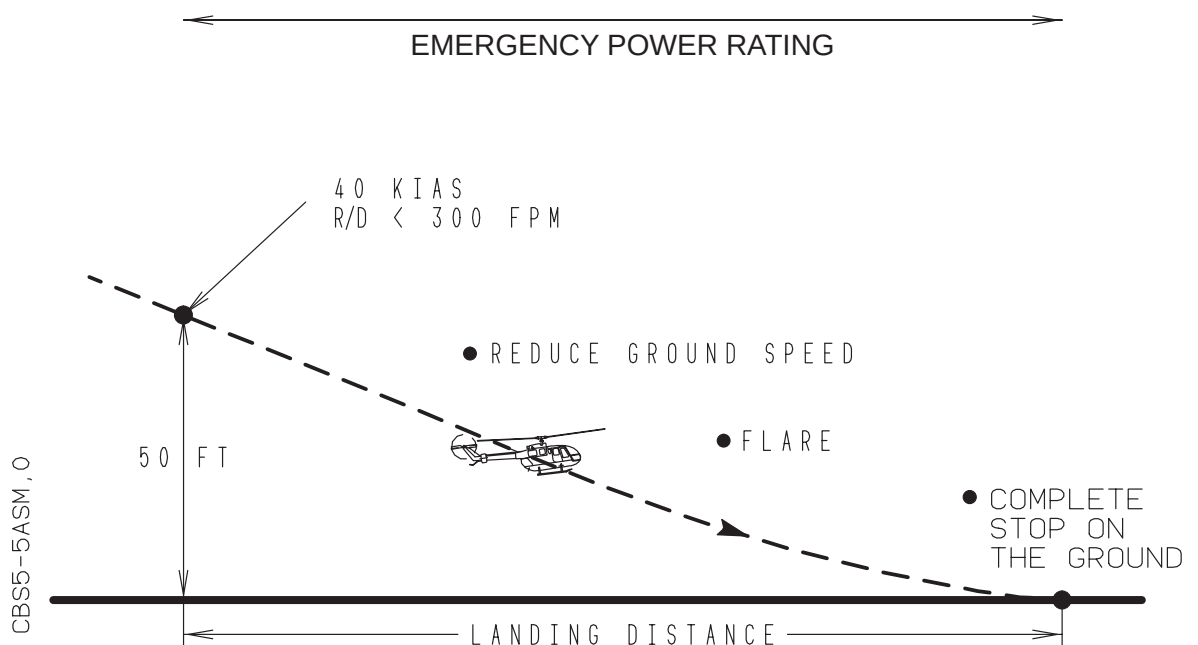


Fig. 5-18 OEI Landing Profile

5.9.2 Landing Distance

The landing distance chart (see Fig. 5-19) provides single-engine landing performance data for various combinations of pressure altitude and outside air temperature. The chart shows the landing distance required for the helicopter to come to a complete stop on a smooth, hard and dry level surface.

To achieve the values derived from the landing distance chart, the landing approach path at a 50-ft height should be established with OEI power required to maintain a 300 fpm rate of descent at 40 KIAS.

NOTE The minimum landing distance is 690 ft (210 m) under winds calm and all combinations of gross mass, pressure altitude and OAT.

EXAMPLE: (see Fig. 5-19)

Determine: **OEI Landing Distance Required to Clear a 50-ft Obstacle, with headwind**

Known:	OAT	25 °C
	Pressure altitude	2000 ft
	Gross mass	2000 kg
	Headwind	12 kt

Solution:

1. Enter chart at known OAT (25 °C).
2. Move vertically up to known pressure altitude (2000 ft).
3. Move horizontally right to known gross mass (2000 kg).
4. Move vertically down to Base Line and trace further down following guide lines.
5. Reenter chart at known headwind (12 kt) and move horizontally right to intersect tracing from above,
6. From point of intersection move vertically down and read landing distance = 512 ft (156 m).

OEI LANDING DISTANCE **(FROM 50-FT HEIGHT TO A COMPLETE STOP ON GROUND)**

1 X ALLISON 250-C20B

OEI – EMERGENCY POWER (810 °C, 95 % TORQUE)

ANTI-ICING – ON OR OFF

BLEED AIR CONSUMERS – OFF

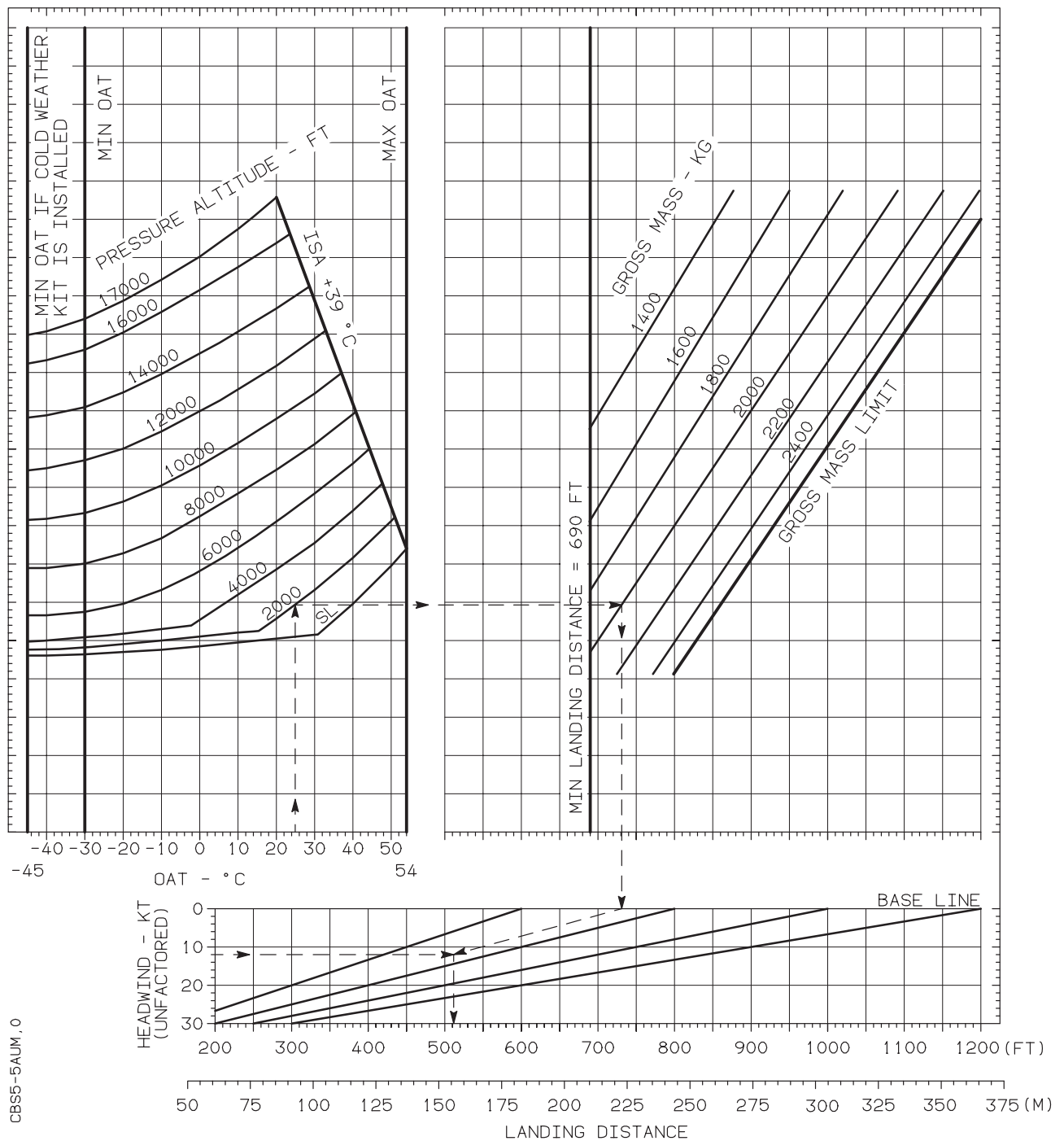


Fig. 5-19 OEI Landing Distance

SECTION 6

MASS AND BALANCE

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APPENDED TO SECTION 6

Form MBR-1, MASS AND BALANCE RECORD

Form EL-1, EQUIPMENT LIST

SECTION 6

MASS AND BALANCE

6.1 GENERAL

This section provides information required for helicopter loading and computing mass and balance.

It shall be the pilot's responsibility to make certain that:

- the helicopter is properly loaded so that the entire flight is conducted within the Center of Gravity Limits,
- all cargo is stowed and tied down properly so that in-flight shifting is impossible,
- proper tie-down equipment (i.e. ropes, belts, etc.) of sufficient strength has to be used. As many tie-down fittings as possible have to be used per single cargo item in order to reduce the individual load per fitting and to avoid inadvertent in-flight shifting.

6.1.1 Mass Definitions

Basic empty mass consists of the basic helicopter with required standard equipment, optional equipment, unusable fuel, and full operating fluids including transmission, gearbox and engine oils, hydraulic fluid, rotor brake oil, and windshield washer fluid.

Gross mass is the sum of the basic empty mass and the pilot/crew, the passengers, the baggage/cargo and the fuel. This value will vary with each mission.

6.1.2 Balance Definitions

Reference datum (RD) is an imaginary vertical plane located 3000 mm (118.1 in) forward of the Leveling Point (LP). For balance purposes, all horizontal distances (ARM) are measured from the RD (see Fig. 6-1).

Arm, for balance purposes, is the horizontal distance from the RD to the center of gravity of a given item. For other purposes, Fuselage Stations may be used. Fuselage stations are vertical planes perpendicular to, and measured along, the longitudinal axis of the helicopter. For the BO 105 helicopter arm and fuselage station are the same.

Moment (Massmoment) is the mass of an item multiplied by its arm.

Moment (kgmm) = Mass (kg) x Arm (mm)

Center of Gravity (CG) is the point about which the helicopter would balance if suspended. Distance from the RD is found by dividing the total moment by the gross mass of the helicopter.

$$\text{Arm (mm)} = \frac{\text{Sum of all moments (kgmm)}}{\text{Sum of all masses (kg)}}$$

6.2 BASIC EMPTY MASS CENTER OF GRAVITY

The procedure for establishing mass and moment (relative to the reference datum) of the empty helicopter is described in the BO 105 Maintenance Manual (MM). The MASS AND BALANCE RECORD (Form MBR-1) appended to this Section is used to maintain a continuous history of changes to the basic “as delivered” helicopter mass and balance data.

6.2.1 Mass and Balance Record

The basic empty mass (BEM) and center of gravity (CG) location are determined through actual weighing carried out by the helicopter manufacturer. This data is then entered on the first line of the MASS AND BALANCE RECORD, Form MBR-1 which then becomes a permanent part of the Flight Manual.

The MASS AND BALANCE RECORD must be updated (normally by transcribing the applicable information from the EQUIPMENT LIST) when necessary as follows:

- When additional equipment is installed on the helicopter necessitating a change in the basic empty mass (as per definition), add the new entry or entries to the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).
- Likewise, when equipment is removed from the helicopter, subtract the new entry or entries from the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).

At all times, the last mass and moment entries are considered the current basic empty mass and balance status of the helicopter.

6.2.2 Equipment List

An EQUIPMENT LIST, Form EL-1 is appended to this section and contains optional equipment of the particular helicopter when delivered. Each item on the list is provided with a number and description for identification, together with its mass, arm and moment.

Those items of equipment that were installed when the particular helicopter was initially weighed are so indicated by a check mark (✓) in the “Initial Weighing” column. Therefore the mass, arm and moment of these items are included in the basic empty mass (BEM) data found on the MASS AND BALANCE RECORD, Form MBR-1.

6.3 LOADING DATA (see Fig. 6-1)

The loading data provides location information necessary in calculating the longitudinal CG of the helicopter. Following is a listing of items with their respective arm (fuselage station location).

6.3.1 Crew and Passenger Stations

	ARM (mm)
Pilot and copilot	1890
Passengers	2690 (2740*)
* Only for Sectioned Bench Seat, FMS 10-21.	

6.3.2 Cargo Tie-down Points

	ARM (mm)
Cabin	2230
	2690
Cargo Compartment	3150
	3380
	3610
	4030

NOTE • Cargo compartment longitudinal center is at ARM 3900 mm.

- Ensure that the cargo is properly secured by restraining it from shifting with tie downs.
- Cargo should be centered in the cabin.
- Cargo shall be loaded within the perimeter defined by the tie-down ring centroids.
- The operator shall be responsible for the proper placement and securing of cargo in accordance with standard operating procedures and practices.

6.3.3 Fuel

	ARM (mm)
Supply Tank	2275
Main Tank	3220

6.3.4 Avionics Equipment Bay

ARM = 4250 mm

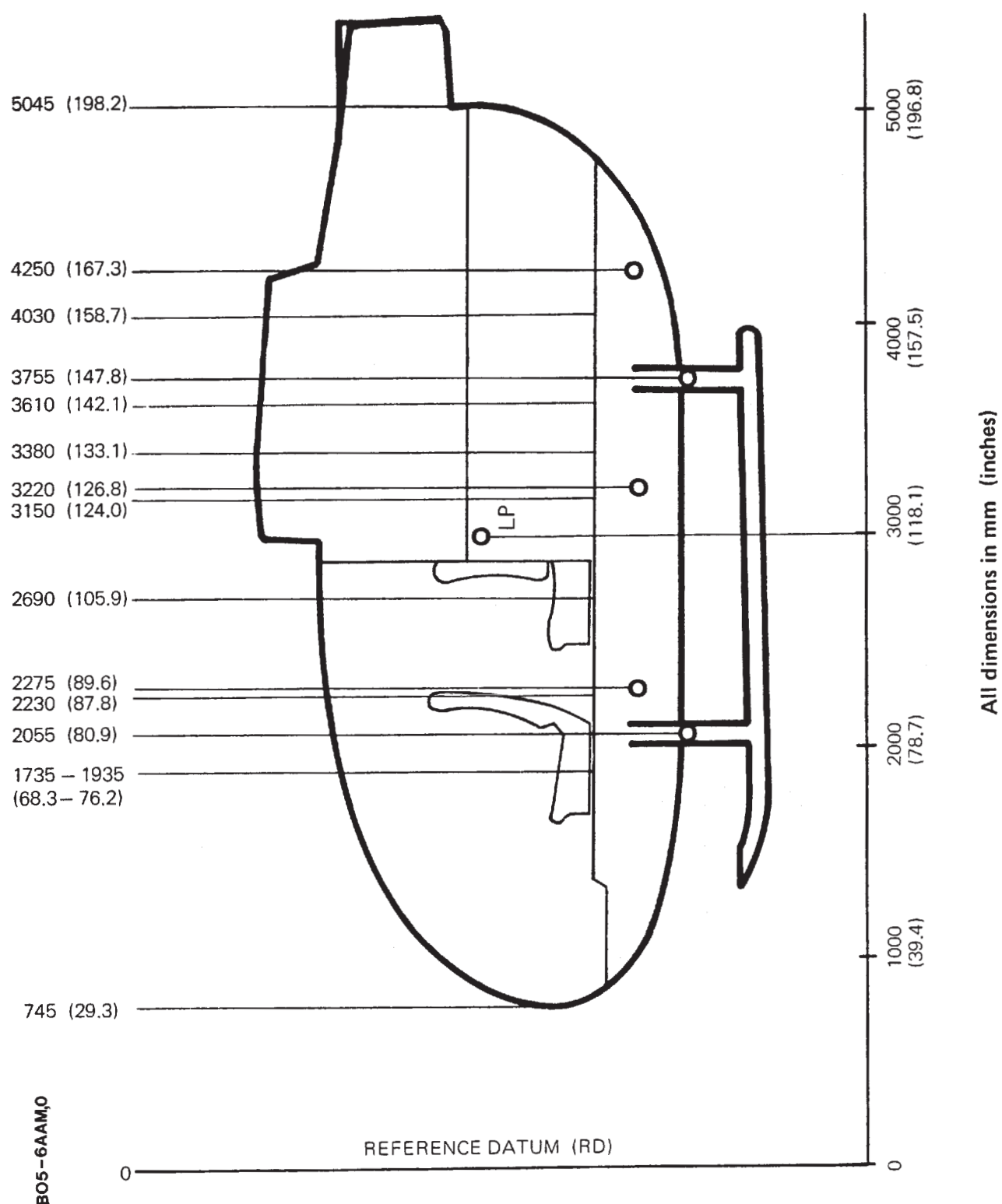


Fig. 6-1 Fuselage Station Diagram

6.4 CALCULATING LONGITUDINAL CG (see Fig. 6-2)

The following examples show a method of calculating the longitudinal center of gravity.

NOTE Basic empty mass data are to be taken from the latest entry on the MASS AND BALANCE RECORD, Form MBR-1 appended to this Section.

LOADING EXAMPLES						
Flight A: 1 pilot (80 kg), bulk cargo load (320 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
				Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1270	3360	4267200	1270	3360	4267200
Pilot	80	1890	151200	1350	3273	4418400
Bulk cargo	320	3900	1248000	1670	3393	5666400 ¹⁾
Fuel, supply tank	74	2275	168350	1744	3346	5834750 ²⁾
Fuel, main tank	382	3220	1230040	2126	3323	7064790 ³⁾
Flight B: 2 pilots (2 x 80 kg), 3 passengers (1 x 80 kg, 2 x 90 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
				Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1270	3360	4267200	1270	3360	4267200
Pilot and copilot	160	1890	302400	1430	3196	4569600
Passengers (3)	260	2740	712400	1690	3125	5282000 ¹⁾
Fuel, supply tank	74	2275	168350	1764	3090	5450350 ²⁾
Fuel, main tank	382	3220	1230040	2146	3113	6680390 ³⁾
NOTE To make certain that the CG does not "wander" outside of the CG envelope during the flight, the helicopter must be so loaded that the above points ¹⁾ (minimum landing mass), ²⁾ (supply tank full), and ³⁾ (supply and main tanks full) lie within the approved limits.						

NOTE The lateral CG position will always remain within limits regardless of the location of internal loads provided that the gross mass remains within limitations.

BO5-6BGM, 0

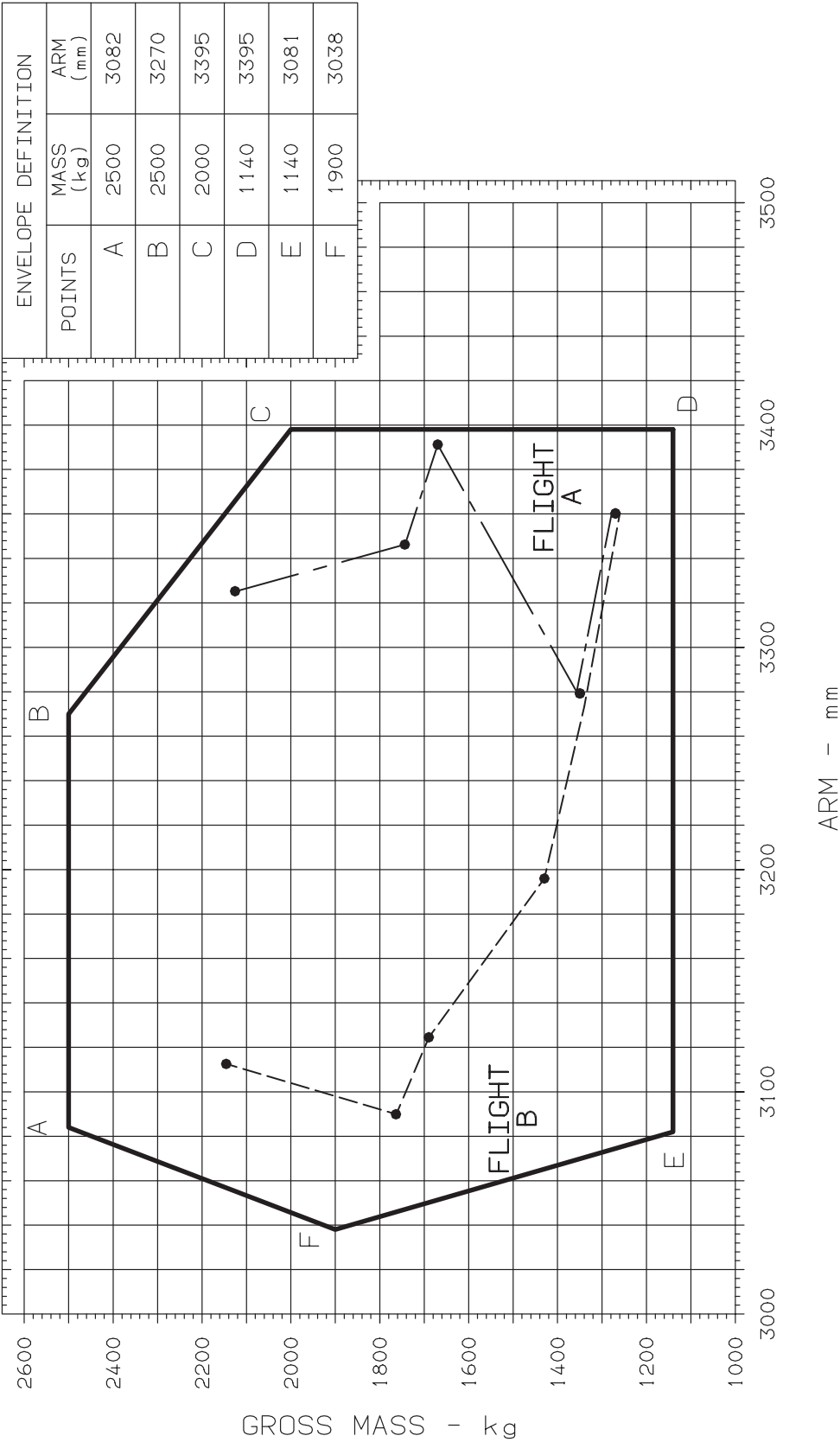


Fig. 6-2 CG Envelope with Loading Examples

NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

6.2 BASIC EMPTY MASS CENTER OF GRAVITY

The procedure for establishing mass and moment (relative to the reference datum) of the empty helicopter is described in the BO 105 Maintenance Manual (MM). The MASS AND BALANCE RECORD (Form MBR-1) appended to this Section is used to maintain a continuous history of changes to the basic “as delivered” helicopter mass and balance data.

6.2.1 Mass and Balance Record

The basic empty mass (BEM) and center of gravity (CG) location are determined through actual weighing carried out by the helicopter manufacturer. This data is then entered on the first line of the MASS AND BALANCE RECORD, Form MBR-1 which then becomes a permanent part of the Flight Manual.

The MASS AND BALANCE RECORD must be updated (normally by transcribing the applicable information from the EQUIPMENT LIST) when necessary as follows:

- When additional equipment is installed on the helicopter necessitating a change in the basic empty mass (as per definition), add the new entry or entries to the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).
- Likewise, when equipment is removed from the helicopter, subtract the new entry or entries from the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).

At all times, the last mass and moment entries are considered the current basic empty mass and balance status of the helicopter.

6.2.2 Equipment List

An EQUIPMENT LIST, Form EL-1 is appended to this section and contains optional equipment of the particular helicopter when delivered. Each item on the list is provided with a number and description for identification, together with its mass, arm and moment.

Those items of equipment that were installed when the particular helicopter was initially weighed are so indicated by a check mark (✓) in the “Initial Weighing” column. Therefore the mass, arm and moment of these items are included in the basic empty mass (BEM) data found on the MASS AND BALANCE RECORD, Form MBR-1.

6.3 LOADING DATA (see Fig. 6-1)

The loading data provides location information necessary in calculating the longitudinal CG of the helicopter. Following is a listing of items with their respective arm (fuselage station location).

6.3.1 Crew and Passenger Stations

	ARM (mm)
Pilot and copilot	1650
Passengers	2690 (2740*)
* Only for Sectioned Bench Seat, FMS 10-21.	

6.3.2 Cargo Tie-down Points

	ARM (mm)
Cabin	2230
	2690
Cargo Compartment	3150
	3380
	3610
	4030

NOTE • Cargo compartment longitudinal center is at ARM 3900 mm.

- Ensure that the cargo is properly secured by restraining it from shifting with tie downs.
- Cargo should be centered in the cabin.
- Cargo shall be loaded within the perimeter defined by the tie-down ring centroids.
- The operator shall be responsible for the proper placement and securing of cargo in accordance with standard operating procedures and practices.

6.3.3 Fuel

	ARM (mm)
Supply Tank	2275
Main Tank	3220

6.3.4 Avionics Equipment Bay

ARM = 4250 mm

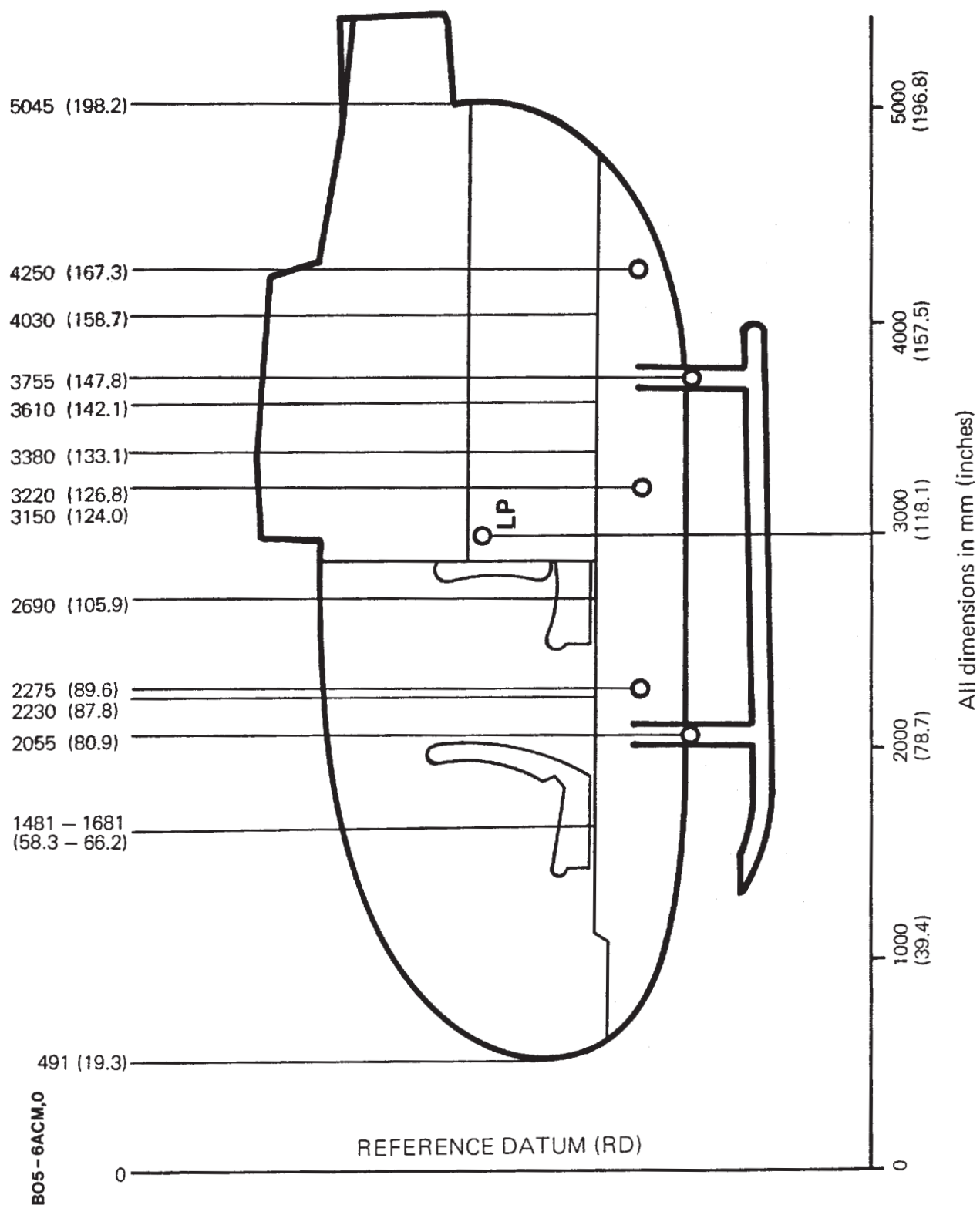


Fig. 6-1 Fuselage Station Diagram

6.4 CALCULATING LONGITUDINAL CG (see Fig. 6-2)

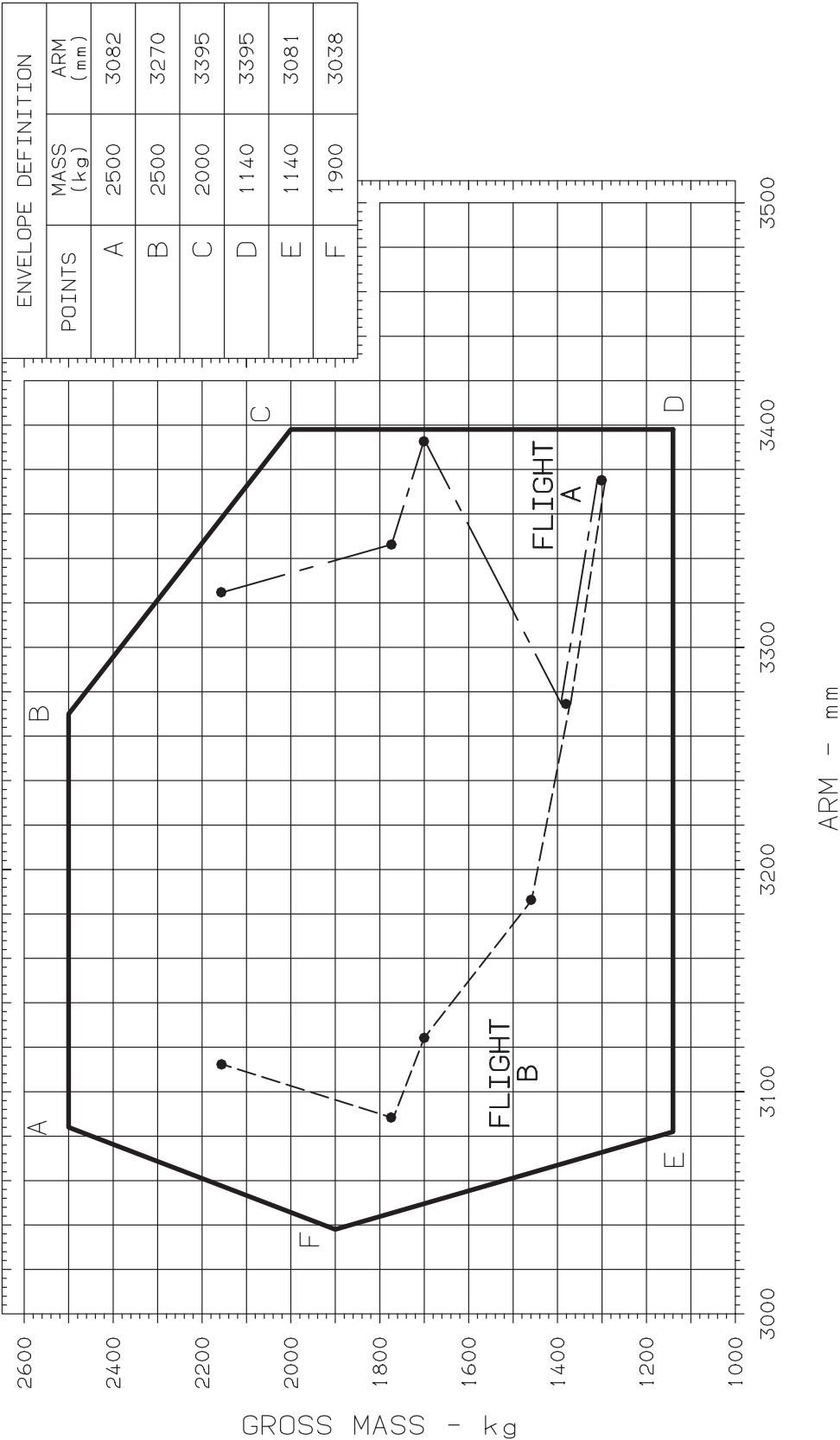
The following examples show a method of calculating the longitudinal center of gravity.

NOTE Basic empty mass data are to be taken from the latest entry on the MASS AND BALANCE RECORD, Form MBR-1 appended to this Section.

LOADING EXAMPLES						
Flight A: 1 pilot (80 kg), bulk cargo load (320 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1300	3375	4387500	1300	3375	4387500
Pilot	80	1650	132000	1380	3275	4519500
Bulk cargo	320	3900	1248000	1700	3393	5767500 ¹⁾
Fuel, supply tank	74	2275	168350	1774	3346	5935850 ²⁾
Fuel, main tank	382	3220	1230040	2156	3324	7165890 ³⁾
Flight B: 2 pilots (2 x 80 kg), 3 passengers (3 x 80 kg), and full fuel (456 kg). The resultant CG is solved as follows:						
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Resultant		
	Mass (kg)	Arm (mm)	Massmoment (kgmm)	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Basic empty mass	1300	3375	4387500	1300	3375	4387500
Pilot and copilot	160	1650	264000	1460	3186	4651500
Passengers (3)	240	2740	657600	1700	3123	5309100 ¹⁾
Fuel, supply tank	74	2275	168350	1774	3088	5477450 ²⁾
Fuel, main tank	382	3220	1230040	2156	3111	6707490 ³⁾
NOTE To make certain that the CG does not "wander" outside of the CG envelope during the flight, the helicopter must be so loaded that the above points ¹⁾ (minimum landing mass), ²⁾ (supply tank full), and ³⁾ (supply and main tanks full) lie within the approved limits.						

NOTE The lateral CG position will always remain within limits regardless of the location of internal loads provided that the gross mass remains within limitations.

BO5-6BLM, 0



NOTE REFER TO TEXT FOR LOADING EXAMPLE DATA.

Fig. 6-2 CG Envelope with Loading Examples

[illegible][illegible][illegible][illegible][illegible][illegible]

(Continuous History of Changes in Structure or Equipment Affecting Mass and Balance)

MODEL/VARIANT:

SERIAL NO.:

Page No.: of

[illegible]

BO105 EQUIPMENT LIST (EL)

MODEL/VARIANT: _____

SERIAL NO.: _____

Page ____ of ____

WEIGHINGS

Initial

Second

Third

Fourth

Date/Sign.

Date/Sign.

Date/Sign.

Date/Sign.

Item
No.

Description of Article or
Modification/Part No.

FMS
No.

Mass
(kg)

Arm
(mm)

Massmoment
(kgmm)

Mark as
required ¹⁾

Mark as
required ¹⁾

Mark as
required ¹⁾

Mark as
required ¹⁾

¹⁾ Required markings: ✓ = installed; 0 = not installed.

BO105 EQUIPMENT LIST (EL)

MODEL/VARIANT: _____

SERIAL NO.: _____

Page ____ of ____

WEIGHINGS

Initial

Second

Third

Fourth

Date/Sign.

Date/Sign.

Date/Sign.

Date/Sign.

Item
No.

Description of Article or
Modification/Part No.

FMS
No.

Mass
(kg)

Arm
(mm)

Massmoment
(kgmm)

Mark as
required ¹⁾

Mark as
required ¹⁾

Mark as
required ¹⁾

Mark as
required ¹⁾

¹⁾ Required markings: ✓ = installed; 0 = not installed.

SECTION 7

SYSTEM DESCRIPTION

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SECTION 7

SYSTEM DESCRIPTION

7.1 GENERAL

This section contains general information that applies specifically to standard equipped helicopters serial number S-161 and subsequent. Refer to the BO 105 Maintenance Manual for information pertaining to earlier serial numbered helicopters.

7.2 LOCATION AREAS (see Fig. 7-1)

7.2.1 Nose Compartment

The nose compartment located forward of the cabin area provides a location for the battery (*Model C Variants* only) and other electrical system components. Access is provided by a nose door fitted with an external power unit (EPU) connector access flap.

7.2.2 Cabin

The cabin area comprises the cockpit and passenger compartment.

A hinged jettisonable door on either side permits direct access to the cockpit. Large bubble-type acrylic windshields and clear acrylic windows in the crew doors and roof allow for good visibility.

A sliding door is located on either side of the fuselage for direct access to the passenger compartment. A large acrylic window is installed in each door.

7.2.3 Cargo Compartment

The cargo compartment may be separated from the cabin by a removable insulating divider. Access to the compartment is from the aft end provided by two clamshell doors. The engine deck forms the roof of the cargo compartment, acts as a firewall and provides mounting for the battery (RH side, *Model S Variants* only).

7.2.4 Avionics Equipment Bay

An avionics equipment bay is located beneath and at the aft end of the cargo compartment. This bay provides for installation of avionics equipment and is usually equipped with a forced-air ventilation system. Access is provided through a hinged and latched panel in the cargo compartment floor.

7.2.5 Tail Boom

The root of the tail boom is attached to the fuselage at an integral extension of the engine deck. The tail boom consists of a conical structure with a vertical fin. The tail rotor system

and components of both the yaw flight control and comm/nav systems are attached to the tail boom.

7.2.6 **Landing Gear**

The landing gear is of the fixed tubular skid-type comprised of two aluminum cross tubes passing transverse through the fuselage structure and to which aluminum longitudinal skid tubes are attached.

7.3 **COCKPIT ARRANGEMENT** (see Fig. 7-2)

7.3.1 **Instrument Panel and Console Layout** (see Fig. 7-3 thru 7-5)

The cockpit-mounted controls, indicators and annunciators shown in Figs. 7-3 thru 7-5 will be discussed later with their associated systems. Annunciator lights and their fault indications are summarized under lighting equipment. Electrical fuses referred to in this section are normally located on a fuse panel discussed later under the electrical system.

7.3.2 **Crew Member Seats**

Both pilot and copilot seats are adjustable fore-aft and fitted with lap safety belts.

7.4 **EMERGENCY EQUIPMENT**

7.4.1 **Engine Overheat and Fire Detection**

Two heat-sensitive detectors are located on each engine. When the ambient temperature in either engine compartment increases beyond limits, the detector contacts open and, via a control unit, energize the respective F warning light located on the instrument panel. Each system is powered from the Main Bus via a fuse.

7.4.2 **Portable Fire Extinguisher**

A portable fire extinguisher is normally mounted on a quick-release support on the forward edge of the pilot's seat. The fire extinguishing agent is released by a hand-operated lever on top of the extinguisher. Inadvertent discharge is prevented by a safety-pin on the actuating lever. Operating instructions are printed on the extinguisher.

7.4.3 **First Aid Kit**

A first aid kit is located either in a mount on the cabin floor or in a retaining net secured on the inside of the cargo compartment RH clamshell door.

EFFECTIVITY *For IAC-AR registered helicopters only*

7.4.4 **Emergency VHF Radio**

An emergency VHF radio P/N 105-897936, located in the cockpit beneath the pilot's collective lever, shall be operated in accordance with its appropriate instructions.

EFFECTIVITY *All*

BO5 - 7 AAM,0

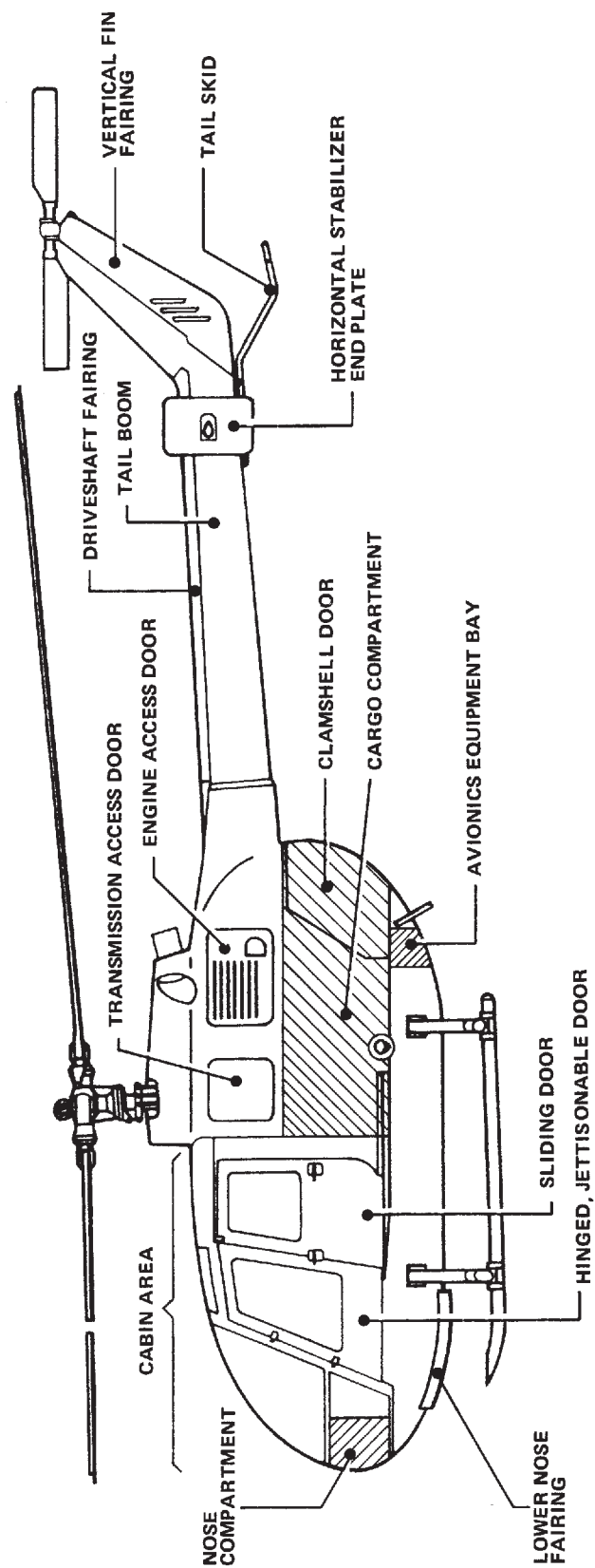


Fig. 7-1 Location Areas and Access

BO5-7 ABM,0

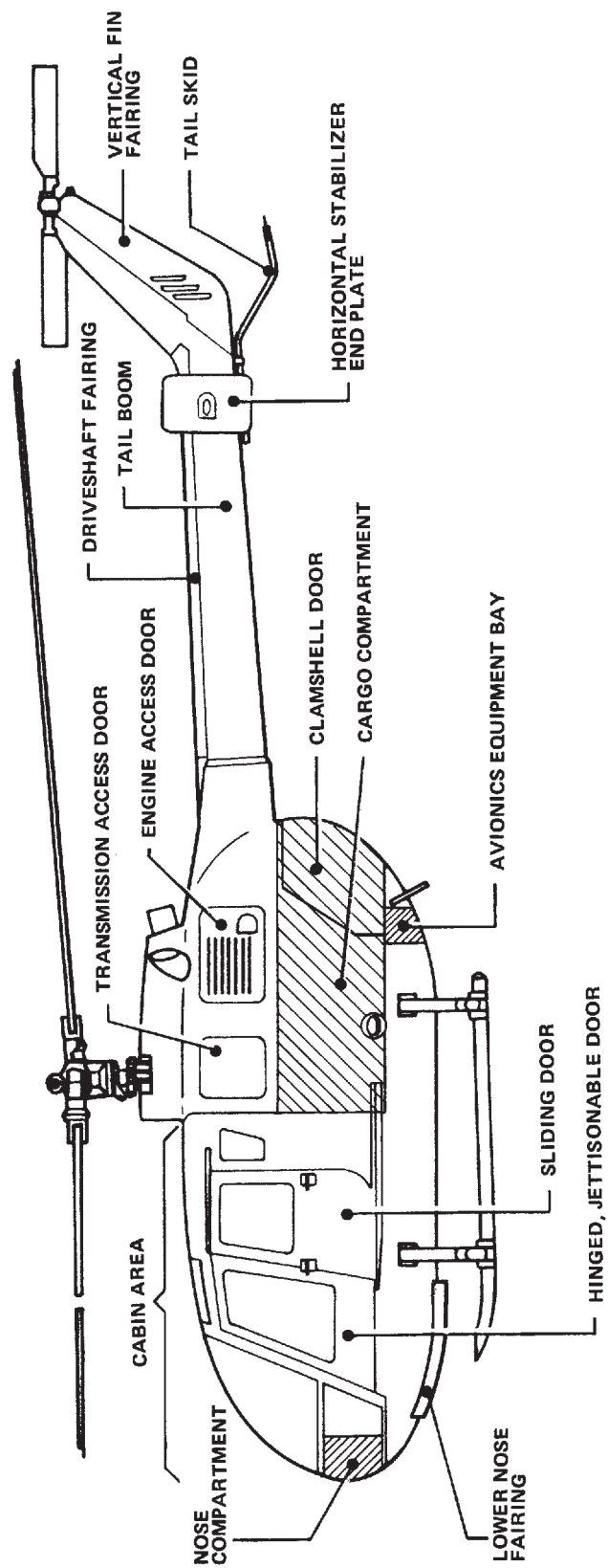
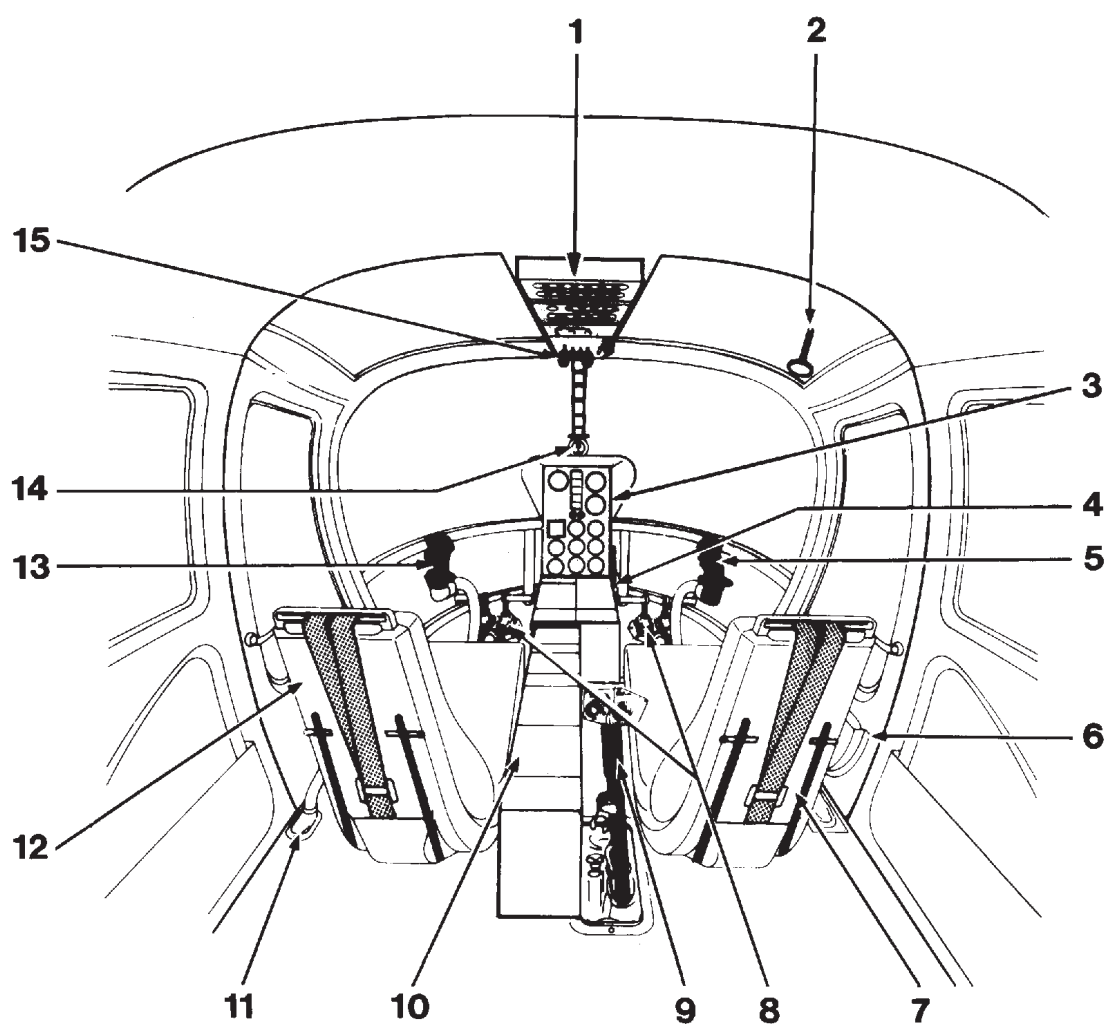


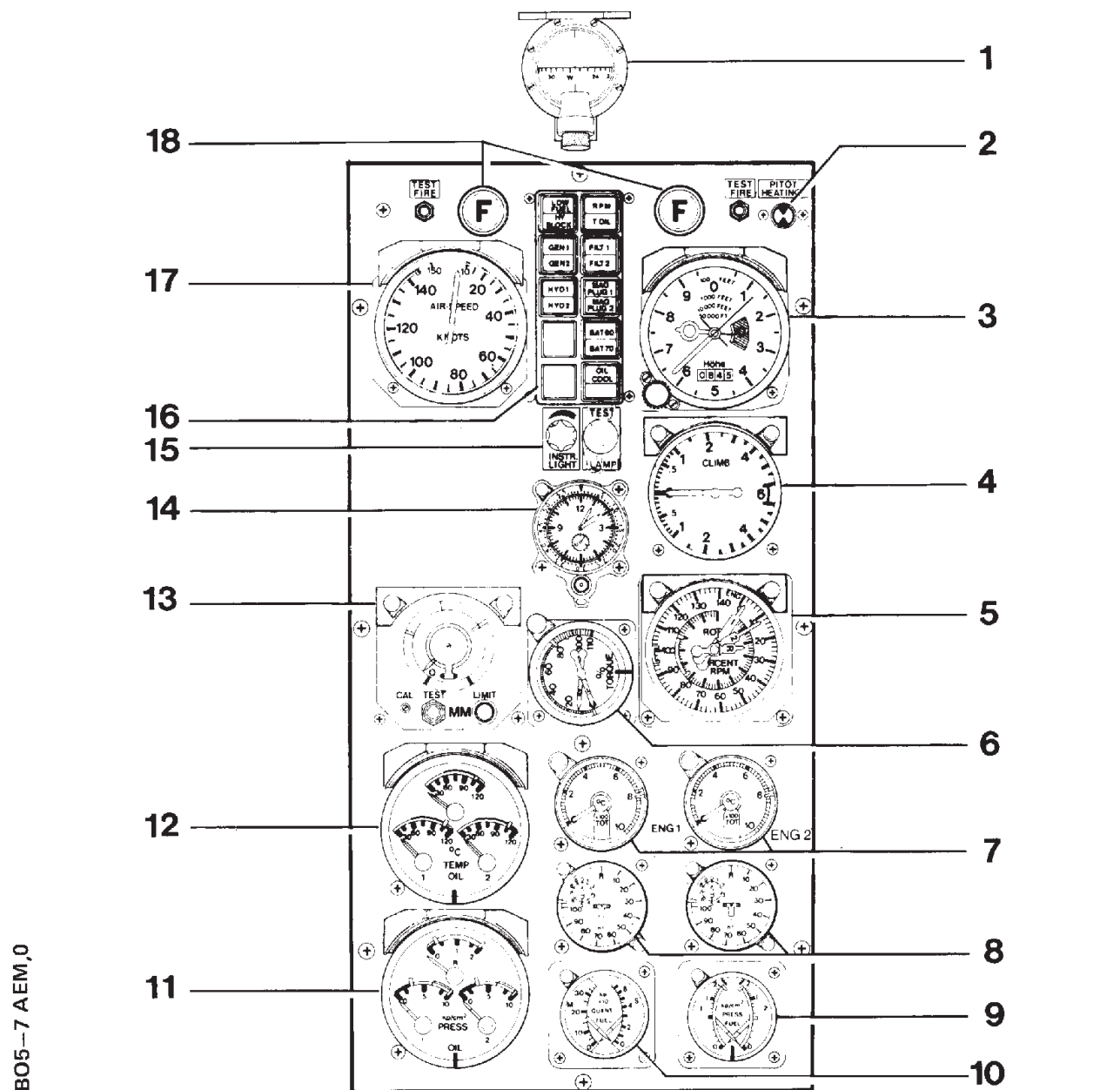
Fig. 7-1 Location Areas and Access



BO5-7 ACM,0

- | | | |
|-----------------------|-------------------------------|------------------------------|
| 1. OVERHEAD CONSOLE | 6. MAP POCKET | 11. COPILOT COLLECTIVE LEVER |
| 2. OAT INDICATOR | 7. PILOT SEAT | 12. COPILOT SEAT |
| 3. INSTRUMENT PANEL | 8. DIRECTIONAL CONTROL PEDALS | 13. COPILOT CYCLIC STICK |
| 4. SWITCH PANEL | 9. PILOT COLLECTIVE LEVER | 14. STANDBY MAGNETIC COMPASS |
| 5. PILOT CYCLIC STICK | 10. CENTER CONSOLE | 15. POWER LEVERS |

Fig. 7-2 Typical Cockpit Layout



- | | | |
|-----------------------------|--------------------------------------|--|
| 1. STANDBY MAGNETIC COMPASS | 8. ENGINE N ₁ INDICATORS | 14. CLOCK |
| 2. PITOT HEAT INDICATOR | 9. FUEL PRESSURE INDICATOR | 15. INSTRUMENT LIGHT RHEOSTAT AND ANNUNCIATOR LIGHT LAMP TEST PUSHBUTTON |
| 3. ALTIMETER | 10. FUEL QUANTITY INDICATOR | 16. WARNING LIGHT SEGMENTS |
| 4. VERTICAL SPEED INDICATOR | 11. TRIPLE OIL PRESSURE INDICATOR | 17. AIRSPEED INDICATOR |
| 5. TRIPLE TACHOMETER | 12. TRIPLE OIL TEMPERATURE INDICATOR | 18. ENGINE FIRE WARNING LIGHTS WITH TEST FIRE PUSHBUTTONS |
| 6. DUAL TORQUE INDICATOR | 13. MAST MOMENT INDICATOR | |
| 7. ENGINE TOT INDICATORS | | |

Fig. 7-3 Typical Instrument Panel

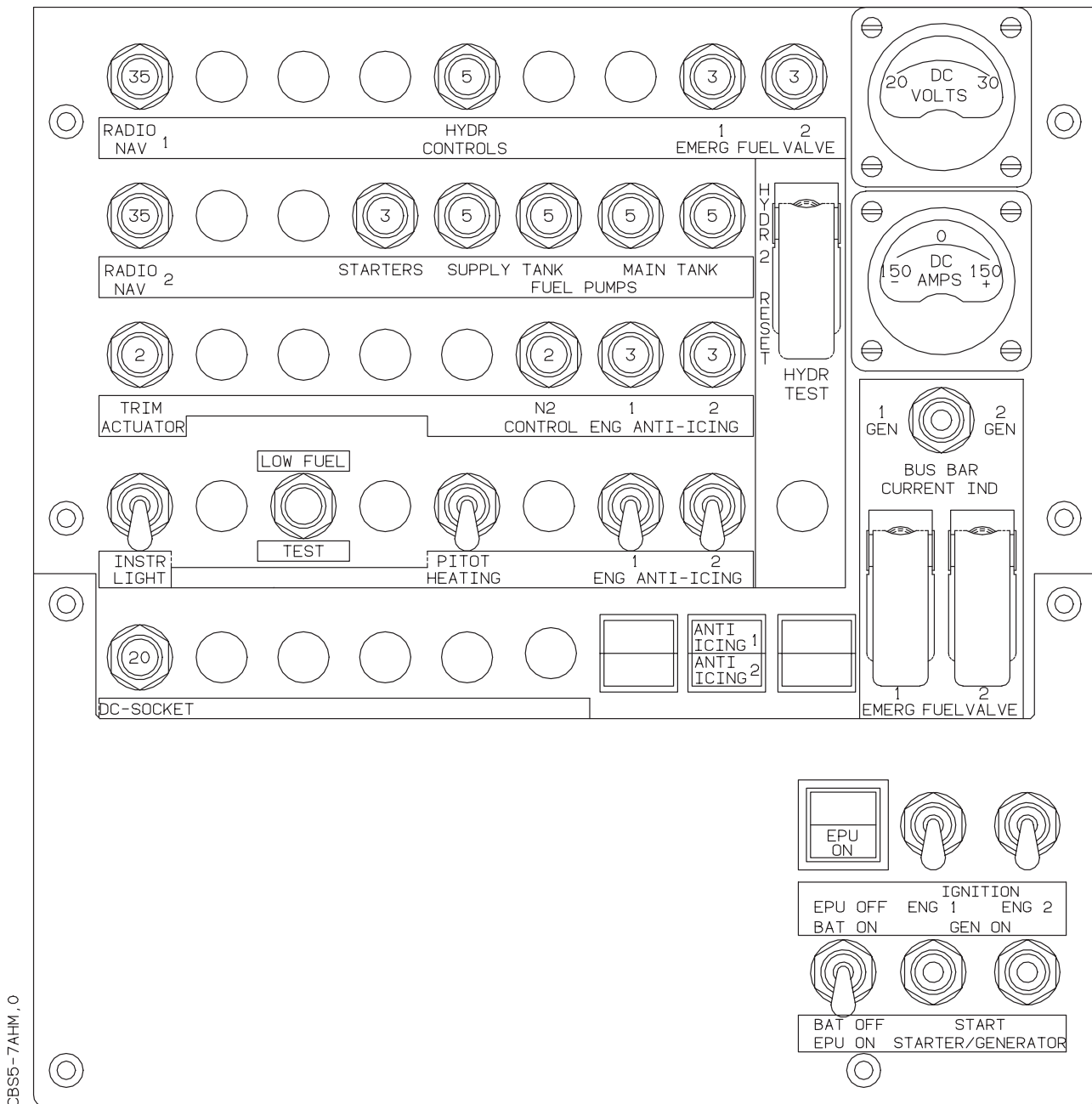


Fig. 7-4 Typical Center Console Switch Panel

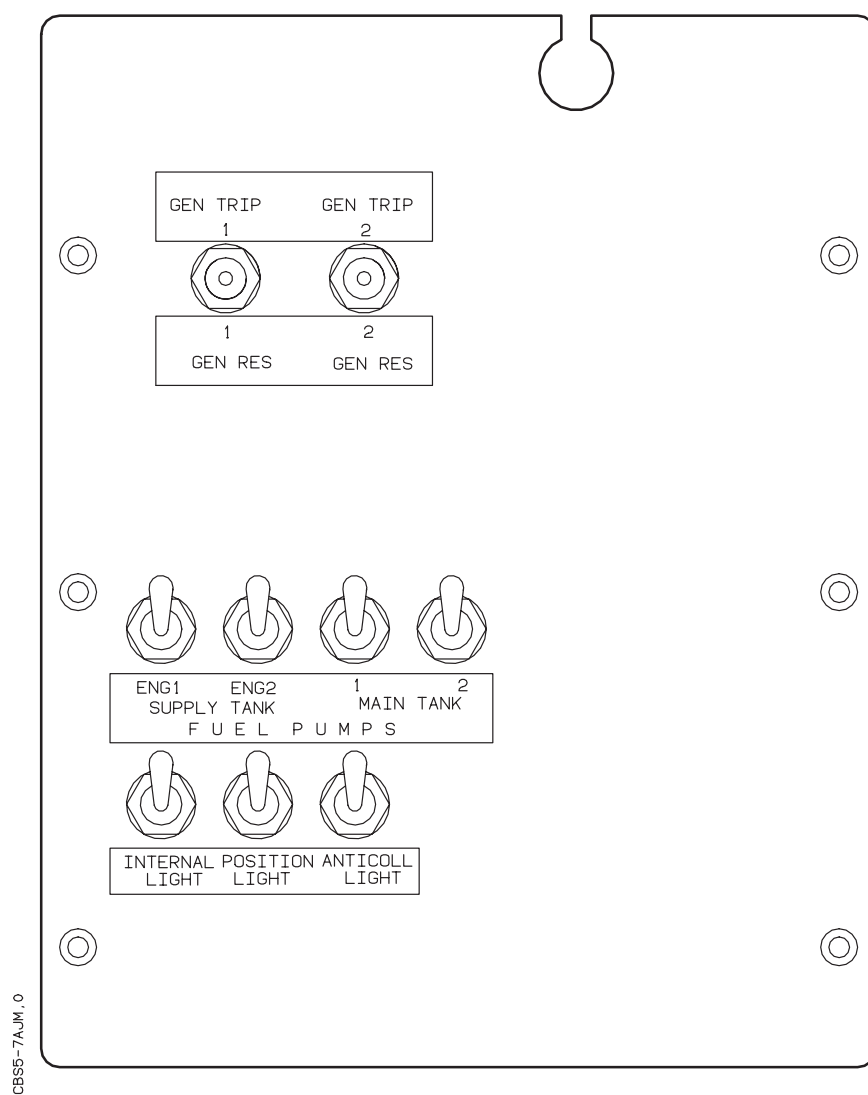


Fig. 7-5 Typical Overhead Console

7.5 POWER PLANT AND RELATED SYSTEMS

7.5.1 Engines (see Fig. 7-6)

Two Allison 250-C20 series gas turbine engines are located in separate fireproof compartments aft of the main transmission and above the cargo compartment. Each engine is divided into four modules:

- Compressor section consisting of a compressor support case assembly, a compressor and a diffuser assembly. The compressor has six axial stages and one centrifugal stage. Components mounted on the compressor section are an anti-icing air valve and a compressor bleed control valve.
- Combustion section consisting of an outer combustion case and a combustion liner. A spark igniter and a fuel nozzle are mounted in the aft end of the outer combustion case. Two external air ducts connect the compressor section with the combustion section.
- Turbine section consisting of turbine supports, collector supports, a gas producer turbine rotor and a power turbine rotor. The two stage gas producer turbine drives the compressor and N₁ accessory gear train. The two-stage power turbine furnishes the output power of the engine via the power and accessory gearbox. Mounted on the turbine section are a thermocouple harness, exhaust ducts and a fire-shield to which are attached the two fire detectors.
- Power and accessory gearbox section housing the main power and accessory gear drive trains in a single gear case. The gear case serves as the structural support of the engine. All engine driven components including the engine-mounted accessories are attached to the case. A two-stage gear set reduces the power turbine rotational speed to the output drive rotational speed. Accessories driven by the power turbine (N₂) are the N₂ tachgenerator and N₂ governor. The gas producer gear train drives the oil pump assembly (internally mounted), the fuel pump, N₁ tachgenerator, gas producer fuel control and starter/generator.

7.5.2 Air Induction

The engines receive air from the main transmission compartment which acts as a plenum chamber. An air intake fairing above the cabin roof forward of the transmission supplies a constant airflow towards the transmission compartment.

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

A deflector shield is installed within the air intake fairing to direct the airflow downwards thus preventing injection of foreign matter into the engines.

EFFECTIVITY ALL

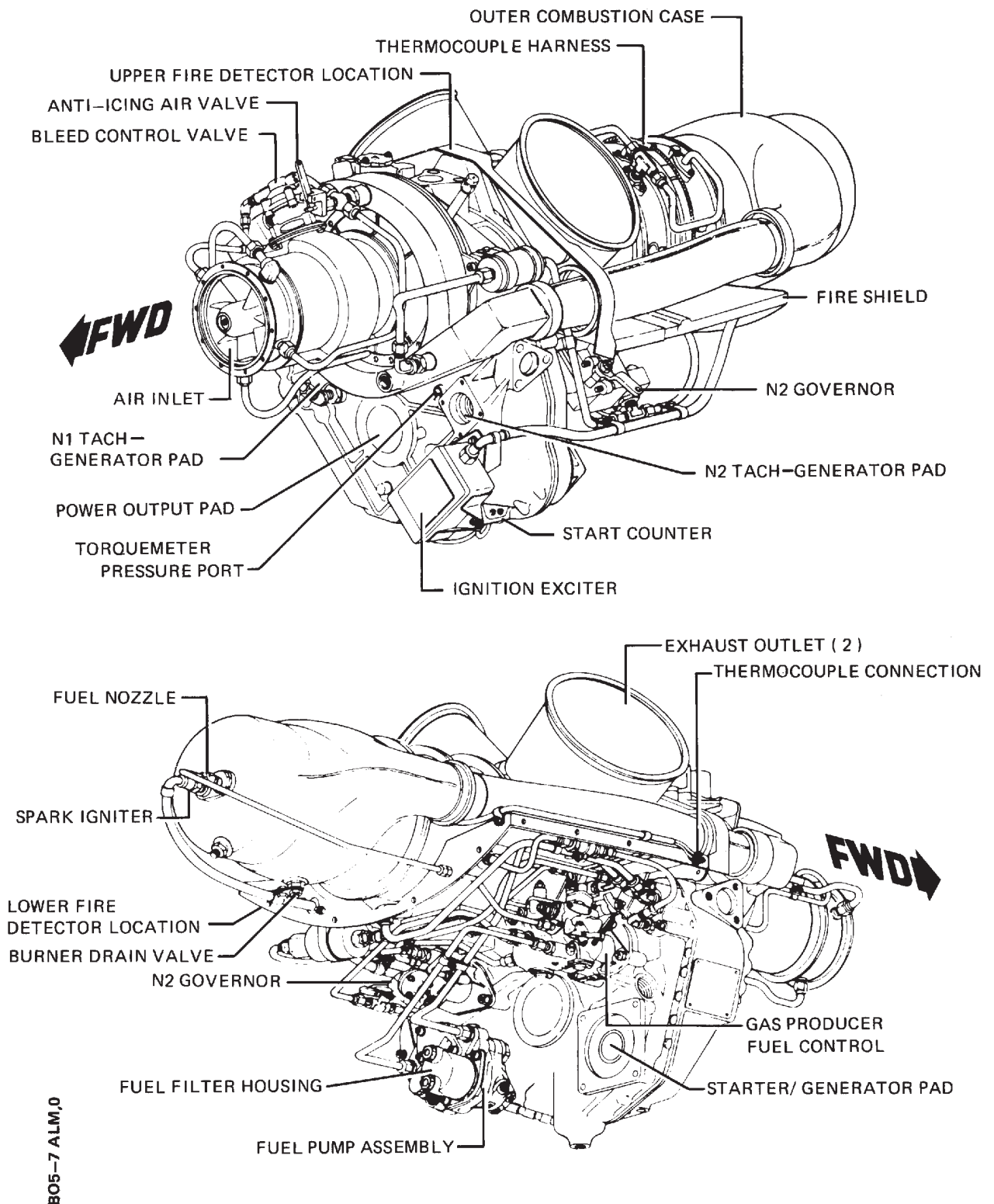


Fig. 7-6 Engine Component Locations

7.5.3 Inlet Anti-icing System

To accomplish engine anti-icing functions, hot compressor discharge air is tapped from the diffuser scroll through an electrical actuator operated anti-icing valve and routed via external lines to the compressor inlet guide vanes and front bearing support hub. The electrical actuators are controlled by the ENG ANTI-ICING - 1, 2 switches and monitored by the **ANTI ICING 1 (2)** caution lights all located on the center console switch panel.

In the ON position, the ENG ANTI-ICING switches supply power from the Main Bus via ENG ANTI-ICING - 1, 2 circuit breakers to the respective actuator motor retracting the actuator arm and opening the anti-icing valve.

A limit switch stops the actuator motor and activates the respective **ANTI ICING** caution light. The caution light remains on until after the system is switched OFF and the actuator has fully closed the anti-icing valve.

7.5.4 Fuel and Control System

Each engine fuel and control system comprises an engine-mounted fuel pump and filter assembly, a gas producer fuel control, a power turbine (N₂) governor, and a fuel nozzle. This system controls engine power output by controlling the gas producer speed. Gas producer speed levels are established by the action of the power turbine fuel governor which senses power turbine (N₂) speed. The power turbine (load) speed is selected by the pilot (ENGINE TRIM switch) and the power required to maintain this speed is automatically maintained by power turbine governor action on metered fuel flow.

A differential pressure switch is connected to the engine fuel filter to detect the beginning of clogging and activate the respective **FILT 1 (2)** warning light. A clogged filter element will be bypassed automatically supplying unfiltered fuel to that engine.

7.5.5 Oil System (see Fig. 7-7)

Each engine is lubricated by an independent pressurized recirculating dry sump system. A gear-type pressure and scavenge pump assembly is mounted within the gearbox. An assembly containing an oil filter element, a filter bypass valve, and a pressure regulating valve is located in the upper RH side of the gearbox housing and is accessible from the top of the engine.

A combined dual engine oil tank and engine/transmission oil cooler assembly is located to the right of the main transmission. The forward half of the tank/cooler assembly is connected to the left engine oil system while the rear half is connected to the right engine oil system. Two engine-mounted indicating chip detectors that activate the respective **MAG PLUG 1 (2)** warning light, are installed in each system.

The operation of each oil system is completely automatic self-regulating and does not require operator control beyond monitoring the indicators.

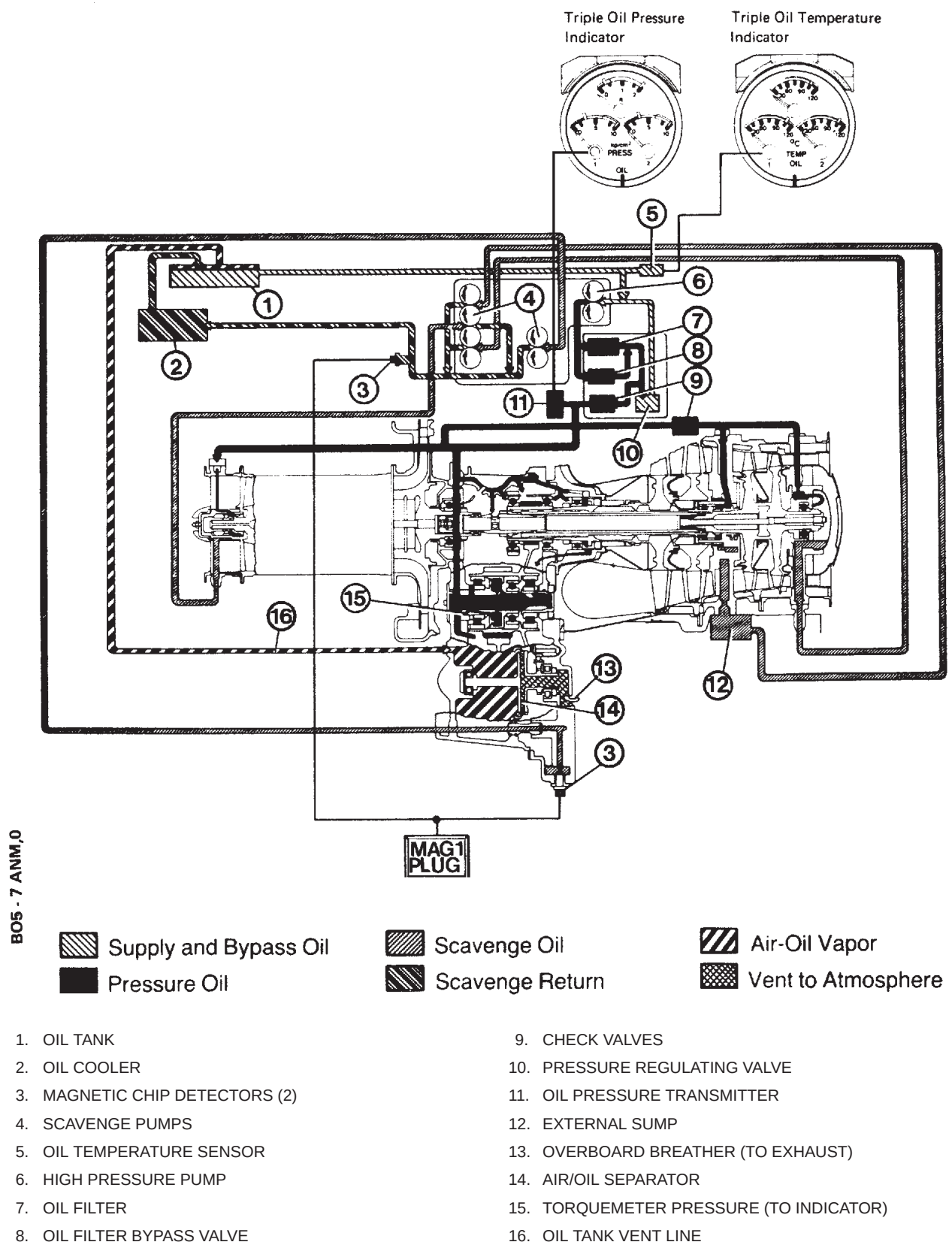


Fig. 7-7 Engine Oil System Schematic

7.5.6 Starting and Ignition System

Each engine is equipped with an independent starting/ignition system for starting the engines individually and so designed that simultaneous starting of both engines is impossible. A starter/generator unit mounted to the accessory gearbox drives the N_1 gear train when the starter motor circuits are activated. An ignition exciter unit and spark igniter comprise the ignition portion of each engine's system.

The STARTER/GENERATOR - ENG 1, ENG 2 switches on the center console switch panel are spring-loaded to the center off position when using the START position. In the START position, the switch supplies electrical power from the Main Bus via a common STARTERS circuit breaker to the respective starter relay. This relay in turn activates the starter motor circuit by supplying power from the Starter Bus and provides power to the respective IGNITION switch.

With the IGNITION switch in the on position, the exciter unit begins sending high power impulses to the spark igniter. Venting the engines is therefore possible by energizing the starter circuit without switching on the ignition circuit.

7.5.7 Control System

The engines have a conventional control system, i.e. power lever position and the degree of collective pitch basically establish the power output demands placed on the engines.

The gas producer (N_1) rpm of both engines is independently controlled by means of separate power levers mounted on the cockpit ceiling forward of the overhead console. Each power lever is mechanically connected to its respective engine gas producer fuel control to permit manual establishment of the gas producer speeds between OFF, IDLE and FLIGHT.

A separate positive lock is provided for each lever in the IDLE position so that repositioning of the levers to a position other than IDLE requires depressing the lock buttons.

Individual and simultaneous dual engine power (N_2) trimming is accomplished by means of a four-way ENGINE TRIM switch on each collective lever switch box (see Fig. 7-14). Pressing the switch to either left or right of center will increase the respective engine torque and simultaneously reduce the torque of the other engine. Pressing the switch either forward or aft will simultaneously increase or decrease respectively the torque of both engines. The pilot's ENGINE TRIM switch has command authority over the copilot's switch if installed. The N_2 trim circuit is powered by the Main Bus via the N_2 CONTROL circuit breaker located on the center console switch panel.

A mechanical connection between the N_2 governor trim actuator and the collective control hydraulic boost actuator provides automatic rpm control (droop compensation) when changes in collective pitch occur.

7.5.8 Indicators (see Fig. 7-3)

Torque indicator (% TORQUE) simultaneously indicates the percent of torque output of both engines. Each engine's torquemeter, incorporated in the accessory gearbox, provides an oil pressure signal directly proportional to the output torque. This oil pressure is piped directly to the respective indicator.

Triple tachometer indicator (ENG-ROTOR PERCENT RPM) simultaneously indicates both engine's power turbine speed (N_2) on an outer scale in percent rpm. The engine % N_2 rpm signals are produced by a tachgenerator mounted on the left front face of the accessory gearbox.

Turbine outlet temperature indicators ($^{\circ}\text{C} \times 100 \text{ TOT}$) indicate individual engine exhaust gas temperature measured by four thermocouple probes placed downstream of the power turbine. The bimetallic thermocouples generate DC voltage directly proportional to the sensed gas temperature.

Gas producer tachometers (% N_1) indicate individual engine gas producer speed (N_1) in percent of rated rpm. The gas producer turbine speed is measured by a tachgenerator mounted on the right front face of the accessory gearbox.

Triple oil temperature indicator ($^{\circ}\text{C TEMP OIL}$) simultaneously indicates engine 1 oil temp (left scale) and engine 2 oil temp (right scale). Each engine's oil temperature sensor is installed in an oil supply line fitting mounted on the top right front face of the accessory gearbox. The triple oil temperature indicator is powered from the Main Bus via a fuse.

Triple oil pressure indicator ($\text{kp/cm}^2 \text{ PRESS OIL}$) simultaneously indicates engine 1 oil pressure (left scale) and engine 2 oil pressure (right scale). Each engine oil pressure transmitter is fitted to a clamp securing the oil lines to the engine compartment floor. The triple oil pressure indicator is powered from the Main Bus via a fuse.

7.5.9 Annunciators

MAG PLUG 1, MAG PLUG 2 warning lights will activate when excessive metallic particles are detected by either engine-mounted chip detector.

ANTI ICING 1, ANTI ICING 2 caution lights will activate when each engine anti-icing system valve is open.

7.6 FUEL SYSTEM (see Fig. 7-8)

The fuel system comprises three subsystems, i.e. storage, supply, and indication. For refueling and defueling procedures, refer to Section 8.

7.6.1 Storage

Fuel is stored in underfloor compartments using three bladder-type fuel cells comprising two tanks; a supply tank and a main tank. The supply tank, located immediately aft of the landing gear forward cross tube, provides fuel for the engines. Two aft cells, located forward and aft of the landing gear aft cross tube, are interconnected with a large diameter tube to form the main tank. The main tank fuel is pumped to the supply tank with excess fuel returning via two overflow ports.

The fuel cells are normally vented overboard and are drained through separate valves accessible on the fuselage underside. The main tank fuel cells are refueled through a common filler neck located on the left side of the fuselage. The filler neck is designed to accept a standard gravity-fill type nozzle.

7.6.2 Supply

Fuel from the main tank is transferred to the supply tank through electric pumps. Fuel is then further pumped from the supply tank to each engine through separate supply systems, each comprised of an electrical supply tank pump and an electrical emergency fuel shutoff valve.

The fuel pump switches, labeled SUPPLY TANK - ENG 1, ENG 2 and MAIN TANK - 1, 2 are located on the overhead console. These switches power the fuel pumps from the Main Bus via separate circuit breakers (FUEL PUMPS - SUPPLY TANK, MAIN TANK) located on the center console switch panel. The EMERG FUEL VALVE - 1, 2 switches are also located on the center console switch panel and power the fuel valves from the Main Bus via separate circuit breakers (EMERG FUEL VALVE - 1, 2). The two EMERG FUEL VALVE switches are guarded to prevent accidental usage.

7.6.3 Indicators

Dual fuel quantity indicator (FUEL QTY) simultaneously indicates both the supply tank (S) and main tank (M) fuel quantities. Two fuel quantity transmitter probes are installed, one in each tank. The indicator is powered from the Main Bus via a fuse.

Dual fuel pressure indicator (FUEL PRESS) simultaneously indicates the left engine (1) supply pressure and the right engine (2) supply pressure in kp/cm^2 . A pressure transmitter is mounted in an adapter fitting installed immediately upstream of each engine-mounted fuel filter. The indicator is powered from the Main Bus via a fuse.

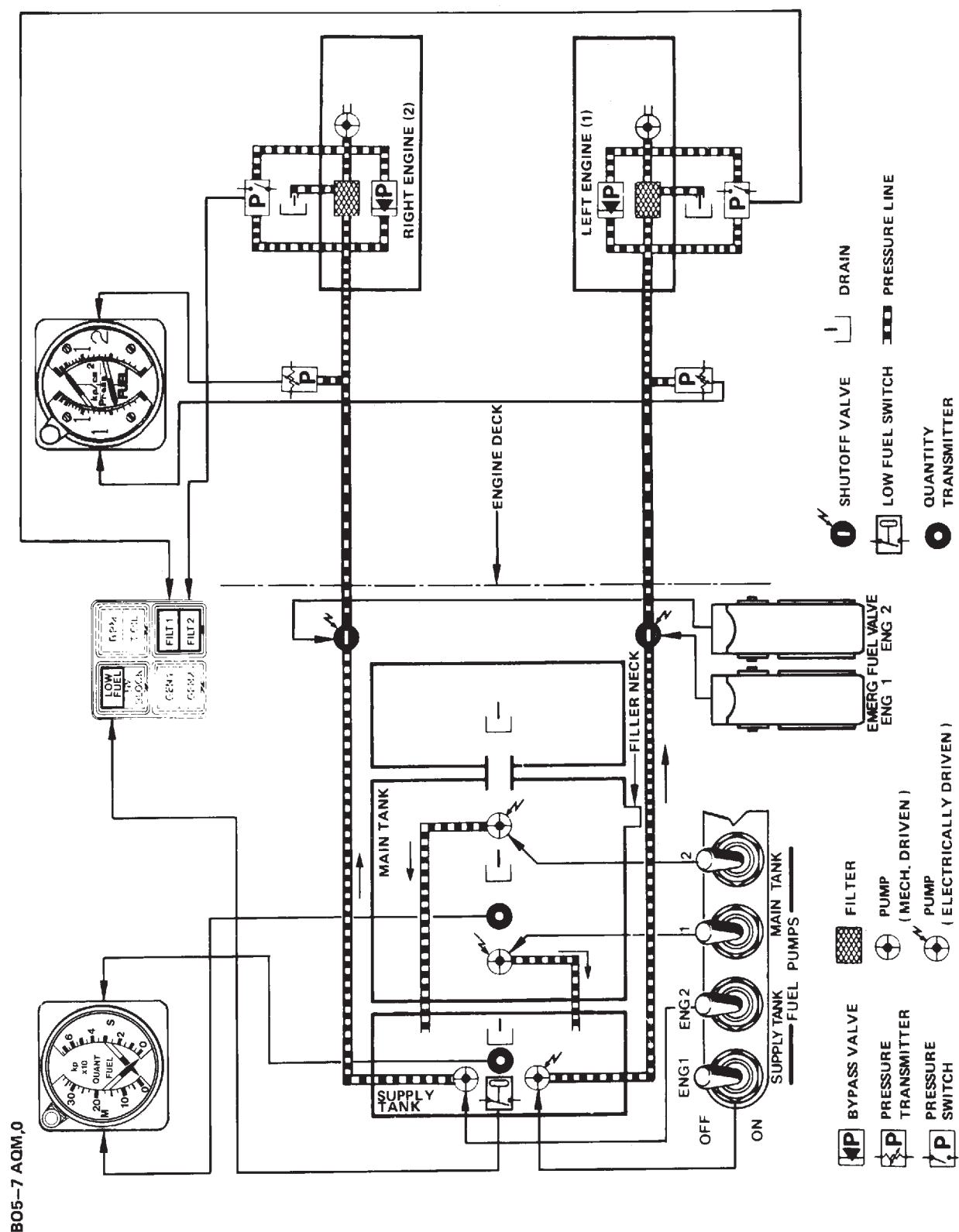


Fig. 7-8 Fuel System Schematic

7.6.4 Annunciators

LOW FUEL warning light is activated by a fuel level float switch, located in the supply tank, when the supply tank usable quantity is below 60 kg. The **LOW FUEL** warning light circuit, powered from the Main Bus via a fuse, can be functionally tested by pressing the LOW FUEL TEST pushbutton located on the center console switch panel.

FILT 1, FILT 2 warning lights will be activated when the engine-mounted filter becomes clogged. The **FILT** warning lights circuit is powered from the Main Bus via a fuse.

7.7 POWER TRAIN (see Fig. 7-9)

The power train transmits engine power to both rotors and the transmission-mounted accessories. The power train comprises two engine drive shafts, a main transmission with accessory drives, tail rotor drive shafts, an intermediate gearbox, and a tail rotor gearbox.

7.7.1 Main Transmission (see Fig. 7-9)

Mounted forward of the engines above the cargo compartment, the main transmission combines the two engine drive shaft inputs and provides drive to the main rotor, tail rotor, accessories, and oil cooler fan. Two freewheeling units at the main transmission inputs permit either engine to be disengaged from the transmission during single-engine operation or both engines during autorotation.

Power from the engines is transmitted through various stages of reduction gearing to obtain the necessary speed for the main rotor and tail rotor. Two hydraulic pumps and a rotor rpm (N_{R0}) tachgenerator are driven by the accessory gearbox mounted on the transmission left side. Mounted on the transmission right side is an oil cooler blower fan and reduction gearbox assembly.

7.7.2 Main Transmission Oil System

The main transmission oil system lubricates and cools transmission components. An oil pump mounted on the bottom right of the transmission, suctions oil from the sump through a filter screen and feeds it then to a fine-mesh filter mounted on the forward side of the oil cooler assembly. The filter incorporates a bypass to ensure oil circulation in the event of filter clogging. Oil from the filter then flows to the oil cooler which is fitted between the two engine oil coolers. The cooled oil is then returned to a transmission case manifold fitting located on the aft side below the tail rotor drive and between the two input drives. An external line from the manifold fitting supplies oil for the pressure sensors. Passages within the transmission casings distribute oil to jets used in spraying critical components.

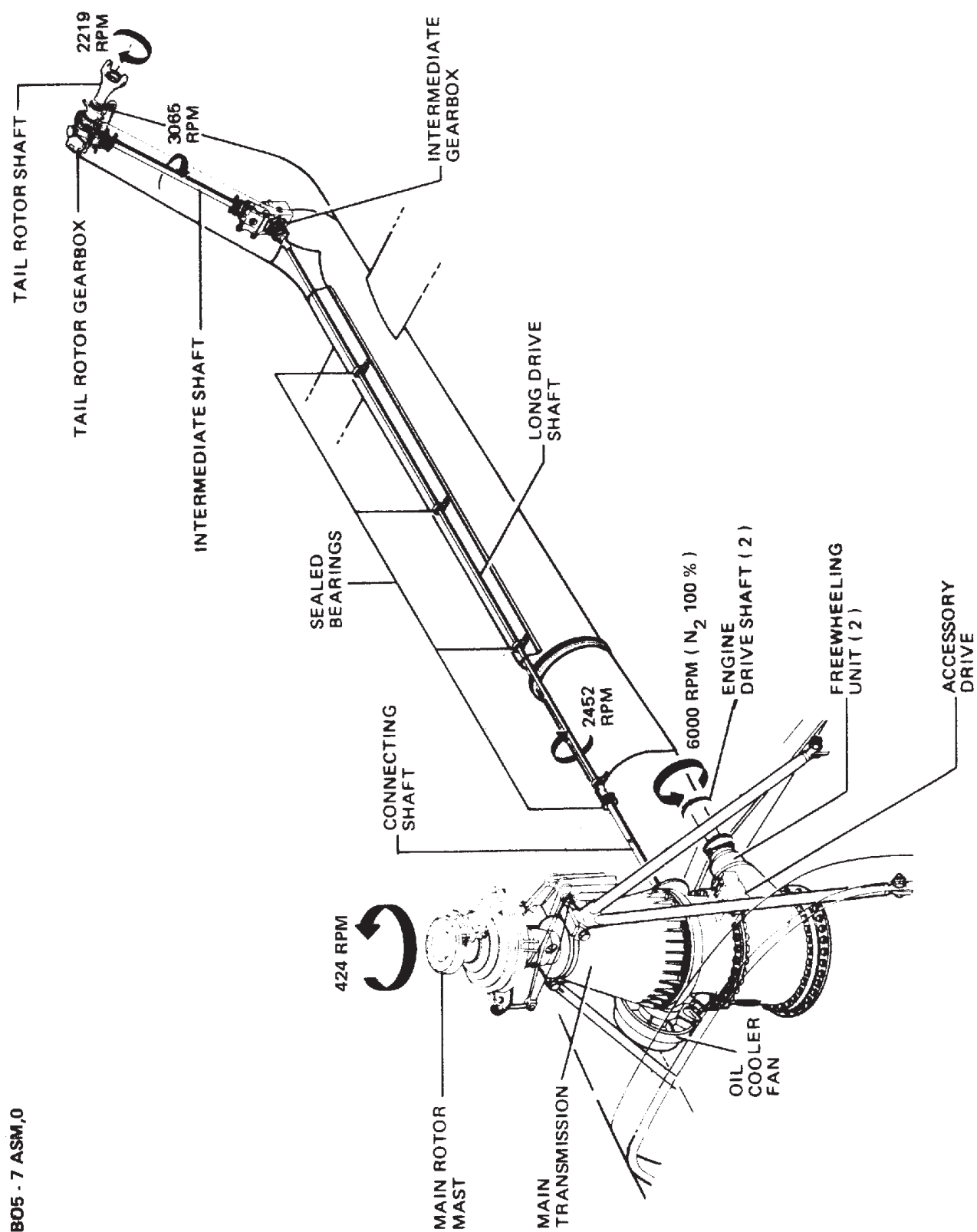


Fig. 7-9 Power Train

7.7.3 Main Transmission Indicators (see Fig. 7-10)

Triple tachometer indicator (ENG-ROTOR PERCENT RPM) indicates rotor speed (N_{R0}) on an inner scale in percent rpm. The rotor rpm signals are produced by a tachgenerator mounted on the aft face of the transmission accessory drive.

Triple oil temperature indicator (°C TEMP OIL) upper scale labeled R, indicates the transmission oil temperature. The temperature sensor is mounted on a transmission oil pump outlet manifold.

Triple oil pressure indicator (kp/cm² PRESS OIL) upper scale labeled R, indicates the transmission oil pressure. The oil pressure transmitter is installed in a T-fitting secured to an engine oil line hose clamp mounted on the transmission compartment floor.

7.7.4 Main Transmission Annunciators (see Fig. 7-10)

T. OIL warning light will be activated by either an overtemperature switch installed in the transmission case below the cooler fan and/or by a pressure switch installed in the T-fitting adjacent to the oil pressure transmitter. The overtemperature switch closes at an oil temperature over 105 °C while the pressure switch closes whenever the oil pressure is below 0.5 kp/cm². The **T. OIL** warning light circuit is powered by the Main Bus via a fuse.

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

OIL COOL warning light monitors operation of the oil cooler blower fan via a differential air pressure switch mounted on the oil cooler air inlet duct above the blower fan. A hose connects the switch to the air outlet shroud between the fan and oil cooler assembly. The pressure switch will activate the warning light whenever the pressure increase caused by the fan drops below a certain value. The **OIL COOL** warning light circuit is powered by the Main Bus via a fuse.

EFFECTIVITY ALL

7.7.5 Mast Moment Indication System

The mast moment indication system measures and indicates bending moments of the main rotor mast. The system is comprised of a strain gauge bridge, a rotary induction transmitter and a mast moment indicator.

Four strain gauges are bonded to the upper inner surface of the hollow rotor mast and are combined to form a bridge circuit. The rotary induction transmitter that conveys signals to and from the bridge circuit is located at the base of the main transmission.

The mast moment (MM) indicator is located on the instrument panel and is comprised of an indicator needle with colored scale, a calibration (CAL) screw, a TEST pushbutton switch and a **LIMIT** warning light.

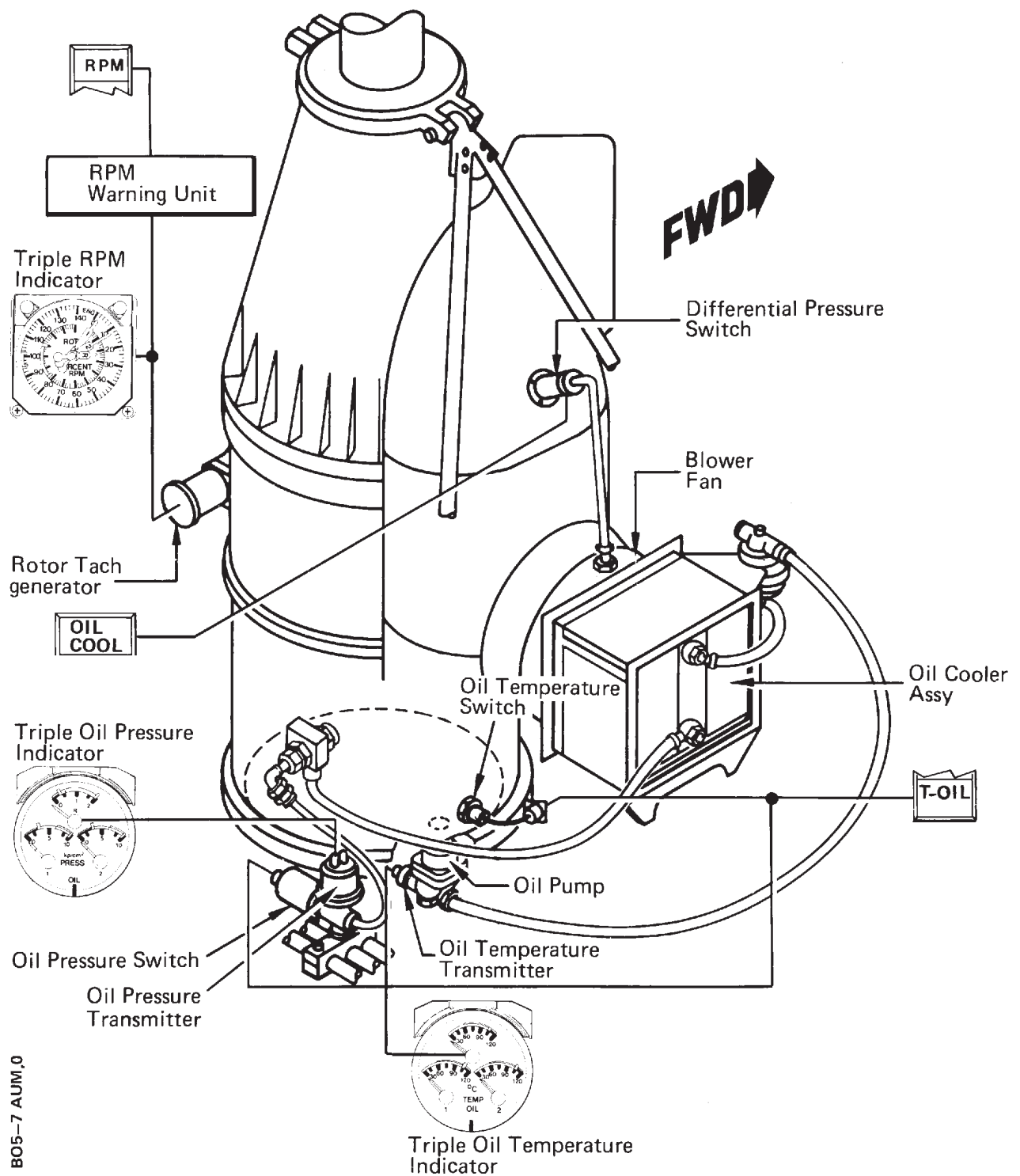


Fig. 7-10 Main Transmission Monitoring Systems

The MM indicator is powered from the Main Bus via a fuse. The indicator generates a carrier frequency which is supplied to the strain gauge bridge via the rotary transmitter. Bending moments of the rotor mast alter the resistance value of the strain gauge bridge. This modified signal is then returned through the rotary transmitter to the MM indicator where it is evaluated and then indicated by a needle deflection.

The system can be checked with the TEST pushbutton provided. When electrical power is initially applied to the system, the **LIMIT** warning light will come on. Pressing the TEST pushbutton should cause the needle to deflect to the extreme limit of the scale and then return to the white calibration line. Releasing the button will cancel the test indications, i.e. the **LIMIT** warning light goes off and the needle returns to the minimum green range.

7.7.6 Tail Rotor Drive Shafts (see Fig. 7-9)

There are three tubular tail rotor drive shaft sections comprising:

- a connecting shaft with flexible couplings,
- a long shaft supported by sealed bearings and fitted with a flexible coupling at the intermediate gearbox end,
- an intermediate shaft with flexible couplings installed at the vertical fin between the intermediate and tail rotor gearboxes.

The connecting shaft is routed through a tunnel in the engine compartment dividing fire-wall. The long drive shaft and the intermediate shaft are covered by fairings which may be opened for maintenance and inspection.

7.7.7 Intermediate Gearbox (see Fig. 7-9)

The intermediate gearbox mounted at the base of the vertical fin increases the main transmission output rpm and changes the angle of drive 50°. The gearbox housing provides for mounting of a tail rotor yaw control bell crank. This gearbox is wet sump lubricated with an oil level sight gauge located on the left side. A magnetic-type drain plug is installed in the base of the gearbox housing.

7.7.8 Tail Rotor Gearbox (see Fig. 7-9)

The tail rotor gearbox mounted on top of the vertical fin reduces the output rpm and changes the angle of drive 90°. The tail rotor shaft passes through and is supported by the gearbox output bevel gear. The gearbox housing provides for mounting of a tail rotor yaw control bell crank. The tail rotor gearbox is also wet sump lubricated with an oil level sight gauge located on the rear face of the gearbox housing. A magnetic-type drain plug is installed at the lower rear of the housing.

7.8 ROTOR SYSTEM

7.8.1 Main Rotor (see Fig. 7-11)

The main rotor ("System Bölkow") comprises a rigid-mounted head with four glassfiber reinforced plastic (GRP) blades.

7.8.2 Main Rotor Head (see Fig. 7-11)

The rigid main rotor head is bolted directly to the main rotor mast and consists of a star hub assembly and four blade-mounting forks with pitch horns. The hub assembly consists of a single-piece cross-shaped drop forged titanium hub housing the blade retaining system with tension-torsion straps and mounting the oil reservoir and feathering hinge bearings. Each blade-mounting fork with inner sleeve assembly is inserted into an arm of the hub and retained by a tension-torsion strap. The blade-mounting forks permit blade pitch changes in response to flight control movements.

7.8.3 Main Rotor Blades (see Fig. 7-11)

Each main-rotor blade has an asymmetrical airfoil having a rectangular plane-form in the inner portion and a trapezoidal plane-form in the outer portion. The blade structure consists of a skin, a C-spar and a core. The fiberglass prepeg skin has additional carbon fiber laminates at the blade tip. For lightning protection, the carbon tip is covered with a copper mesh. The C-spar is made of a unidirectional fiberglass rovings which form a loop around the blade root to carry the centrifugal forces. The core of the blade is made of a hard foam.

The blade leading edge is protected against erosion and foreign object damage by nickel strips. A blade fitting assembly is bolted over the blade root and provides mounting of the blade to the head. Two trim tabs are provided on the trailing edge of the outer portion of each blade to allow for tracking adjustment.

Aerodynamic motions of lead/lag and flapping are absorbed by the blade inherent elasticity. Each blade is fitted with pendulum absorbers near the root to reduce inflight vibrations.

7.8.4 Tail Rotor Assembly (see Fig. 7-12)

The tail rotor is of the semi-rigid teetering design. The two-bladed rotor fastened in a delta hinged hub provides antitorque action and directional control. A titanium shaft having a fork head and splined end houses two spherical teetering bearings and drives the rotating part of the yaw control system sliding sleeve. The tail rotor shaft is splined to and driven by the tail rotor output bevel gear.

Blade centrifugal forces are carried by laminated steel tension straps that attach outboard to the blade mounting forks and inboard to a strap shoe which is held centered by the tail rotor assembly mounting bolt. The blade mounting forks rotate about the hub on Teflon bushings. Dynamic weights are installed on the control yoke to reduce control pedal forces. These weights are adjusted to compensate for main rotor torque required at a hover.

The tail rotor blades are constructed similar to the main rotor blades. The blades are retained in the mounting forks by two bolts.

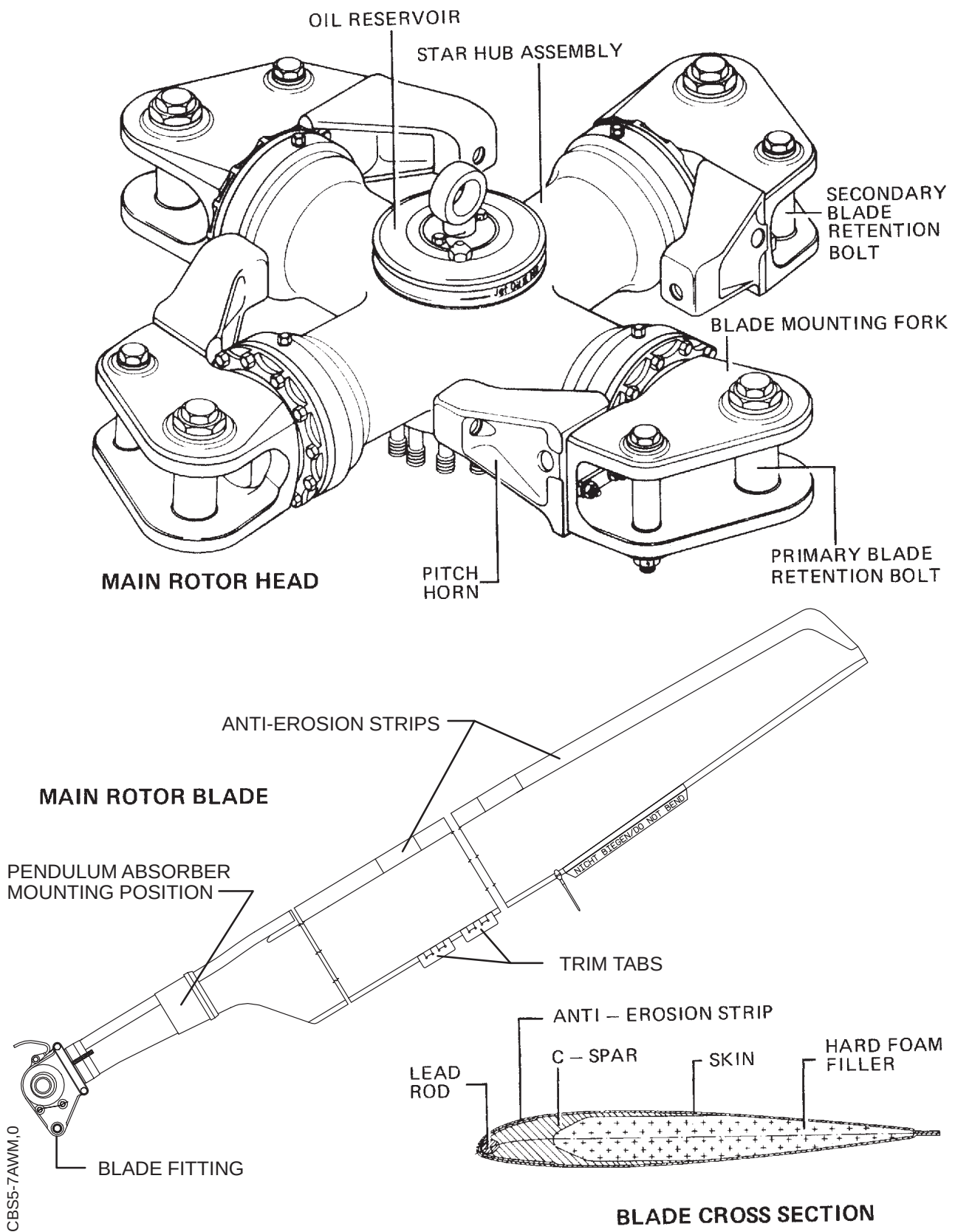
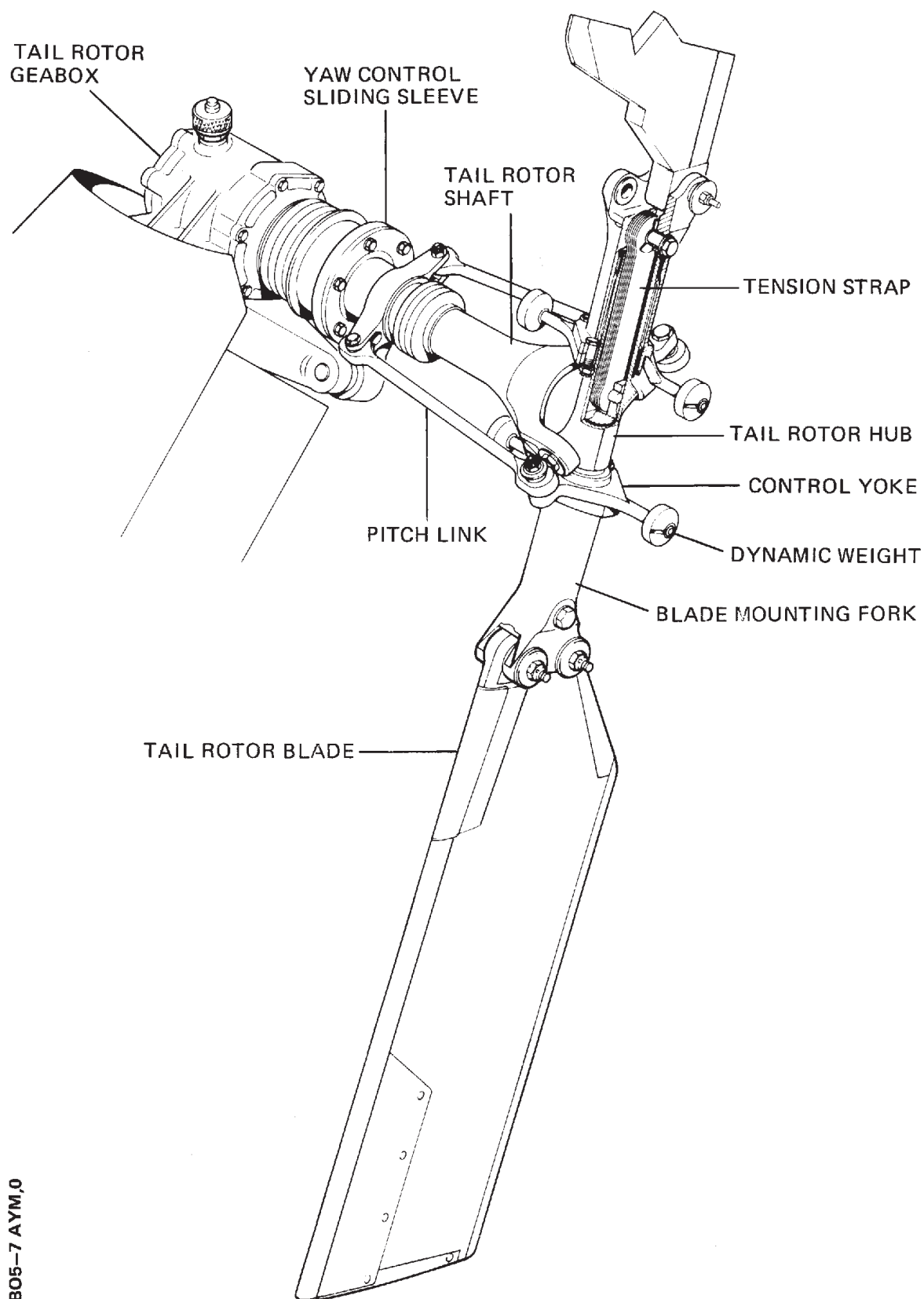


Fig. 7-11 Main Rotor

MANUFACTURER'S DATA



BO5-7 AYM,0

Fig. 7-12 Typical Tail Rotor Assembly

7.9 **FLIGHT CONTROL SYSTEM** (see Fig. 7-13)

The flight control system comprises a cyclic stick, a collective lever, and directional pedals to produce control inputs to the main rotor hydraulic servo actuators and tail rotor pitch change links.

Cyclic and collective servo actuator control inputs are combined by mixing levers and further transmitted to a swashplate. The swashplate changes the linear motion from the mixing levers to rotating motion and conveys pitch changes to the four main rotor blades.

Yaw control inputs are transmitted to the tail rotor sliding sleeve. The pitch of each blade changes when directional control inputs cause the nonrotating portion of the sliding sleeve to act upon the rotating portion. Pitch links between the sliding sleeve rotating portion and the pitch horn of each blade cause collective pitch angle changes of the tail rotor blades.

7.9.1 **Cyclic Stick** (see Fig. 7-13 and 7-14)

The cyclic stick is used to induce helicopter movements about the pitch and roll axes. The basic cyclic stick grip has switches for RPM-warning audio cutoff, cyclic control force trimming, and radio and internal communications microphone keying.

7.9.2 **Collective Lever** (see Fig. 7-13 and 7-14)

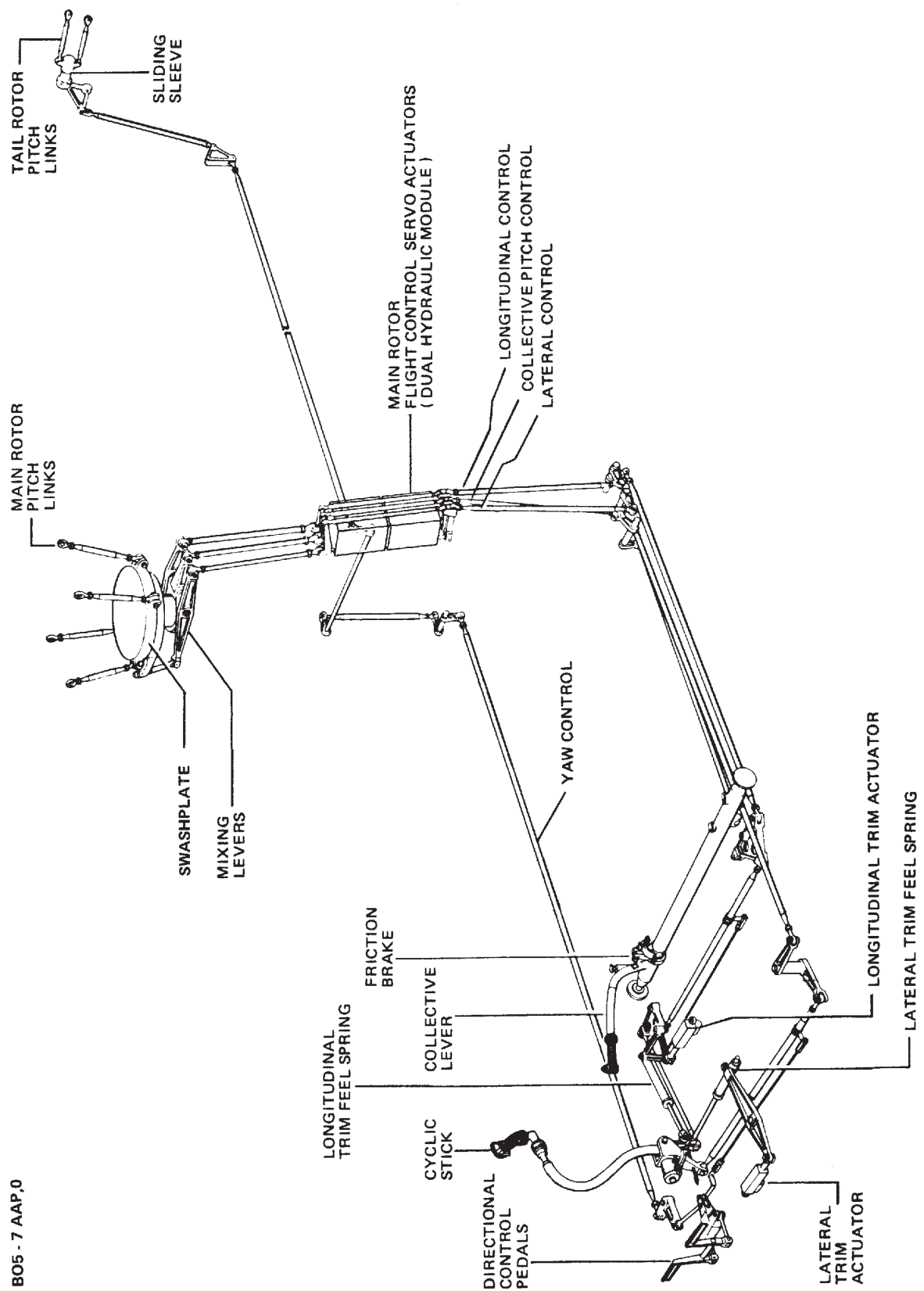
The collective lever provides a means of adjusting pitch angle of the main rotor blades and fuel flow metering requirements of the power turbine. A switch panel at the end of the lever has several switches installed, however, only the 4-way ENGINE TRIM switch and, on S-751 and subsequent, the LANDING LIGHT switches are active on the standard model. A friction adjustment at the base of the lever is used to prevent the collective pitch setting from moving when the lever is released.

7.9.3 **Directional Control Pedals** (see Fig. 7-13)

The directional control pedals are used to induce helicopter movement about the yaw axis. The pedals are adjustable fore-aft by use of a knob located aft of and between the pedals.

7.9.4 **Cyclic Trim System** (see Fig. 7-13 and 7-14)

The position of the cyclic stick may be electrically trimmed using the cyclic trim system. A lateral and a longitudinal trim feel spring assembly, each with an electrical actuator, are incorporated to produce artificial feel and cyclic control system trim. The spring assemblies will hold the cyclic stick in a neutral-feel trimmed position. Movement of the cyclic stick will cause the spring assemblies to compress and provide feel to the controls. When control pressure is released the cyclic will return to the neutral-feel position.



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Fig. 7-13 Flight Control System

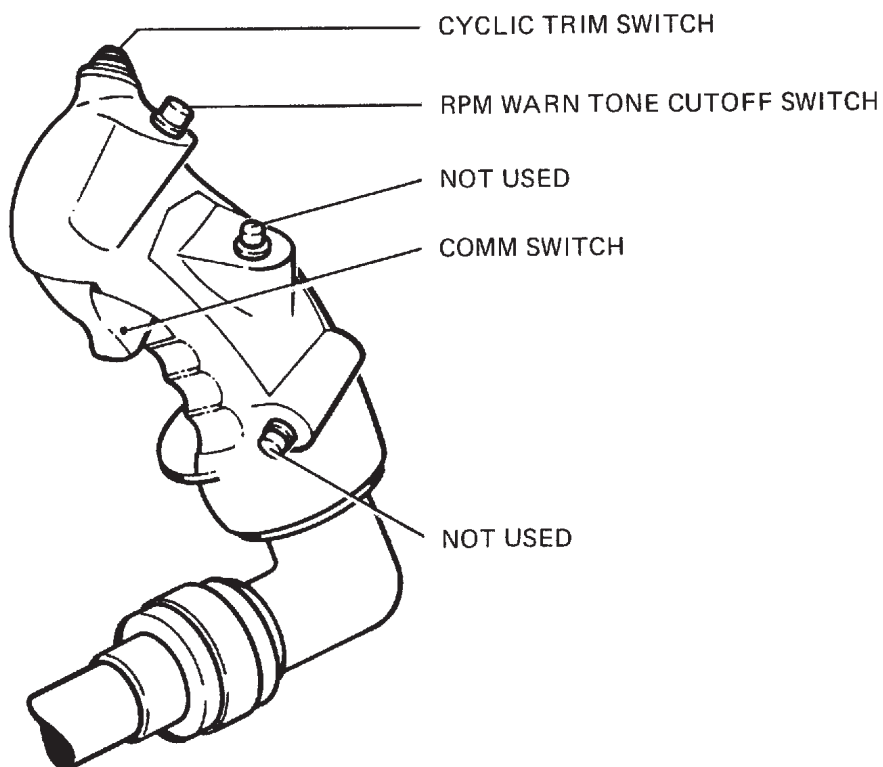
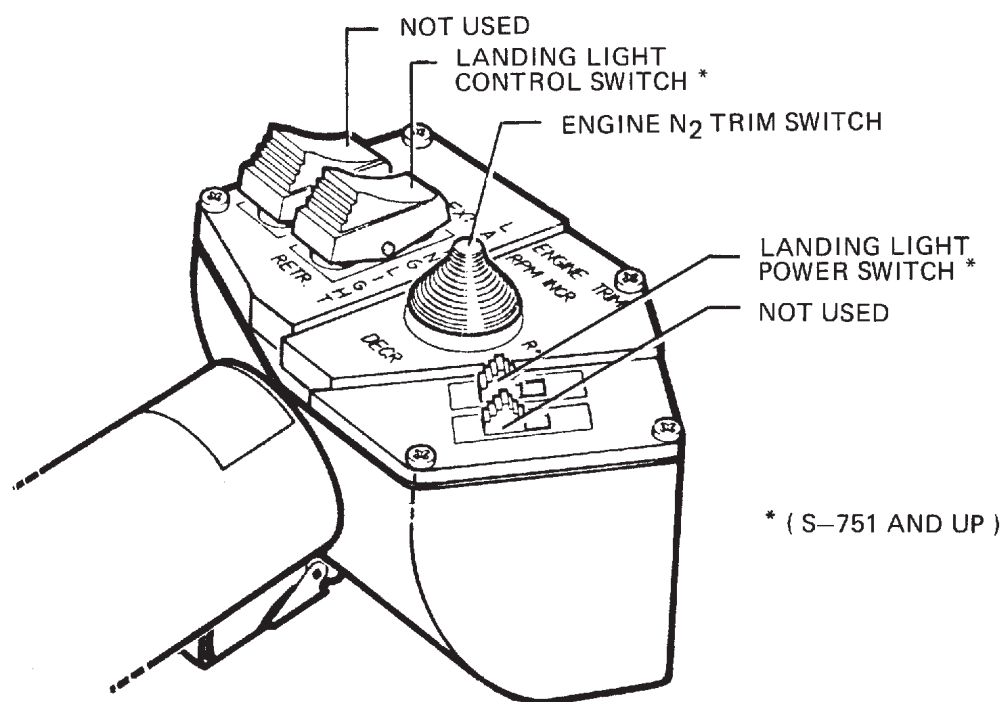
Manual retrimming of the neutral-feel position is accomplished by use of the 4-way TRIM switch on top of the cyclic stick grip. Pressing the switch in the desired direction will activate the appropriate trim actuator motor to reposition its respective spring assembly to establish a new neutral-feel position.

The system is powered direct from the Main Bus via the TRIM ACTUATOR circuit breaker on the center console switch panel.

7.9.5 **Flight Control Servos** (see Fig. 7-13 and 7-16)

The flight control servos are utilized to amplify the flight control forces applied to the main rotor system and thereby ease pilot workload.

Each of the three main rotor control axes has a tandem hydraulic servo actuator comprised of a primary servo and a standby servo. These servos are independent of each other to provide redundancy in the flight control system. Should one hydraulic system fail the remaining system can drive the main rotor flight control servos.



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Fig. 7-14 Cyclic Stick Grip and Collective Lever Controls

7.10 HYDRAULIC SYSTEM (see Fig. 7-15 and 7-16)

The hydraulic system, designed to reduce the main-rotor control forces, is constructed of modular components and mounted on the left side of the transmission compartment. It consists of two independent but mainly identical systems, of which System 1 is the primary operating system while System 2 is the stand-by reserve system. If necessary, operational control crosses over from the primary system to the stand-by system automatically and without delay. Both systems offer 100 % specific power support.

Pressure supply is provided by two hydraulic pumps which are permanently driven by the main transmission and which normally deliver full system pressure. The control of pressurization for the three actuators in each system is provided by two selector valves (one each system) which are linked by an interconnecting plate to ensure that only one system is active. The selector valves are mechanically coupled with and controlled by a hydraulic shuttle valve that works on the basis of pressure differentiation. Should the pressure of the primary system (System 1) fall below approx. 50% of the pressure in the stand-by system, the shuttle valve will position the selector valves to cause the stand-by system (System 2) to take over control (cross-over) and System 1 to become inactive.

The pressure of each hydraulic system is monitored by a pressure switch that, if system pressure falls below approx 70 %, will activate the respective **HYDR 1** or **HYDR 2** warning light. The **HYDR 1** warning light will also come on (together with the **HY BLOCK** warning light) whenever System 2 is activated.

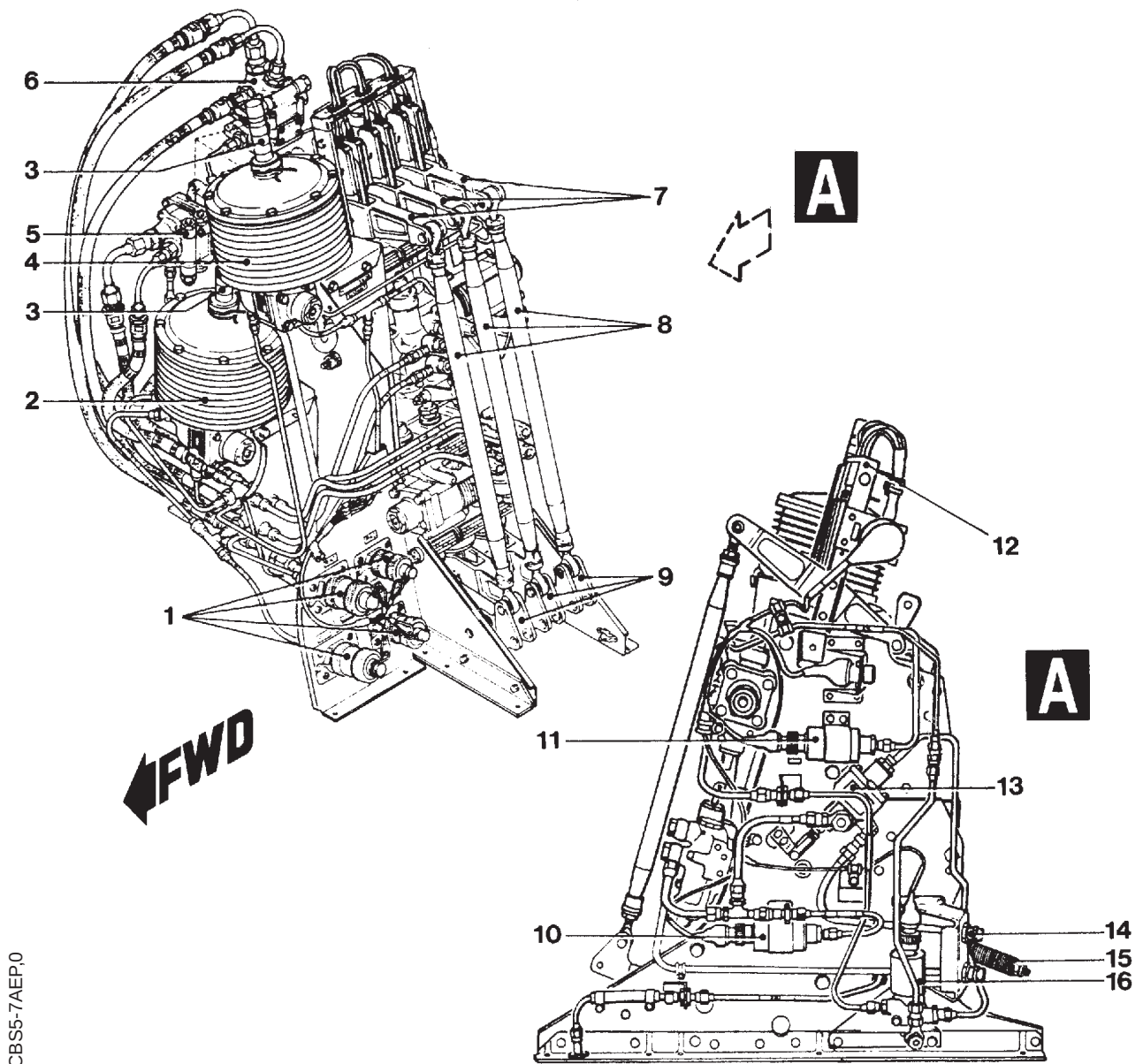
A jammed servo valve in System 1 will cause an electric signal to be released by micro switches which, by means of a relay, trigger a solenoid valve. This solenoid valve causes an artificial loss of pressure by blocking pressure in System 1 upstream of the shuttle valve so that System 2 is activated by the shuttle valve and System 1 is then deactivated (cross-over). The **HY BLOCK** warning light comes on whenever the relay controlling the solenoid valve is activated.

The micro switches, noted above, are designed to recognize a jammed servo valve and are only integrated in System 1. In System 2, however, a connection provided by springs will by-pass a jammed servo valve noticed by increased control forces about the affected axis.

NOTE Should a servo valve jam in System 2, an automatic cross-over to System 1 is impossible. Therefore, normal operation in System 2 is not allowed as the then lack of redundancy would cause an inflight system failure to be a safety risk.

A cross-over functional check can be performed using a guarded three-position HYDR TEST switch located on the center console switch panel. Placing the switch from the spring-loaded center (off) position to the TEST position activates the relay and, as a result, the solenoid valve to cause control to cross-over to System 2. Placing the switch to the RESET position deactivates the self-holding solenoid valve causing control to revert back to System 1.

All system control and monitoring functions are powered direct from the Main Bus via the HYDR. CONTROLS circuit breaker on the center console switch panel.



1. SERVICE COUPLINGS
2. RESERVOIR - SYSTEM 2
3. FLUID LEVEL AND TRAPPED AIR INDICATORS
4. RESERVOIR - SYSTEM 1
5. HYDRAULIC PUMP - SYSTEM 2
6. HYDRAULIC PUMP - SYSTEM 1
7. INPUT LEVERS - SYSTEM 1
8. INTERCONNECTING CONTROL RODS

9. INPUT LEVERS - SYSTEM 2
10. PRESSURE SWITCH - SYSTEM 2
11. PRESSURE SWITCH - SYSTEM 1
12. MICRO SWITCHES
13. SHUTTLE VALVE
14. OPTIONAL EQUIPMENT COUPLINGS
15. DROOP COMPENSATION ASSEMBLY
16. SOLENOID VALVE

Fig. 7-15 Dual Hydraulic Module

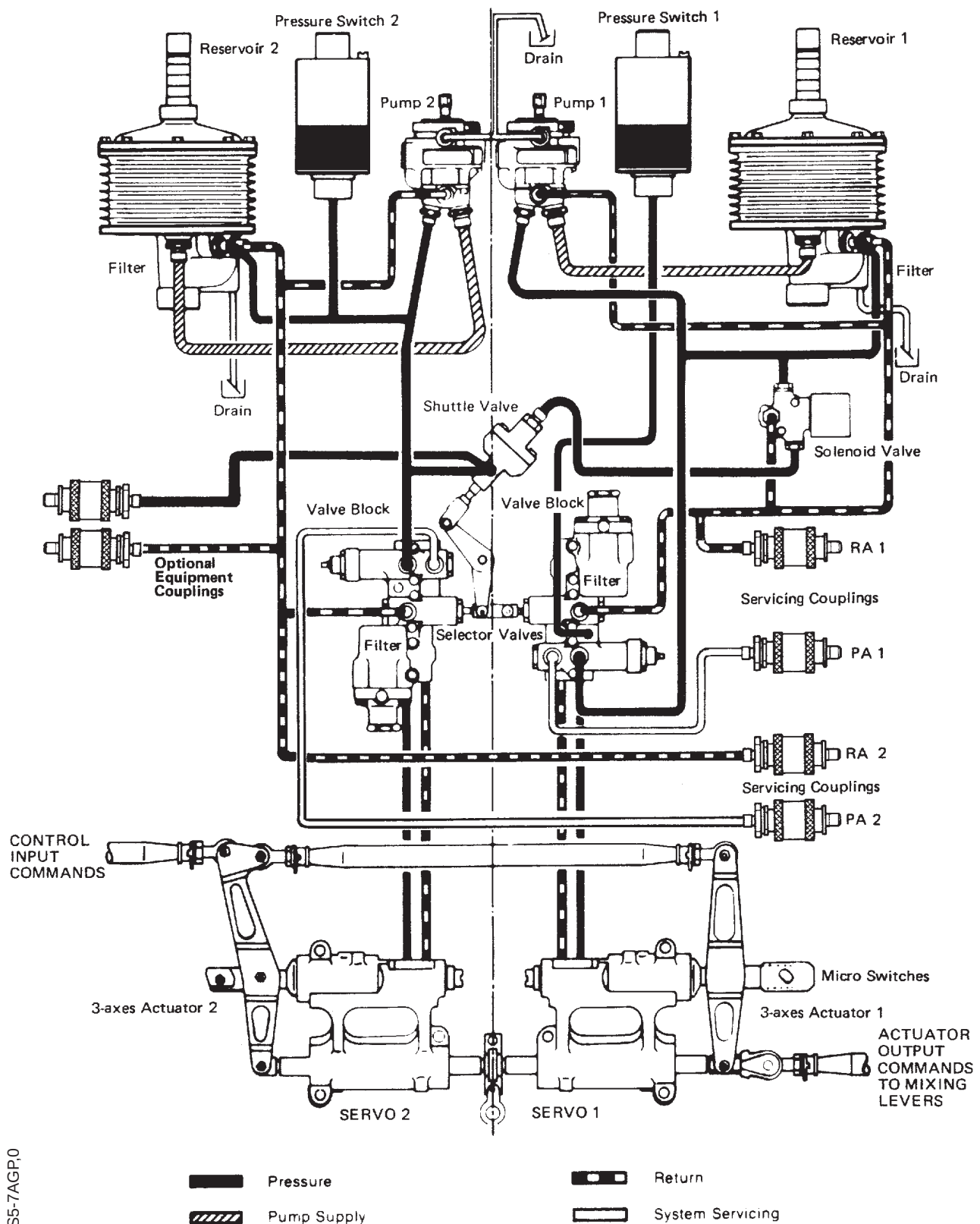


Fig. 7-16 Hydraulic System Schematic

7.11 ELECTRICAL SYSTEM

The electrical system provides 28V DC electrical power to the helicopter Starter, Main and Auxiliary Buses. An onboard 24V DC nickelcadmium battery supplies engine starting power and serves as an inflight emergency power source should both generators fail.

7.11.1 Power Supply (see Fig. 7-17)

The power supply comprises two engine-driven 150-ampere DC starter/generators. The two generators, each with a voltage regulator and a differential reverse current relay, operate in parallel to supply 28V DC single-pole negative ground electrical power. Each voltage regulator controls the output voltage of its generator and equalizes the load between the two generators. The differential reverse current relay monitors generator output and will disconnect the generator from the system if its output voltage is less than the battery voltage and will cause the respective **GEN** warning light to come on.

7.11.2 Distribution (see Fig. 7-17)

Electrical power is distributed through use of three primary buses:

- Starter Bus primary power distributor that supplies battery and/or external power for engine starting, generator power for equipment requiring high current loads and all sources of power for further distribution to other buses.
- Main Bus supplies power to all equipment and systems essential for normal helicopter operation. This bus further supplies two Radio Nav Buses that power the COMM/NAV equipment.
- Auxiliary Bus supplies power to all non-essential consumers only when both generators are operational or when an external power source is connected.

7.11.3 Controls, Indicators and Annunciators (see Fig. 7-17)

STARTER/GENERATOR, ENG 1/ENG 2 switches, when placed in the GEN ON position, allow control voltages from the respective generator to flow to the reverse current relays. When the control voltage is greater than the battery voltage the relay will switch the generator on-line and the appropriate **GEN** warning light will go off. During external power assisted starts, the generators are not brought on-line regardless of STARTER/GENERATOR switch position as long as the external power source is providing power to the system.

Generator (1, 2) field switches, located on the overhead panel, energize (GEN RES position) or de-energize (GEN TRIP position) the generator field via the respective voltage regulator.

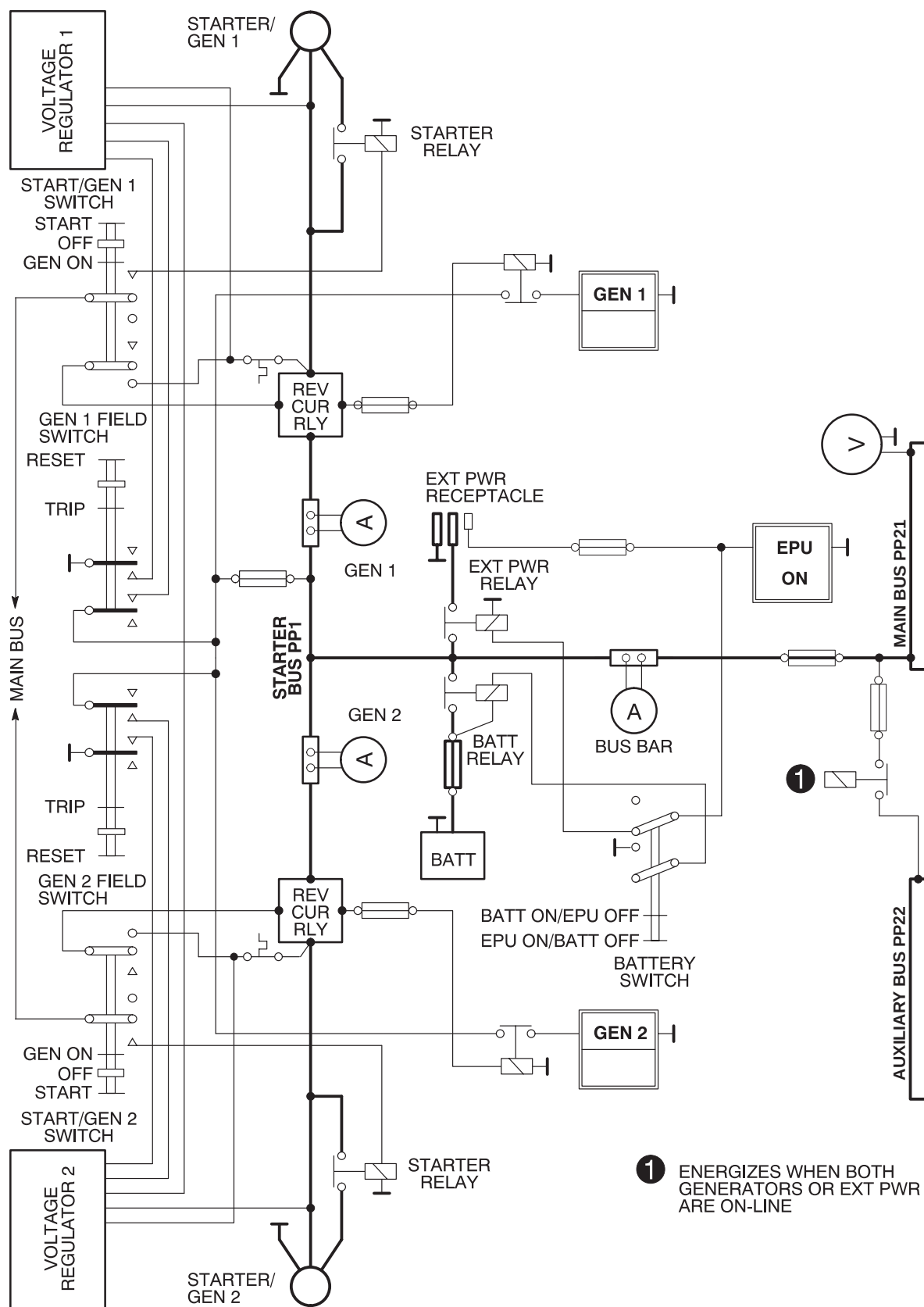
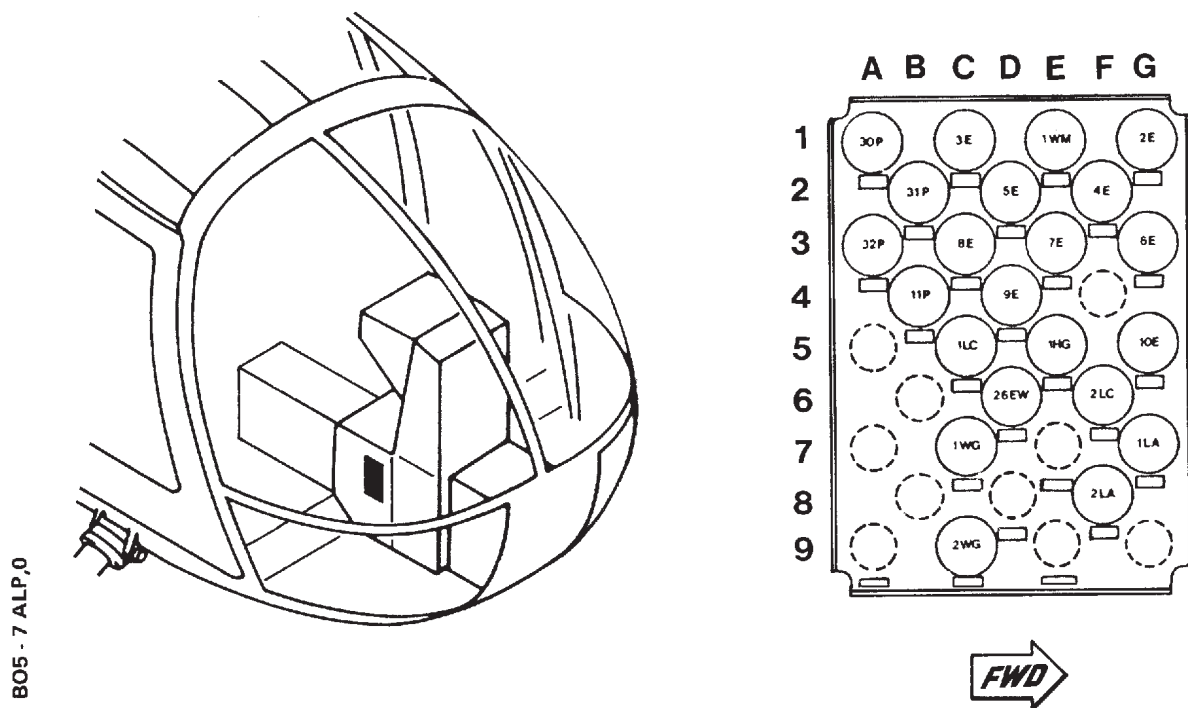


Fig. 7-17 Electrical Power Supply Schematic



FUSE IDENT	PANEL LOCATION	RATING	CIRCUIT(S) PROTECTED
2E	1G	1A	LOW FUEL warning
3E	1C	1A	Fuel quantity indicator
4E	2F	1A	RPM warning system
5E	2D	1A	Triple oil pressure indicator
6E	3G	1A	Triple oil temperature indicator
7E	3E	1A	Fuel pressure indicator
8E	3C	4A	Warn It test, T PLUG and OIL COOL warnings (optional)
9E	4D	1A	Fuel FILT 1/FILT 2 warnings
10E	5G	1A	Engine MAG PLUG 1/MAG PLUG 2 warnings
26EW	6D	1A	Transmission T OIL warning
1HG	5E	3A	Pitot tube heating
1LA	7G	10A	Anticollision lights
2LA	8F	4A	Position lights
1LC	5C	3A	Instrument lights
2LC	6F	4A	Interior lights and utility light
11P	4B	4A	Battery control relay
30P	1A	4A	EPU control and EPU ON caution light
31P	2B	4A	Ammeter main bus shunt - negative pole
32P	3A	4A	Ammeter main bus shunt - positive pole
1WG	7C	1A	Fire warning - Engine 1
2WG	9C	1A	Fire warning - Engine 2
1WM	1E	2A	Mast moment indicator system

Fig. 7-18 Typical Fuse Panel Layout

Battery switch, when placed in the BAT ON/EPU OFF position, connects the onboard battery to the Starter and Main Buses. Battery charging is then supplied by the generators when they are brought on-line. When the battery switch is placed in the BAT OFF/EPU ON position a 28V DC external power unit (EPU) may then supply power to the Starter Bus which in turn powers both the Main and Auxiliary Buses. An **EPU ON** caution light comes on when external power is connected and switched on. The EPU receptacle in the nose compartment allows for a starting current of 700 A.

BAT 60 and BAT 70 warning lights are provided to indicate an overheating condition of the battery. These lights operate independently being activated by separate battery-mounted temperature sensors.

Voltmeter (DC VOLTS) indicator indicates the voltage output supplied by the Main Bus during generator, battery or external power operation.

Ammeter (DC AMPS) indicator with the CURRENT IND switch placed in the BUS BAR position indicates current consumption (– side) measured between the Starter Bus and Main Bus, and with the CURRENT IND switch in either 1 GEN or 2 GEN position will indicate the current supplied (+ side) by the respective generator.

7.11.4 Circuit Protection

All electrical power circuits are protected by either a circuit breaker or fuse. Most circuit breakers are located on the center console switch panel. Most low amperage fuses are found on a fuse panel (see Fig. 7-18) located behind a cover on the instrument panel lower RH structure. Fuses and circuit breakers used to protect high amperage consumers and the generators are found in a main relay box mounted to the fuselage RH structure in the cargo compartment.

7.12 LIGHTING EQUIPMENT

The lighting equipment comprises exterior and interior lighting systems and the annunciator lights. The exterior lighting system includes the position lights, the landing light and the anticollision lights. The interior lighting system includes the instrument lights, a utility light, and the internal compartment lights.

7.12.1 Position Lights

The position lights are located on the horizontal stabilizer end-plates (RH - green, LH - red) and on the top aft end of the vertical fin (white). Power for operation of the lights is provided by the Main Bus via a fuse. Operation of the lights is controlled by the POSITION LIGHT switch on the overhead console.

7.12.2 Anticollision Lights

Two high-intensity flashing red anticollision lights are located, one on top of the tail rotor gearbox, the other on the underside of the fuselage. Power for operation of the lights is provided by the Main Bus via a fuse. Operation of the lights is controlled by the ANTI-COLL LIGHT switch on the overhead console.

7.12.3 Instrument Lights

Power to the instrument lights is controlled by the INSTR LIGHT switch on the center console switch panel. With this switch on power is supplied from the Main Bus via a fuse to a dim control unit. The light intensity is then controlled by the INSTR LIGHT rheostat switch located on the instrument panel.

A whip light at the left of the center console switch panel is also powered when the INSTR LIGHT switch is on. The whip light, however, has its own on-off switch and the light intensity is not effected by the rheostat switch.

7.12.4 Utility Light

A utility light is provided as a detachable light with coiled extension cord located at the left of the overhead console. This light is operated by an OFF-BRT rheostat switch integral with the light unit and is independently powered by the Main Bus via a fuse.

7.12.5 Interior Lights

A cabin light and two cargo compartment lights are powered from the Main Bus via the same fuse provided for the utility light. Operation of the three ceiling lights is controlled by the INTERNAL LIGHT switch on the overhead console.

7.12.6 Annunciator Lights (see Fig. 7-19)

When the TEST WARN. LIGHTS pushbutton on the instrument panel is pressed, all warning/caution lights (except the two **F** [Fire] and the **LIMIT** [MM ind] lights) will come on. Each warning/caution light segment has two lamps so that failure of a single lamp will produce only a dimming effect. If a segment fails to light, it indicates either the failure of both lamps or a defective circuit.

7.12.7 Rpm Warning System

Visual and in most cases audio signals are provided as a warning whenever:

- N_{R0} is above or below the normal operating range,
- an engine N_1 differential equal to or greater than 12 % occurs (after either power lever is in the FLIGHT position).

The engine N_1 differential warning may be cancelled by pressing the RPM WARN pushbutton on the cyclic stick grip.

WORD SEGMENT	COLOR	ILLUMINATION PARAMETER OR FAULT
ANTI ICING 1	Amber	No. 1 engine anti-icing actuator in open position.
ANTI ICING 2	Amber	No. 2 engine anti-icing actuator in open position.
BAT 60	Red	Battery temperature exceeds 60 °C.
BAT 70	Red	Battery temperature exceeds 70 °C.
EPU ON	Amber	External power unit connected and switched on.
F (LH)	Red	Fire detected in the No. 1 engine compartment.
F (RH)	Red	Fire detected in the No. 2 engine compartment.
FILT 1	Red	No. 1 fuel filter is clogged and bypass has begun.
FILT 2	Red	No. 2 fuel filter is clogged and bypass has begun.
GEN 1	Red	No. 1 generator is not on-line.
GEN 2	Red	No. 2 generator is not on-line.
HY BLOCK	Red	Jammed servo valve in hydraulic system no. 1 and cross-over to system no. 2 has occurred.
HYD 1	Red	No. 1 hydraulic system pressure is low and cross-over to system no. 2 has occurred.
HYD 2	Red	No. 2 hydraulic system pressure is low.
LIMIT	Red	Max mast moment limit exceeded.
LOW FUEL	Red	Fuel supply tank usable quantity is below 60 kg.
MAG PLUG 1	Red	No. 1 engine oil contains metal fragments.
MAG PLUG 2	Red	No. 2 engine oil contains metal fragments.
OIL COOL	Red	(After SB 60-37) Oil cooler fan failure.
RPM	Red	N_{R0} below 95% or above 103%; N_1 differential greater than 12%.
T OIL	Red	Transmission oil pressure below 0.5 kp/cm ² ; or oil temperature above 105 °C.

Fig. 7-19 Alphabetical Listing of Warning/Caution Lights

The visual signal will always be indicated by a flashing **RPM** warning light. The audio signals heard through the pilot headset earphones vary depending on the fault condition as follows:

FAULT CONDITION	AUDIO SIGNAL
N_1 Differential $\geq \times 12\%$	Beeping tone
N_{R0} Low - 75 to 95 %	Beeping tone
N_{R0} High - 103 %	None
N_{R0} High - 108 %	Steady, high-pitched tone

NOTE The N_{R0} High - 108 % RPM warning applies only to helicopters equipped with a two-stage warning unit.

7.13 FLIGHT INSTRUMENTS

The instruments discussed under this subject are, for the most part, those that directly measure flight performance.

7.13.1 Pitot-static System

An electrically heated pitot tube is installed at the fuselage lower RH nose area. Two associated static ports are installed flush with the fuselage; one on the right and one on the left. The pitot tube supplies ram air to the airspeed indicator. The pitot heater is powered from the Main Bus via a fuse. The PITOT HEATING switch on the center console switch panel controls the heating element as indicated by the PITOT HEATING indicator on the instrument panel. The static ports provide static air pressure to the barometric altimeter, vertical speed indicator and airspeed indicator.

7.13.2 Barometric Altimeter

The barometric altimeter reacts to changes in barometric pressure delivered via the static ports. The altimeter scale is graduated in fifty-foot increments and numbered at hundred-foot intervals. The long needle indicates 100 feet intervals, the short needle indicates 1000 feet intervals and the rotating scale to the right of center indicates 10,000 feet intervals. The scale window at the lower center of the instrument face indicates barometric pressure setting in hectopascals (millibars) adjustable by use of the knob on the lower left corner of the instrument case.

7.13.3 Vertical Speed Indicator

The vertical speed indicator presents the rate of climb or descent from 0 to 6000 feet per minute. The instrument scale is graduated in 100 feet per minute increments from 0 up to 1000 feet per minute, and in 500 feet per minute increments from 1000 to 6000 feet per minute.

7.13.4 Airspeed Indicator

The airspeed indicator presents airspeed from 0 to 150 knots. The indicator dial is scaled in five-knot increments above 10 knots.

7.13.5 **Clock**

The mechanical clock on the instrument panel is wound and set with a knob at the lower center of the clock face. Below the knob is a lever used for starting, stopping and resetting (zeroing) an elapsed-time counter (stopwatch). A small lever to the right of the knob is used for stopping the sweep second-hand.

7.13.6 **Standby Magnetic Compass**

The standby magnetic compass is attached to the windshield center frame above the instrument panel. A compass correction card is affixed to the windshield frame above the compass.

7.13.7 **Outside Air Temperature (OAT) Indicator**

The OAT indicator is a self-contained unit mounted to the RH cabin roof window. A probe extends through the window to sense outside free air temperature. The dial has two scales to indicate temperature in degrees Celsius and degrees Fahrenheit.

SECTION 8

HANDLING, SERVICING, AND MAINTENANCE

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SECTION 8

HANDLING, SERVICING, AND MAINTENANCE

8.1 GENERAL

This section describes ways an operator can ensure that the necessary handling, servicing and maintenance of the helicopter are accomplished.

NOTE It is the operator's responsibility to ensure that all airworthiness directives are complied with and that the handling, servicing and maintenance of the helicopter are accomplished when required and in accordance with the applicable Federal Aviation Regulations.

In order to meet the above requirements, the helicopter operator should establish contact with the helicopter manufacturer or certified service station for service and information; and that all correspondence regarding the helicopter include the helicopter serial number found on the data plate secured to the RH fuselage structure. Helicopter and component maintenance manuals, and parts catalogs are available from the helicopter manufacturer.

For definition of terms, abbreviations and symbols used in this section, refer to Section 1.

8.2 GROUND HANDLING

8.2.1 Towing (see Fig. 8-1)

CAUTION TO AVOID DAMAGING GYRO INSTRUMENTS, DO NOT MOVE HELICOPTER DURING THE TIME SPAN OF 10 TO 30 MINUTES AFTER SWITCHING OFF INSTRUMENT ELECTRICAL POWER.

1. Position ground handling wheels onto skid tubes making certain that lugs on skid tubes engage with flanges on wheel units.
2. Push downwards on latch lever and check that latches are locked over lugs.
3. Stabilize helicopter during raising/lowering and towing operations by grasping tail skid.
4. Close hydraulic jack valve and, using pump handle, raise helicopter.

NOTE On jacks fitted with a safety stop, lower safety stop against jack housing then slowly open jack valve.

CAUTION HANDLE HELICOPTER USING ONLY THE FOLLOWING AUTHORIZED HARD POINTS:

- TAIL SKID
- REAR EDGE OF CABIN CEILING (BELOW TAIL BOOM)
- WINDSHIELD FRAME (ABOVE BATTERY ACCESS DOOR)

5. Maneuver helicopter into desired position.

NOTE To ease handling, maneuver helicopter in a rearward direction.

CAUTION TOW HELICOPTER AT SPEEDS LESS THAN 10 KM/HOUR. MINIMUM TURNING RADIUS WHEN TOWING IS 10 METERS.

6. Secure towbar to forward end of each skid tube and using guide, tow helicopter as required.

7. Upon reaching desired area, remove towbar.

8. Slowly open jack valve and lower helicopter to ground.

NOTE On jacks fitted with a safety stop, close jack valve and, using pump handle, release safety stop. Raise safety stop then slowly open jack valve to lower helicopter to ground.

9. Pull latch lever and remove ground handling wheels.

8.2.2 **Parking**

NOTE For parking periods over 3 hours or when winds are above 15 kt, main and tail rotors should be secured as a minimum.

1. Connect electrical grounding cable.

2. Make certain that all electrical power is switched off.

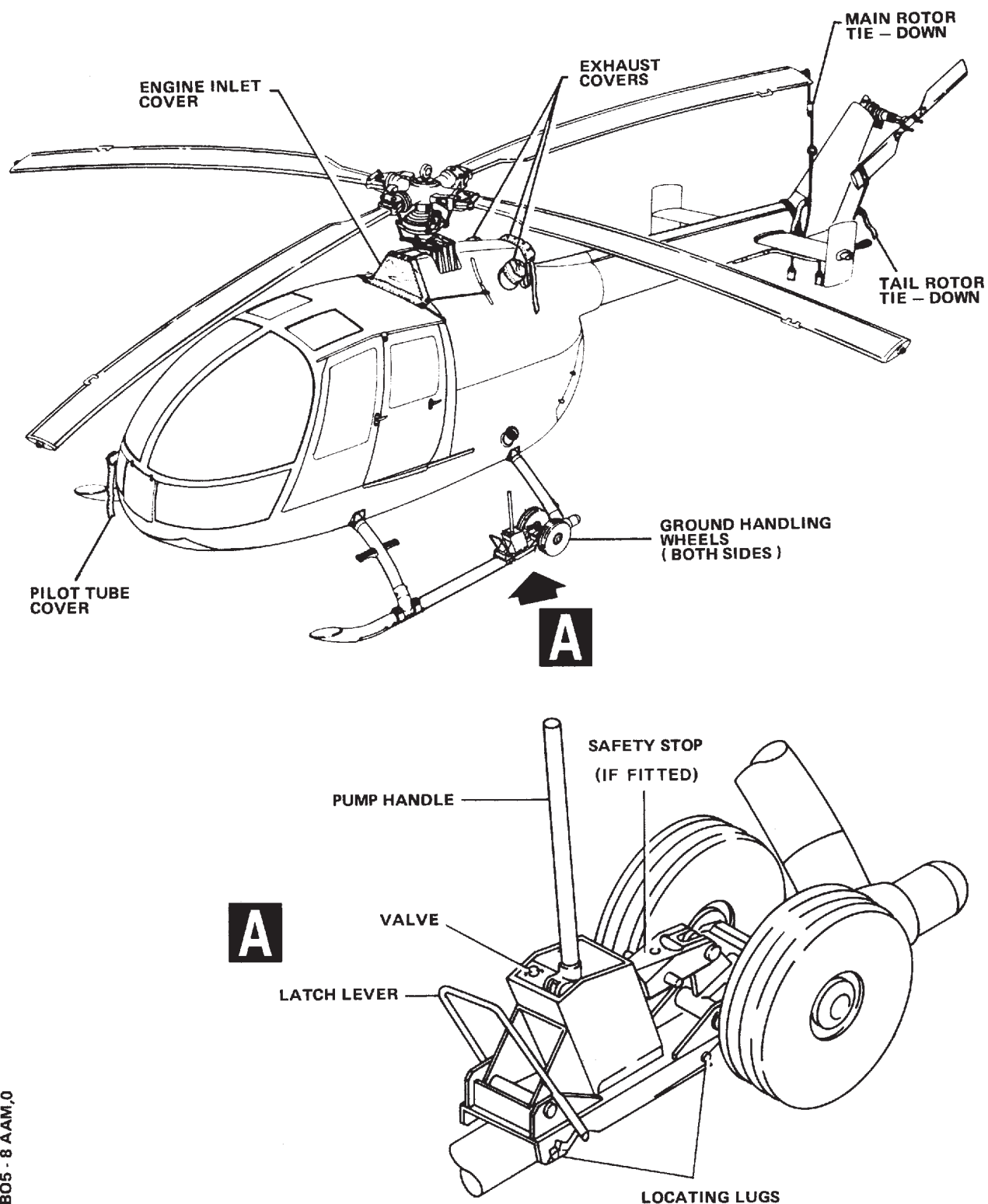
NOTE Disconnect and, if necessary, remove battery when helicopter is to be parked for long periods.

3. Secure main and tail rotors using tie-downs provided.

4. Install engine inlet, exhaust and pitot tube protective covers.

NOTE Tie-downs and protective covers are part of the helicopter flyaway kit.

5. Make certain that all doors and windows are closed and that all access panels and cowlings are secured.



BO5 - 8 AAM,0

Fig. 8-1 Ground Handling

8.2.3 Mooring

When parking the helicopter in the open during periods of forecast high winds of 50 kt or higher, the helicopter should be headed in the direction from which the forecast winds are expected and then moored (anchored) as follows:

- Secure helicopter to ramp tie-downs or mooring anchors using ropes or cables wrapped around the landing gear cross tubes at both the fuselage and skid tube junctures.
- Secure rotor blades and install protective covers.
- Fill fuel tanks to max capacity and place sand bags on the skid tubes to add weight to the helicopter.

8.3 SERVICING

8.3.1 Refueling

Park helicopter on level, even surface and perform engine shutdown. Keep fire extinguisher ready at hand.

WARNING • ALWAYS ESTABLISH ELECTRICAL GROUND CONNECTION BETWEEN HELICOPTER, FUELING VEHICLE AND EXTERNAL POWER UNIT (AS REQUIRED) BEFORE STARTING REFUELING OPERATION AND DISCONNECT ONLY WHEN OPERATION IS COMPLETED.

- DO NOT CONNECT, DISCONNECT OR REMOVE BATTERY DURING OR IMMEDIATELY FOLLOWING REFUELING.

1. Establish electrical ground connection.
2. Apply power (battery or EPU) to the helicopter electrical system and check supply tank (S) fuel quantity.
3. Unlock and remove fuel filler cap.
4. Electrically ground fuel nozzle to helicopter and then insert nozzle in filler neck.
5. Service fuel tanks to required level.

NOTE If supply tank is not full (less than 75 kg), switch on the main tank fuel pumps during refueling until supply tank is full.

6. Switch off electrical power.
7. Remove fuel nozzle and disconnect its ground cable.
8. Install fuel filler cap and check locked.
9. Remove electrical ground connection.

8.3.2 Defueling

For defueling purposes, the helicopter fuel pumps are used. An external power unit should be used whenever possible to conserve battery power. Position external power unit at a safe distance (approx 10m) and connect to helicopter. If a container is used to collect the fuel during the defueling operation, that container must be electrically grounded. Keep a fire extinguisher ready at hand.

WARNING • ALWAYS ESTABLISH ELECTRICAL GROUND CONNECTION BETWEEN HELICOPTER, DEFUELING VEHICLE, CONTAINERS AND EXTERNAL POWER UNIT (AS REQUIRED) BEFORE STARTING DEFUELING OPERATION AND DISCONNECT ONLY WHEN OPERATION IS COMPLETED.

- DO NOT CONNECT, DISCONNECT OR REMOVE BATTERY DURING OR IMMEDIATELY FOLLOWING DEFUELING.

1. Establish electrical ground connection.
2. Release latches and open LH/RH engine compartment access panels.
3. Disconnect fuel supply line at the engine deck coupling of one or both engines.
4. Connect defueling line(s) of suitable length to the coupling(s) and route free-end of line to defueling vehicle or container.
5. Power helicopter electrical system (battery or EPU) and switch on respective fuel pumps.

NOTE If EPU is not available, operate only one fuel pump at a time alternating between supply and main tank pumps.

6. Monitor fuel quantity indicator. When both tanks are emptied switch off fuel pumps and de-energize helicopter electrical system.
7. Disconnect and remove defueling line(s) and connect fuel supply line(s).
8. Close LH/RH engine compartment access panels.
9. Position containers and open fuel tank drain valves.
10. Disconnect electrical ground connection.
11. Remove containers and dispose of fuel accordingly.

8.3.3 Engine Oil System

Both engine oil tanks are located in the RH transmission compartment beneath the oil cooler assembly. The left engine oil tank is located forward of the right engine oil tank. Each tank is serviced through a gravity filler neck having a screen-type filter. An oil level sight gage is located on the outer face of each tank. When the oil level falls to the ADD OIL mark, replenish with correct oil (see Section 2) until the level reaches the MAX mark on the sight gage or until an internal float blocks off the filler neck opening.

8.3.4 Main Transmission

Access to the main transmission oil filler neck is best achieved from the engine air inlet at the cabin top. The oil level sight gage is visible from the RH transmission compartment. When the oil level falls to the MIN mark, replenish with correct oil (see Section 2) until the level reaches the MAX mark on the sight gage.

8.3.5 Rotor Hub Reservoir

The oil level of the transparent rotor hub reservoir must be between the MIN and MAX markings when viewed from the nose of the helicopter. Replenishing instructions are in the BO 105 Maintenance Manual.

8.3.6 Intermediate Gearbox

The oil level sight gage for the intermediate gearbox is viewed on the left side at the base of the vertical fin fairing. The oil level is full when the bottom half of the sight gage is filled. The oil filler cap is accessible after opening a flap above and to the right of the sight gage. Replenish with correct oil (see Section 2).

NOTE When the sight gage glass is stained, the oil level may be best verified by grasping the tail skid and shortly shaking the tail boom while carefully observing the sight gage.

8.3.7 Tail Rotor Gearbox

The oil level sight gage for the tail rotor gearbox is located on the aft end of the gearbox housing. The oil level is full when the bottom half of the sight gage is filled. The oil filler cap is located above the sight gage. Replenish with correct oil (see Section 2).

NOTE When the sight gage glass is stained, the oil level may be best verified by grasping the tail skid and shortly shaking the tail boom while carefully observing the sight gage.

8.3.8 Hydraulic Systems

The hydraulic systems should only be serviced with approved fluids (see Section 2). Since the systems are of the closed-circuit type, only trained personnel are authorized to replenish the fluids following servicing instructions in the BO 105 Maintenance Manual.

8.3.9 Engine Compressor Rinsing

- CAUTION**
- SALT-LADEN HUMIDITY AND CHEMICALS WILL CORRODE COMPRESSOR BLADES AND VANES.
 - SALT-LADEN AIR MAY BE ENCOUNTERED UP TO 280 KM INLAND UNDER CERTAIN WEATHER CONDITIONS. IF ANY DOUBT EXISTS ABOUT THE CONDITIONS IN WHICH THE ENGINES WERE OPERATED, PERFORM A DAILY COMPRESSOR WATER RINSE.

Engines subjected to salt water contamination or other chemically laden atmosphere (including pesticides and herbicides) shall undergo water rinsing after shutdown following the last flight of the day.

The rinse operation (see BO 105 Maintenance Manual) should be performed as soon as practical after flight, but not before the engines have cooled to near ambient temperature.

8.4 COLD WEATHER OPERATION - GENERAL PRACTICES

8.4.1 Fuels

When operating in ambient temperatures below +4 °C (+40 °F) the fuel must contain an anti-icing additive meeting the requirements of MIL-I-27686.

If available use JP-4 or commercial Jet B fuel. As an alternate use an AVGAS/Jet fuel mixture following limitations described in Section 2, Fuel limitations.

8.4.2 Fuel/Additive Blending

- CAUTION** IF IT CANNOT BE DETERMINED THAT THE FUEL IN THE HELICOPTER CONTAINS AN ANTI-ICING ADDITIVE OR IF THE CONCENTRATION IS NOT KNOWN, THE HELICOPTER MUST BE DEFUELED AND REFUELED WITH FUEL CONTAINING THE PROPER ADDITIVE.

The required additive concentrations are in Section 2, Fuel limitations.

Blending the additive with the fuel is to be carried out using equipment normally supplied by the additive manufacturer. Follow manufacturer's recommended procedure when operating the blending equipment.

The flow of the additive must commence simultaneously with the flow of fuel. Refueling flow rates are to be between 95 and 227 liters per minute (25 and 60 US gal/min).

A differential refractometer may be used to accurately measure the concentration of additive in the fuel.

8.4.3 Mixing Alternate Fuel

CAUTION JP-4 OR COMMERCIAL JET B FUELS SHALL NOT BE MIXED WITH AVGAS.

As an alternate cold weather fuel when JP-4 or Commercial Jet B fuels cannot be obtained, use the following mixture:

- One part by volume AVGAS grade 80/87 or AVGAS 100L and,
- Two parts by volume Aviation Kerosene; JP-5; Commercial Jet A; or Commercial Jet A-1.

Refer to Fuel limitations in Section 2.

NOTE There is no engine operational time limit when using the AVGAS/Jet fuel mixture as long as 80/87 grade AVGAS is used and the 1:2 ratio is followed.

When operating on AVGAS/Jet fuel mixture, the anti-icing additive is required only for the jet fuel portion of the mixture.

CAUTION WHEN MIXING FUELS AVOID HIGH FLOW RATES. DO NOT EXCEED A FLOW RATE OF 189 LITERS PER MINUTE (50 US GAL/MIN). ALSO MAKE SURE THE FUEL NOZZLE AND FUEL TANK ARE ELECTRICALLY GROUNDED TO THE HELICOPTER.

Observe the following guidelines when making the alternate fuel mixture:

- Make the fuel mix in the helicopter tank. The AVGAS can be added before or after the jet fuel, the only restriction is that the final mix (including fuel in the tank before starting the mix) be two parts by volume jet fuel to one part by volume AVGAS.
- When the AVGAS/Jet fuel mixture is used in the engine the lead from the gasoline accumulates on the turbine and exhaust collector outlet ducts. The normal appearance after operation on lead fuel is a pale yellow powder deposit on the exhaust surfaces.
- If low temperature starting problems are encountered and the AVGAS/Jet fuel mixture is made in the fuel tank it may be necessary to bleed the unmixed fuel from the fuel lines in order to obtain a light-off. Refer to Allison Operation and Maintenance Manual.

8.4.4 Oils

CAUTION DO NOT MIX MIL-L-23699 AND MIL-L-7808 OILS.

When temperatures are below -40°C (-40°F) OAT or MIL-L-23699 oils are not available, use MIL-L-7808G and subsequent oils.

8.4.5 Preheating

WARNING DO NOT USE AN OPEN FLAME HEATER TO PREHEAT THE ENGINES OR BATTERY.

At temperatures below -18°C OAT, preheat the engines concentrating heat application on the fuel controls.

CAUTION WHEN ENGINE AND MAIN TRANSMISSION OIL TEMPERATURES ARE BELOW -25°C PREHEAT THE OILS TO A TEMPERATURE OF -25°C OR HIGHER.

At temperatures below -30°C OAT, preheat the cockpit to a temperature of -10°C or higher for a duration of at least 15 minutes.

8.5 MAINTENANCE

The component retirement life limitations in the Airworthiness Limitations Section of the BO 105 Maintenance Manual shall be complied with.

Refer to the BO 105 Maintenance Manual for information on the helicopter inspection periods and special inspections for helicopter systems.

SECTION 9

OPERATIONAL INFORMATION

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SECTION 9

OPERATIONAL INFORMATION

9.1 GENERAL

This section deals with additional information to supplement Section 5 (Performance Data), provide assistance in preflight planning, and be helpful in estimating the remaining flight time or range if, for some reason, the originally intended flight plan must be changed during flight.

For definition of terms, abbreviations and symbols used in this section, refer to Section 1.

9.1.1 Standard Performance Conditions

The data presented cover the maximum range of performance that can be expected and are based on the following conditions:

- Engine power does not exceed limits (see Section 2)
- Rotor RPM - 100%

In each area of performance, the effects of altitude, temperature, gross mass, and other parameters relating to that phase of flight are presented. In addition to the presented data, judgment and experience will be necessary to accurately determine performance under a given set of circumstances.

9.1.2 Conditions Affecting Performance

Conditions affecting performance are listed in the heading of the respective charts. The data presented are accurate only for these specific conditions.

Other general conditions applicable to the performance data are:

- Airframe and engine control rigging (assumed within allowable tolerances)
- Pilot technique (assumed normal, control movements smooth and continuous)
- Variations in performance between individual helicopters
- Instrument variation (inaccuracies or malfunctions are not accounted for)
- Clean configuration (externally mounted optional equipment are not accounted for).

9.1.3 Reading the Charts

None of the curves presented should be extrapolated, however, interpolation between given data is permissible.

9.2 FUEL CONSUMPTION

The fuel consumption diagrams (see Fig. 9-1 thru 9-2) are based on the following conditions:

- Fuel density is 0.8 kg/l (6.682 lb/U.S. gal.)
- The fuel used meets one of the approved fuel types as specified in Section 2
- Engine anti-icing and optional bleed air consumers (if fitted) are switched OFF

Fuel consumption as a function of LEVEL SPEED, pressure altitude, OAT, and gross mass is presented in Fig. 9-1.

Fuel consumption as a function of TORQUE, pressure altitude, speed, and OAT is presented in Fig. 9-2.

EXAMPLE: (see Fig. 9-1)

Determine: **Fuel consumption, level flight**

Known:	Speed (level flight)	25 KCAS
	Gross mass	2000 kg
	OAT	30 °C
	Pressure altitude	7500 ft

- Solution:
1. Enter chart at known speed (25 KCAS).
 2. Move horizontally right to known gross mass (2000 kg).
 3. Move vertically down to temperature Base Line (0 °C) and trace up (or down for OAT below 0 °C) following parallel to the guide lines.
 4. Reenter chart at known OAT (30 °C) and move horizontally right to intersect tracing from previous step.
 5. From point of intersection move vertically down to altitude Base Line (SL) and trace further down following parallel to the guide lines.
 6. Reenter chart at known pressure altitude (7500 ft) and move horizontally right to intersect tracing from previous step.
 7. From this point of intersection move vertically down and read fuel consumption = 135 kg/h.

FUEL CONSUMPTION - LEVEL FLIGHT

2 X ALLISON 250-C20B

POWER – AS REQUIRED

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

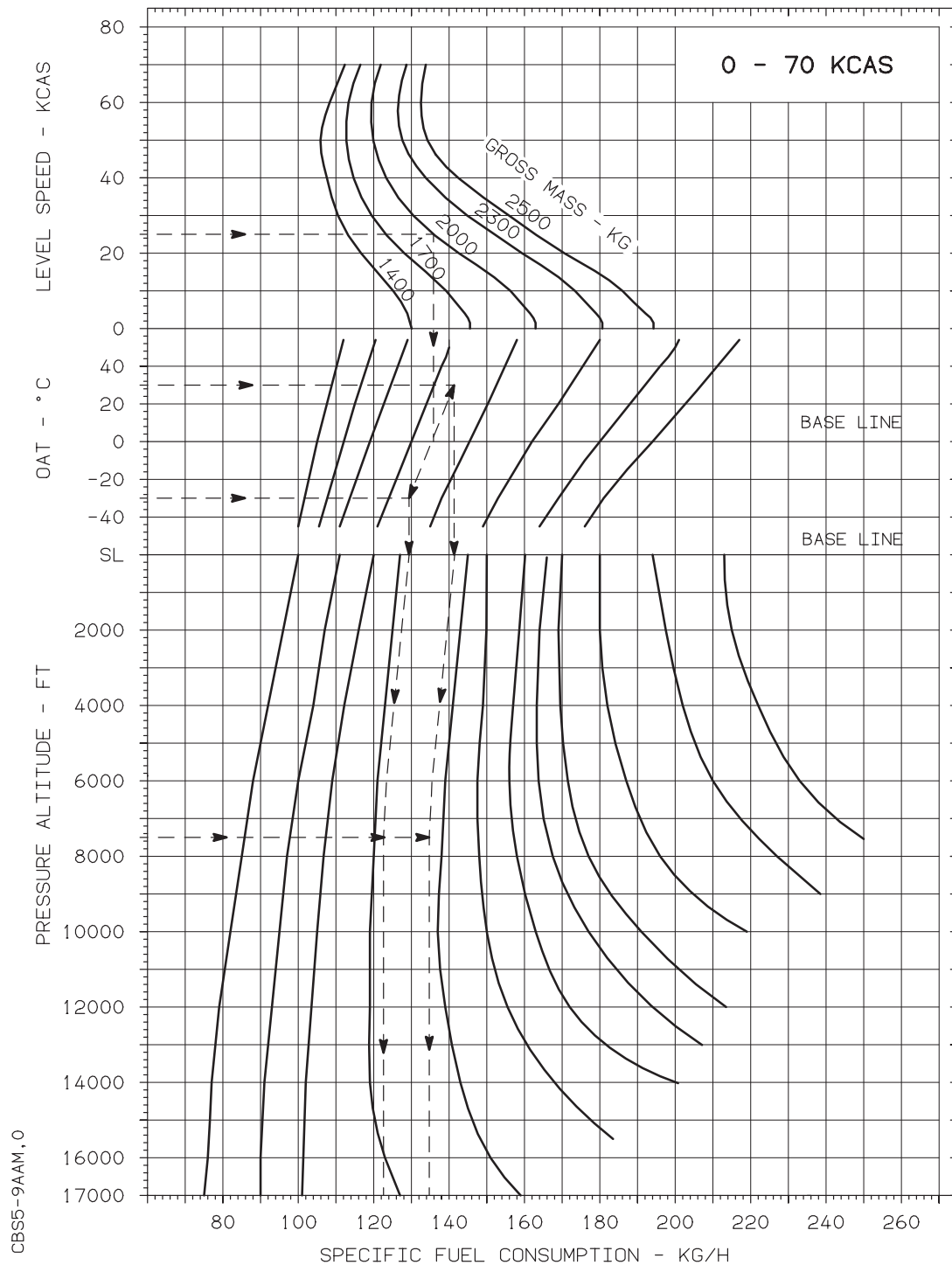
NOTE WITH ENGINE ANTI-ICING ON, SPECIFIC FUEL CONSUMPTION INCREASES BY 4.5 %.

Fig. 9-1 Fuel Consumption - Level Flight (Sheet 1 of 2)

FUEL CONSUMPTION - LEVEL FLIGHT

2 X ALLISON 250-C20B

POWER – AS REQUIRED

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

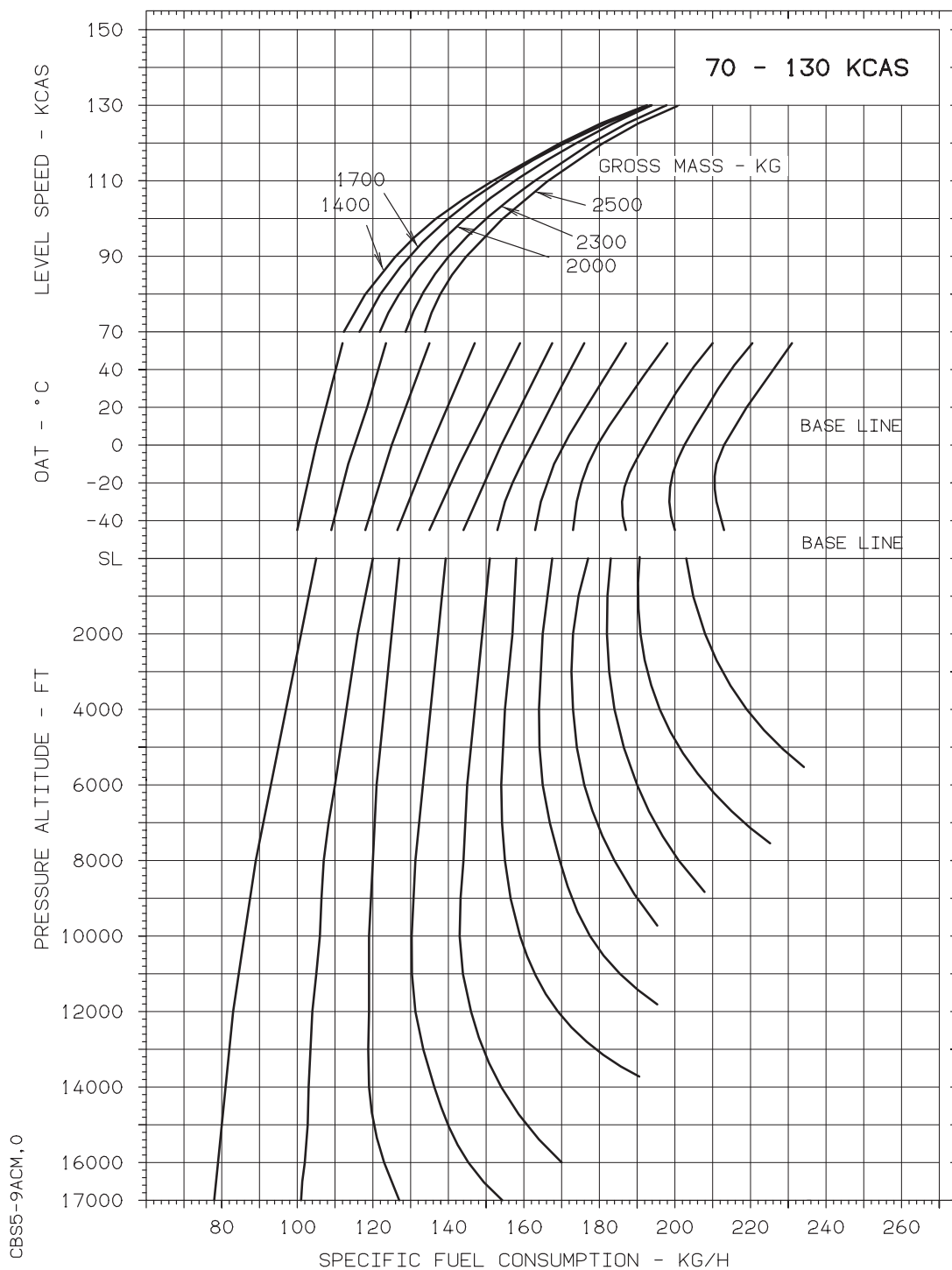
NOTE WITH ENGINE ANTI-ICING ON, SPECIFIC FUEL CONSUMPTION INCREASES BY 4.5 %.

Fig. 9-1 Fuel Consumption - Level Flight (Sheet 2 of 2)

MANUFACTURER'S DATA

FUEL CONSUMPTION - TORQUE

2 X ALLISON 250-C20B

AIRSPEED – AS REQUIRED

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

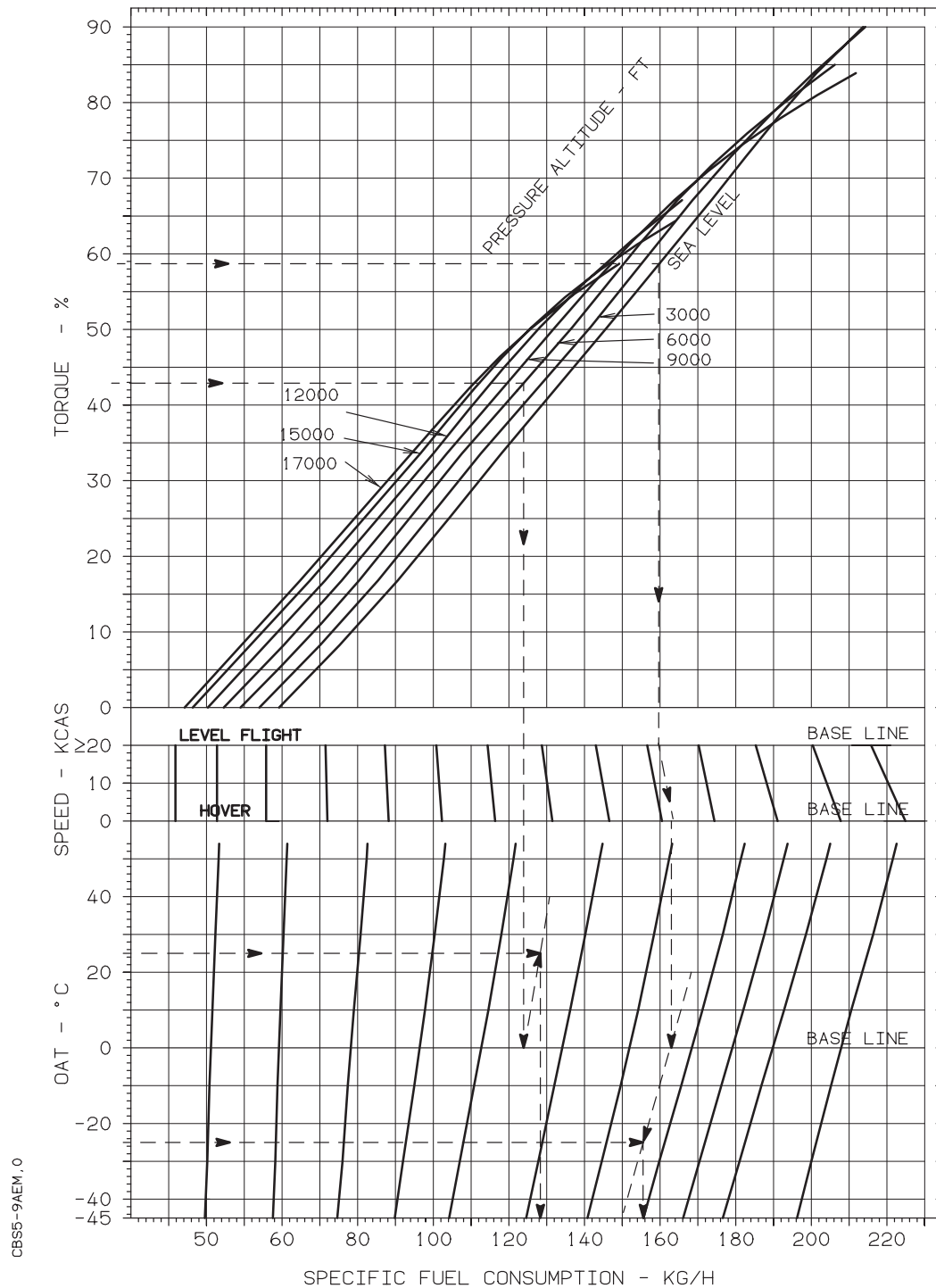
NOTE WITH ENGINE ANTI-ICING ON, SPECIFIC FUEL CONSUMPTION INCREASES BY 4.5 %.

Fig. 9-2 Fuel Consumption, TORQUE

9.3 MAXIMUM RANGE

The maximum range diagrams (see Fig. 9-3) are based on the following conditions:

- International Standard Atmosphere (ISA)
- Usable fuel - 456 kg (no reserve)
- Level flight (no climb, descent, or other maneuver)
- Engine anti-icing and optional bleed air consumers (if fitted) are switched OFF

NOTE With engine anti-icing and/or optional bleed air consumers (if fitted) ON, the maximum range will be reduced.

The maximum range charts are presented as a function of level speed and pressure altitude at various takeoff gross masses.

EXAMPLE: (see Fig. 9-3)

Determine: **Maximum range (without reserve)**

Known: Takeoff gross mass 2500 kg
 Level speed 90 KCAS
 Pressure altitude Sea Level

Solution: 1. Using known takeoff gross mass chart (2500 kg), enter at known level speed (90 KCAS).
 2. Move vertically up to known pressure altitude (Sea Level).
 3. Move horizontally left and read maximum range = 286 NM.

MAXIMUM RANGE

2 X ALLISON 250-C20B

FUEL – 456 KG, NO RESERVE

ANTI-ICING – OFF

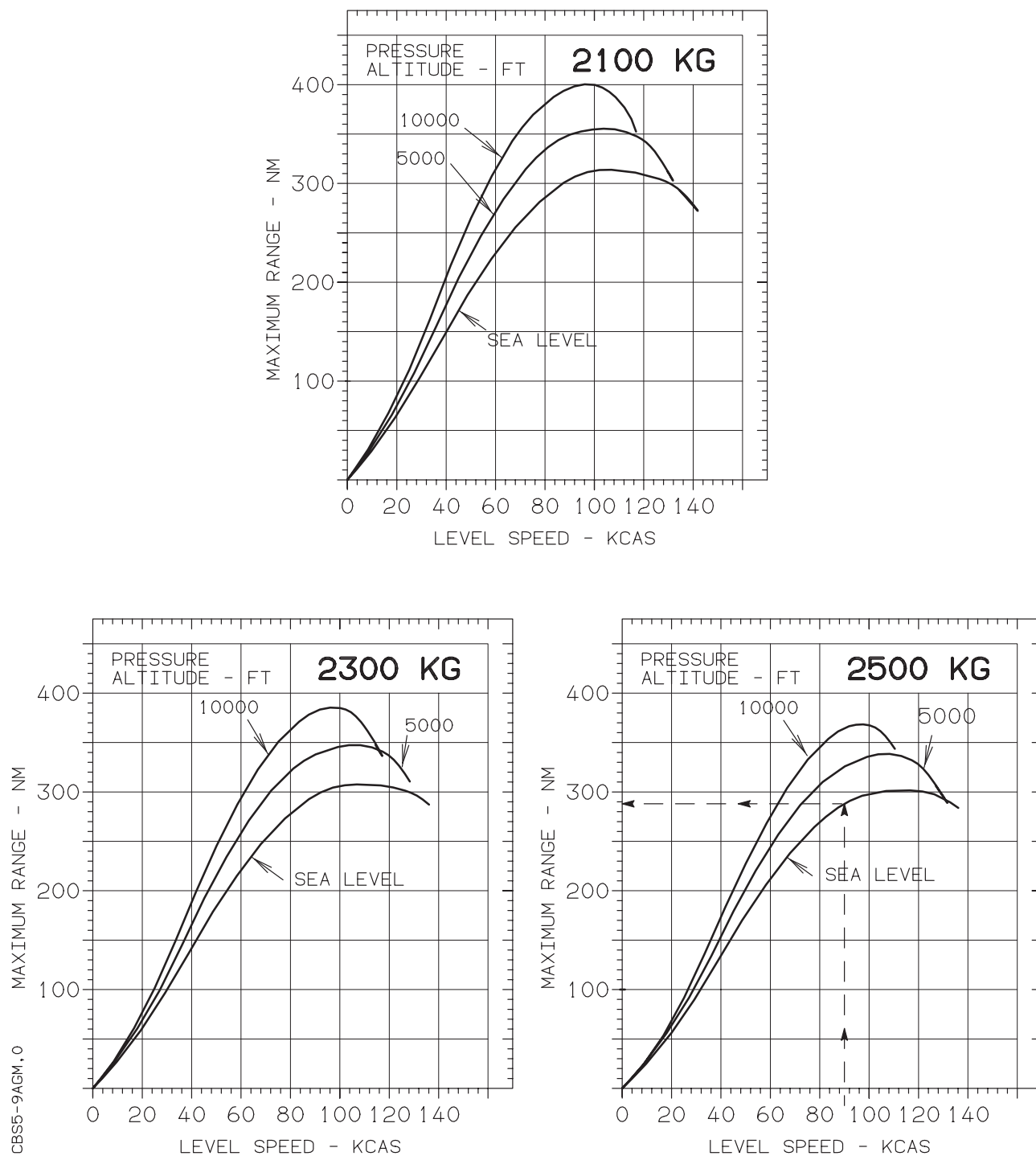
NOTE WITH ENGINE ANTI-ICING ON, MAX RANGE DECREASES BY 4.5 %.

Fig. 9-3 Maximum Range

9.4 ENDURANCE

The endurance diagrams (see Fig. 9-4) are based on the following conditions:

- International Standard Atmosphere (ISA)
- Usable fuel - 456 kg (no reserve)
- Level flight (no climb, descent, or other maneuver)
- Engine anti-icing and optional bleed air consumers (if fitted) are switched OFF

NOTE With engine anti-icing and/or optional bleed air consumers (if fitted) ON, the maximum endurance will be reduced.

The endurance charts are presented as a function of level speed and pressure altitude at various takeoff gross masses.

EXAMPLE: (see Fig. 9-4)

Determine: **Endurance (without reserve)**

Known: Takeoff gross mass 2500 kg
 Level speed 90 KCAS
 Pressure altitude Sea Level

Solution: 1. Using known takeoff gross mass chart (2500 kg), enter at known level speed (90 KCAS).
 2. Move vertically up to known pressure altitude (Sea Level).
 3. Move horizontally left and read endurance = 3.2 hours.

ENDURANCE

2 X ALLISON 250-C20B

FUEL – 456 KG, NO RESERVE

ANTI-ICING – OFF

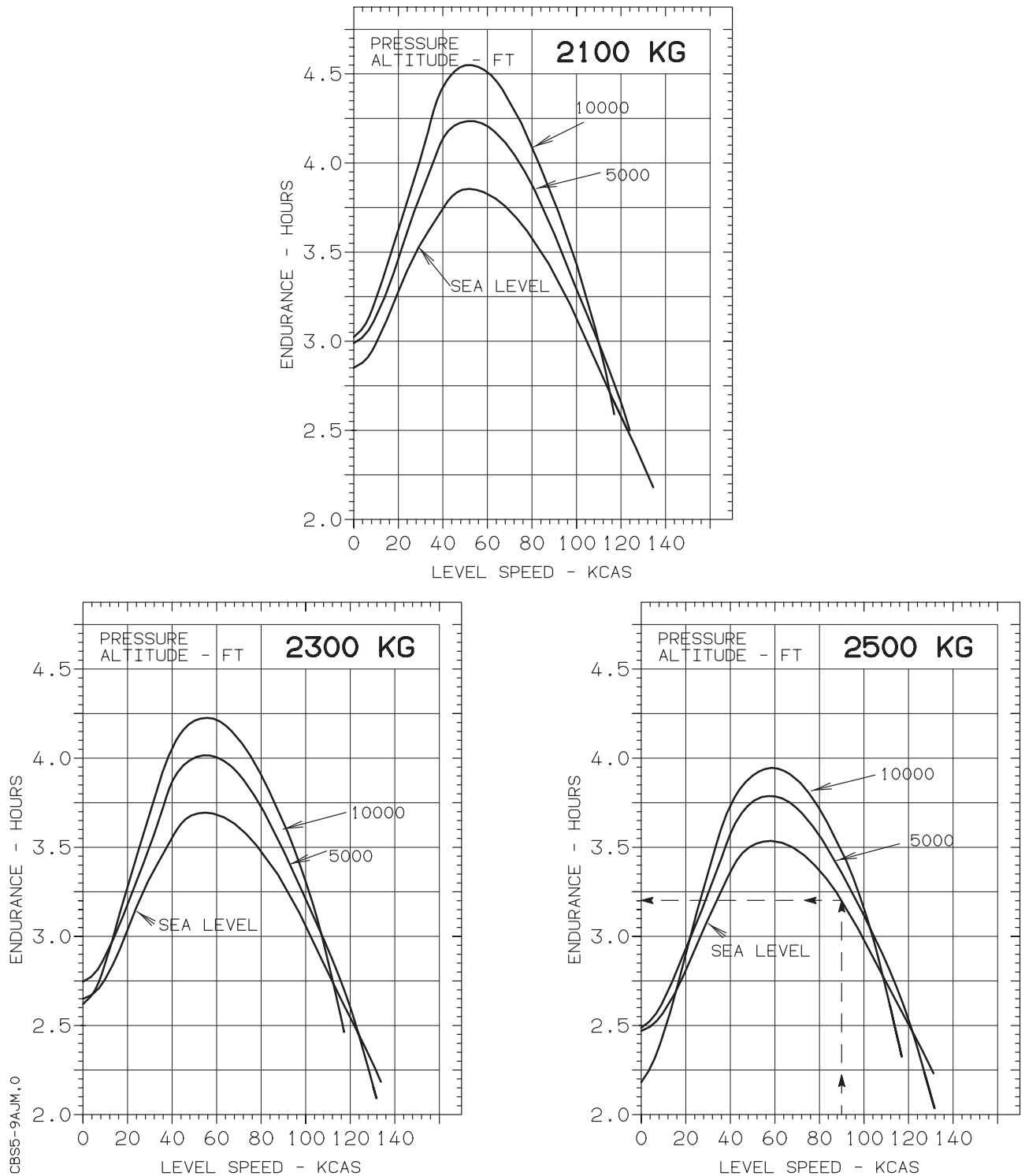
NOTE WITH ENGINE ANTI-ICING ON, ENDURANCE TIME DECREASES BY 4.5 %.

Fig. 9-4 Endurance

9.5 MAXIMUM CRUISING SPEED

The maximum cruising speed diagrams (see Fig. 9-5 thru 9-7) are based on both engines operating at maximum continuous power (779 °C TOT, 86 % torque) and presented as a function of helicopter gross mass and pressure altitude for the following various atmospheric conditions:

- ISA –20 °C (see Fig. 9-5)
- ISA (see Fig. 9-6)
- ISA +20 °C (see Fig. 9-7)

NOTE With engine anti-icing and/or optional bleed air consumers (if fitted) ON, the maximum cruising speed will be reduced.

EXAMPLE: (see Fig. 9-5)

Determine: **Maximum cruising speed**

Known:	Atmospheric condition	ISA –20 °C
	Pressure altitude	11,500 ft
	Gross mass	2500 kg

Solution:

1. Enter chart at known pressure altitude (11,500 ft).
2. Move horizontally right to known gross mass (2500 kg).
3. Move vertically down and read maximum cruising speed = 119 KTAS.

MAXIMUM CRUISING SPEED - ISA -20 °C

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

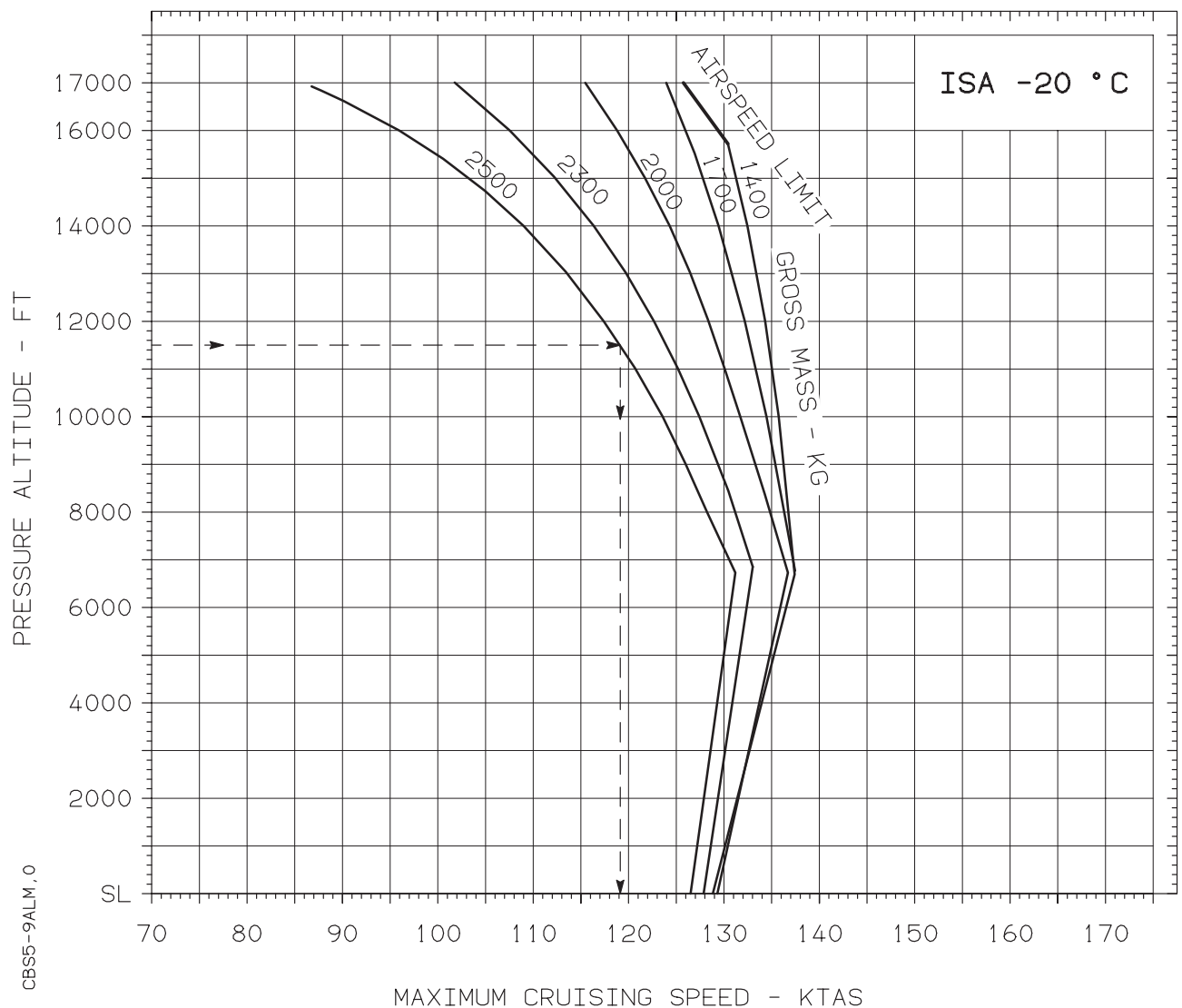
NOTE WITH ENGINE ANTI-ICING ON, MAX CRUISING SPEED DECREASES BY 6 KTAS.

Fig. 9-5 Maximum Cruising Speed - ISA -20 °C

MAXIMUM CRUISING SPEED - ISA

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

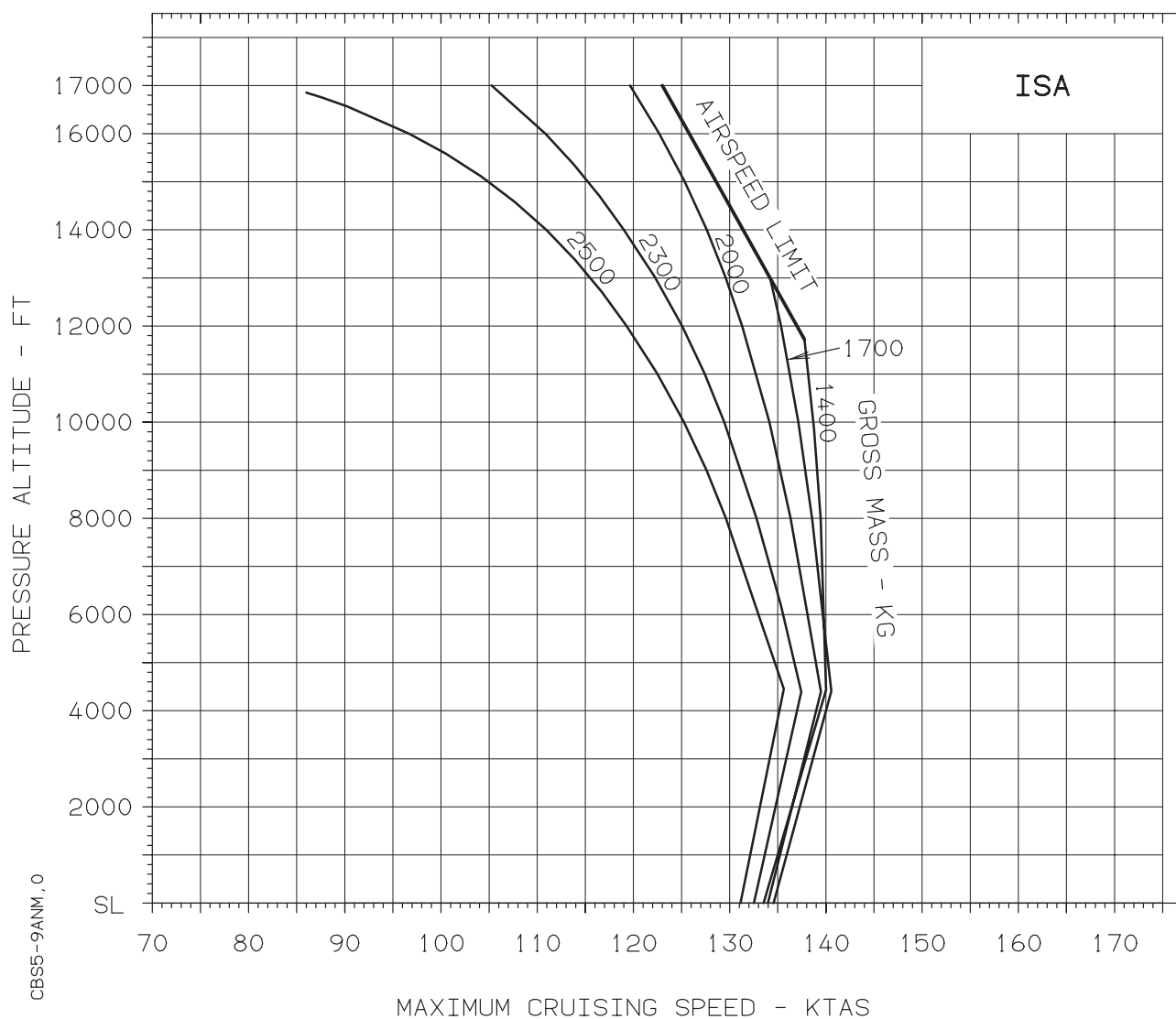
NOTE WITH ENGINE ANTI-ICING ON, MAX CRUISING SPEED DECREASES BY 6 KTAS.

Fig. 9-6 Maximum Cruising Speed - ISA

MANUFACTURER'S DATA

MAXIMUM CRUISING SPEED - ISA +20 °C

2 X ALLISON 250-C20B

AEO – MAX CONTINUOUS POWER (779 °C TOT, 86% TORQUE)

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

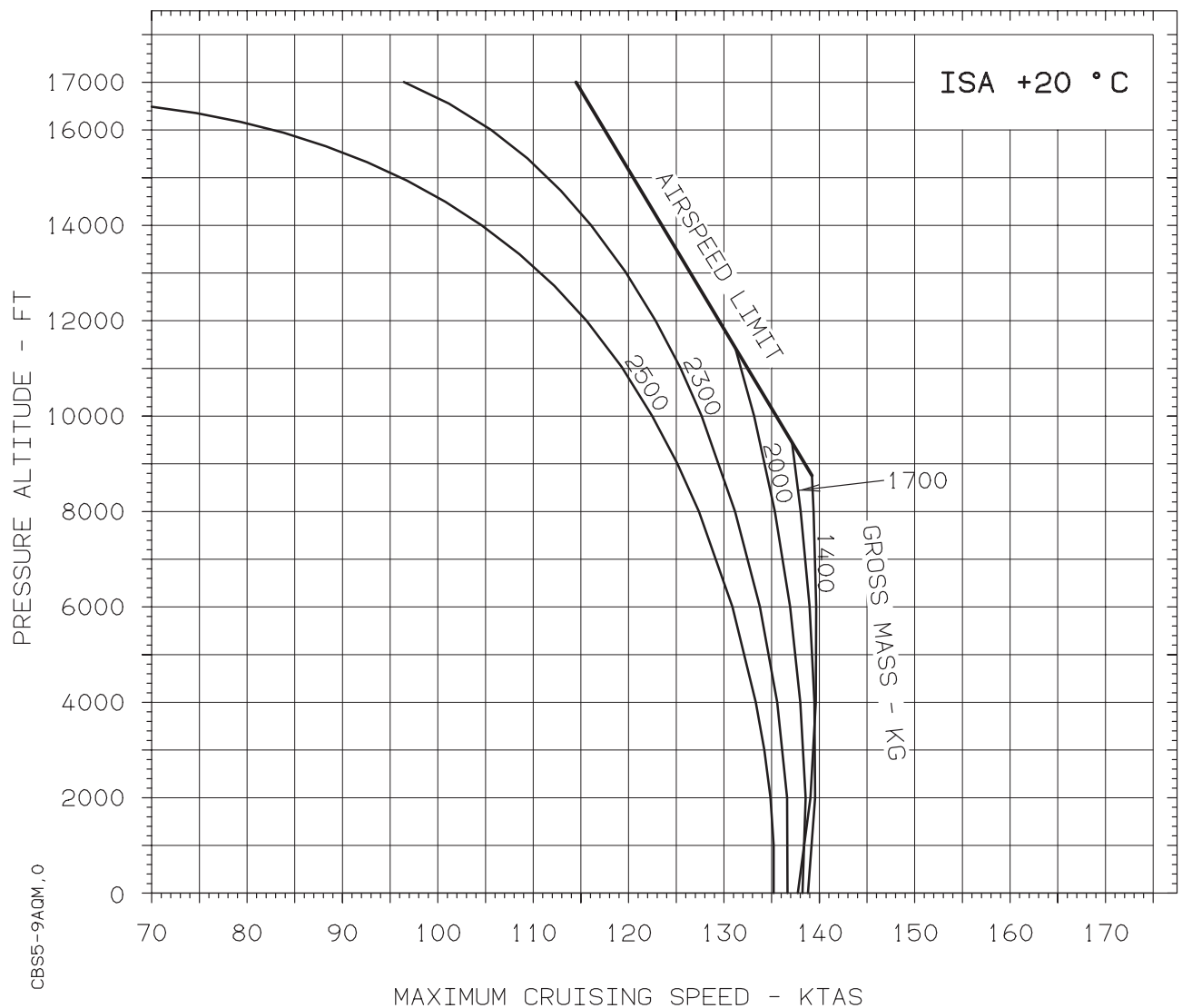
NOTE WITH ENGINE ANTI-ICING ON, MAX CRUISING SPEED DECREASES BY 6 KTAS.

Fig. 9-7 Maximum Cruising Speed - ISA +20 °C

SECTION 10**OPTIONAL EQUIPMENT SUPPLEMENTS****LOG OF SUPPLEMENTS**

TITLE	FMS NUMBER	APPROVING AUTHORITY/DATE	STD EQPT EFFECTIVITY
AC Power Supply System	10-56	LBA JUN 30, 2000 FAA PENDING	
Aerial Flare System	10-45	LBA APR 10, 1995 FAA MAR 28, 1996	
Automatic Flight Control System	10-41	LBA APR 10, 1995 FAA MAR 28, 1996	
Bleed Air Heating/Ventilation System	10-44	LBA APR 10, 1995 FAA MAR 28, 1996	
Dual Controls	10-5	LBA APR 10, 1995 FAA MAR 28, 1996	0751-9999
Emergency DC-Bus System	10-36	LBA APR 10, 1995 FAA MAR 28, 1996	
Emergency Flotation System	10-22	LBA APR 10, 1995 FAA MAR 28, 1996	
Engine Air Cleaner System	10-37	LBA APR 10, 1995 FAA MAR 28, 1996	
Engine Air Cleaner System - Blower Purge	10-50	LBA APR 10, 1995 FAA MAR 28, 1996	
Engine Fire Extinguishing System	10-25	LBA APR 10, 1995 FAA MAR 28, 1996	
External Cargo Hook - 900 KG	10-12	LBA JAN 05, 1996 FAA SEP 30, 1997	
External Cargo Hook - 1200 KG	10-53	LBA JAN 05, 1996 FAA SEP 30, 1997	
External Loudspeakers	10-23	LBA APR 10, 1995 FAA MAR 28, 1996	
FLIR System - BSS 400	10-54	LBA JUL 12, 1996 FAA SEP 30, 1997	
Follow-up Trim System	10-40	LBA APR 10, 1995 FAA MAR 28, 1996	

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Engine Air Cleaner System	10-37	LBA APR 10, 1995 FAA MAR 28, 1996	
Engine Air Cleaner System - Blower Purge	10-50	LBA APR 10, 1995 FAA MAR 28, 1996	
Engine Fire Extinguishing System	10-25	LBA APR 10, 1995 FAA MAR 28, 1996	
External Cargo Hook - 900 KG	10-12	LBA JAN 05, 1996 FAA SEP 30, 1997	
External Cargo Hook - 1200 KG	10-53	LBA JAN 05, 1996 FAA SEP 30, 1997	
External Loudspeakers	10-23	LBA APR 10, 1995 FAA MAR 28, 1996	
FLIR System - BSS 400	10-54	LBA JUL 12, 1996 FAA SEP 30, 1997	
Follow-up Trim System	10-40	LBA APR 10, 1995 FAA MAR 28, 1996	

TITLE	FMS NUMBER	APPROVING AUTHORITY/DATE		STD EQPT EFFECTIVITY
Fuel Jettison System	10-10	LBA FAA	APR 10, 1995 MAR 28, 1996	
Fuel Microfilters	10-39	LBA FAA	APR 10, 1995 MAR 28, 1996	
Fuel Microfilters with Back-Up Supply System	10-28	LBA FAA	APR 10, 1995 MAR 28, 1996	
Heating/Ventilation System	10-11	LBA FAA	APR 10, 1995 MAR 28, 1996	0751-9999
High Clearance Landing Gear	10-42	LBA FAA	APR 10, 1995 MAR 28, 1996	
LIGHTS				
Interior Night Lights	10-7	LBA FAA	APR 10, 1995 MAR 28, 1996	0751-9999
Landing Light	10-6	LBA FAA	APR 10, 1995 MAR 28, 1996	
Landing and Search Light	10-34	LBA FAA	APR 10, 1995 MAR 28, 1996	
Landing and Search Light	10-51	LBA FAA	APR 10, 1995 MAR 28, 1996	
NVG Compatible Instrument Lighting	10-52	LBA FAA	JAN 18, 1996 SEP 30, 1997	
Searchlight (SX 16) with CTR Controls	10-38	LBA FAA	APR 10, 1995 MAR 28, 1996	
Searchlight (SX 16) with LH Controls	10-35	LBA FAA	APR 10, 1995 MAR 28, 1996	
Searchlight (SX 16) with IR Filter	10-60	LBA FAA	APR 10, 1995 MAR 28, 1996	
Searchlight 450W	10-32	LBA FAA	APR 10, 1995 MAR 28, 1996	
Litter Installation	10-8	LBA FAA	APR 10, 1995 MAR 28, 1996	
Long Range Fuel System	10-24	LBA FAA	APR 10, 1995 MAR 28, 1996	
Magnetic Cyclic Trim	10-9	LBA FAA	APR 10, 1995 MAR 28, 1996	

TITLE	FMS NUMBER	APPROVING AUTHORITY/DATE		STD EQPT EFFECTIVITY
Main Rotor Folding System	10-15	LBA FAA	APR 10, 1995 MAR 28, 1996	
Main Tank Pump Off Advisory Light System	10-30	LBA FAA	APR 10, 1995 MAR 28, 1996	
Medical Equipment Board	10-16	LBA FAA	APR 10, 1995 MAR 28, 1996	
Modified Engine Starting/Ignition System Controls	10-33	LBA FAA	APR 10, 1995 MAR 28, 1996	
Pulsed Chip Detector System	10-55	LBA FAA	APR 10, 1995 MAR 28, 1996	
Rescue Hoist System	10-17	LBA FAA	APR 10, 1995 MAR 28, 1996	
Rotor Brake System	10-1	LBA FAA	APR 10, 1995 MAR 28, 1996	0481-9999
Scavenge Oil Filters	10-57	LBA FAA	APR 10, 1995 MAR 28, 1996	
SEATS				
Copilot Reversible Seat	10-13	LBA FAA	APR 10, 1995 MAR 28, 1996	
Passenger Bench	10-2	LBA FAA	APR 10, 1995 MAR 28, 1996	
Passenger Centerline Bench	10-14	LBA FAA	APR 10, 1995 MAR 28, 1996	
Passenger Foldable Bench	10-27	LBA FAA	APR 10, 1995 MAR 28, 1996	
Passenger Sectioned Bench	10-21	LBA FAA	APR 10, 1995 MAR 28, 1996	
Self-Sealing Fuel Tanks	10-26	LBA FAA	APR 10, 1995 MAR 28, 1996	
Settling Protectors	10-20	LBA FAA	APR 10, 1995 MAR 28, 1996	
Shoulder Harness	10-4	LBA FAA	APR 10, 1995 MAR 28, 1996	0651-9999
Snow Skids	10-18	LBA FAA	APR 10, 1995 MAR 28, 1996	

TITLE	FMS NUMBER	APPROVING AUTHORITY/DATE	STD EQPT EFFECTIVITY
Transmission/Gearbox Chip Warning	10-49	LBA APR 10, 1995 FAA MAR 28, 1996	
Two-axis Stability Augmentation System	10-29	LBA APR 10, 1995 FAA MAR 28, 1996	
Windshield Wiper	10-3	LBA APR 10, 1995 FAA MAR 28, 1996	
Windshield Wiper/Washer System	10-47	LBA APR 10, 1995 FAA MAR 28, 1996	
Wire Strike Protection System	10-46	LBA APR 10, 1995 FAA MAR 28, 1996	

F M S 11-4

SUPPLEMENT FOR EQUIVALENT CATEGORY A OPERATIONS

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F M S 11 - 4**SUPPLEMENT FOR**
EQUIVALENT CATEGORY A OPERATIONS

NOTE Due to the nature of its content, this supplement is divided into separate, related subsections that first present General Data applicable to all Equivalent Category A Operations and then specific data applicable to approved types of operation.

A. GENERAL DATA**A.1 GENERAL**

This supplement shall be attached to the BO 105 Flight Manual (Section 11) when the VTOL Retrofit Kit has been installed.

System/Equipment Designation	Part No.	Effectivity
VTOL Retrofit Kit	105-80037	All
NOTE • Helicopters S/N 0001-0450 must have SB 60-37 (ram air ventilation) installed. • The Engine Fire Extinguishing System (FMS 10-25) must be installed and operational.		

The information contained herein supplements or supersedes the information of the basic Flight Manual; for limitations, procedures, and performance data not contained in this supplement, refer to the basic Flight Manual.

NOTE For approving authorities and original issue date of this supplement, refer to the Log of Supplements.

A.1.1 INTRODUCTION

Although the BO-105 helicopter is certified basically for operation according to FAR 27, this supplement provides standardized takeoff and landing procedures and performance data which are certified according to JAR PART 29, Second Draft, Category A operations.

This supplement only applies to helicopters having main transmission ZF FS72E (Retrofit kit, P/N 105-80037) installed.

The information presented in this supplement is required for the following approved types of operations:

- Clear Heliport
- Elevated or Restricted Helipad (VTOL)
- Short Field

A.1.2 ABBREVIATIONS

AHE	- Above Helipad Elevation
LDP	- Landing Decision Point
MAT	- Mass/Altitude/Temperature
TDP	- Takeoff Decision Point
TOGM	- Takeoff Gross Mass
V _{TOSS}	- Takeoff Safety Speed
VTOL	- Vertical Takeoff and Landing

A.1.3 DEFINITIONS

Category A Takeoff is the operation of the helicopter in such a manner that, if one engine fails at any time after the start of a takeoff, the helicopter can

- prior to TDP return to and safely land and stop on the takeoff area, or
- after TDP continue the takeoff and climb out from the point of failure and attain single-engine forward flight.

Category A Landing is the operation of the helicopter in such a manner that, if one engine fails at any time after the start of a landing approach, the helicopter can

- prior to LDP climb-out from the point of engine failure and attain single-engine forward flight or continue the approach and safely land and stop on the landing area, or
- after LDP continue the approach and safely land and stop on the landing area.

Takeoff Decision Point (TDP) is the last point on the takeoff path at which, in the event of failure of one engine, a rejected takeoff is assured within the distances determined according to the procedure used, and is the first point at which a continued takeoff is assured.

If one engine fails prior to TDP the takeoff must be rejected.

If one engine fails after TDP the takeoff may be continued.

Landing Decision Point (LDP) is the only point on the landing profile at which, in the event of failure of one engine, the combination of height and airspeed permits the decision to proceed with the initiated landing or to accelerate to takeoff safety speed (V_{TOSS}) and to accomplish a safe climb.

If one engine fails prior to LDP the pilot may elect to go-around or continue the landing approach.

If one engine fails after LDP the landing must be continued.

Takeoff Distance is the horizontal distance along the takeoff flight path from the start of the takeoff to the point at which the helicopter attains and remains at least 35 feet above the takeoff surface, attains and maintains a speed of at least V_{TOSS} , and establishes a positive rate of climb.

Rejected Takeoff Distance Required is the horizontal distance necessary to safely land and stop the helicopter when one engine becomes inoperative prior to TDP.

Continued Takeoff Distance Required is the horizontal distance necessary to achieve a takeoff flight path at a height of 35 feet or higher at V_{TOSS} and with a 100 fpm rate of climb, when one engine becomes inoperative after TDP.

Landing Distance Required is the horizontal distance necessary to safely land and stop the helicopter with one engine inoperative from a 50 ft height to a complete stop on the ground.

Takeoff Segment I Distance is the horizontal distance necessary to climb with V_{TOSS} from the end of the takeoff distance to 200 ft AGL.

Takeoff Segment II Distance is the horizontal distance necessary to climb with V_Y from 200 ft to 1000 ft AGL.

Takeoff Safety Speed is the airspeed that will assure a required steady rate of climb of at least 100 fpm with 2.5-min power (OEI) and with a helicopter maximum gross mass specified for the ambient condition.

$$V_{TOSS} = 50 \text{ KIAS}$$

Best Rate-of-climb Speed is the airspeed which leads to the greatest gain in altitude in the shortest possible time.

$$V_Y = 60 \text{ KIAS}$$

A.2 LIMITATIONS**A.2.1 ROTOR RPM LIMITATIONS**

	Power ON	Power OFF
Minimum Transient	85%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	104%
Maximum Transient (15-sec limited)	105%	110%

A.2.2 ENGINE AND TRANSMISSION POWER LIMITATIONS

Engines: Allison 250 - C 20 B

Transmission: Zahnradfabrik Friedrichshafen AG (ZF) FS 72E

WARNING IF TRANSIENT TORQUE/TOT VALUES ARE NOT OBSERVED DUE TO WORK LOAD THEN IT MUST BE ASSUMED THAT THE MAX TORQUE/TOT VALUES HAVE BEEN REACHED AND ONE USAGE OF THE TRANSIENT TORQUE/TOT SHALL BE RECORDED IN THE LOGBOOK.

- CAUTION**
- OEI LIMITS ARE RESTRICTED TO EMERGENCY USE ONLY.
 - FOR POWER CHECK THE ONE ENGINE INOPERATIVE TRANSMISSION LIMIT IS MAX 95% TORQUE.
 - FOR OEI TRAINING PURPOSES SEE PARAGRAPH 9 THIS SUPPLEMENT SUBSECTION.

CONDITION	TRANSMISSION (HELICOPTER) LIMITS MAX TORQUE %	ENGINE OPERATING LIMITS		
		MAX TOT °C	MAX N ₁ %	MAX N ₂ %
Start/shutdown Transient (Max 10 sec above 810 °C)		927		
Transient (Max 6 sec above 810 °C)		843		
Transient (Max 12 sec above 810 °C)		899 ¹⁾		
Transient (Max 15 sec)			106	105
All Engines Operating				
Max Continuous Power	2 x 86	779	105	102
Takeoff Power	2 x 86	810 ²⁾	105	102
One Engine Inoperative				
Max Continuous Power	105	810	105	102
2.5-min Power	110	810	105	102
Transient (Max 16 sec above 110%)	123 ³⁾			
NOTE ¹⁾ Max three occurrences per life of each turbine wheel. Logbook entry required. ²⁾ Takeoff Power TOT range of 779-810 °C has a 5-min limit. ³⁾ After a max of ten occurrences above 110% maintenance action is required (see MM). Logbook entry required.				

A.2.3 WIND LIMITS

Takeoff and landing with a tailwind is prohibited.

A.2.4 APPROVED LUBRICANTS

Main rotor transmission *) Restricted to EXXON T.O. 25 and BP Turbo Oil 25

*) To be placarded or marked on the equipment.

A.2.5 INSTRUMENT MARKINGS

The pointers and scales of the instruments listed below are marked as follows:

Start limits	red triangle
Transient limits	red dot
Minimum and maximum limits	red radial
OEI 2.5-min Power	red dashed radial
OEI Max Continuous Power	yellow dashed radial
Takeoff power or caution range	yellow arc
Normal or continuous range	green arc

A.2.5.1 Turbine Outlet Temperature (TOT) Indicator

0 to 779 °C	green arc
779 to 810 °C	yellow arc
810 °C	red radial
810 °C	yellow dashed radial
843 °C	red dot
899 °C	red dot
927 °C	red triangle

A.2.5.2 Dual Torque Indicator

0 to 86%	green arc
86%	red radial
105%	yellow dashed radial
110%	red dashed radial
123%	red dot

A.2.6 OPTIONAL EQUIPMENT INSTALLATION LIMITATIONS

A.2.6.1 General

All optional bleed air consumers (e.g. Engine Air Cleaner System, FMS 10-37; Bleed Air Heating, FMS 10-44; etc) must be switched off during takeoff and landing.

A.2.6.2 Clear Heliport

When the Emergency Flotation System is installed operation at Clear Heliports is with reduced performance (see FMS 10-22).

When the cross-tube mounted External Loudspeakers are installed operation at Clear Heliports is with reduced performance (see FMS 10-23).

The SX 16 Searchlight (FMS 10-35, 10-38 and 10-60) shall be removed when performing Clear Heliport operations.

Clear Heliport operations are prohibited when flying with opened/removed doors (FMS 11-1).

A.2.6.3 **VTOL and Short Field**

VTOL and Short Field operations are prohibited when flying with opened/removed doors (FMS 11-1).

A.3 **EMERGENCY AND MALFUNCTION PROCEDURES**

A.3.1 **OEI DURING TAKEOFF**

If one engine fails before TDP the takeoff must be rejected.

If one engine fails after TDP the takeoff must be continued.

Rejecting/continuing takeoff: Refer to the respective subsection this supplement.

A.3.2 **OEI DURING LANDING**

If one engine fails before LDP the pilot may elect to go-around or to continue the approach.

If one engine fails after LDP the landing must be continued.

Go-around/single-engine landing: Refer to the respective subsection this supplement.

A.4 **NORMAL PROCEDURES**

Refer to the respective subsection this supplement.

A.5 **PERFORMANCE DATA**

A.5.1 **HEIGHT/VELOCITY DIAGRAMS**

The height/velocity diagrams in the basic FLM Section 5, do not apply when performing VTOL and Short Field operations.

A.5.2 **OEI CLIMB PERFORMANCE**

OEI climb performance charts (see Fig. A1 thru A5) are presented for OEI/MCP and 2.5-min power ratings at different airspeeds (V_Y and V_{TOSS}). These charts show rate of climb and height gain data as functions of pressure or takeoff altitude, outside air temperature and gross mass.

NOTE Chart data are based on clean configuration.

EXAMPLE: (see Fig. A1)

Determine: **OEI rate of climb**

Known:	Pressure altitude	7650 ft
	OAT	20 °C
	Gross mass	1650 kg
	Airspeed	V_{TOSS}

- Solution:
1. Enter appropriate chart at known pressure altitude (7650 ft).
 2. Move horizontally right to known OAT (20 °C).
 3. Move vertically down to the Base Line and trace further down following parallel to the guide lines.
 4. Reenter chart at known gross mass (1650 kg).
 5. Move horizontally right to intersect tracing from above.
 6. From point of intersection move vertically down and read rate of climb = 490 ft.

EXAMPLE: (see Fig. A4)

Determine: **OEI height gain over a horizontal distance of 100 ft**

Known:	Takeoff altitude	5700 ft
	OAT	20 °C
	Gross mass	1875 kg
	Headwind	12 kt
	Airspeed	V_{TOSS}

- Solution:
1. Enter appropriate chart at known takeoff altitude (5700 ft).
 2. Move horizontally right to known OAT (20 °C).
 3. Move vertically down to the Base Line and trace further down following parallel to the gross mass guide lines.
 4. Reenter chart at known gross mass (1875 kg).
 5. Move horizontally right to intersect tracing from above.
 6. From point of intersection move vertically down to Base Line and trace further down following parallel to the headwind guide lines.
 7. Reenter chart at known headwind (12 kt).
 8. Move horizontally right to intersect tracing from above.
 9. From point of intersection move vertically down and read height gain over a horizontal distance of 100 ft = 8.5 ft.

OEI RATE OF CLIMB - FLIGHT PATH SEGMENT I

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE)

ANTI-ICING – OFF

TAKEOFF SAFETY SPEED – 50 KIAS

BLEED AIR CONSUMERS – OFF

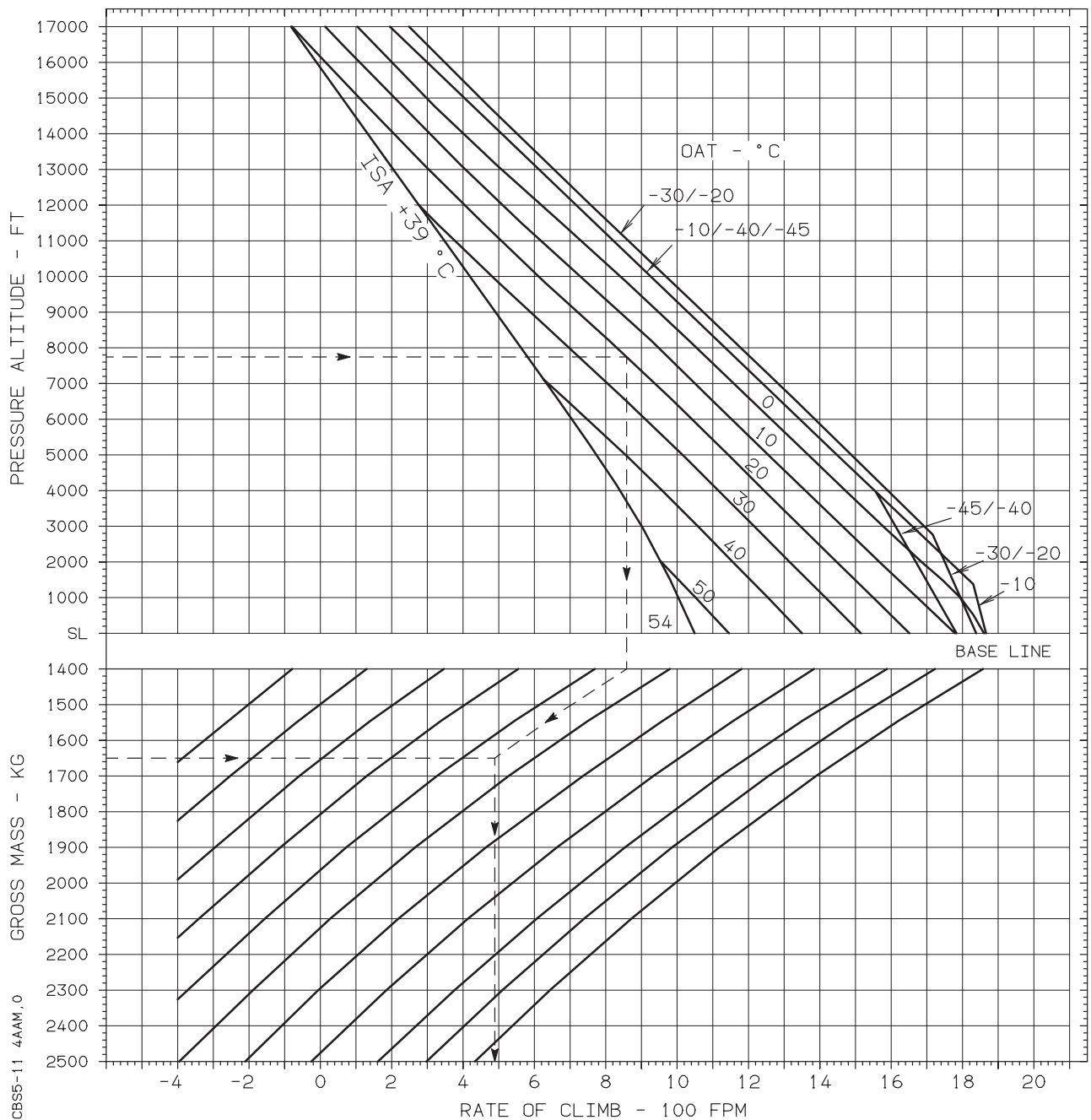


Fig. A1 OEI Rate of Climb - Flight Path Segment I

OEI RATE OF CLIMB - FLIGHT PATH SEGMENT II

1 X ALLISON 250-C20B

OEI – MAX CONTINUOUS POWER (810 °C TOT, 105% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

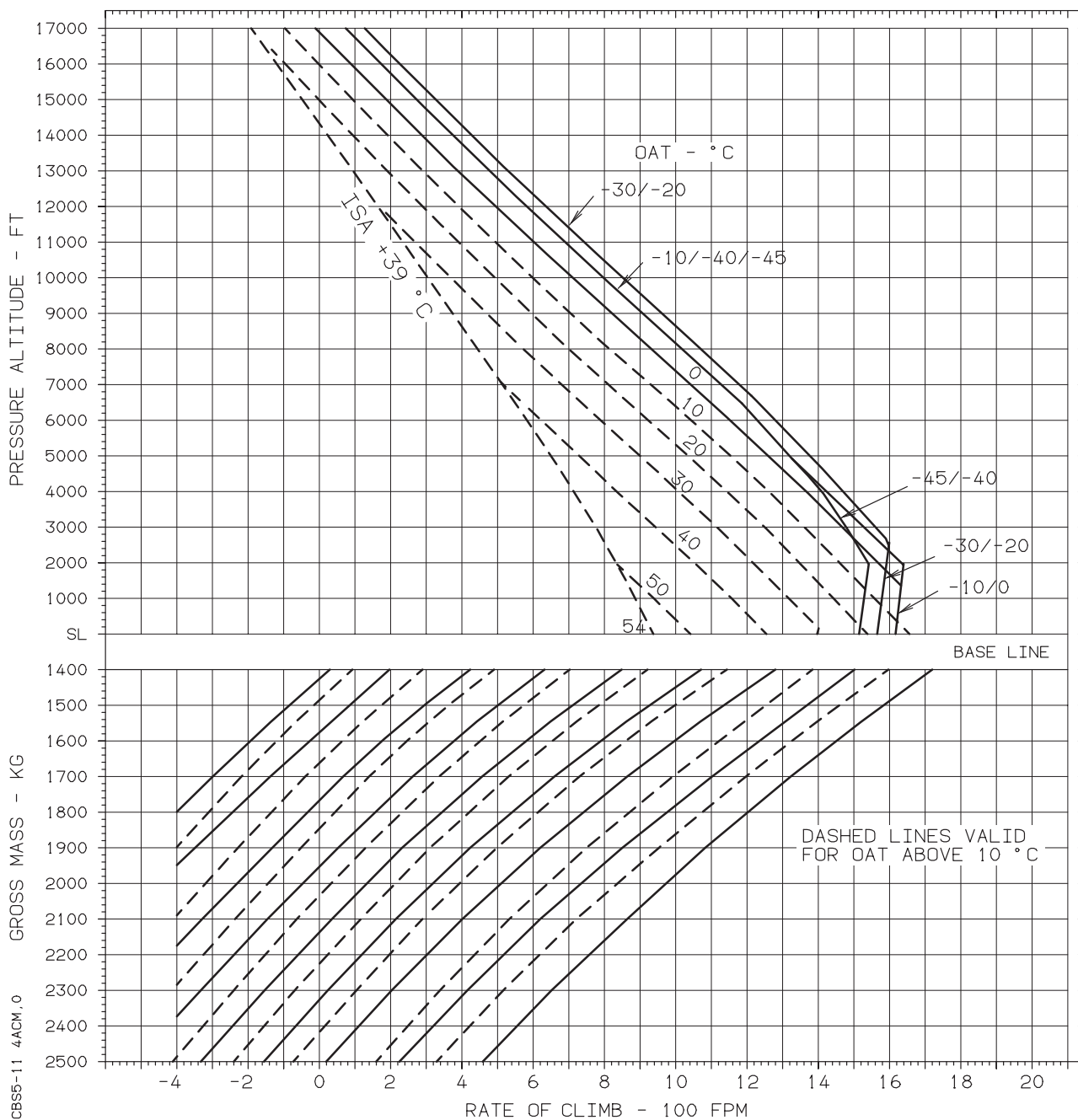


Fig. A2 OEI Rate of Climb - Flight Path Segment II

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OEI RATE OF CLIMB

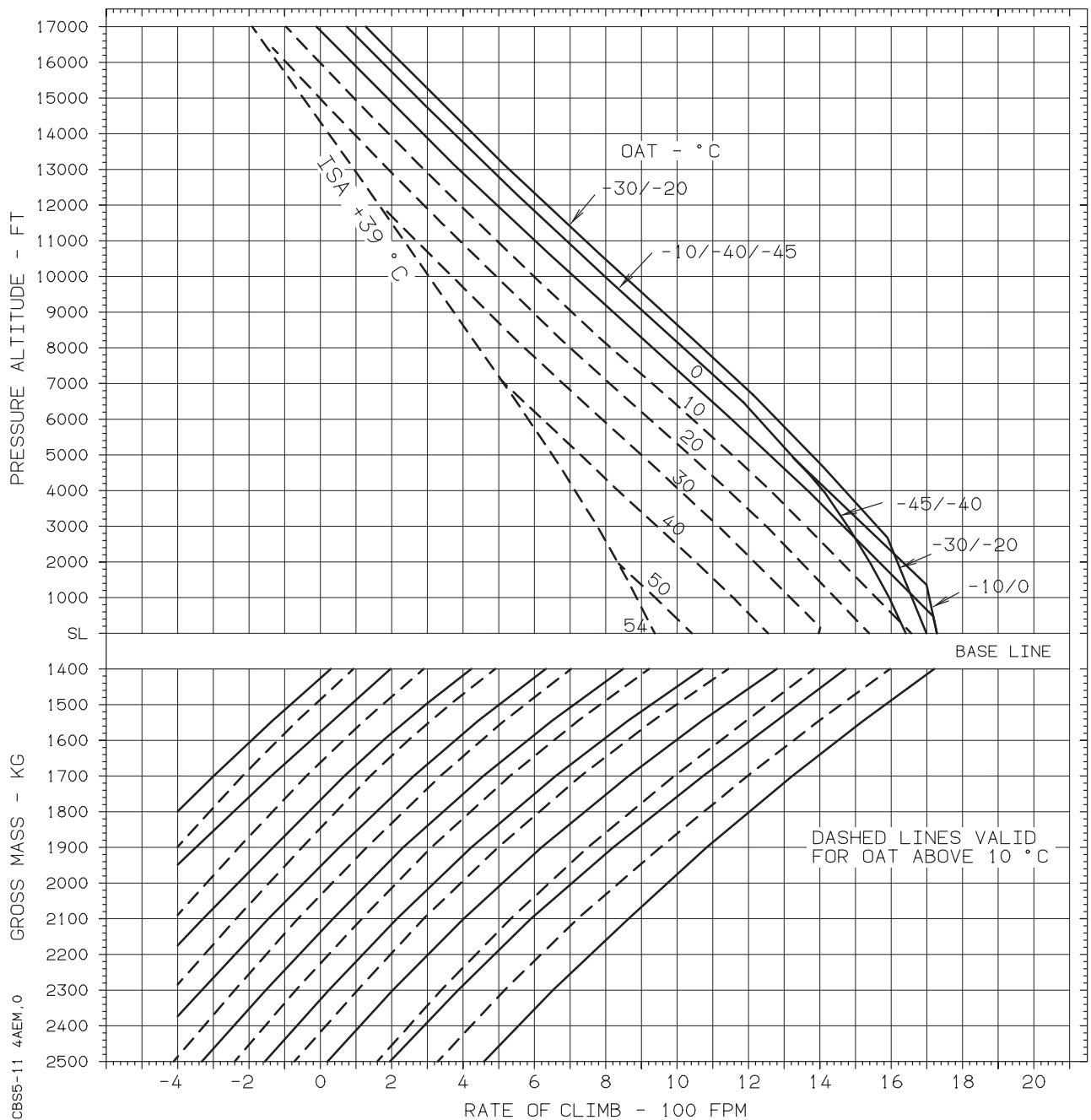
1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

Fig. A3 OEI Rate of Climb - 2.5-min Power, V_Y

OEI HEIGHT GAIN - FLIGHT PATH SEGMENT I

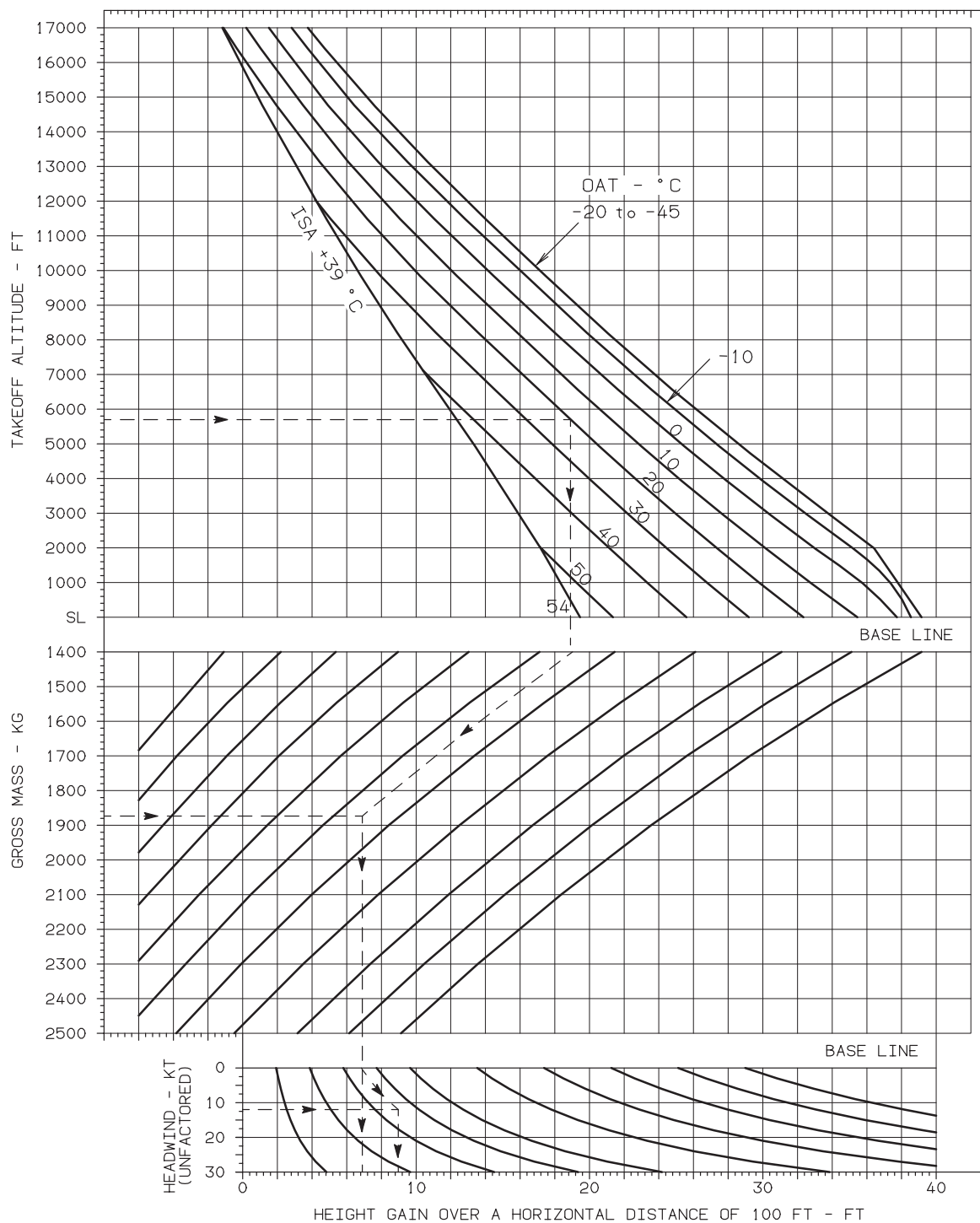
1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE)

ANTI-ICING – OFF

TAKEOFF SAFETY SPEED – 50 KIAS

BLEED AIR CONSUMERS – OFF



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Fig. A4 OEI Height Gain - Flight Path Segment I

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OEI HEIGHT GAIN - FLIGHT PATH SEGMENT II

1 X ALLISON 250-C20B

OEI – MAX CONTINUOUS POWER (810 °C TOT, 105% TORQUE)

ANTI-ICING – OFF

BEST RATE-OF-CLIMB SPEED – 60 KIAS

BLEED AIR CONSUMERS – OFF

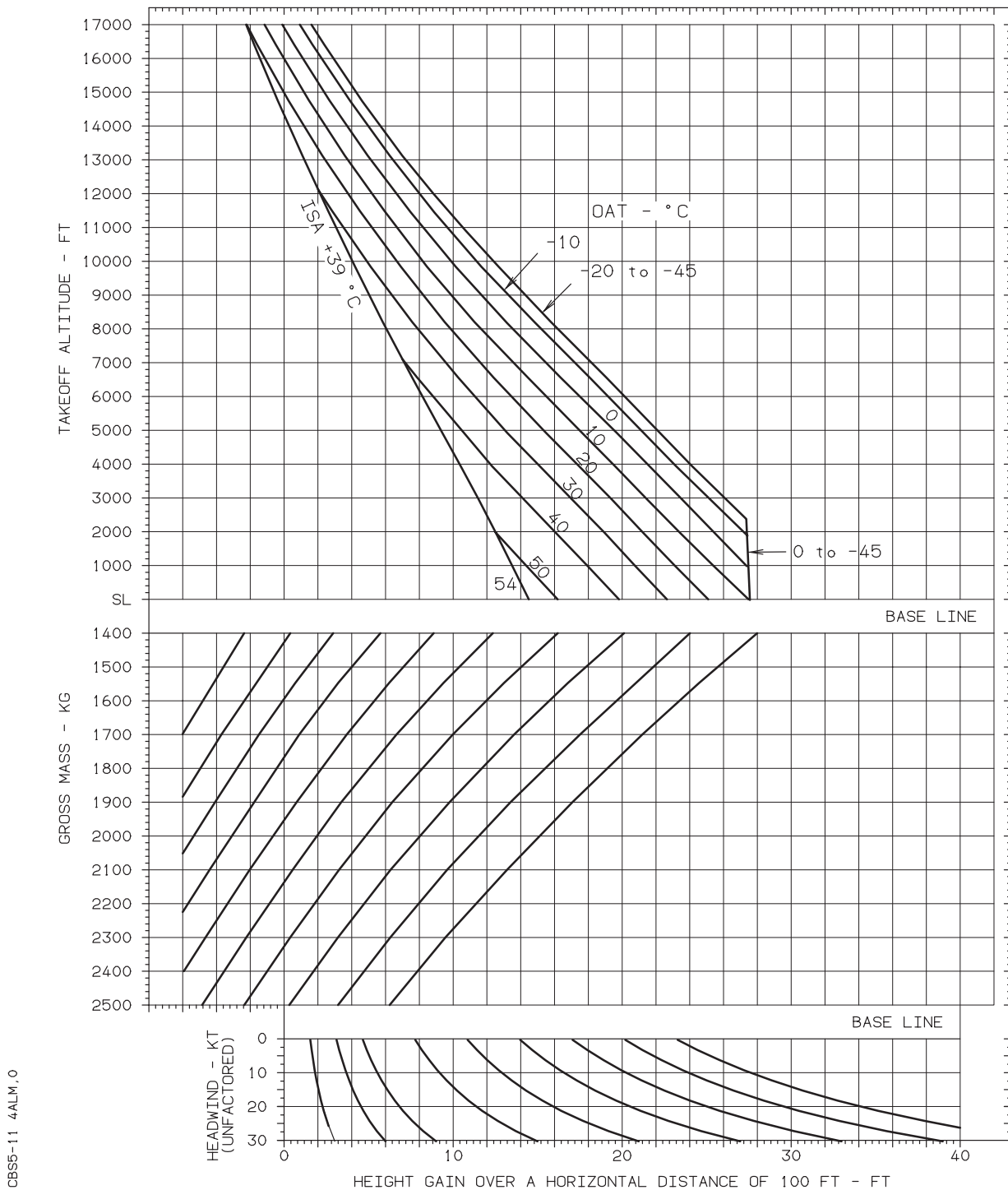


Fig. A5 OEI Height Gain - Flight Path Segment II

A.5.3 CATEGORY B LANDING PERFORMANCE

A.5.3.1 Landing Procedure

No change in the basic Flight Manual data.

A.5.3.2 Landing Distance

The landing distance chart (see Fig. A6) provides single-engine landing performance data for various combinations of pressure altitude and outside air temperature. The chart shows the landing distance required for the helicopter to come to a complete stop on a smooth, hard and dry level surface.

To achieve the values derived from the landing distance chart, the landing approach path should be established at a 50-ft height with OEI power required to maintain a 300-fpm rate of descent at 40 KIAS.

NOTE The minimum landing distance is 620 ft (189 m) under winds calm and all combinations of gross mass, pressure altitude and OAT.

EXAMPLE: (see Fig. A6)

Determine: **OEI landing distance required to clear a 50-ft obstacle, with headwind**

Known:	OAT	25 °C
	Pressure altitude	SL
	Gross mass	2000 kg
	Headwind	12 kt

Solution:

1. Enter chart at known OAT (25 °C).
2. Move vertically up to known pressure altitude (SL).
3. Move horizontally right to known gross mass (2000 kg).
4. Move vertically down to Base Line and trace further down following guide lines.
5. Reenter chart at known headwind (12 kt) and move horizontally right to intersect tracing from above.
6. From point of intersection move vertically down and read landing distance = 485 ft (148 m).

CATEGORY B OEI LANDING DISTANCE (FROM 50-FT HEIGHT TO A COMPLETE STOP ON GROUND)

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C, 110 % TORQUE)

ANTI-ICING – ON OR OFF

BLEED AIR CONSUMERS – OFF

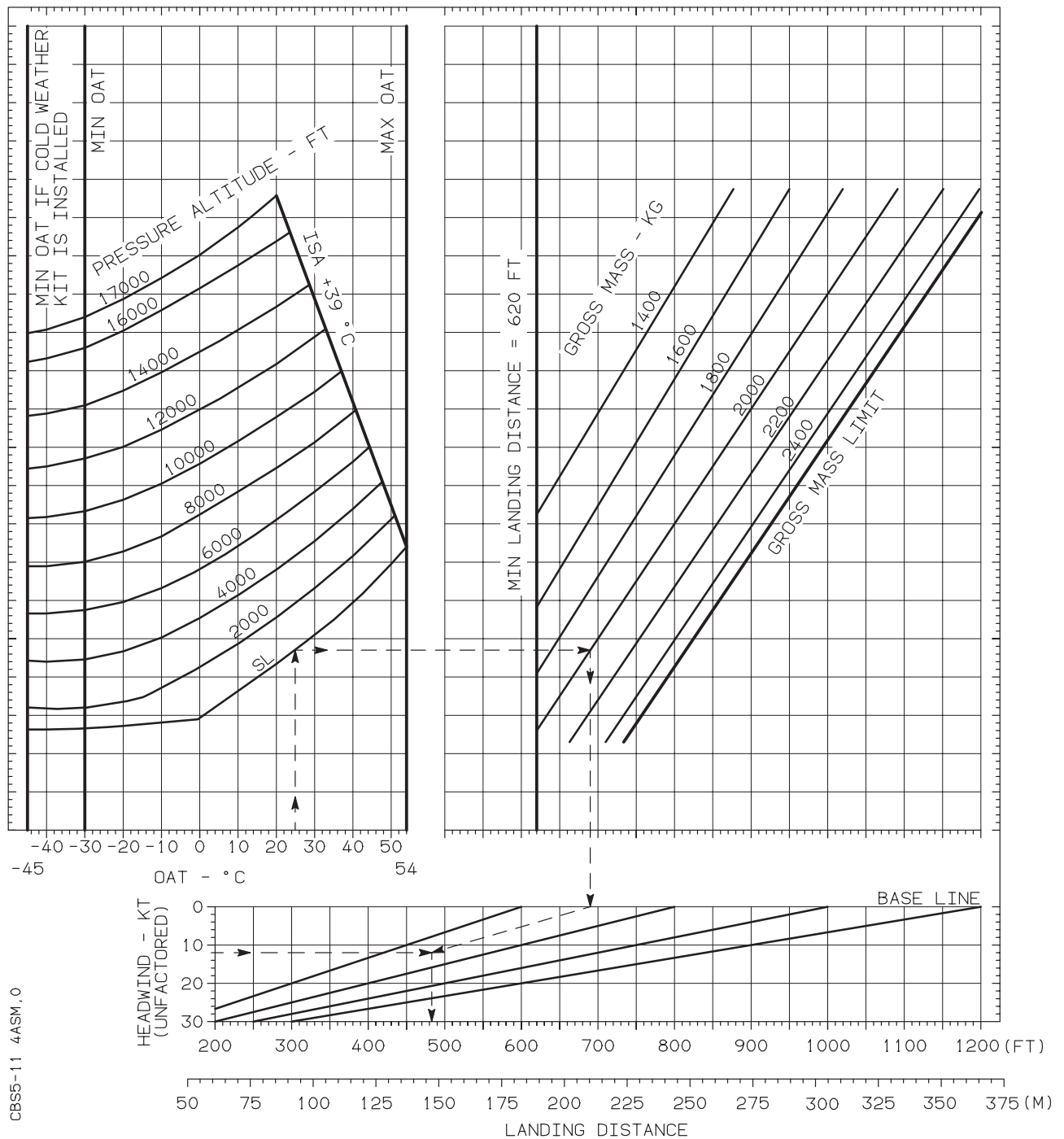


Fig. A6 OEI Landing Distance - Category B

A.6 MASS AND BALANCE

No change in the basic Flight Manual data.

A.7 SYSTEM DESCRIPTION

No change in the basic Flight Manual data.

A.8 HANDLING, SERVICING, AND MAINTENANCE

No change in the basic Flight Manual data.

A.9 OPERATIONAL INFORMATION**A.9.1 TRAINING**

The following is intended for use when performing Special Takeoff and Landing Operations training.

A.9.1.1 Engine and Transmission Power Training Limits

Engines: Allison 250 - C 20 B

Transmission: ZF FS 72E

NOTE The following limits are for OEI training purposes only.

CONDITION	TRANSMISSION (HELICOPTER) LIMITS MAX TORQUE %	ENGINE OPERATING LIMITS		
		MAX TOT °C	MAX N ₁ %	MAX N ₂ %
One Engine Inoperative				
Max Continuous Power	86	800	105	102
2.5-min Power	95	800	105	102
Transient (Max 6 sec above 800 °C)		843		
Transient (Max 16 sec)	110			

A.9.1.2 OEI Training Procedures

NOTE For training purposes, OEI emergency procedures will normally be simulated with both engines operating.

A.9.1.2.1 OEI Simulation with All Engines Operating

To simulate OEI operation using both engines at reduced power, follow the emergency procedures of the respective subsection in conjunction with the corresponding max take-off/landing gross mass.

For determining the training AEO-torque values, use the OEI Simulation 2.5-min power diagram (see Fig. A7). Transient torque limits as required, however, not more than Takeoff Power.

A.9.1.2.2 OEI Simulation with One Engine at IDLE

For determining the training takeoff and landing gross mass in order to simulate OEI conditions during Category A operations with one engine at IDLE, refer to "Training" paragraph in the applicable subsection.

EXAMPLE: (see Fig. A7)

Determine: **2.5-min AEO-torque when simulating OEI takeoffs and landings**

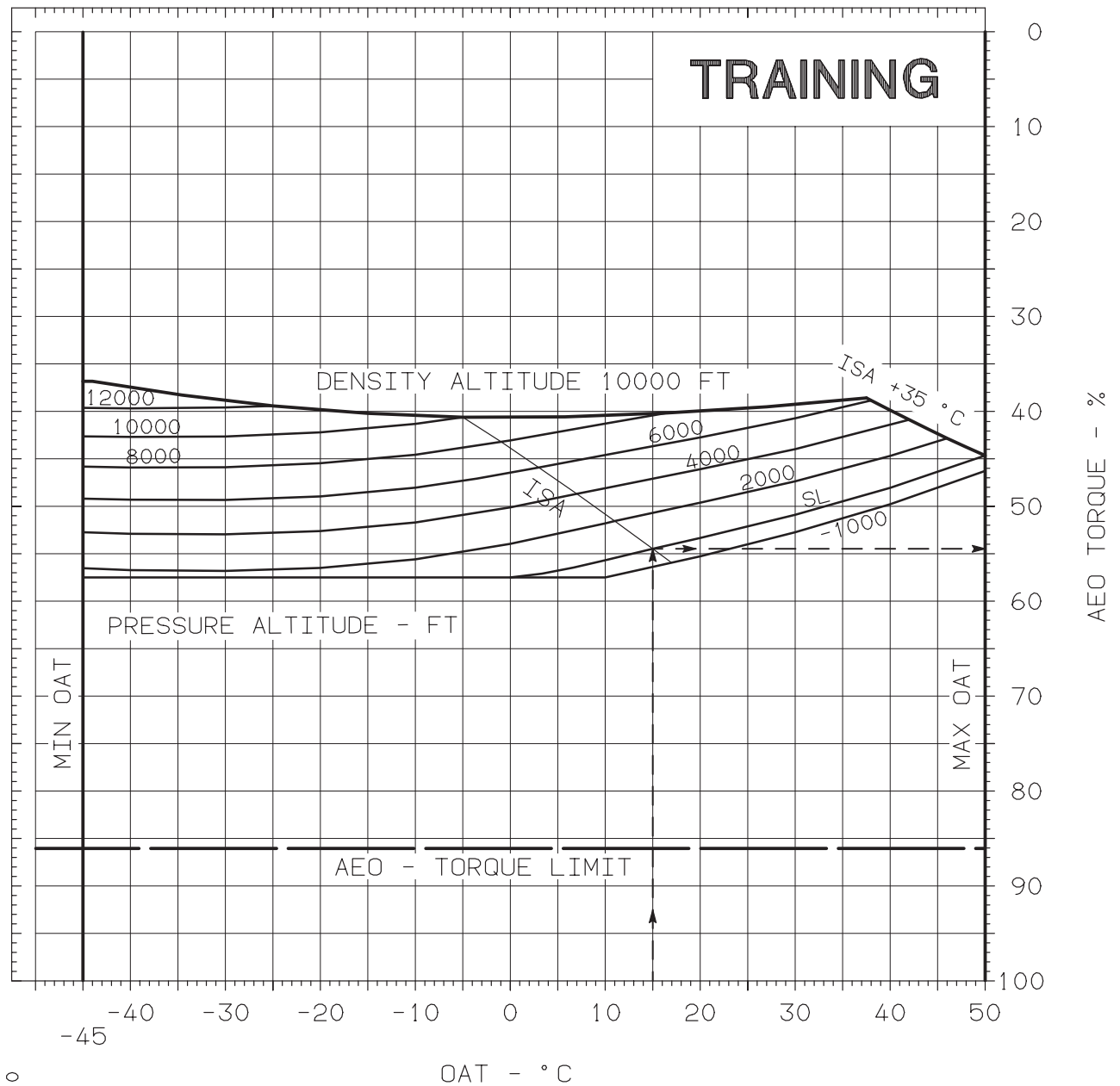
Known: OAT 15 °C
Pressure Altitude SL

Solution: 1. Enter chart at known OAT (15 °C).
2. Move up to known pressure altitude (SL).
3. Move right and read indicated torque = 54.5 %.

NOTE Resultant torque value is the 2.5-min power for simulating OEI conditions when using the applicable subsection Maximum Takeoff or Landing Gross Mass diagram.

OEI-SIMULATION 2.5-MIN POWER

ALLISON 250-C20B



B05-11.4AWM,0

Fig. A7 OEI-simulation 2.5-min Power

A.9.2 OEI OUT-OF-GROUND-EFFECT HOVER CEILING

The OEI Hover Ceiling chart (see Fig. A8) has been established for flight planning purposes only and is to be used in calculating OEI hover gross mass operational performance requirements.

NOTE OEI hover with tailwind component shall be avoided.

EXAMPLE: (see Fig. A8)

Determine: **Max gross mass for OEI hover, wind calm**

Known:	OAT	20 °C
	Pressure altitude	1000
	Wind	Calm

Solution:

1. Enter chart at known OAT (20 °C).
2. Move vertically up to known pressure altitude (1000).
3. Move horizontally right and read gross mass = 1675 kg.

Determine: **Max gross mass for OEI hover, with headwind**

Known:	OAT	20 °C
	Pressure altitude	1000
	Headwind	10 kt

Solution:

1. Enter chart at known OAT (20 °C).
2. Move vertically up to known pressure altitude (1000).
3. Move horizontally right to Base Line and trace further down following parallel to guide lines.
4. Reenter chart at known headwind (10 kt) and move vertically up to intersect tracing from above.
5. From point of intersection move horizontally right and read gross mass = 1750 kg.

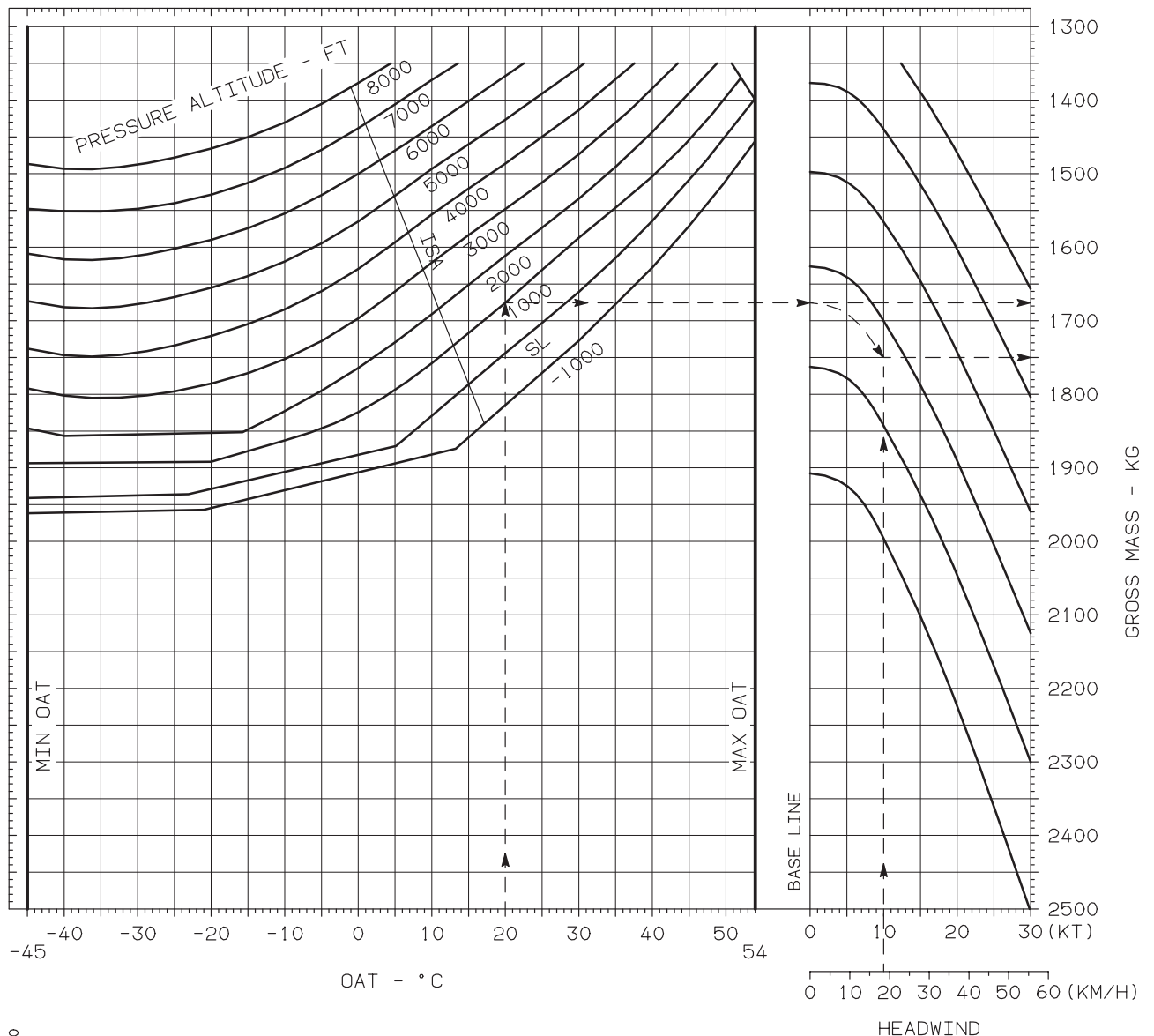
OEI HOVER CEILING OUT OF GROUND EFFECT

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE)

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF



NOTE WIND SPEEDS ARE UNFACTORED.
APPLY FACTOR AS REQUIRED BY
OPERATIONAL RULES.

Fig. A8 OEI Hover Ceiling Out of Ground Effect

B. CLEAR HELIPORT

B.1 GENERAL

This subsection provides information necessary for Category A operations at clear heliports.

B.2 LIMITATIONS

B.2.1 OPERATIONAL LIMITS

B.2.1.1 Bleed Air Consumers

Bleed air consumers shall be off during takeoff and landing.

B.2.1.2 Altitude/Ambient Air Temperature Limitations

The altitude and OAT limitations are established for a reduced envelope:

- Max altitude 6000 ft
- Max OAT ISA +25 °C, but not more than 40 °C
- Min OAT –30 °C

B.2.1.3 Decision Point Height Determination

The TDP height and the LDP height shall be measured using a radar altimeter.

B.2.2 MASS LIMITS

For maximum takeoff and landing gross mass refer to Fig. B1 and B2.

EXAMPLE: (see Fig. B1)

Determine: **Maximum takeoff/landing gross mass**

Known: OAT 22 °C
Pressure altitude 3000 ft

- Solution:
1. Enter chart at known OAT (22 °C).
 2. Move vertically up to known pressure altitude (3000 ft).
 3. Move horizontally left and read max gross mass = 2220 kg.

Determine: **Max takeoff/landing gross mass with external mounted optional equipment installed, e.g. External Loudspeakers FMS 10-23**

Known: OAT 10 °C
Pressure altitude 1000 ft

- Solution:
1. Enter chart at known OAT (10 °C).
 2. Move vertically up to known pressure altitude (1000 ft).
 3. Move horizontally left and read gross mass = 2545 kg.
 4. Subtract 60 kg (FMS 10-23) from the chart-derived gross mass (2545 kg) to obtain the maximum gross mass with the skid-mounted loudspeakers installed = 2485 kg.

MAXIMUM TAKEOFF AND LANDING GROSS MASS

1 X ALLISON 250-C20B

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

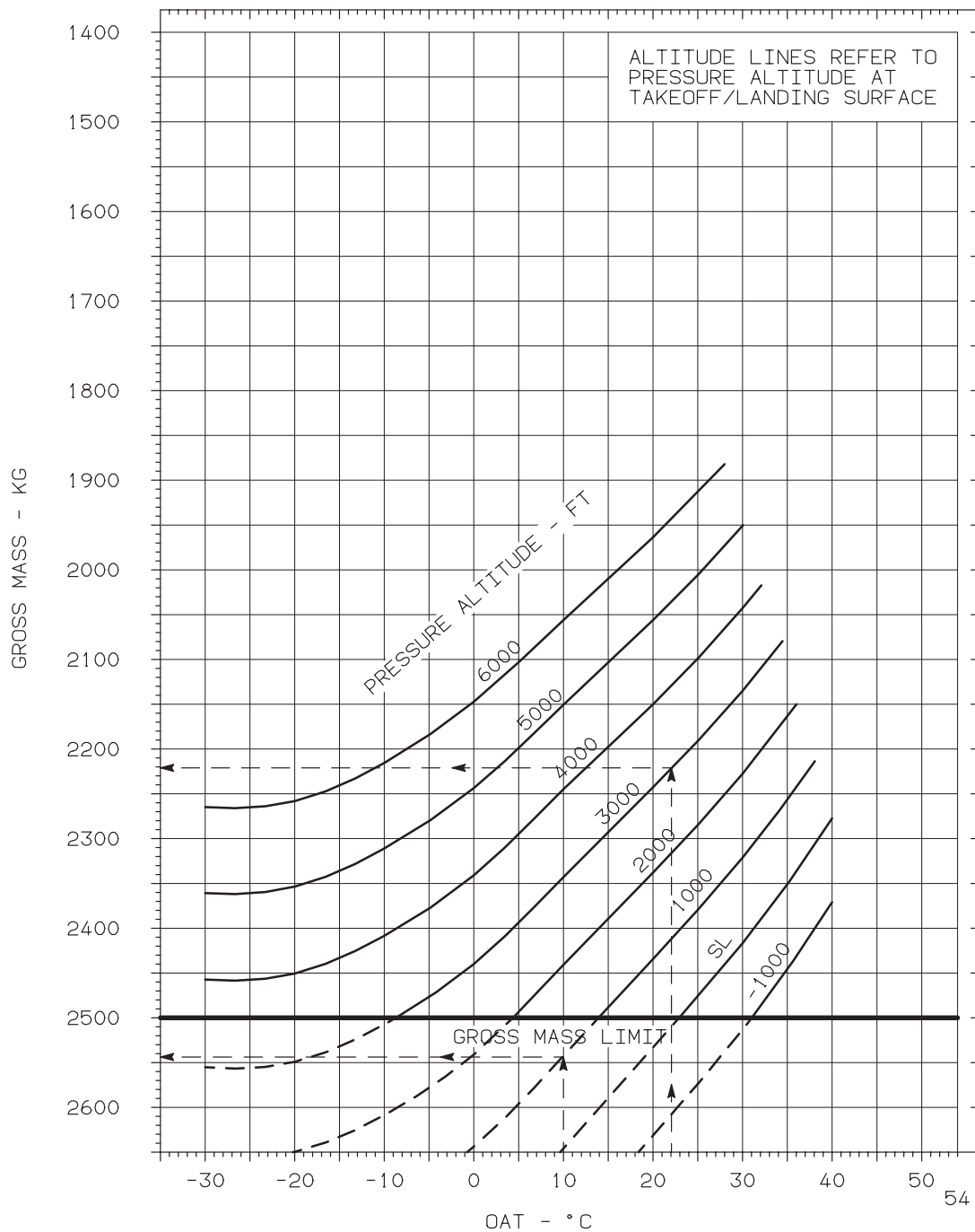


Fig. B1 Max Takeoff and Landing Gross Mass - Anti-Icing OFF

MAXIMUM TAKEOFF AND LANDING GROSS MASS

1 X ALLISON 250-C20B

ANTI-ICING – ON

BLEED AIR CONSUMERS – OFF

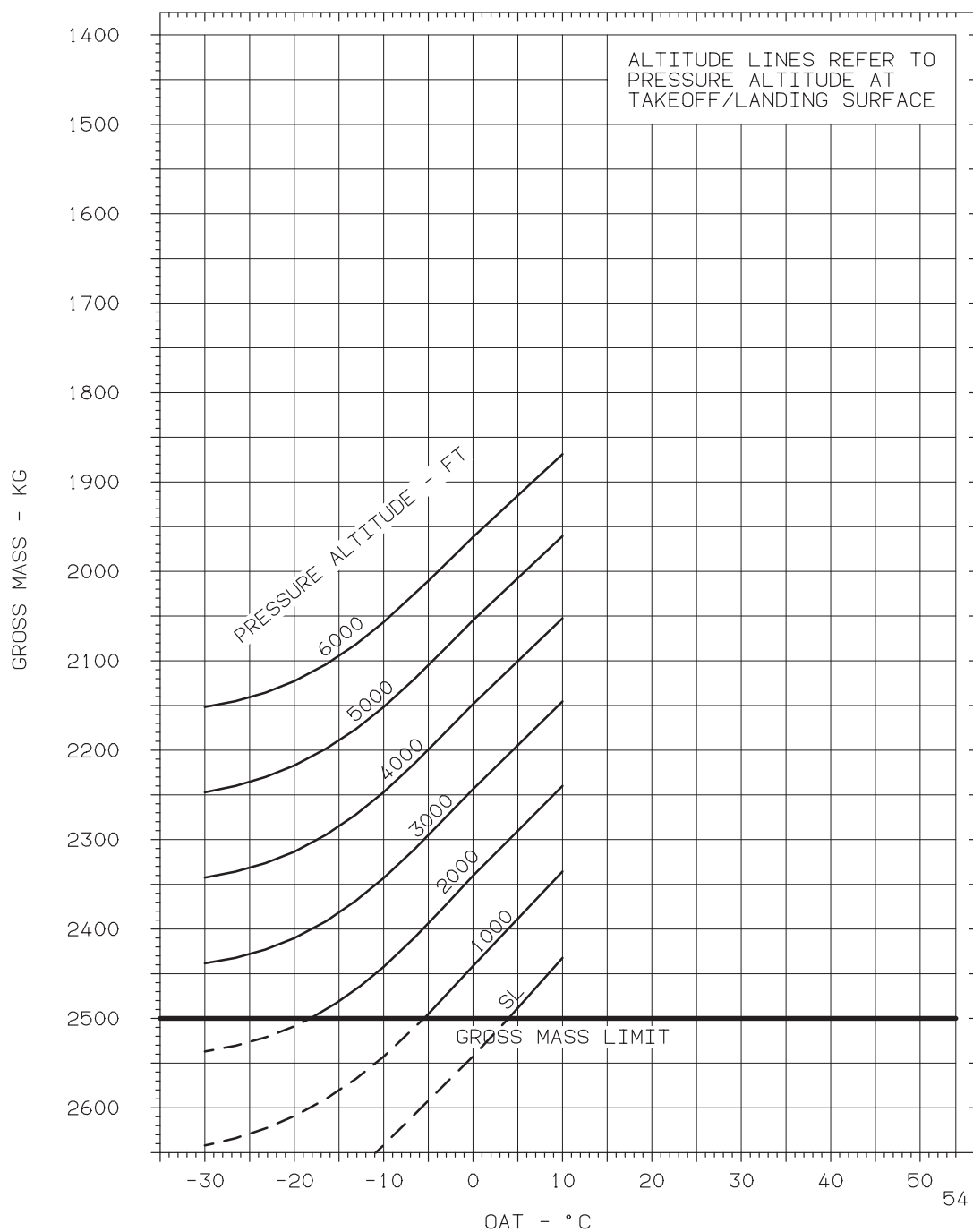


Fig. B2 Max Takeoff and Landing Gross Mass - Anti-Icing ON

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B.3 EMERGENCY AND MALFUNCTION PROCEDURES

B.3.1 OEI DURING TAKEOFF PRIOR TO TDP (see Fig. B3)

Procedure

- | | |
|---------------------|---------------------------------|
| 1. Attitude | – Adjust as required |
| 2. Collective lever | – Adjust to OEI limits or below |
| 3. Ground speed | – Reduce (near-to-ground flare) |

Prior to touchdown:

- | | |
|---------------------|---|
| 4. Landing attitude | – Assume |
| 5. Collective lever | – Raise to cushion landing (observe OEI limits) |

After touchdown:

- | | |
|---------------------|--------------------|
| 6. Collective lever | – Lower slowly |
| 7. Cyclic stick | – Neutral position |

NOTE Observe mast moment limits.

- | | |
|-------------------------------------|-----------|
| 8. Single engine emergency shutdown | – Perform |
|-------------------------------------|-----------|

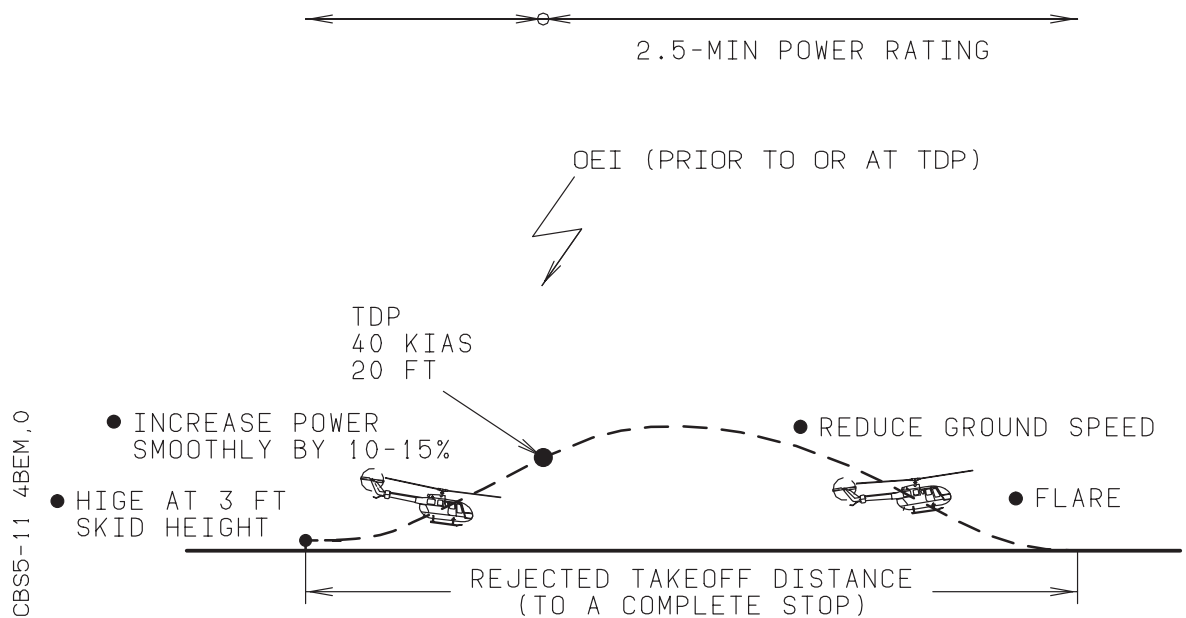


Fig. B3 OEI Rejected Takeoff Profile

B.3.2 OEI DURING TAKEOFF AFTER TDP (see Fig. B4)

Procedure

- | | |
|---------------------|----------------------------|
| 1. Collective lever | – Adjust to 2.5-min Power |
| 2. Airspeed | – Accelerate to V_{TOSS} |
| 3. Climb | – Initiate with V_{TOSS} |

When 200 ft AGL is reached:

- | | |
|-------------------------------------|--------------------------------------|
| 4. Airspeed | – Accelerate to V_Y |
| 5. Collective lever | – Adjust to Max Continuous Power |
| 6. Climb | – Continue with V_Y to 1000 ft AGL |
| 7. Single engine emergency shutdown | – Perform |
| 8. Bleed air consumers | – As required |
| 9. LAND AS SOON AS POSSIBLE | |

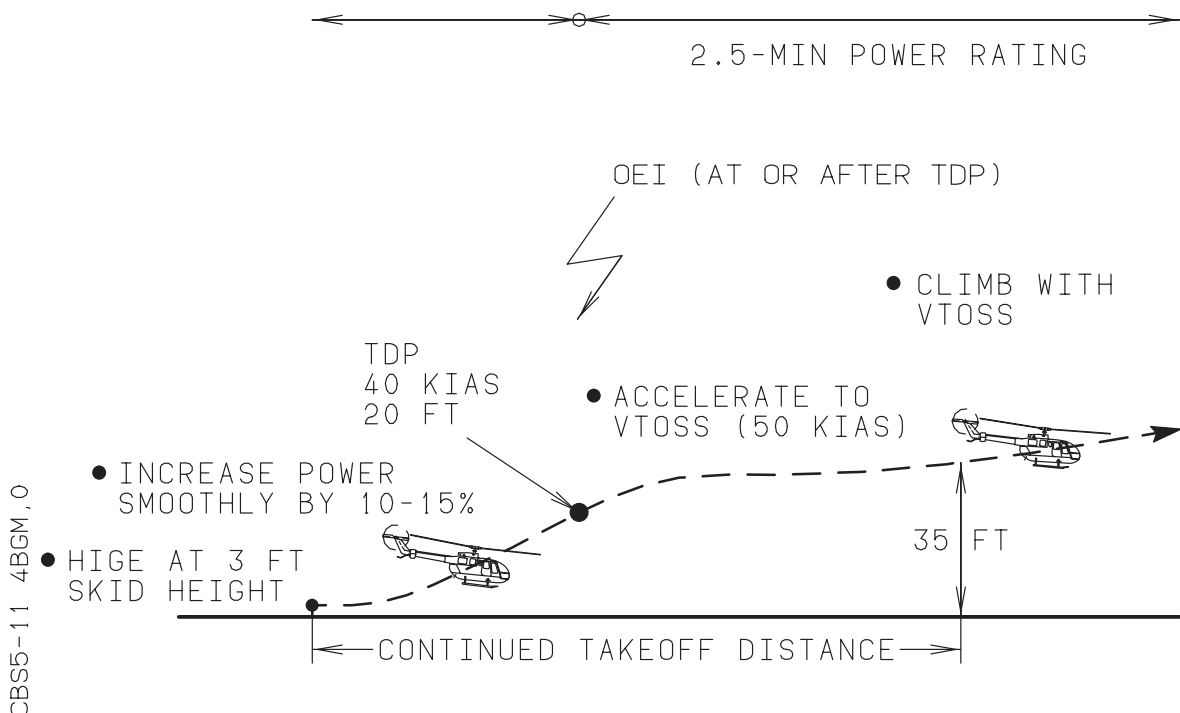


Fig. B4 OEI Continued Takeoff Profile

B.3.3 OEI DURING LANDING PRIOR TO LDP (see Fig. B5)

NOTE The pilot may elect to go-around or continue the approach. If the decision is to continue the approach, use procedures for **OEI DURING LANDING AFTER LDP**.

Procedure

- | | |
|---------------------|----------------------------|
| 1. Collective lever | – Adjust to 2.5-min Power |
| 2. Airspeed | – Accelerate to V_{TOSS} |

NOTE If airspeed is 60 KIAS or above use V_Y .

When 200 ft AGL is reached:

- | | |
|-------------------------------------|--------------------------------------|
| 3. Airspeed | – Accelerate to V_Y |
| 4. Collective lever | – Adjust to Max Continuous Power |
| 5. Climb | – Continue with V_Y to 1000 ft AGL |
| 6. Single engine emergency shutdown | – Perform |
| 7. Bleed air consumers | – As required |
| 8. LAND AS SOON AS POSSIBLE | |

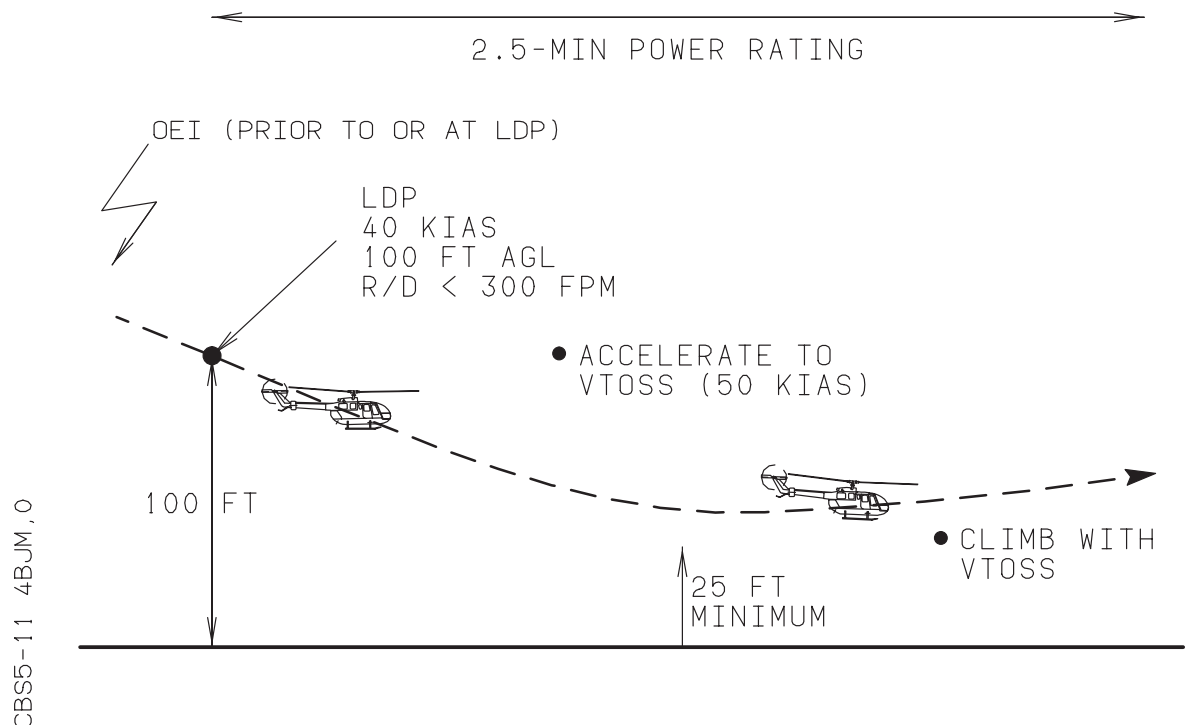


Fig. B5 OEI Go-around Profile

B.3.4 OEI DURING LANDING AFTER LDP (see Fig. B6)

Procedure

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI limits or below |
| 2. Ground speed | – Reduce (near-to-ground flare) |

Prior to touchdown:

- | | |
|---------------------|---|
| 3. Landing attitude | – Assume |
| 4. Collective lever | – Raise to cushion landing (observe OEI limits) |

After touchdown:

- | | |
|---------------------|--------------------|
| 5. Collective lever | – Lower slowly |
| 6. Cyclic stick | – Neutral position |

NOTE Observe mast moment limits

- | | |
|-------------------------------------|-----------|
| 7. Single engine emergency shutdown | – Perform |
|-------------------------------------|-----------|

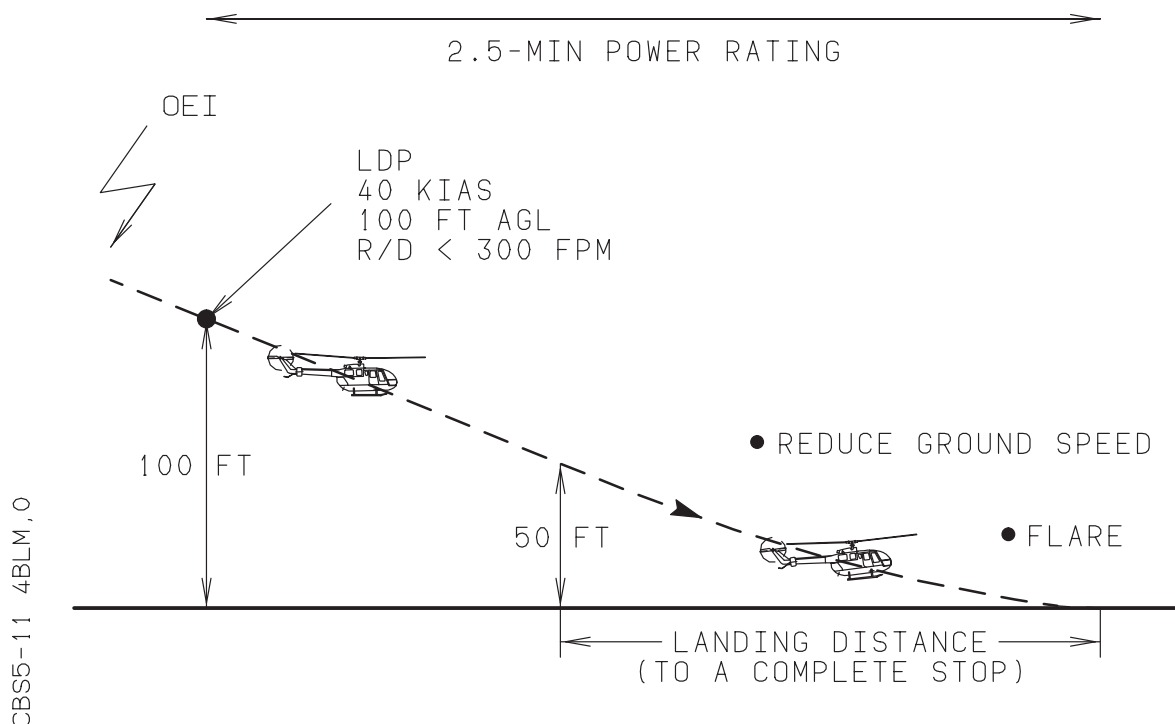


Fig. B6 OEI Continued Landing Profile

B.4 NORMAL PROCEDURES

B.4.1 TAKEOFF (see Fig. B7)

- | | |
|--|--|
| Pre-takeoff check | – Perform |
| Altimeter | – Set |
| Bleed air consumers other than engine anti-icing | – Off |
| Hover | – 3-ft skid height |
| Rotor RPM | – 100 % set with matched torque |
| All instruments | – Normal operating ranges |
| Acceleration and climb | – Start nose-down pitch rotation and simultaneously increase power smoothly by approx 10-15 % to accelerate in a shallow climb. Adjust pitch attitude at about 25 KIAS to achieve 40 KIAS at TDP height. |

When TDP height is reached:

- | | |
|------------------|----------------------------------|
| Collective lever | – Adjust to Takeoff Power |
| Airspeed | – Accelerate to V_Y |
| Collective lever | – Adjust to Max Continuous Power |
| Climbout | – Continue to desired altitude |

After reaching 1000 ft AGL:

- | | |
|---------------------|---------------|
| Bleed air consumers | – As required |
|---------------------|---------------|

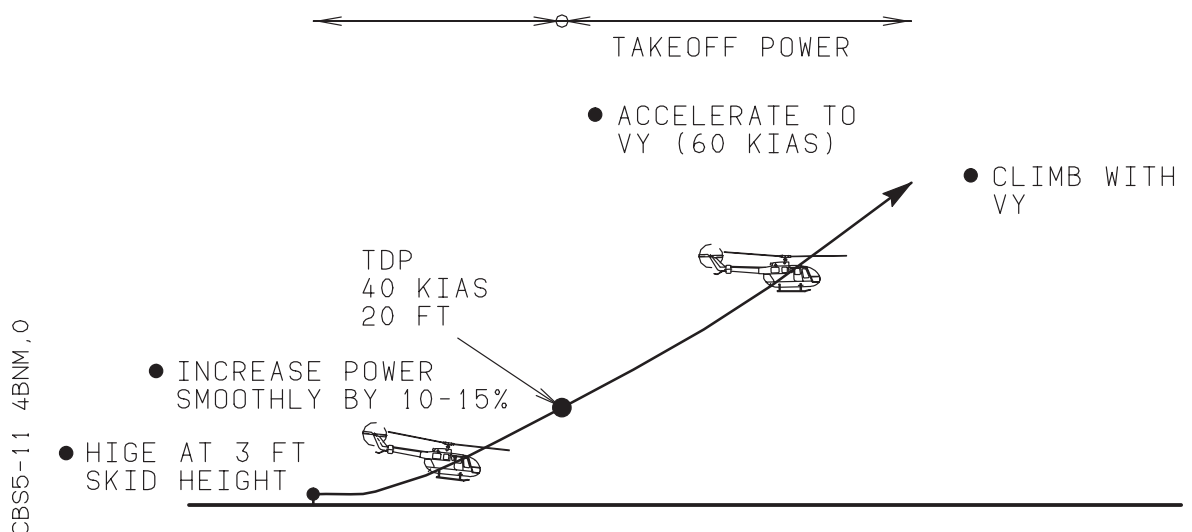


Fig. B7 Takeoff Profile

B.4.2 LANDING

- | | |
|-------------------|--|
| Pre-landing check | – Perform |
| Landing approach | – Initiate to arrive at LDP with a speed of 40 KIAS and a rate of descent of not more than 300 fpm |

NOTE Approach airspeed can be increased by half of the wind speed.

After reaching LDP:

- | | |
|------------------|--|
| Collective lever | – Adjust to maintain desired rate of descent, and terminate approach |
|------------------|--|

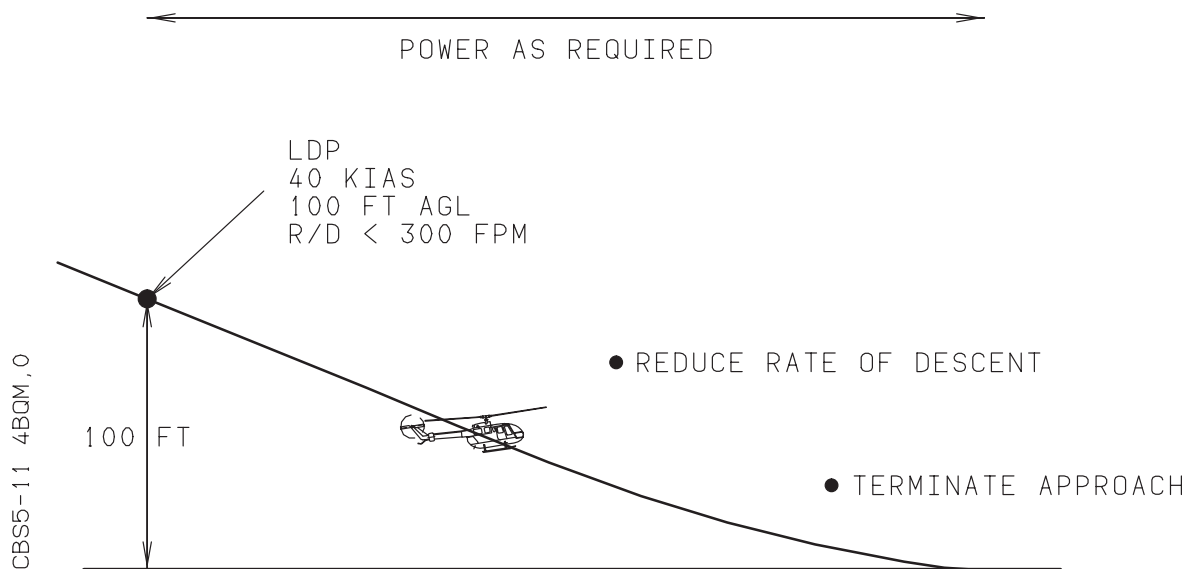


Fig. B8 Landing Profile

B.5 PERFORMANCE DATA

B.5.1 TAKEOFF PERFORMANCE

NOTE The performance data presented herein apply to climbs without any significant bank angle. Turning during climbing will reduce climb performance.

B.5.1.1 Maximum Takeoff Gross Mass (see Fig. B1 and B2)

The takeoff performance is determined and limited by gross mass/altitude/OAT limits according to Fig. B1 or B2. The gross mass is limited by takeoff segment II (climb with V_Y).

B.5.1.2 Assured Minimum Flight Conditions

Flight Condition	Power	Speed	Rate of Climb	Altitude (AGL)
HIGE	TOP (AEO)	0	0	3 ft skid height
CLIMB	2.5-min (OEI)	V_{TOSS}	100 fpm	200 ft
CLIMB	MCP (OEI)	V_Y	150 fpm	1000 ft

NOTE The rejected takeoff distance determines the length of the required takeoff surface.

B.5.1.3 Rejected Takeoff Distance

The rejected takeoff distance (to a complete stop on the ground)
under zero wind conditions 290m (950ft)

NOTE Above distance is valid for engine anti-icing ON or OFF and for all combinations of gross mass, OAT and altitude.

To calculate the rejected takeoff distance (to a complete stop on the ground)
with headwind (see Fig. B9)

EXAMPLE: (see Fig. B9)

Determine: **Rejected takeoff distance with headwind**

Known: Headwind 18 kt

- Solution:
1. Enter chart at known headwind (18 kt).
 2. Move horizontally right to intersect the diagonal line.
 3. Move vertically down and read rejected takeoff distance 160m (520ft)

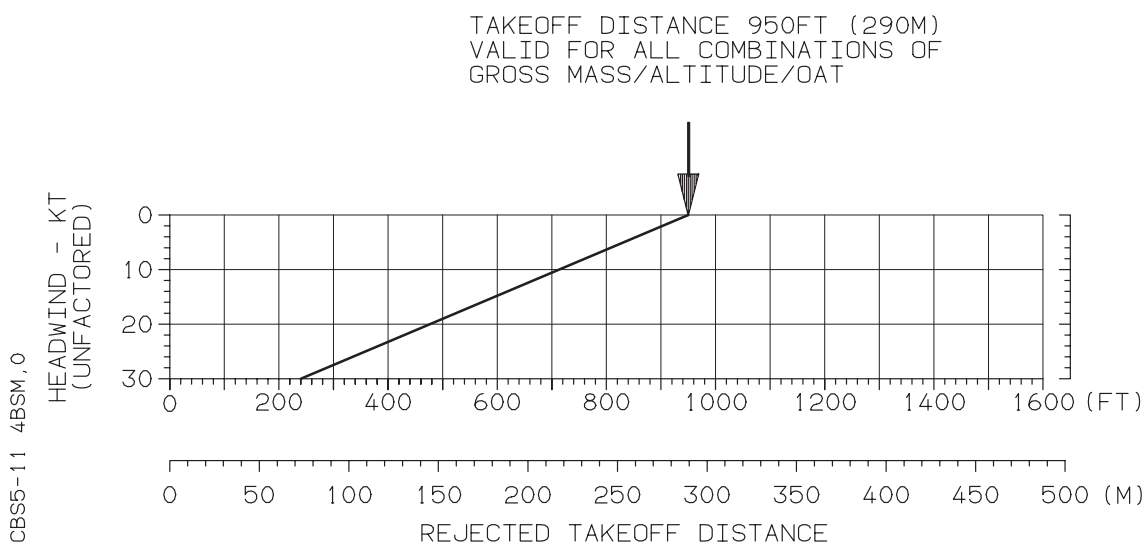


Fig. B9 Rejected Takeoff Distance with Headwind

B.5.1.4 Continued Takeoff Distance

The continued takeoff distance over a 35-ft obstacle
under zero wind conditions 230 m (750 ft)

NOTE Above distance is valid for engine anti-icing ON or OFF and for all combinations of gross mass, OAT and altitude.

The continued takeoff distance over a 35-ft obstacle
with headwind (see Fig. B10)

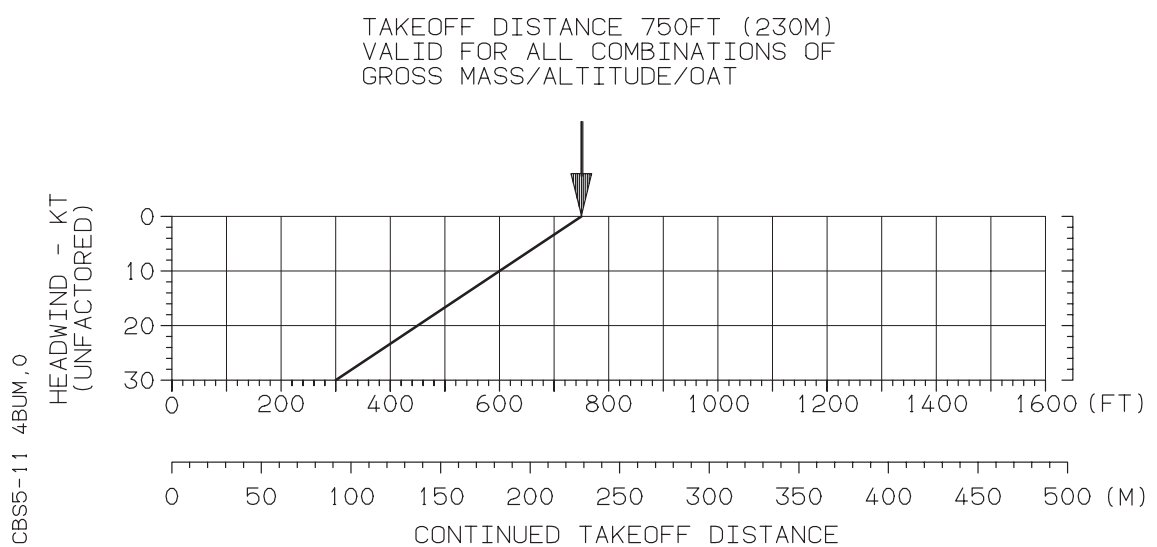


Fig. B10 Continued Takeoff Distance with Headwind

B.5.1.5 OEI Takeoff Flight Path (see Fig. B11)

The Standard Procedure OEI takeoff flight path begins at the end of the "Continued Takeoff Distance" at 35 feet above the takeoff surface, with a positive rate of climb at V_{TOSS} , and is divided into two segments:

Segment I OEI climb through segment I has to be accomplished with V_{TOSS} and 2.5-min Power until reaching 200 ft AGL. Segment I climb performance data (OEI Height Gain Over a Horizontal Distance of 100 ft) are presented in Subsection A, Fig. A4.

Segment II OEI climb through segment II has to be accomplished with V_Y and Max Continuous Power until reaching 1000 ft AGL. Segment II climb performance data (OEI Height Gain Over a Horizontal Distance of 100 ft) are presented in Subsection A, Fig. A5.

The horizontal distance required for acceleration from V_{TOSS} to V_Y is 600 m (1980 ft) under calm wind conditions.

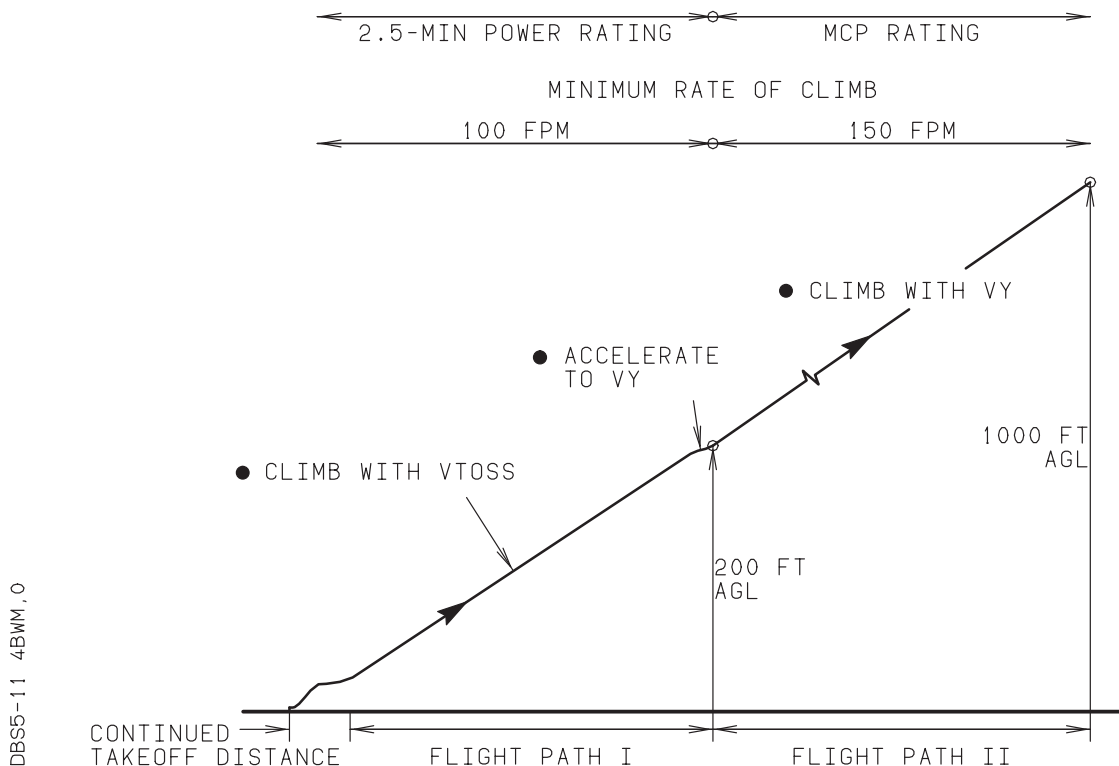


Fig. B11 OEI Takeoff Profile

B.5.2 LANDING PERFORMANCE

B.5.2.1 Landing Distance from a 50-ft Height to a Complete Stop on the Ground

Refer to Para A.5.3.2, Landing Distance.

B.5.2.2 Maximum Landing Gross Mass

For maximum landing gross mass refer to Fig. B1 and B2.

B.6 MASS AND BALANCE

No change in the basic Flight Manual data.

B.7 SYSTEM DESCRIPTION

No change in the basic Flight Manual data.

B.8 HANDLING, SERVICING, AND MAINTENANCE

No change in the basic Flight Manual data.

B.9 OPERATIONAL INFORMATION**B.9.1 TRAINING****B.9.1.1 OEI Simulation with One Engine at IDLE**

To simulate OEI operation with one engine at IDLE, follow the emergency procedures this supplement subsection, however, use the GENERAL DATA, Engine and Transmission Power Training Limitations (see Subsection A) and the gross mass value from the TRAINING MAT curve this subsection (see Fig. B12).

EXAMPLE: (see Fig. B12)

Determine: **Training takeoff and landing gross mass**

Known:	OAT	−7 °C
	Pressure Altitude	6000 ft

Solution: 1. Enter chart at known OAT (−7 °C).
2. Move vertically up to known pressure altitude (6000 ft).
3. Move horizontally left and read gross mass = 2175 kg.

TRAINING TAKEOFF AND LANDING GROSS MASS

1 X ALLISON 250-C20B

OEI – TRAINING RATINGS

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

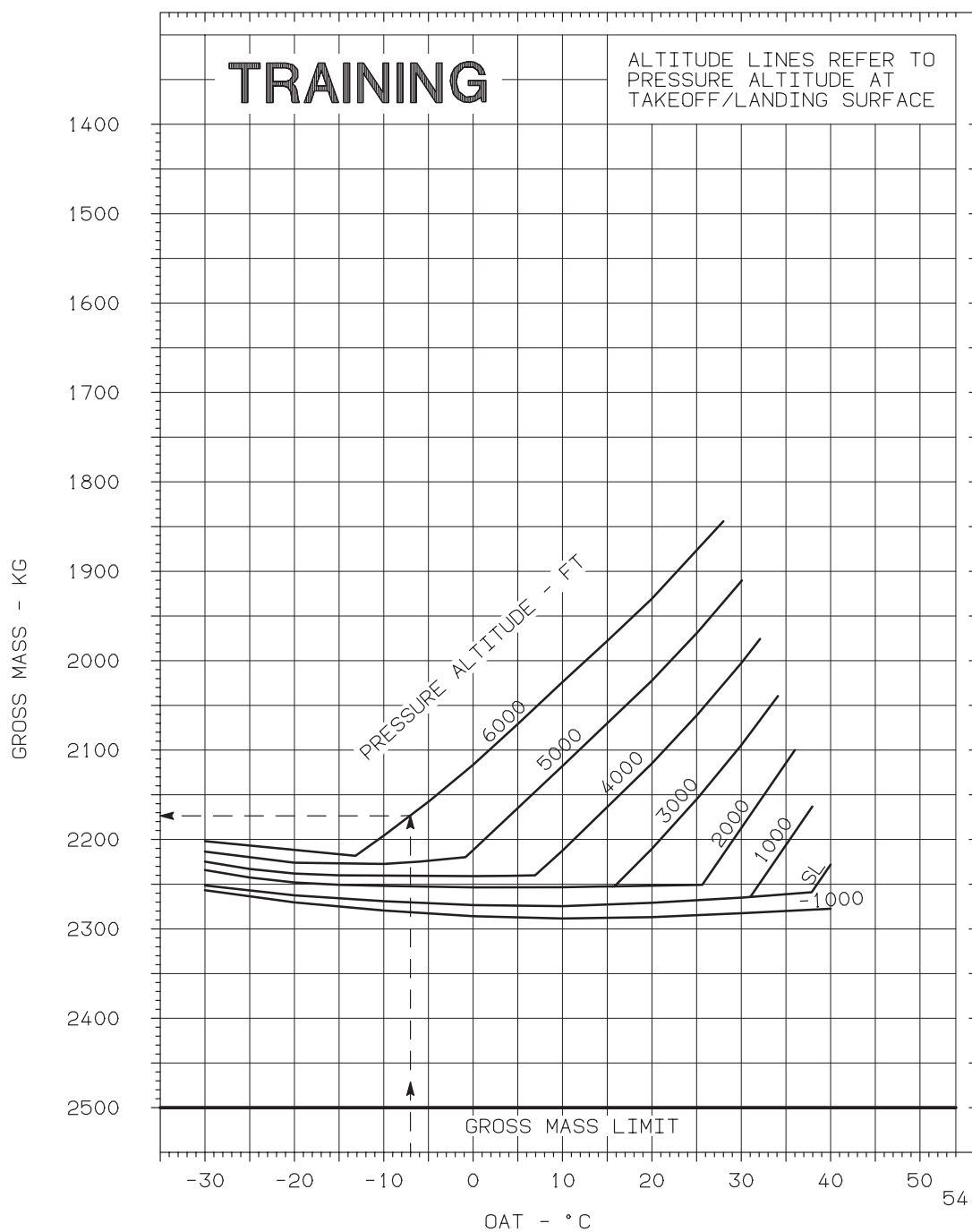
NOTE WITH ENGINE ANTI-ICING ON, THE MAX GROSS MASS DECREASES BY 200 KG.

Fig. B12 Training Takeoff and Landing Gross Mass

MANUFACTURER'S DATA

C. ELEVATED OR RESTRICTED HELIPADS (VTOL)

C.1 GENERAL

This subsection provides standardized vertical takeoff and landing (VTOL) procedures for Category A operations at elevated or restricted helipads.

C.2 LIMITATIONS

C.2.1 CERTIFICATION CRITERIA

The emergency and normal procedures apply to:

- restricted helipads having dimensions of at least 15 x 15 meters under day and night conditions, and
- elevated helipads under day and night conditions. The minimum helipad size is defined as 20 x 20 meters for rectangular helipads or 20-meter diameter for circular helipads.

NOTE Operation on rectangular elevated helipads smaller than 20 x 20 meters but not less than 15 x 15 meters are possible, however, require individual operational approval and enhanced attention during takeoff and landing.

C.2.2 OPERATIONAL LIMITATIONS

C.2.2.1 Helipad Surface

Surface shall be solid to generate ground effect.

C.2.2.2 Wind Limits

Takeoff and landing with tailwind component is prohibited.

Max sidewind component (unfactored) 17 KTAS

C.2.2.3 Bleed Air Consumers

Engine anti-icing and other bleed air consumers must be off during takeoff and landing.

C.2.2.4 Icing Prevention Measures

Category A operations at elevated or restricted helipads are prohibited whenever the ambient temperature is below +4 °C and visible moisture is present.

C.2.2.5 Altitude/Ambient Air Limitations

The altitude and OAT limitations for vertical takeoff and landing procedures are defined as:

- Max Density Altitude 10,000 ft
- Max OAT ISA +35 °C, but not more than 50 °C
- Min OAT –45 °C

C.2.3 MASS LIMITS

For maximum takeoff and landing gross mass refer to Fig. C1.

EXAMPLE: (see Fig. C1)

Determine: **Max takeoff and landing gross mass, wind calm**

Known:	OAT	15 °C
	Pressure altitude	SL
	Wind	Calm

- Solution:
1. Enter chart at known OAT (15 °C).
 2. Move vertically up to known pressure altitude (SL).
 3. Move horizontally right and read gross mass = 2262 kg.

Determine: **Max takeoff and landing gross mass, with headwind**

Known:	OAT	15 °C
	Pressure altitude	SL
	Headwind	10 kt

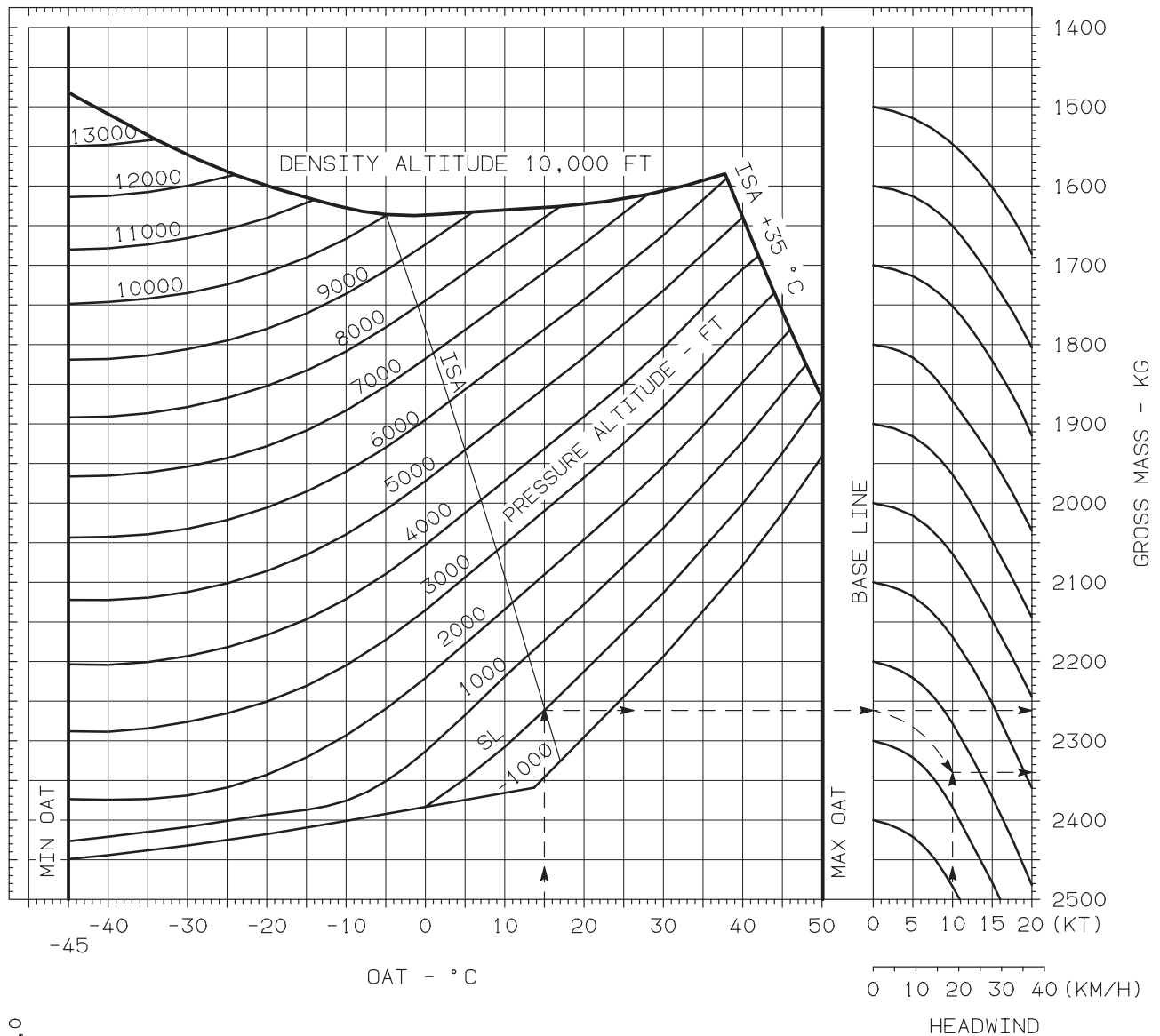
- Solution:
1. Enter chart at known OAT (15 °C).
 2. Move vertically up to known pressure altitude (SL).
 3. Move horizontally right to Base Line and trace further right following curve of guide lines.
 4. Reenter chart at known headwind (10 kt) and move vertically up to intersect tracing from above.
 5. From point of intersection move horizontally right and read gross mass = 2340 kg.

MAXIMUM TAKEOFF AND LANDING GROSS MASS

1 X ALLISON 250-C20B

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF



NOTE WIND SPEEDS ARE UNFACTORED.
APPLY FACTOR AS REQUIRED BY
OPERATIONAL RULES.

Fig. C1 Maximum Takeoff and Landing Gross Mass

C.3 EMERGENCY AND MALFUNCTION PROCEDURES

C.3.1 OEI DURING TAKEOFF PRIOR TO TDP (see Fig. C2)

Procedure

- | | |
|---------------------|--|
| 1. Attitude | – Nosedown -3° to -4° ; maintain starting point in sight |
| 2. Collective lever | – Adjust to 2.5-min Power |

Before touchdown:

- | | |
|---------------------|--------------------------------------|
| 3. Landing attitude | – Establish for minimum ground speed |
| 4. Collective lever | – Increase to cushion landing |

NOTE Plan for zero ground speed touchdown.

After touchdown:

- | | |
|-------------------------------------|--|
| 5. Collective lever | – Lower to full down |
| 6. Cyclic stick | – Neutral position; observe mast moment limits |
| 7. Single engine emergency shutdown | – Perform |

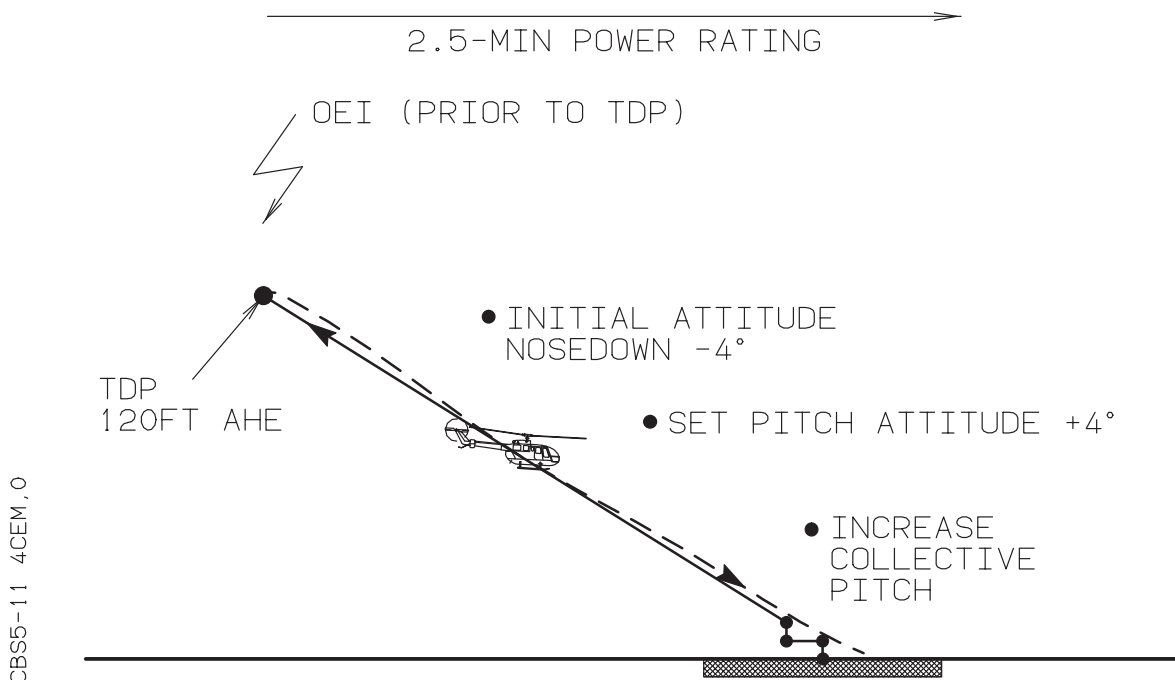


Fig. C2 OEI Rejected Takeoff Profile

C.3.2 OEI DURING TAKEOFF AFTER TDP (see Fig. C3)

Procedure

- | | |
|---------------------|---|
| 1. Attitude | – Nosedown -15° |
| 2. Collective lever | – Adjust to 2.5-min Power |
| 3. Attitude | – Slowly adjust to near level attitude while accelerating to V_{TOSS} |

When 200 ft AHE is reached:

- | | |
|-------------------------------------|---|
| 4. Airspeed | – Accelerate to V_Y |
| 5. Collective lever | – Adjust to Max Continuous Power |
| 6. Climbout | – Continue with V_Y to 1000 ft AHE |
| 7. Single engine emergency shutdown | – Perform |
| 8. Bleed air consumers | – As required, when at or above 1000 ft AHE |

9. LAND AS SOON AS POSSIBLE

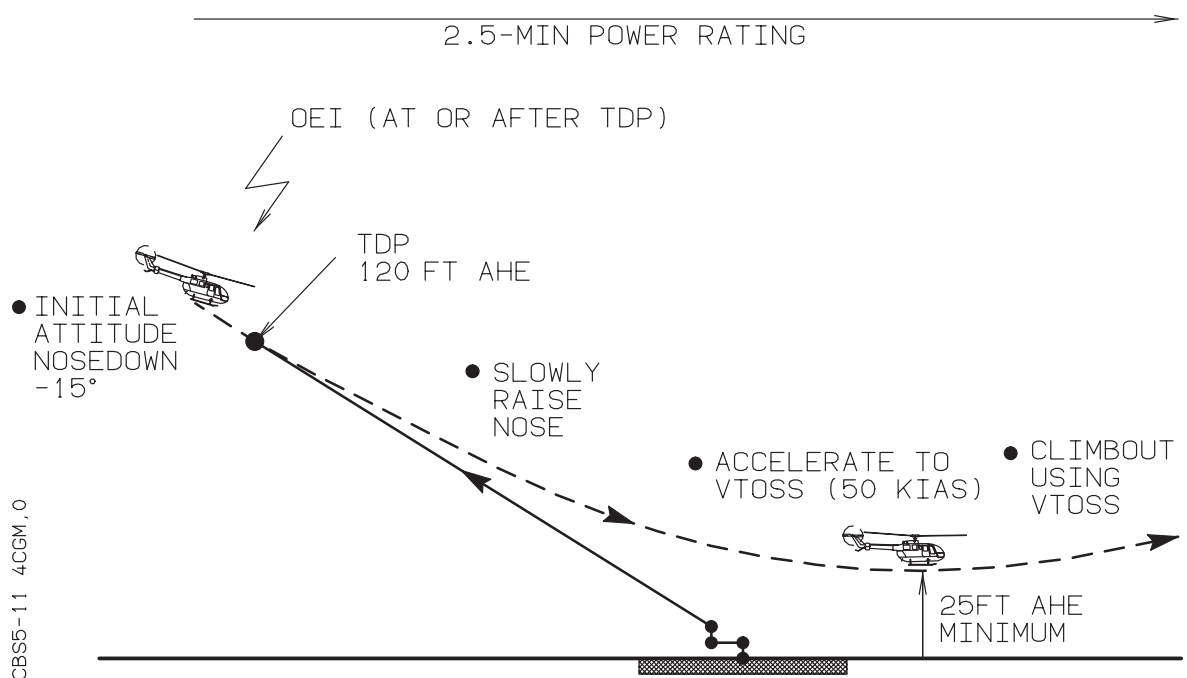


Fig. C3 OEI Continued Takeoff Profile

C.3.3 OEI DURING LANDING PRIOR TO LDP (see Fig. C4)

NOTE The pilot may elect either to go-around or to continue the approach. If the decision is to continue the approach, use procedures for **OEI DURING LANDING AFTER LDP**.

Procedure

1. Collective lever – Adjust to 2.5-min Power
2. Airspeed – V_{TOSS}

NOTE If airspeed is 60 KIAS or above use V_Y .

When 200 ft AHE is reached:

3. Airspeed – V_Y
4. Collective lever – Adjust to Max Continuous Power
5. Climbout – Continue with V_Y to 1000 ft AHE
6. Single engine emergency shutdown – Perform
7. Bleed air consumers – As required, when at or above 1000 ft AHE
8. LAND AS SOON AS POSSIBLE

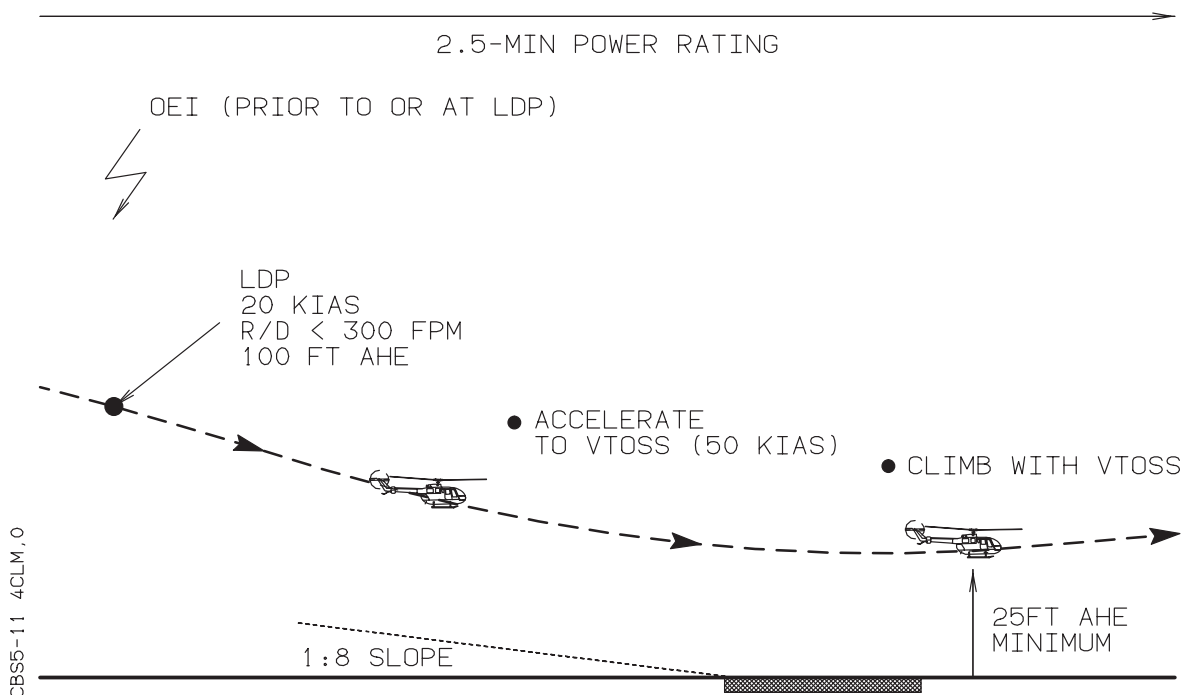


Fig. C4 OEI Go-around Profile

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C.3.4 OEI DURING LANDING AFTER LDP (see Fig. C5)

Procedure

1. Collective lever — Adjust to 2.5-min Power

Before touchdown:

2. Landing attitude — Establish for minimum ground speed
3. Collective lever — Raise to cushion landing

NOTE Plan for zero ground speed touchdown.

After touchdown:

4. Collective lever — Lower to full down
5. Cyclic stick — Neutral position; observe mast moment limits
6. Single engine emergency shutdown — Perform

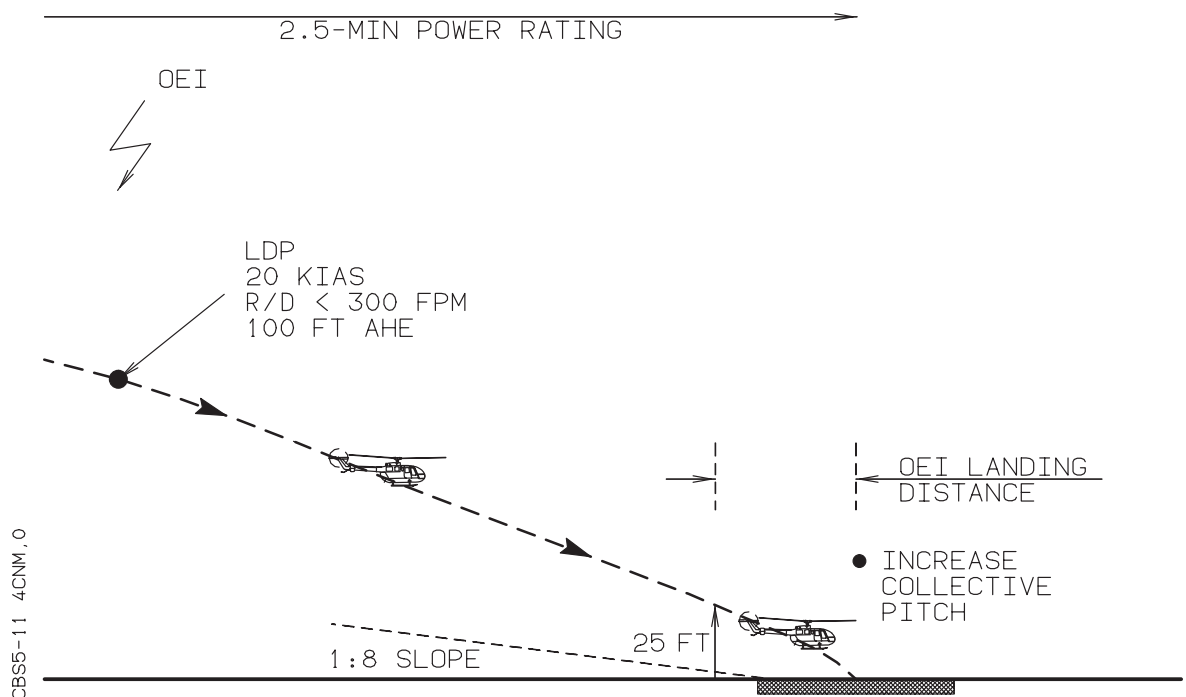


Fig. C5 OEI Continued Landing Profile

C.4 NORMAL PROCEDURES

C.4.1 HELIPAD VERTICAL TAKEOFF (see Fig. C6 and C7)

- | | |
|---------------------|---|
| Takeoff flight path | – Select as nearly into wind as obstacles will permit |
|---------------------|---|

NOTE The helipad maneuvering area and takeoff flight path shall be clear of obstacles.

- | | |
|--|---|
| Pre-takeoff check | – Perform |
| Altimeter | – Set |
| Bleed air consumers other than engine anti-icing | – Off |
| Hover | – 3-ft skid height above helipad center |
| Rotor RPM | – 100 % set with matched torque |
| All instruments | – Normal operating ranges, note torque indication |
| Rearward hover | – Hover aft approx 3 m |
| Slow vertical climb | – Establish until reaching 10 ft AHE |
| Rearward climb | – Initiate (use hover power plus 10-15 % to maintain 300 fpm rate of climb); maintain takeoff area in sight |
| When TDP is reached: | |
| Collective lever | – Adjust to Takeoff Power |
| Airspeed | – Accelerate to V_Y |
| Collective lever | – Adjust to Max Continuous Power |
| Bleed air consumers | – As required, when at or above 1000 ft AHE |
| Climbout | – Continue to desired altitude |

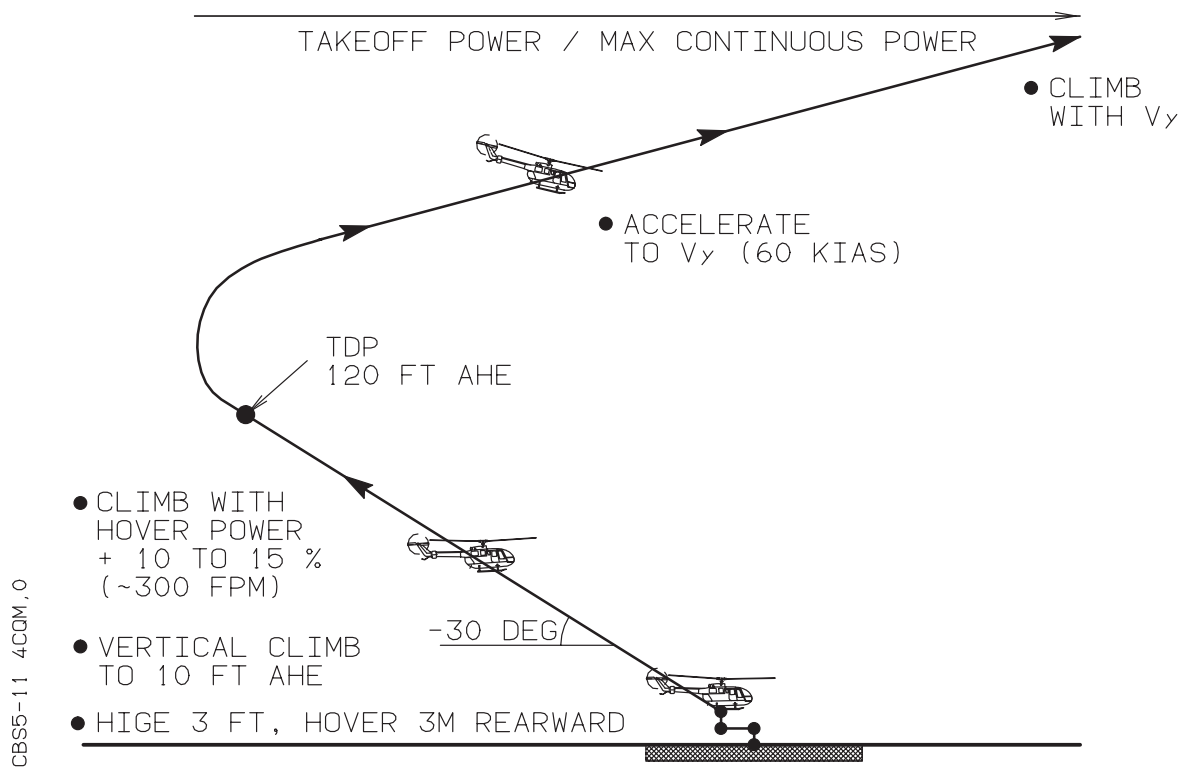
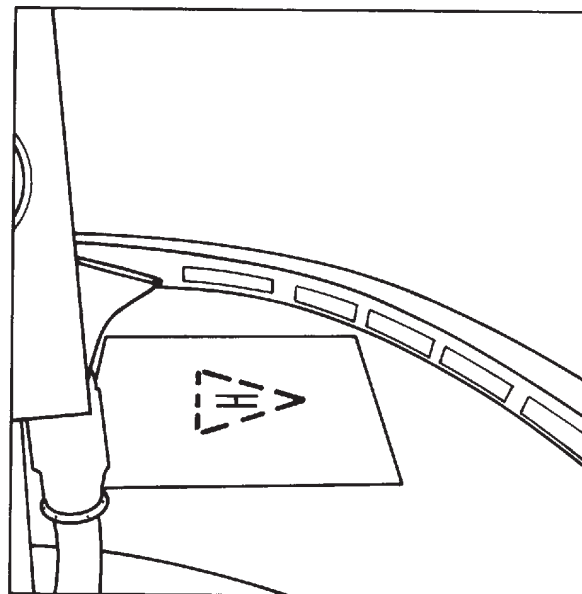


Fig. C6 Vertical Takeoff Profile



BO5-11.4AQM,0

Fig. C7 Takeoff Decision Point Sight Picture

C.4.2 HELIPAD VERTICAL LANDING (see Fig. C8 and C9)

- | | |
|------------------------------------|--|
| Prelanding check | – Perform |
| Anti-icing and bleed air consumers | – Off |
| Landing approach | – Initiate with 60 KIAS and 300 ft AHE, plan to arrive at: <ul style="list-style-type: none"> – <u>200 ft AHE</u> with 40 KIAS and R/D < 300 fpm – <u>150 ft AHE</u> with 30 KIAS and R/D < 300 fpm – <u>LDP (100 ft AHE)</u> with 20 KIAS and R/D < 300 fpm |

NOTE Approach airspeeds can be increased by half of the wind speed.

After passing LDP:

- | | |
|-----------------------|---|
| Speed | – Decrease slowly to arrive at a 3-ft hover above landing point |
| Slow vertical descent | – Initiate to touchdown |

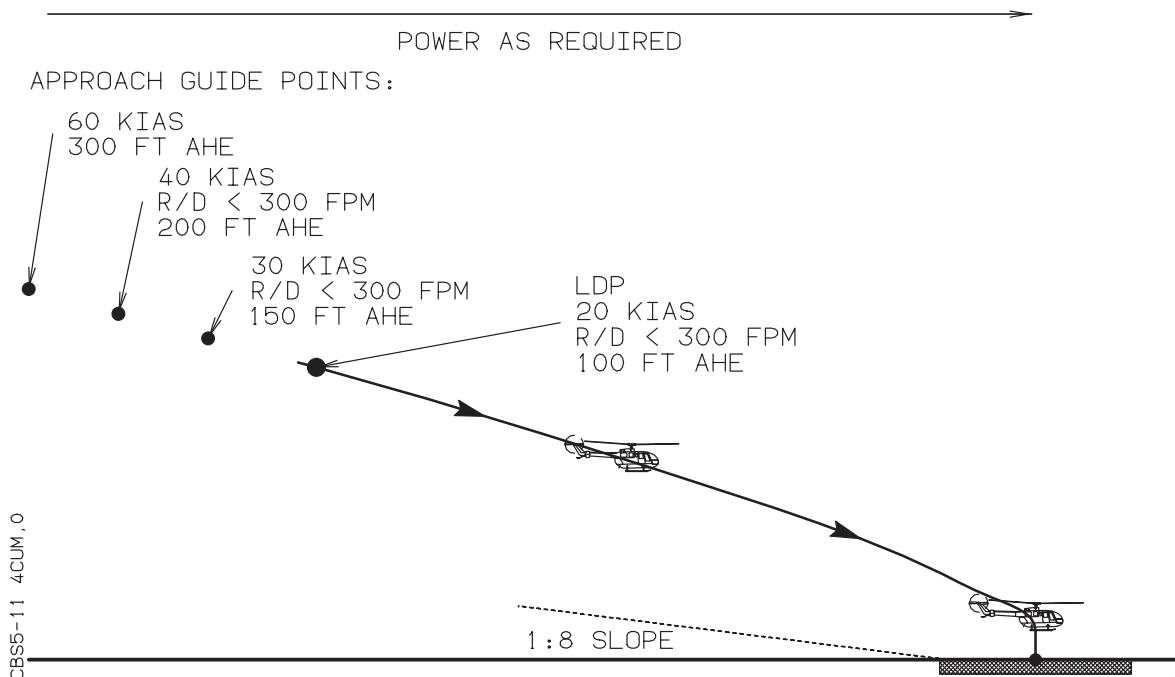


Fig. C8 Vertical Landing Profile

BO5-11.4AUM,0

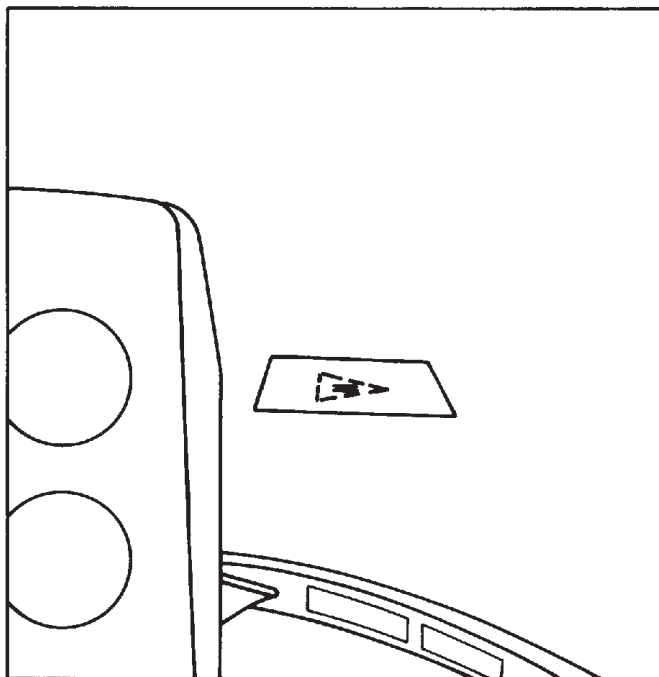


Fig. C9 Landing Decision Point Sight Picture

C.5 PERFORMANCE DATA

C.5.1 TAKEOFF PERFORMANCE

NOTE Presented performance data apply to climbs without any significant bank angle. Turning during climbing will reduce climb performance.

C.5.1.1 Maximum Takeoff Gross Mass

The maximum takeoff gross mass is determined using Fig. C1.

C.5.1.2 OEI Takeoff Distance (see Table C1)

Distance for standard 120 ft TDP height 250 m

C.5.1.3 Takeoff Flight Path Segments I and II

The OEI standard takeoff flight path (see Fig. C10) begins at the end of the “takeoff distance”, at 35 feet above the takeoff surface, with a positive rate of climb and V_{TOSS} , and is divided into two segments:

Segment I OEI climb through segment I has to be accomplished with V_{TOSS} and 2.5-min Power until reaching 200 ft AGL. Segment I climb performance data (OEI Height Gain Over a Horizontal Distance of 100 ft) are presented in Subsection A, Fig. A4.

Segment II OEI climb through segment II has to be accomplished with V_Y and Max Continuous Power until reaching 1000 ft AGL. Segment II climb performance data (OEI Height Gain Over a Horizontal Distance of 100 ft) are presented in Subsection A, Fig. A5.

The horizontal distance required for acceleration from V_{TOSS} to V_Y is 600 m (1980 ft) under calm wind conditions.

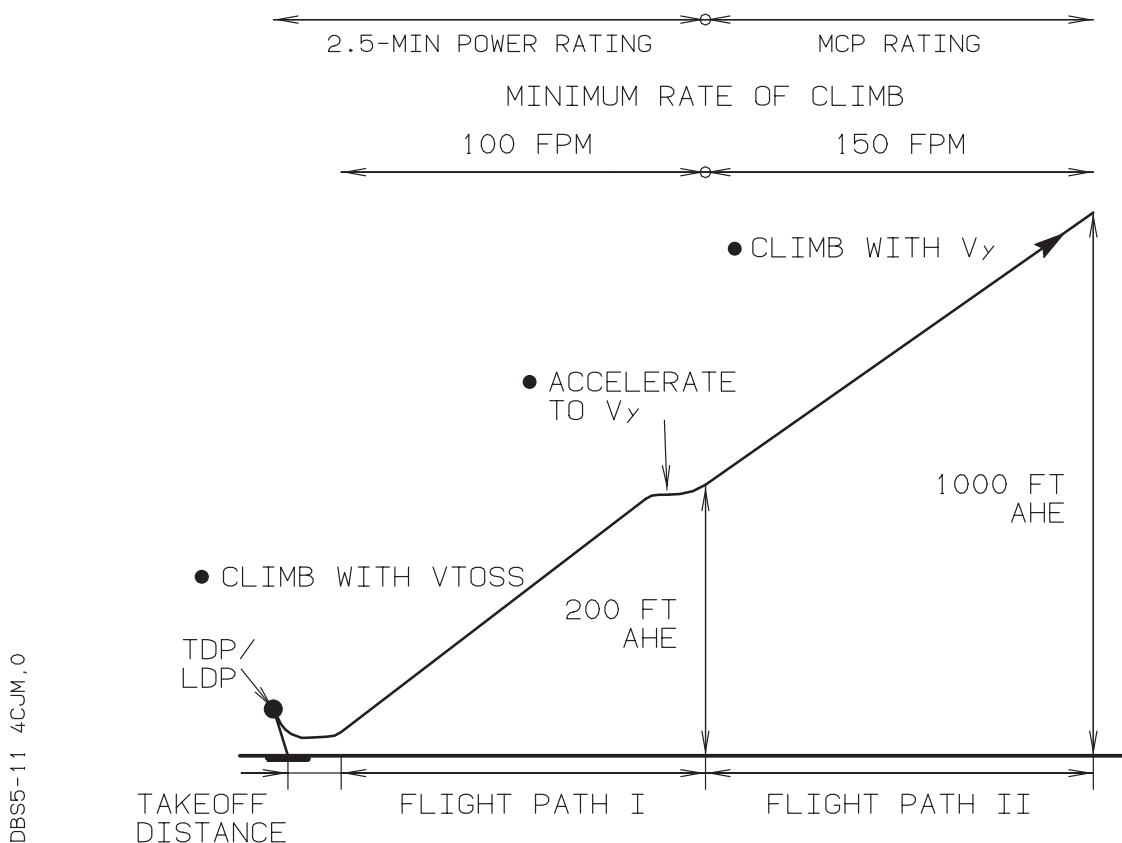


Fig. C10 OEI Takeoff Profile

C.5.2 LANDING PERFORMANCE

C.5.2.1 Maximum Landing Gross Mass

The maximum landing gross mass is determined using Fig. C1.

C.5.2.2 OEI Landing Distance

Distance to clear a 25-ft obstacle 45 m

C.5.3 MODIFIED FLIGHT PATH TO CLEAR HIGH OBSTACLES

C.5.3.1 Modified Takeoff Flight Path

Some heliports may require an increase in the standard TDP height due to obstacles in the close surroundings of the site. For this purpose, the following procedure for varying the TDP standard height is permissible.

Depending on the location of the obstacle relative to the heliport, two different procedures have been established:

C.5.3.1.1 Distance From Heliport To Obstacle is Less Than the Takeoff Distance Required (see Fig. C11)

Establish the TDP height such that the minimum height presented in Table C1 is not lower than the obstacle height plus the minimum clearance as defined by the operational rules.

EXAMPLE: (see Fig. C11 and Table C1)

Determine: **TDP height for vertical takeoff**

Known: Obstacle height 40 ft
Obstacle distance 200 m (in direction of departure)

- Solution:
1. Add minimum clearance as defined by operational rules (e.g. 35 ft) to known obstacle height (40 ft) to obtain minimum height for takeoff (75 ft).
 2. Using Table C1, select for the derived minimum height (75 ft) the corresponding TDP height = 180 ft.

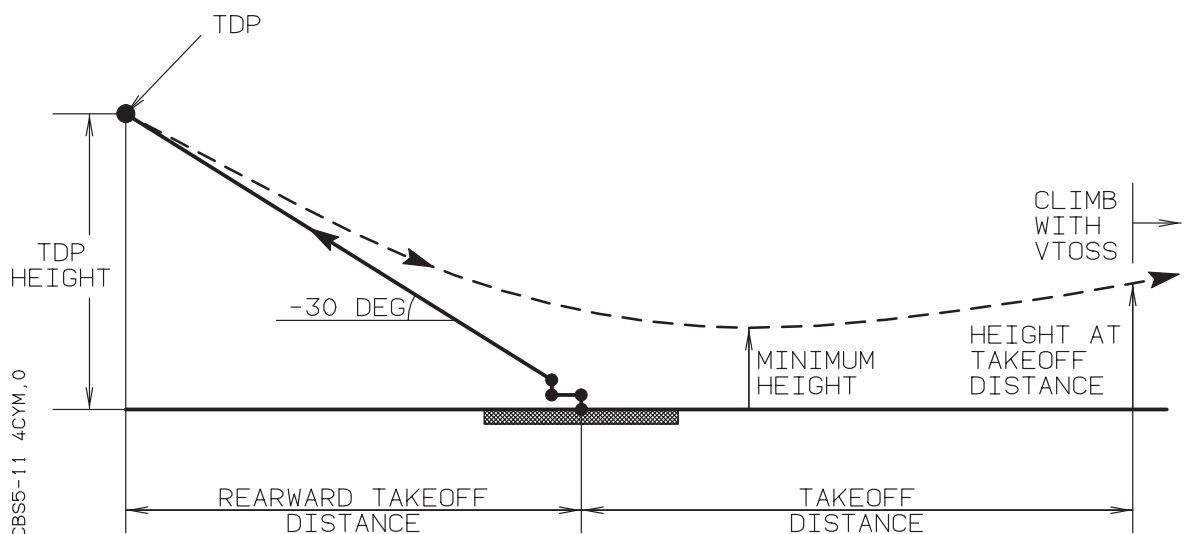


Fig. C11 Modified Flight Path to Clear High Obstacles

TDP Height (ft)	Rearward Take-off Distance (m)	Minimum Height (ft)	Height at Takeoff Distance ¹⁾ (ft)	Maximum Take-off Distance (m)
120	76	25	50	250
140	87	45	70	240
160	98	65	90	230
180	108	85	110	220
200	119	105	130	210
¹⁾ Height at which V_{TOSS} and positive rate of climb are achieved.				

Table C1 Distances and Heights with Variable TDP

C.5.3.1.2 Distance From Heliport To Obstacle is Greater Than the Takeoff Distance Required

First, from the "OEI Height Gain Over a Horizontal Distance of 100 ft" charts in Subsection A, determine the climbout height attainable over the distance from the heliport to the obstacle less the takeoff distance required. The height thus obtained plus the height for the takeoff distance shown in Table C1 is the attainable height above the takeoff surface when the helicopter reaches the obstacle. Select a TDP height such that this attainable height is not lower than the obstacle height plus the minimum clearance as defined by operational rules. Since the takeoff distance required is reduced by an appropriate amount (see Table C1), the calculation must be repeated until an acceptable TDP height is obtained.

EXAMPLE: (see Fig. C11 and Table C1)

Determine: **TDP height for Vertical Takeoff**

Known: Pressure Altitude SL
OAT 15 °C
Gross Mass 2200 kg
Obstacle Height: 40 ft
Obstacle Distance: 350 m (in direction of departure)
Takeoff Distance Required 250 m (Table C1)

- Solution:
1. With the known MAT data (SL, 15 °C, 2200 kg), plot the climb performance using the OEI Height Gain - Segment I Chart (see Subsection A, Fig. A4) and calculate for a horizontal distance of 100 m (approx 11.5 ft per 100 ft, i.e. approx 38 ft).
 2. Calculate minimum height required at the obstacle = 82 ft:
Obstacle height 40 ft
Minimum clearance 35 ft
 3. Calculate minimum height required at takeoff distance (82 ft minus 38 ft, i.e. 44 ft).
 4. Using Table C1, round up to the next highest line (Height at Takeoff Distance - 50 ft) giving TDP height of 120 ft.

C.5.3.2 Modified Landing Flight Path

For varying the LDP height in order to clear obstacles when carrying out a go-around, the same procedure may be used accordingly. In this case, the speed should be increased linearly; i. e.:

200 ft/40 KIAS

150 ft/30 KIAS

100 ft/20 KIAS

NOTE Approach airspeeds can be increased by half of the wind speed.

C.6 MASS AND BALANCE

No change in the basic Flight Manual data.

C.7 SYSTEM DESCRIPTION

No change in the basic Flight Manual data.

C.8 HANDLING, SERVICING, AND MAINTENANCE

No change in the basic Flight Manual data.

C.9 OPERATIONAL INFORMATION**C.9.1 TRAINING****C.9.1.1 OEI Simulation with One Engine at IDLE**

To simulate OEI operation with one engine at IDLE, follow the emergency procedures this supplement subsection, however, use the GENERAL DATA, Engine and Transmission Power Training Limitations (see subsection A) and the gross mass value from the TRAINING MAT curve this subsection (see Fig. C12).

EXAMPLE: (see Fig. C12)

Determine: **Training takeoff and landing gross mass, wind calm**

Known:	OAT	15 °C
	Pressure altitude	SL
	Wind	Calm

Solution: 1. Enter chart at known OAT (15 °C).
 2. Move vertically up to known pressure altitude (SL).
 3. Move horizontally right and read gross mass = 1878 kg.

Determine: **Training takeoff and landing gross mass, with headwind**

Known:	OAT	15 °C
	Pressure altitude	SL
	Headwind	10 kt

Solution: 1. Enter chart at known OAT (15 °C).
 2. Move vertically up to known pressure altitude (SL).
 3. Move horizontally right to Base Line and trace further right following parallel to headwind guide lines.
 4. Reenter chart at known headwind (10 kt) and move up to intersect tracing from above.
 5. From point of intersection move horizontally right and read gross mass = 1942 kg.

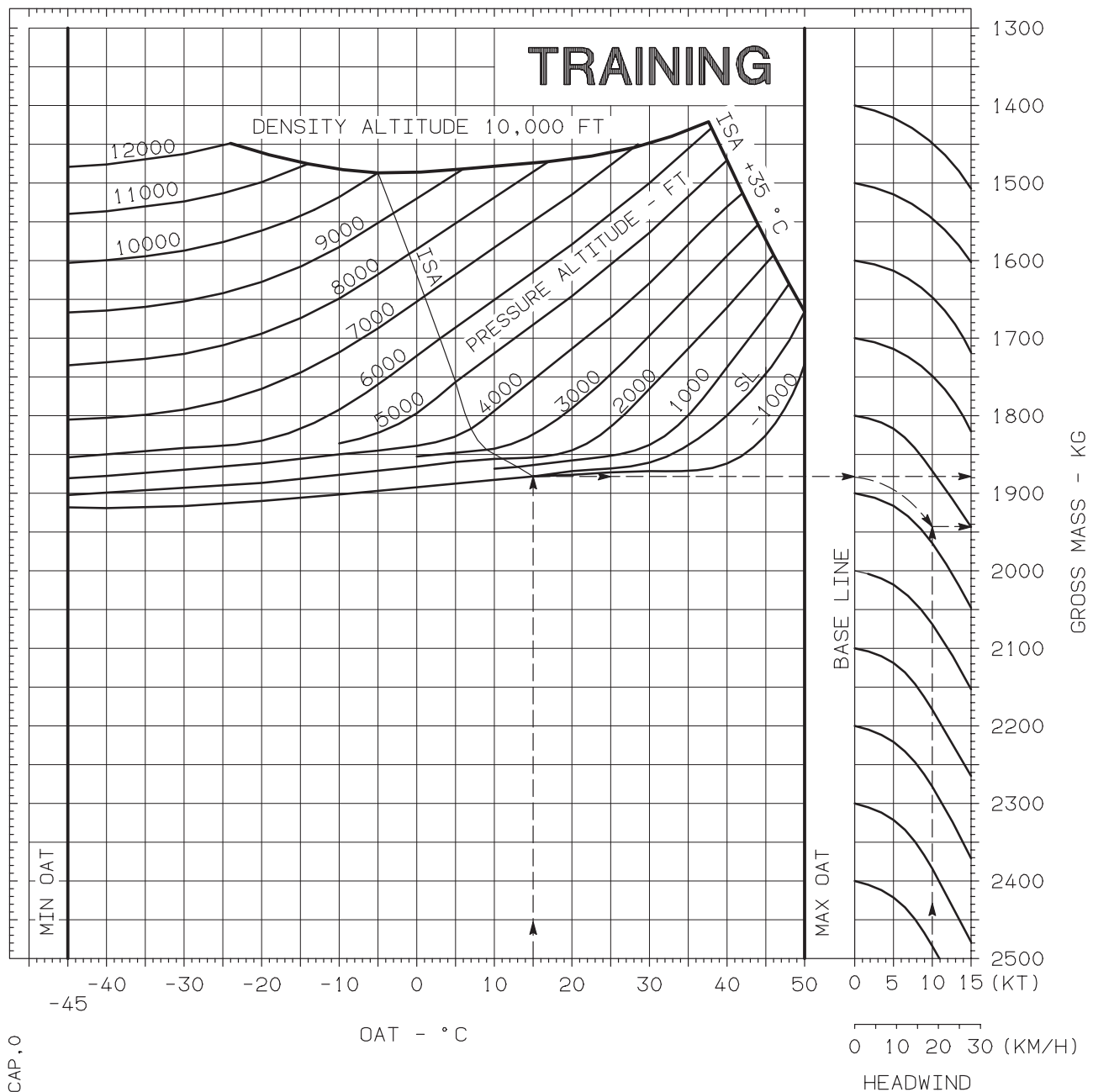
TRAINING TAKEOFF AND LANDING GROSS MASS

1 X ALLISON 250-C20B

OEI – TRAINING RATINGS

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF



NOTE WIND SPEEDS ARE UNFACTORED.
APPLY FACTOR AS REQUIRED BY
OPERATIONAL RULES.

Fig. C12 Training Takeoff and Landing Gross Mass

MANUFACTURER'S DATA

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D. SHORT FIELD

D.1 GENERAL

The information contained herein is intended for use when performing Category A operations at Short Fields.

Short Field operations permit an increase in the maximum Takeoff and Landing Gross Mass by 100 kg over that specified for Elevated/Restricted Helipad operations (see Sub-section C), provided that a flat clear area of at least 50 meters in length (measured from the downwind edge of the helipad or landing area) is available and suitable for a running landing in the event of a Rejected Takeoff.

D.2 LIMITATIONS

D.2.1 CERTIFICATION CRITERIA

The emergency and normal procedures apply to:

- Short Fields having dimensions of at least 50 x 15 meters under day and night conditions.

D.2.2 OPERATIONAL LIMITATIONS

D.2.2.1 Wind Limitations

Takeoff and landing with a tailwind component is prohibited.

Max sidewind component (unfactored) 17 KTAS

D.2.2.2 Bleed Air Consumers

Engine anti-icing and other bleed air consumers must be off during takeoff and landing.

D.2.2.3 Icing Prevention Measures

Category A operations at Short Fields are prohibited whenever the ambient temperature is below +4 °C and visible moisture is present.

D.2.2.4 Altitude/Ambient Air Limitations

The altitude and OAT limitations for vertical takeoff and landing procedures are defined as:

- Max Density Altitude 10,000 ft
- Max OAT ISA +35 °C, but not more than 50 °C
- Min OAT –45 °C

D.2.3 MASS LIMITATIONS

For maximum takeoff and landing gross mass refer to Fig. D1.

EXAMPLE: (see Fig. D1)

Determine: **Max takeoff and landing gross mass, winds calm**

Known:	OAT	15 °C
	Pressure Altitude	SL
	Headwind	Calm

- Solution:
1. Enter chart at known OAT (15 °C).
 2. Move vertically up to known pressure altitude (SL).
 3. Move horizontally right and read max gross mass = 2362 kg.

Determine: **Max takeoff and landing gross mass, with headwind**

Known:	OAT	15 °C
	Pressure Altitude	SL
	Headwind	20 kt
	Factored Headwind	10 kt (based on factoring of 50 %)

NOTE Wind speed must be factored in accordance with operational rules.

- Solution:
1. Enter chart at known OAT (15 °C).
 2. Move vertically up to known pressure altitude (SL).
 3. Move horizontally right to Base Line and trace further right following parallel to headwind guide lines.
 4. Reenter chart at factored headwind (10 kt) and move vertically up to intersect tracing from above.
 5. From point of intersection move horizontally right and read max gross mass = 2445 kg.

MAXIMUM TAKEOFF AND LANDING GROSS MASS

1 X ALLISON 250-C20B

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

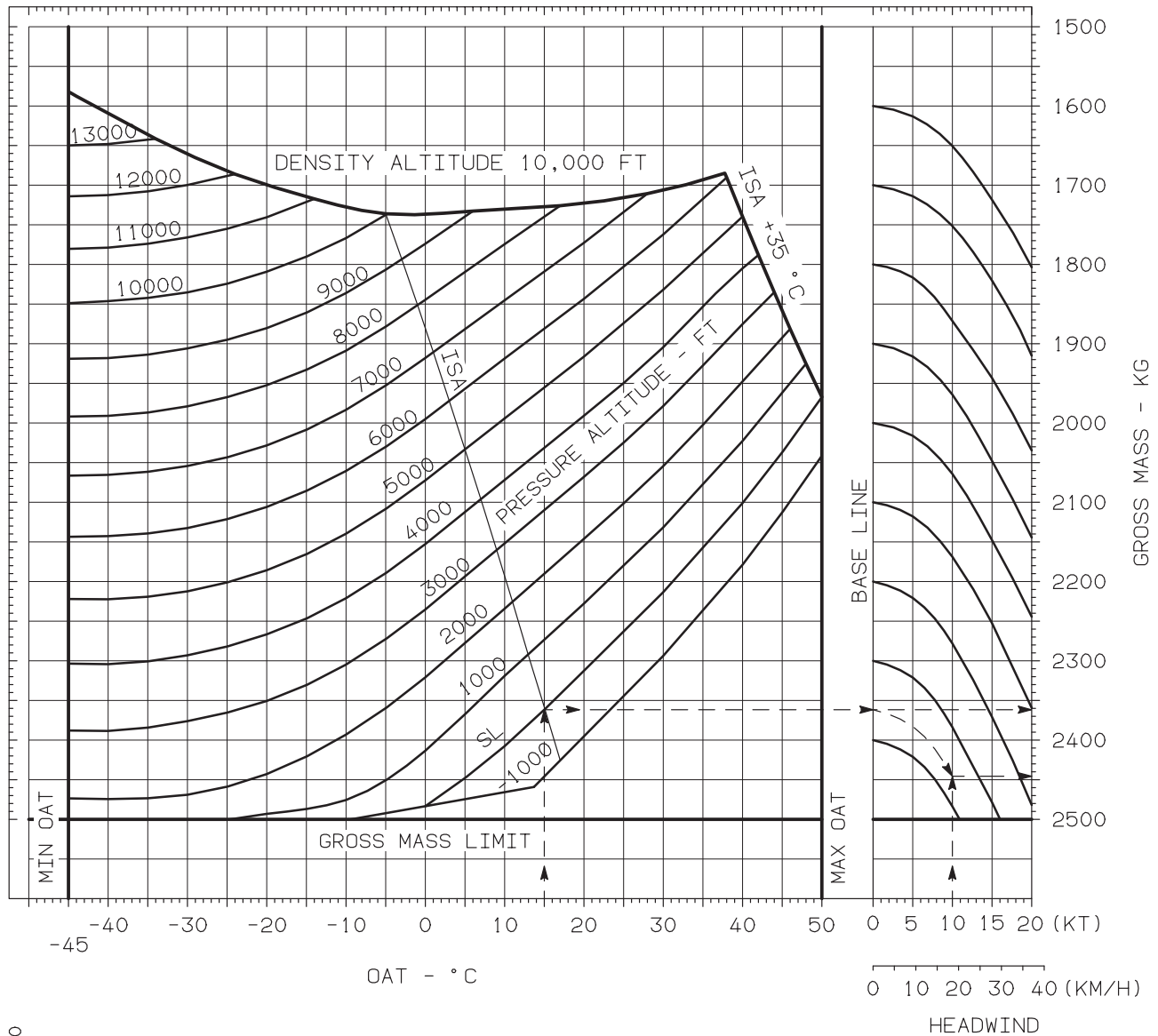


Fig. D1 Maximum Takeoff and Landing Gross Mass

D.3 EMERGENCY AND MALFUNCTION PROCEDURES

D.3.1 OEI DURING TAKEOFF PRIOR TO TDP (see Fig. D2)

Procedure

- | | |
|---------------------|--|
| 1. Attitude | – Nosedown -6° to -10° (below 40 ft) to -10° to -15° (above 100 ft) to increase speed. Maintain starting point in sight. |
| 2. Collective lever | – Adjust to 2.5-min Power |

Before touchdown:

- | | |
|---------------------|----------------------------|
| 3. Landing attitude | – Adjust to slight nose-up |
| 4. Collective lever | – Raise to cushion landing |

NOTE Plan for 10 kt ground speed touchdown.

After touchdown:

- | | |
|-------------------------------------|--|
| 5. Collective lever | – Lower to full down |
| 6. Cyclic stick | – Neutral position; observe mast moment limits |
| 7. Single engine emergency shutdown | – Perform |

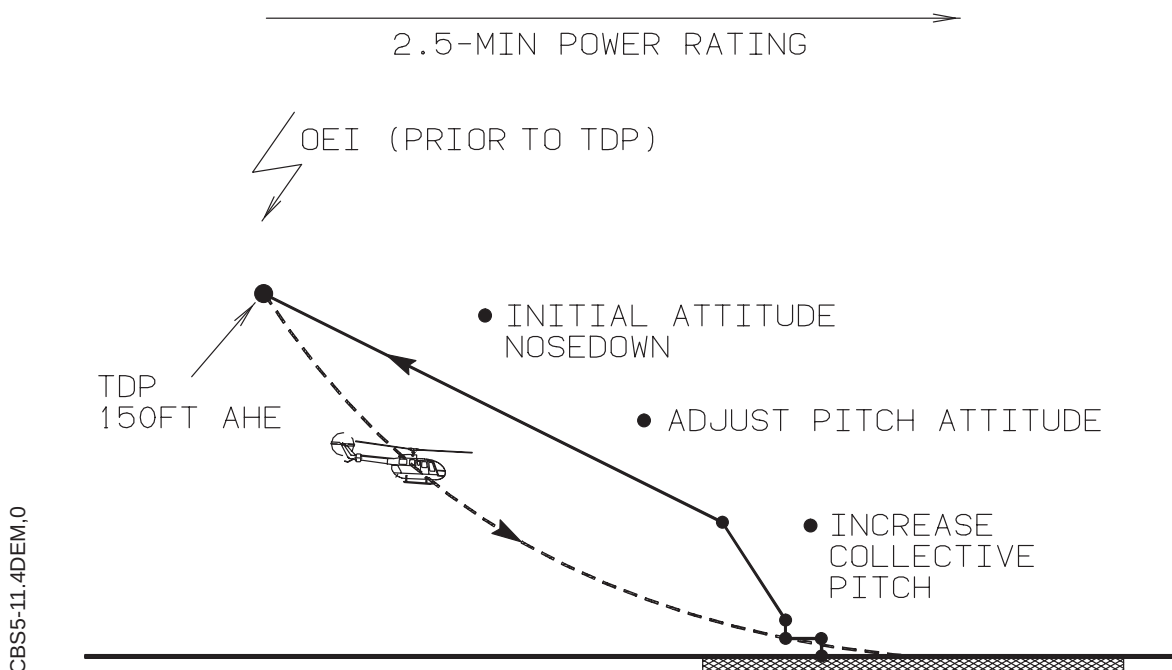


Fig. D2 OEI Rejected Takeoff Profile

D.3.2 OEI DURING TAKEOFF AFTER TDP (see Fig. D3)

Procedure

- | | |
|---------------------|---|
| 1. Attitude | – Nosedown -15° to -20° |
| 2. Collective lever | – Adjust to 2.5-min Power |
| 3. Attitude | – Slowly adjust to near level attitude while accelerating to V_{TOSS} |

When 200 ft AHE is reached:

- | | |
|-------------------------------------|--------------------------------------|
| 4. Airspeed | – Accelerate to V_Y |
| 5. Collective lever | – Adjust to Max Continuous Power |
| 6. Bleed air consumers | – As required |
| 7. Climbout | – Continue with V_Y to 1000 ft AHE |
| 8. Single engine emergency shutdown | – Perform |
| 9. LAND AS SOON AS POSSIBLE | |

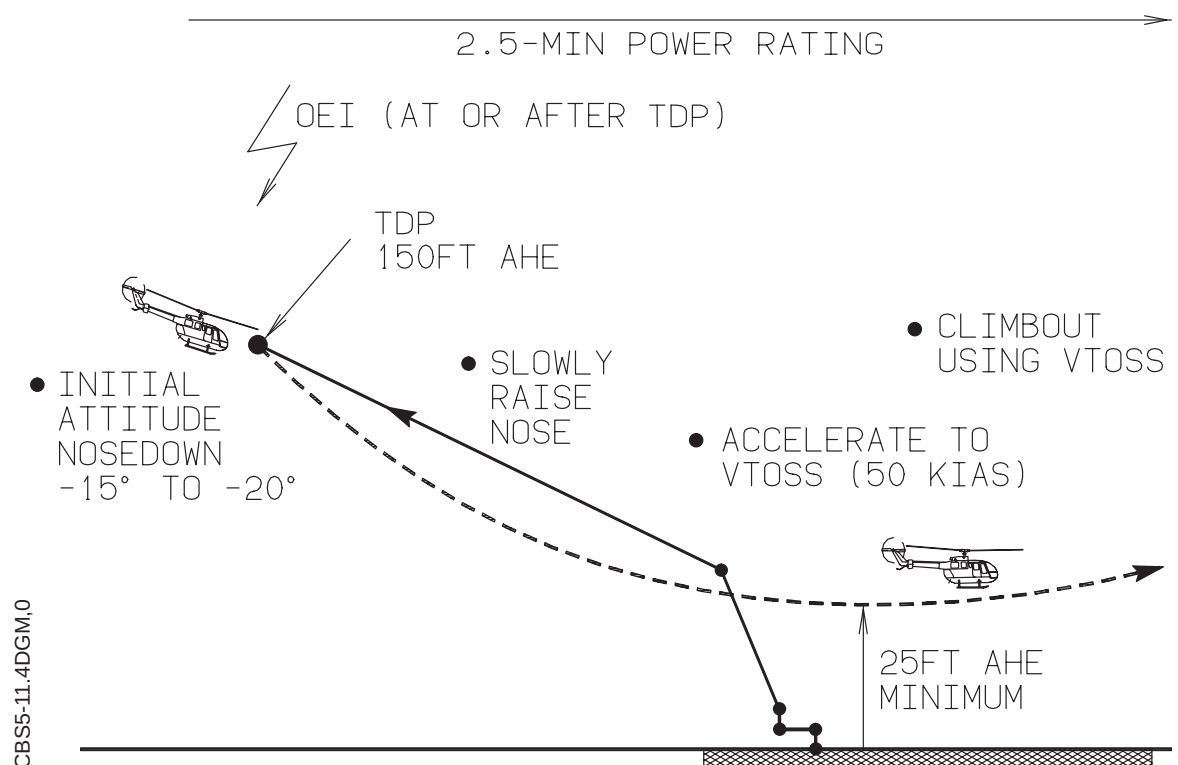


Fig. D3 OEI Continued Takeoff Profile

D.3.3 OEI DURING LANDING PRIOR TO LDP (see Fig. D4)

Procedure

1. Collective lever – Adjust to 2.5-min Power
2. Airspeed – V_{TOSS}

NOTE If airspeed is 60 KIAS or above use V_Y (60 KIAS).

When 200 ft AHE is reached:

3. Airspeed – V_Y
4. Collective lever – Adjust to Max Continuous Power
5. Climbout – Continue with V_Y to 1000 ft AHE
6. Single engine emergency shutdown – Perform
7. Bleed air consumers – As required, when at or above 1000 ft AHE
8. LAND AS SOON AS POSSIBLE

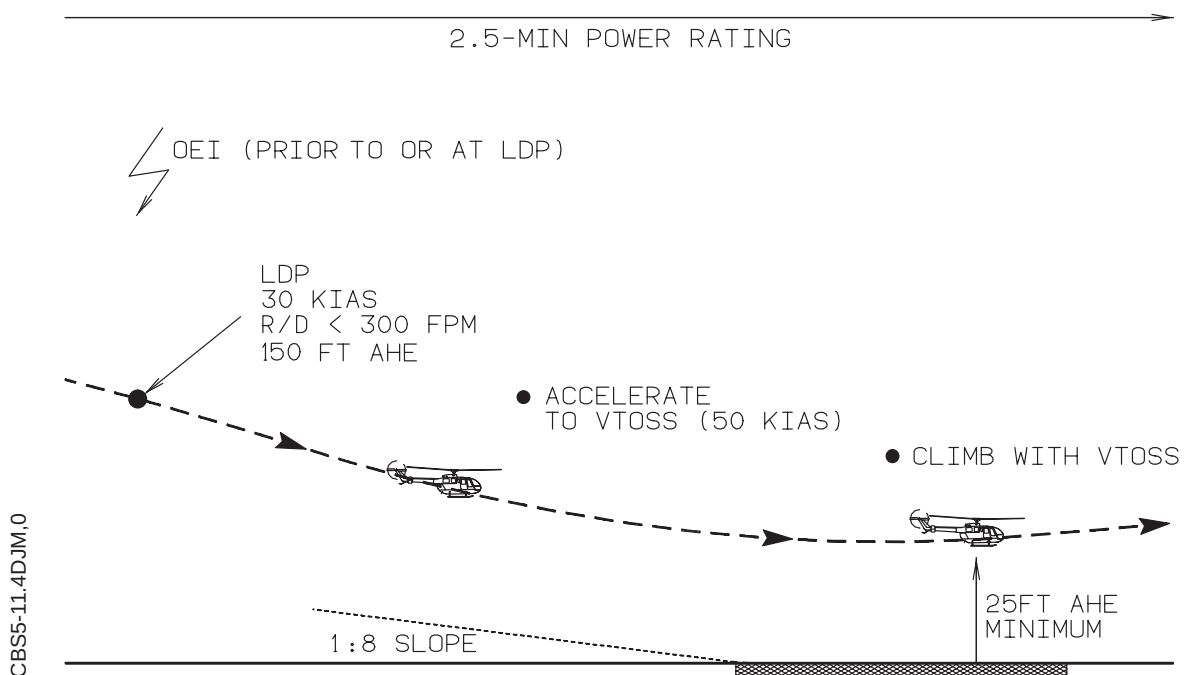


Fig. D4 OEI Go-around Profile

D.3.4 OEI DURING LANDING AFTER LDP (see Fig. D5)

Procedure

1. Collective lever — Adjust to 2.5-min Power

Before touchdown:

2. Landing attitude — Adjust to slight nose-up
3. Collective lever — Raise to cushion landing

NOTE Plan for 10 kt ground speed touchdown.

After touchdown:

4. Collective lever — Lower to full down
5. Cyclic stick — Neutral position; observe mast moment limits
6. Single engine emergency shutdown — Perform

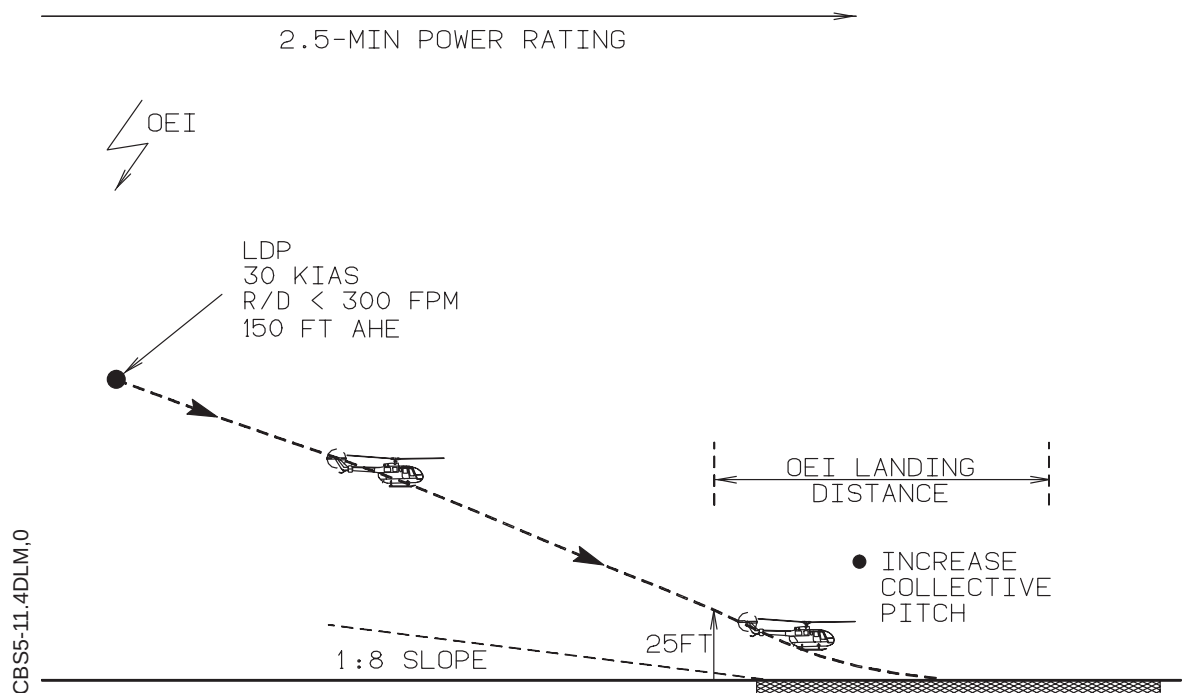


Fig. D5 OEI Continued Landing Profile

D.4 NORMAL PROCEDURES

D.4.1 SHORT FIELD VERTICAL TAKEOFF (see Fig. D6 thru D8)

- | | |
|---------------------|---|
| Takeoff flight path | – Select as nearly into wind as obstacles will permit |
|---------------------|---|

NOTE The helipad maneuvering area and takeoff flight path shall be clear of obstacles.

- | | |
|---|--|
| Pre-takeoff check | – Perform |
| Altimeter | – Set |
| Artificial horizon | – Set horizon bar to zero degrees |
| Engine anti-icing and bleed air consumers | – Off |
| Hover | – 3-ft skid height above aiming point, 15 meters from downwind edge of the take-off area |
| Rotor RPM | – 100 % set with matched torque |
| All instruments | – Normal operating ranges, note torque indication |
| Rearward hover | – Hover aft approx 3 m |
| Slow vertical climb | – Establish until reaching 10 ft AHE |

CAUTION DO NOT EXCEED 86% TORQUE EACH ENGINE.

- | | |
|----------------|--|
| Rearward climb | – Initiate (use hover power plus 10 % to maintain 200-fpm rate of climb). Maintain takeoff area in sight. At 40 ft AHE allow aircraft to drift back so that the aiming point moves up to the windshield, to be just above the frame by 100 ft AHE (see Fig. D7 and D8) |
|----------------|--|

When TDP is reached:

- | | |
|---------------------|---|
| Collective lever | – Adjust to Takeoff Power |
| Airspeed | – Accelerate to V_Y |
| Collective lever | – Adjust to Max Continuous Power |
| Bleed air consumers | – As required, when at or above 1000 ft AHE |
| Climbout | – Continue to desired altitude |

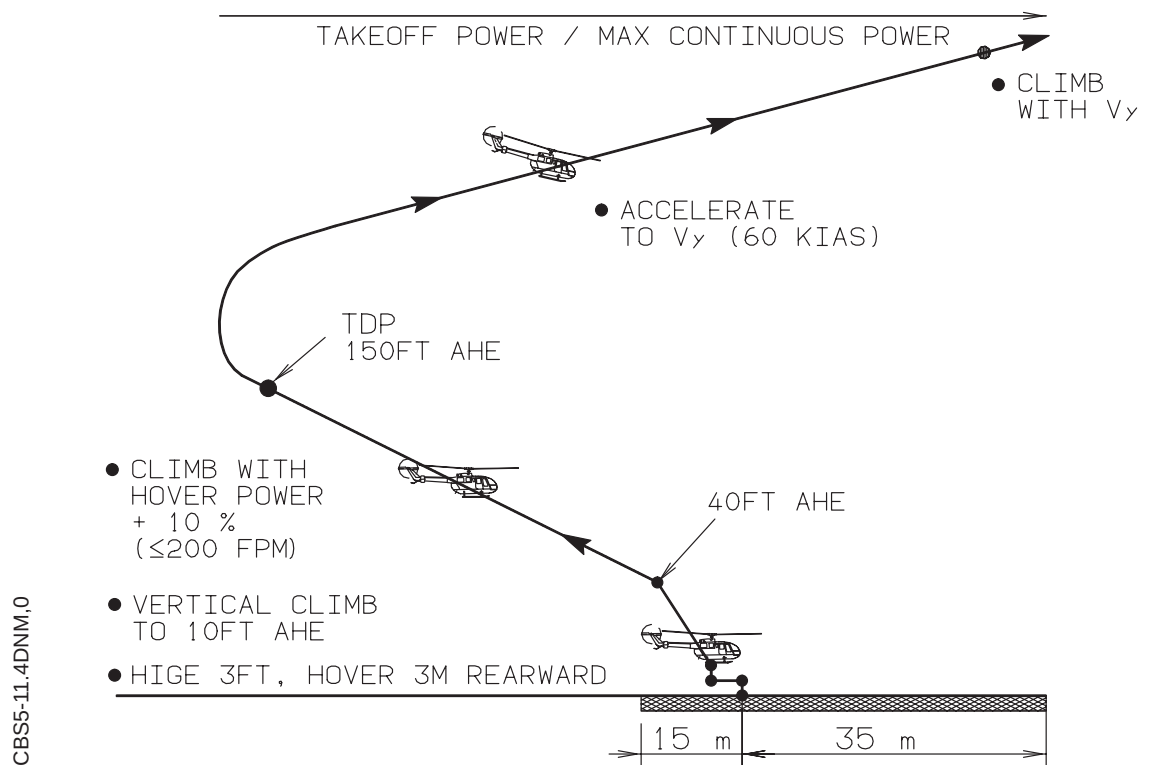


Fig. D6 Vertical Takeoff Profile

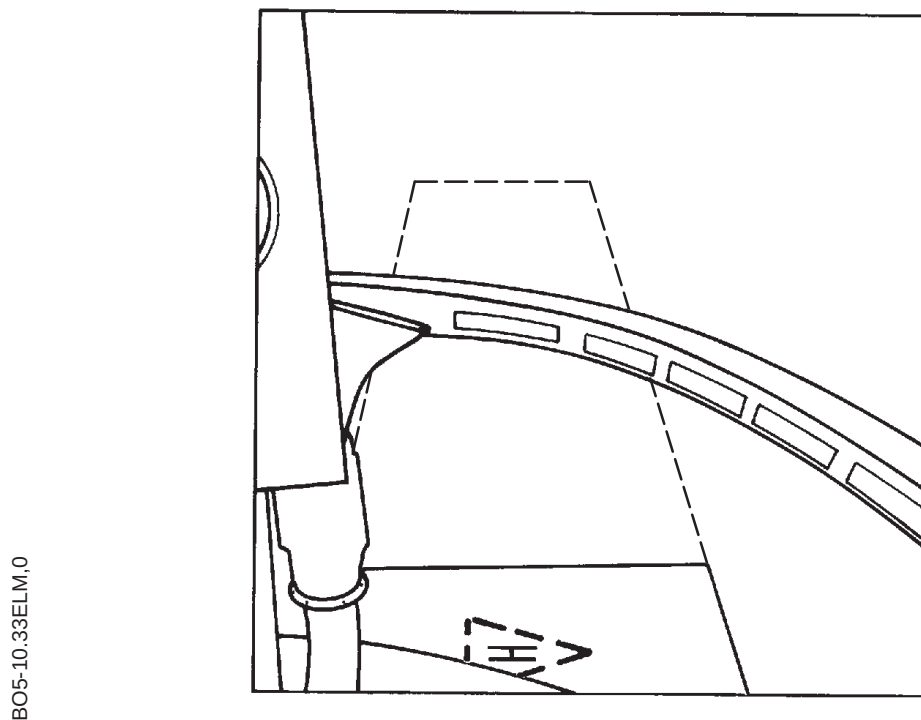


Fig. D7 Sight Picture at 40 ft AHE

BO5-10.33ENM,0

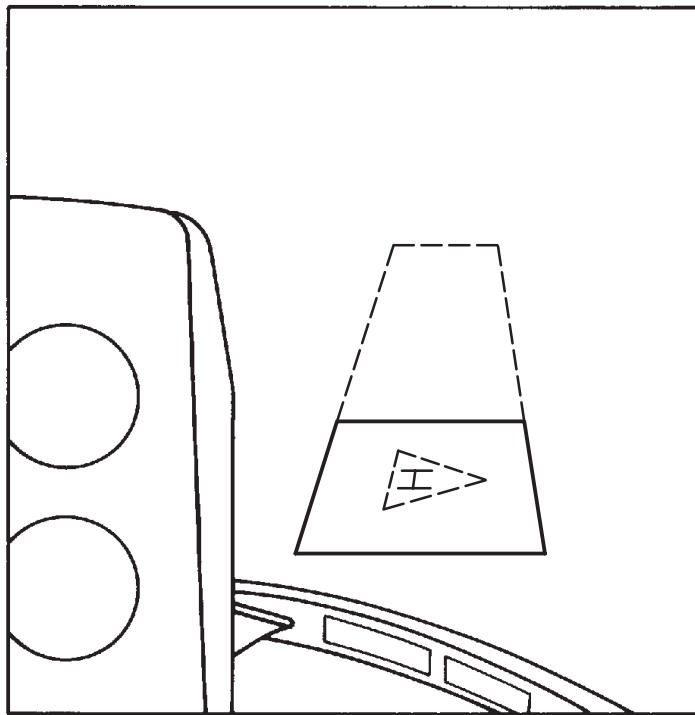


Fig. D8 Sight Picture from 100 ft AHE to Takeoff Decision Point

D.4.2 SHORT FIELD LANDING (see Fig. D9 and D10)

- | | |
|---|--|
| Pre-landing check | – Perform |
| Engine anti-icing and bleed air consumers | – Off |
| Landing approach | – Initiate with 60 KIAS and 300 ft AHE, plan to arrive at: |
| | <u>200 ft AHE</u> with 40 KIAS and R/D < 300 fpm |
| | <u>LDP (150 ft AHE)</u> with 30 KIAS and R/D < 300 fpm |

NOTE Approach airspeeds can be increased by half of the wind speed.

After passing LDP:

- | | |
|------------------|--|
| Speed | – Decrease slowly to arrive at landing point with about 10 kt ground speed |
| Collective lever | – Raise to cushion landing |

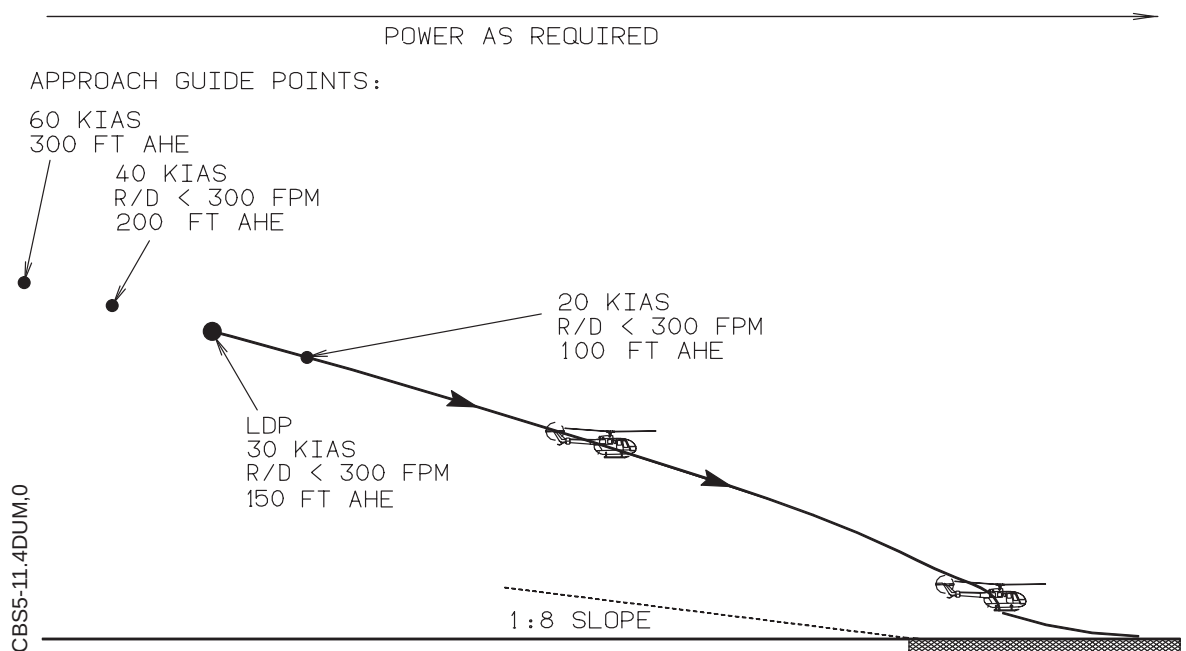


Fig. D9 Landing Profile

BO5-10.33ESM,0

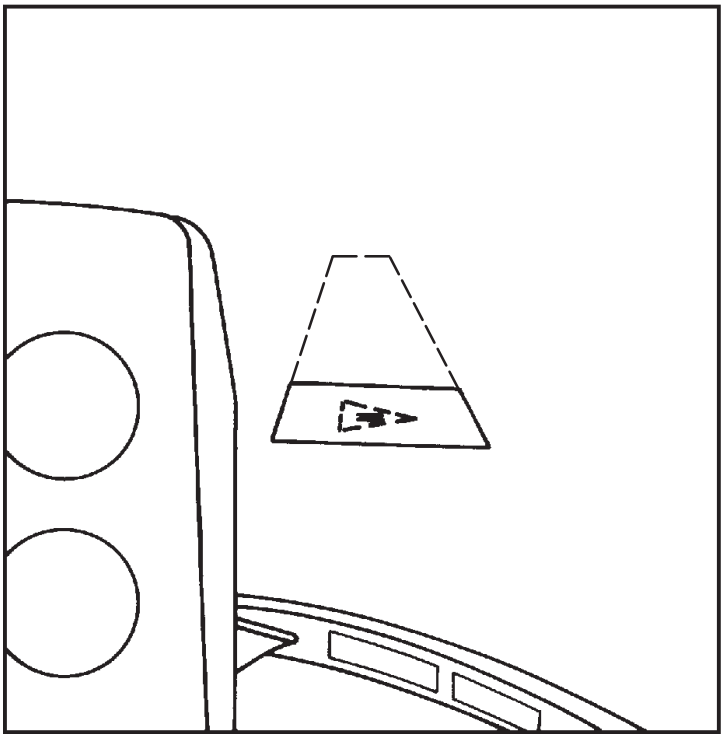


Fig. D10 Landing Decision Point Sight Picture

D.5 PERFORMANCE DATA

D.5.1 TAKEOFF PERFORMANCE

D.5.1.1 Maximum Takeoff Gross Mass

The maximum takeoff gross mass is determined using Fig. D1.

D.5.1.2 Takeoff Flight Paths Segments I and II

Refer to Subsection B, Clear Heliport, Takeoff Performance this supplement.

D.5.2 LANDING PERFORMANCE

D.5.2.1 Maximum Landing Gross Mass

The maximum landing gross mass is determined using Fig. D1.

D.5.2.2 OEI Landing Distance

Distance to clear 25-ft obstacle 75 m

D.5.3 MODIFIED FLIGHT PATH TO CLEAR HIGH OBSTACLES

D.5.3.1 Modified Takeoff Flight Path

Some heliports may require an increase in the standard TDP height due to obstacles in close proximity of the site. For this purpose, the following procedure for varying the TDP standard height is permissible.

Depending on the location of the obstacle relative to the heliport, two different procedures have been established:

D.5.3.1.1 Distance From Heliport To Obstacle is Less Than the Takeoff Distance Required (see Fig. D11)

Establish the TDP height such that the minimum height presented in Table D1 is not lower than the obstacle height plus the minimum clearance as defined by operational rules.

EXAMPLE: (see Fig. D11 and Table D1)

Determine: **TDP height for Short Field takeoff**

Known: Obstacle height 40 ft
 Obstacle distance 200 m (in direction of departure)

- Solution: 1. Add minimum clearance as defined by operational rules (e.g. 35 ft) to known obstacle height (40 ft) to obtain minimum height for takeoff (75 ft).
 2. Using Table D1, select for the derived minimum height (75 ft) the corresponding TDP height = 200 ft.

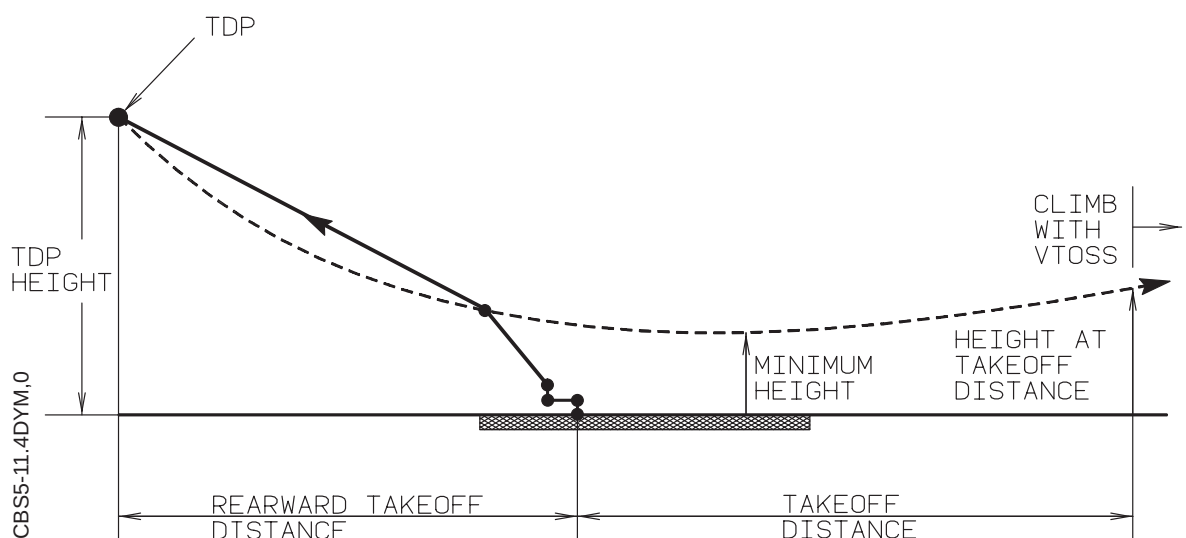


Fig. D11 Modified Flight Path to Clear High Obstacles

TDP Height (ft)	Rearward Take-off Distance (m)	Minimum Height (ft)	Height at Takeoff Distance ¹⁾ (ft)	Maximum Take-off Distance (m)
150	130	25	50	250
160	140	35	60	240
170	150	45	70	230
180	160	55	80	220
190	170	65	90	210
200	180	75	100	200
¹⁾ Height at which V_{TOSS} and positive rate of climb are achieved.				

Table D1 Distances and Heights with Variable TDP

D.5.3.1.2 Distance From Heliport To Obstacle is Greater Than the Takeoff Distance Required

First, from the "OEI Height Gain Over a Horizontal Distance of 100 ft" charts in Subsection A, determine the climbout height attainable over the distance from the heliport to the obstacle less the takeoff distance required. The height thus obtained plus the height for the takeoff distance shown in Table D1 is the attainable height above the takeoff surface when the helicopter reaches the obstacle. Select a TDP height such that this attainable height is not lower than the obstacle height plus the minimum clearance as defined by operational rules. Since the takeoff distance required is reduced by an appropriate amount (see Table D1), the calculation must be repeated until an acceptable TDP height is obtained.

EXAMPLE: (see Fig. D11 and Table D1)

Determine: **TDP height for Short Field Takeoff**

Known: Pressure Altitude SL
OAT 15 °C
Gross Mass 2200 kg
Obstacle Height: 60 ft
Obstacle Distance: 350 m (in direction of departure)
Takeoff Distance Required 250 m (Table D1)

- Solution:
1. With the known MAT data (SL, 15 °C, 2200 kg), plot the climb performance using the OEI Height Gain - Segment I Chart (see Subsection A, Fig. A4) and calculate for a horizontal distance of 100 m (approx 11.5 ft per 100 ft, i.e. approx 38 ft).
 2. Calculate minimum height required at the obstacle = 95 ft:
Obstacle height 60 ft
Minimum clearance 35 ft
 3. Calculate minimum height required at takeoff distance (95 ft minus 38 ft, i.e. 57 ft).
 4. Using Table D1, round up to the next highest line (Height at Takeoff Distance - 60 ft) giving TDP height of 160 ft.

D.5.3.2 Modified Landing Flight Path

For varying the LDP height in order to clear obstacles when carrying out a go-around, the LDP speed must be increased linearly following the guide points shown in Fig. D9, i.e.:

200 ft/40 KIAS

150 ft/30 KIAS

NOTE Approach airspeeds can be increased by half of the wind speed.

D.6 MASS AND BALANCE

No change in the basic Flight Manual data.

D.7 SYSTEM DESCRIPTION

No change in the basic Flight Manual data.

D.8 HANDLING, SERVICING, AND MAINTENANCE

No change in the basic Flight Manual data.

D.9 OPERATIONAL INFORMATION**D.9.1 TRAINING****D.9.1.1 OEI Simulation with One Engine at IDLE**

To simulate OEI operation with one engine at IDLE, follow the emergency procedures this subsection, however, use the GENERAL DATA, Engine and Transmission Power Training Limits (see Subsection A) and the gross mass value from the TRAINING MAT curve, Subsection C.

SECTION 11

SPECIAL OPERATIONS SUPPLEMENTS

LOG OF SUPPLEMENTS

TITLE	FMS NUMBER	APPROVING AUTHORITY/DATE	STD EQPT EFFECTIVITY
Flights with Opened/Removed Doors	11-1	LBA APR 10, 1995 FAA MAR 28, 1996	
IFR Operations	11-2	LBA APR 10, 1995	
Operation in Snow	11-3	LBA APR 10, 1995 FAA MAR 28, 1996	
Equivalent Category A Operations	11-4	LBA FEB 27, 2003 FAA Pending	